

APPENDIX 1

Part 1 of 5

8646332



THE UNITED STATES OF AMERICA

TO ALL TO WHOM THESE PRESENTS SHALL COME:

UNITED STATES DEPARTMENT OF COMMERCE

United States Patent and Trademark Office

October 20, 2025

**THIS IS TO CERTIFY THAT ANNEXED IS A TRUE COPY FROM THE
RECORDS OF THIS OFFICE OF THE FILE WRAPPER AND CONTENTS
OF:**

APPLICATION NUMBER: 16/218,418

FILING DATE: *December 12, 2018*

PATENT NUMBER: 10903734

ISSUE DATE: *January 26, 2021*



Certified by

DIRECTOR OF THE UNITED STATES PATENT AND TRADEMARK OFFICE



December 12, 2018

Attorney Docket No.: 00614-0183001 / 0269-US-1

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Street Address
Fish & Richardson P.C.
One Marina Park Drive
Boston, MA 02210-1878

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P.O. Box 1022
Minneapolis, MN 55440-1022
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Presented for filing is a new continuation-in-part patent application of:

Inventor(s): PATRIZIO VINCIARELLI

Title: DELIVERING POWER TO SEMICONDUCTOR LOADS

Enclosed are the following papers, including those required to receive a filing date under 37 CFR 1.53(b):

	<u>Pages</u>
Specification	24
Claims	11
Abstract	1
Drawing(s)	13

Enclosures:

- Application Data Sheet, 8 pages.
- Non Publication Request.

Basic Filing Fee	\$300
Search Fee	\$660
Examination Fee	\$760
Total Claims 45 over 20 25 x \$100	\$2500
Independent Claims 6 over 3 3 x \$460	\$1380
Fee for Multiple Dependent claims	\$0
Application size fee for each 50 pages over 100	
Total Sheets: 36 x .75 - 100/50 = 0x	\$0
 Total Filing fee	 \$5600



Commissioner for Patents
December 12, 2018
Page 2

Please apply any necessary charges or credits to Deposit Account 06-1050, referencing the attorney docket number shown above.

I hereby authorize the USPTO to communicate in response to e-mail communications from the undersigned and practitioners under Customer No. 26214 in accordance with 37 CFR 1.33 and 37 CFR 1.34 concerning any subject matter of this application via e-mail. I understand that a copy of these communications will be made of record in the application file. (MPEP 502.03).

If this application is considered to be incomplete, or if a telephone conference would otherwise be helpful, call the undersigned at (617) 542-5070.

Send all correspondence to:

26214
Customer Number

Respectfully submitted,

/Rex I. Huang/

Rex I. Huang
Registration No. 57,661

Enclosures

RIH/acr
23843102.doc

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

**NONPUBLICATION REQUEST
UNDER
35 U.S.C. 122(b)(2)(B)(i)**

First Named Inventor

Patrizio Vinciarelli

Title

Delivering Power To Semiconductor Loads

Attorney Docket Number

00614-0183001

I hereby certify that the invention disclosed in the attached application **has not and will not be** the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

I hereby request that the attached application not be published under 35 U.S.C. 122(b).

/Rex I. Huang/

Signature

December 12, 2018

Date

Rex I. Huang

Typed or printed name

57,661

Registration Number, if applicable

(617) 542-5070

Telephone Number

This request must be signed in compliance with 37 CFR 1.33(b) and submitted with the application **upon filing**.

Applicant may rescind this nonpublication request at any time. If applicant rescinds a request that an application not be published under 35 U.S.C. 122(b), the application will be scheduled for publication at eighteen months from the earliest claimed filing date for which a benefit is claimed.

If applicant subsequently files an application directed to the invention disclosed in the attached application in another country, or under a multilateral international agreement, that requires publication of applications eighteen months after filing, the applicant **must** notify the United States Patent and Trademark Office of such filing within forty-five (45) days after the date of the filing of such foreign or international application. **Failure to do so will result in abandonment of this application (35 U.S.C. 122(b)(2)(B)(iii)).**

This collection of information is required by 37 CFR 1.213(a). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 6 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 (1-800-786-9199) and select option 2.



Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00614-0183001
		Application Number	
Title of Invention	Delivering Power To Semiconductor Loads		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--------------------------	---

Inventor Information:

Inventor	1				Remove	
Legal Name						
Prefix	Given Name	Middle Name	Family Name	Suffix		
	Patrizio		Vinciarelli			
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service						
City	Boston	State/Province	MA	Country of Residence	US	
Mailing Address of Inventor:						
Address 1	25 Frontage Road					
Address 2						
City	Andover	State/Province	MA			
Postal Code	01810	Country	US			
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button. Add						

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).	
☐ An Address is being provided for the correspondence information of this application.	
Customer Number	26214
Email Address	Add Email Remove Email

Application Information:

Title of the Invention	Delivering Power To Semiconductor Loads		
Attorney Docket Number	00614-0183001	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	13	Suggested Figure for Publication (if any)	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00614-0183001
		Application Number	
Title of Invention	Delivering Power To Semiconductor Loads		

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

☒ **Request Not to Publish.** I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	26214		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
	Continuation in part of	15616288	2017-06-07

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00614-0183001
		Application Number	
Title of Invention	Delivering Power To Semiconductor Loads		
Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
15616288	Continuation in part of	15091346	2016-04-05
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.			Add

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

			Remove
Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)
Additional Foreign Priority Data may be generated within this form by selecting the Add button.			Add

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00614-0183001
		Application Number	
Title of Invention	Delivering Power To Semiconductor Loads		

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00614-0183001
		Application Number	
Title of Invention	Delivering Power To Semiconductor Loads		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant	1	Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section. Clear		
Assignee	Legal Representative under 35 U.S.C. 117	Joint Inventor
☐ Person to whom the inventor is obligated to assign.		☐ Person who shows sufficient proprietary interest
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:		
Name of the Deceased or Legally Incapacitated Inventor:		
If the Applicant is an Organization check here. ☒		
Organization Name	VLT, Inc.	
Mailing Address Information For Applicant:		
Address 1	25 Frontage Road	
Address 2		
City	Andover	State/Province
Country	US	Postal Code
Phone Number		Fax Number
Email Address		
Additional Applicant Data may be generated within this form by selecting the Add button. Add		

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00614-0183001
		Application Number	
Title of Invention	Delivering Power To Semiconductor Loads		

Assignee	1			
Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.				
				Remove
If the Assignee or Non-Applicant Assignee is an Organization check here.				☐
Prefix	Given Name	Middle Name	Family Name	Suffix
Mailing Address Information For Assignee including Non-Applicant Assignee:				
Address 1				
Address 2				
City		State/Province		
Country ⁱ		Postal Code		
Phone Number		Fax Number		
Email Address				
Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button.				Add

Signature:

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the **INITIAL** filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Rex I. Huang/		Date (YYYY-MM-DD)	2018-12-12
First Name	Rex	Last Name	Huang	Registration Number
				57661
Additional Signature may be generated within this form by selecting the Add button.				Add

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00614-0183001
		Application Number	
Title of Invention	Delivering Power To Semiconductor Loads		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

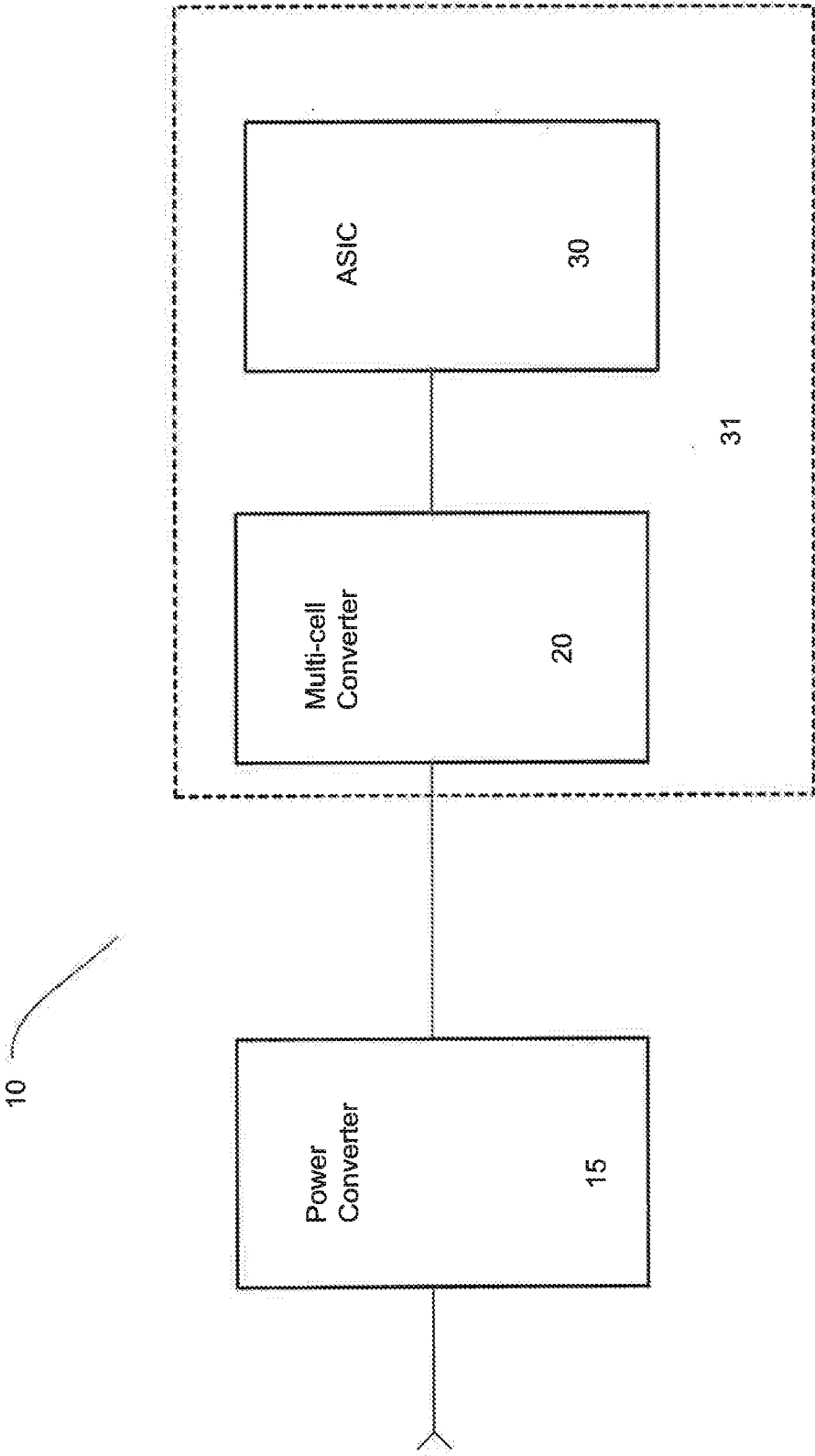
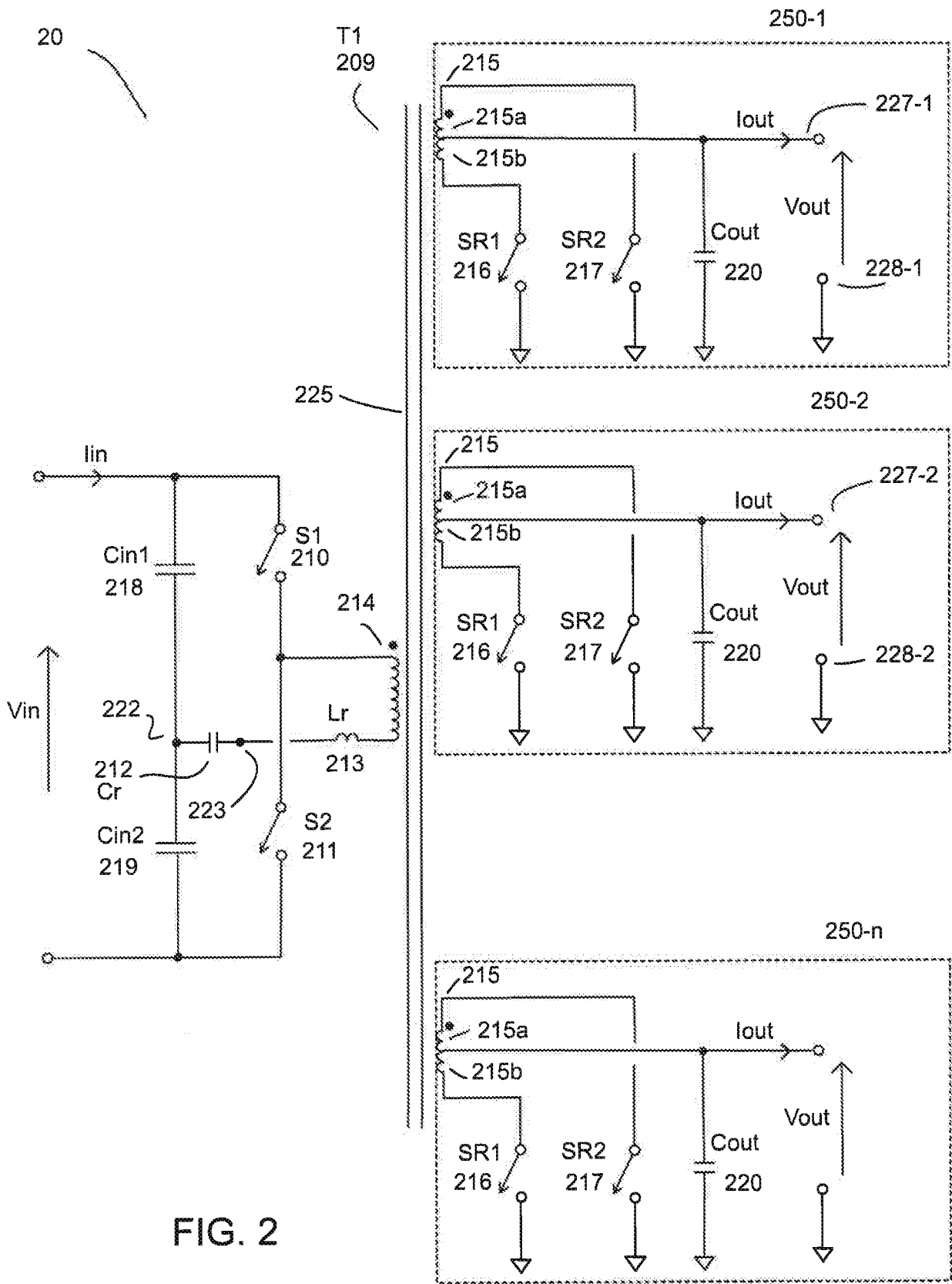


FIG. 1



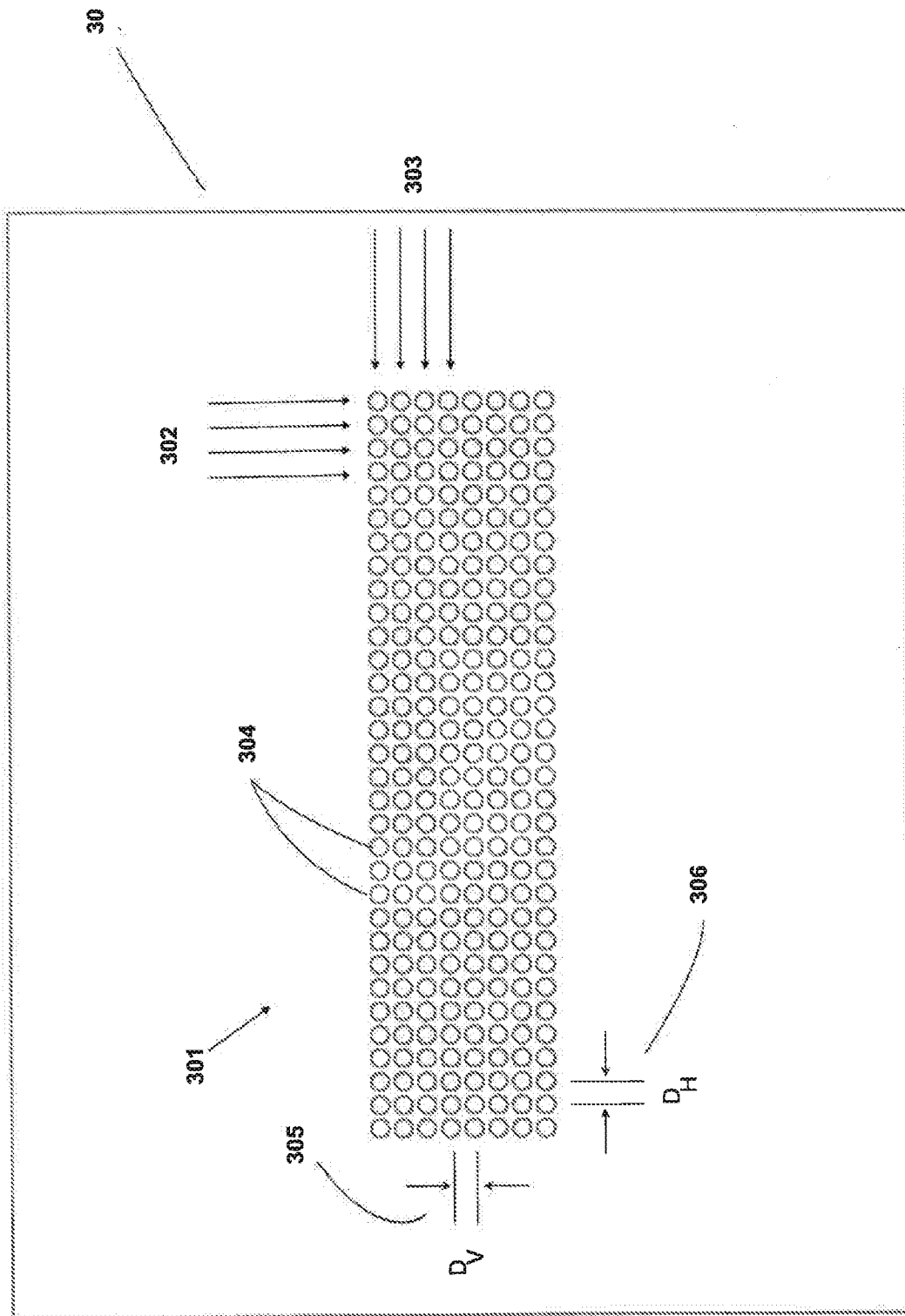
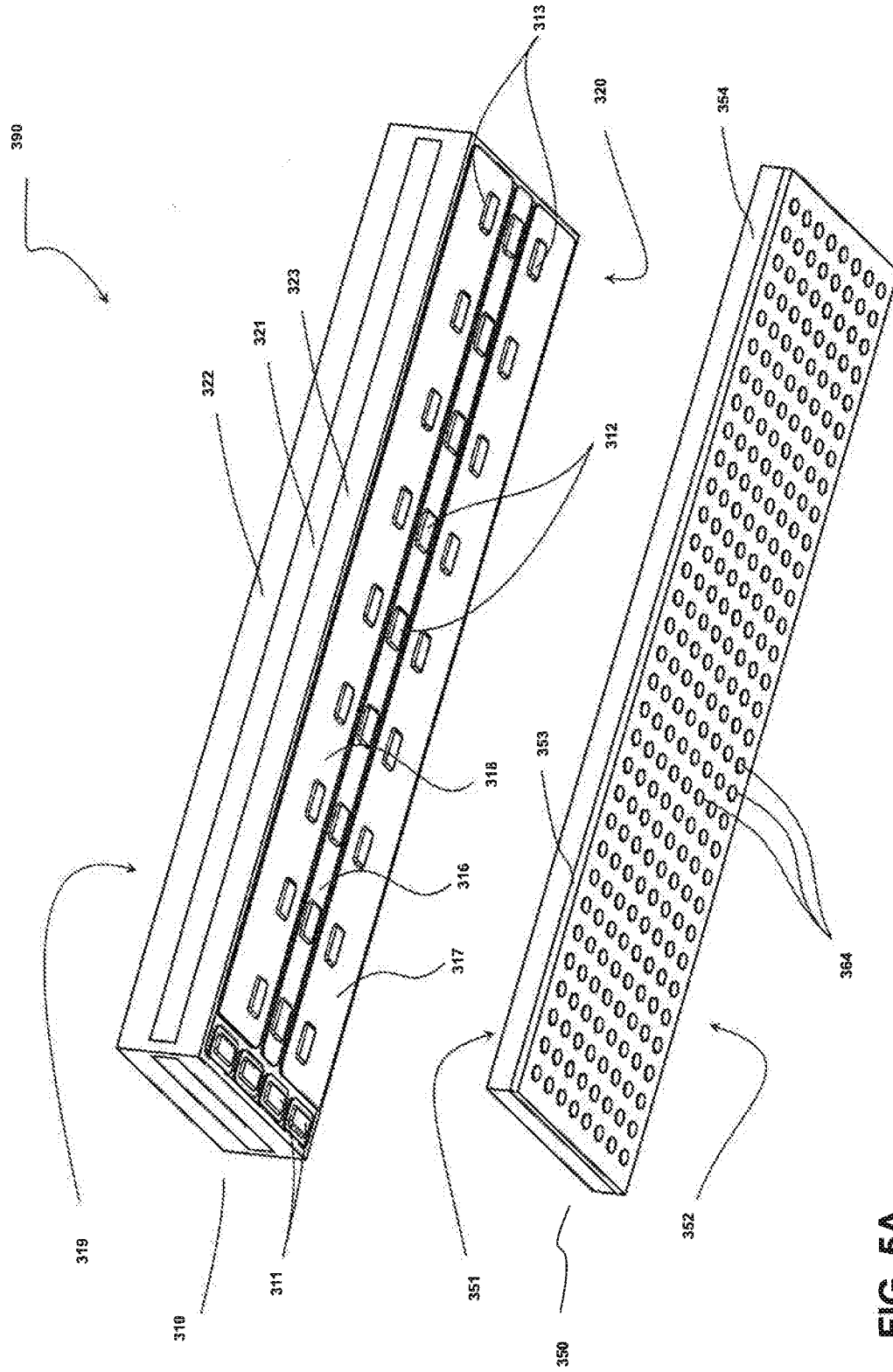


FIG. 4



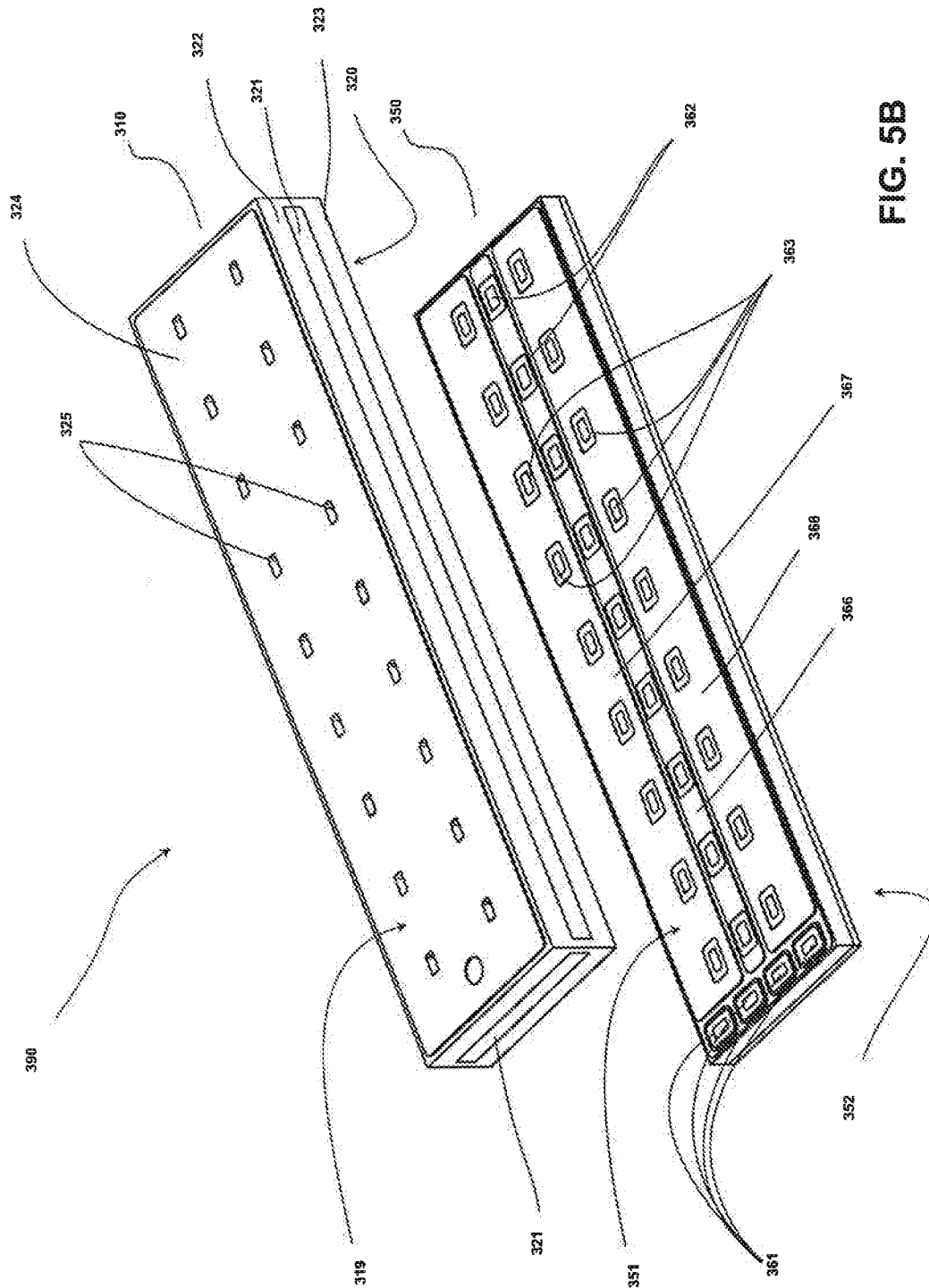


FIG. 5B

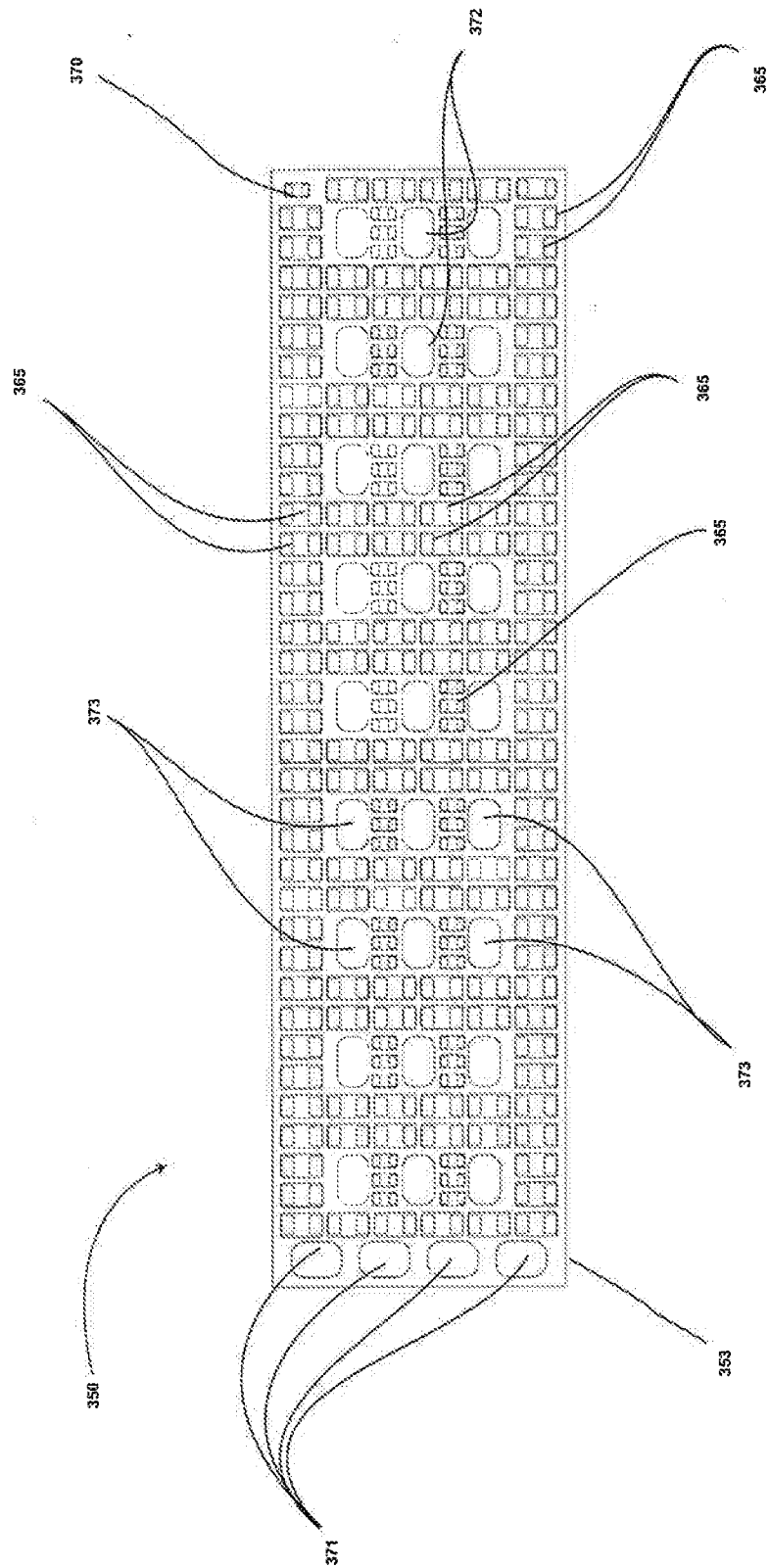
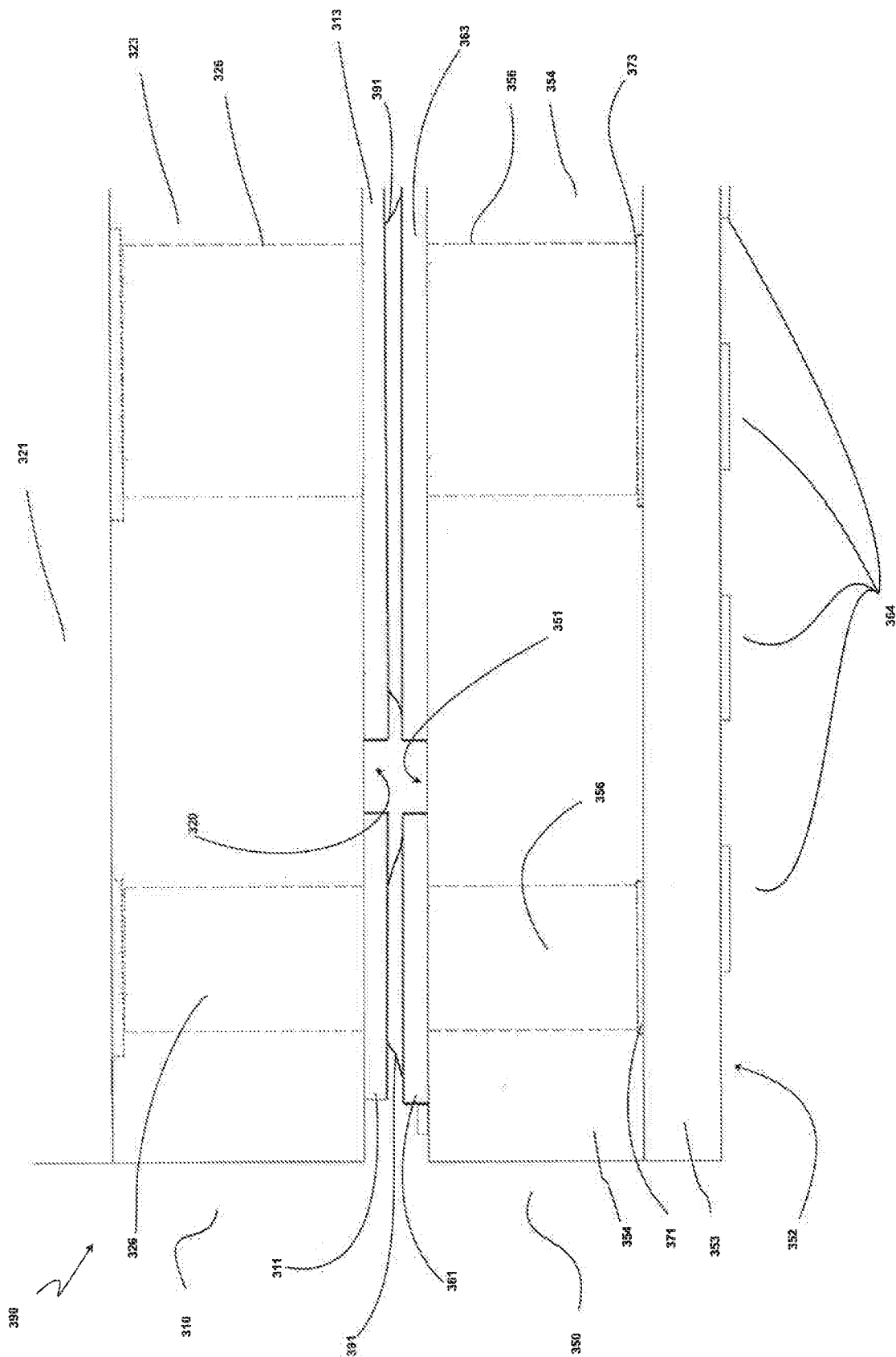


FIG. 6



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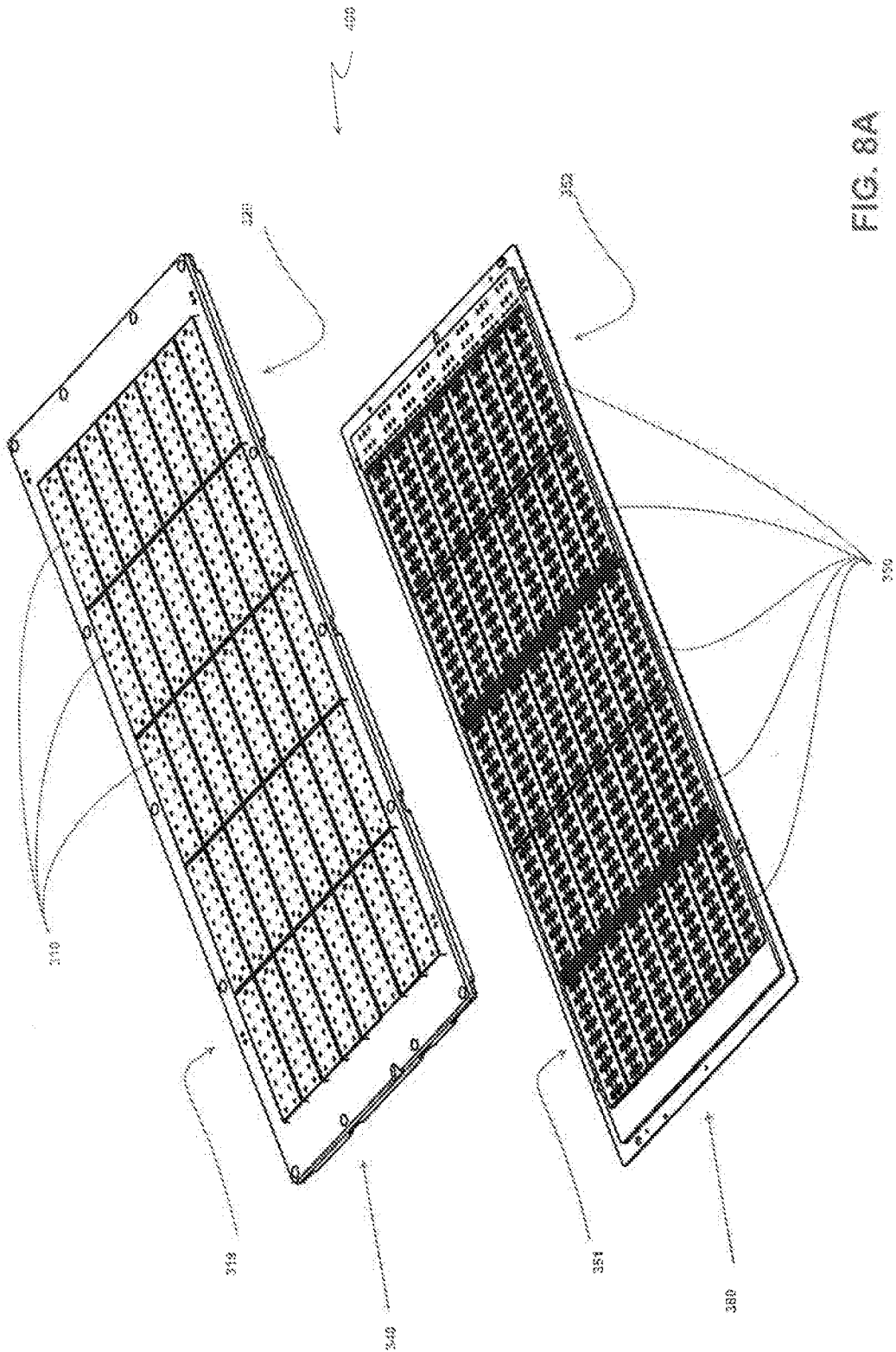


FIG. 8A

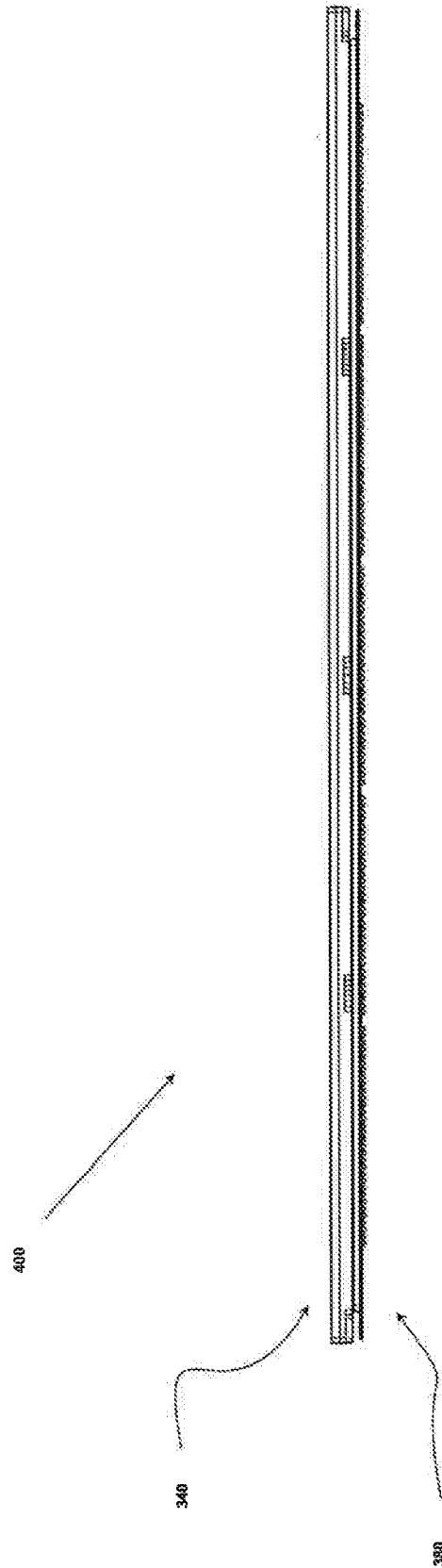


FIG. 8B

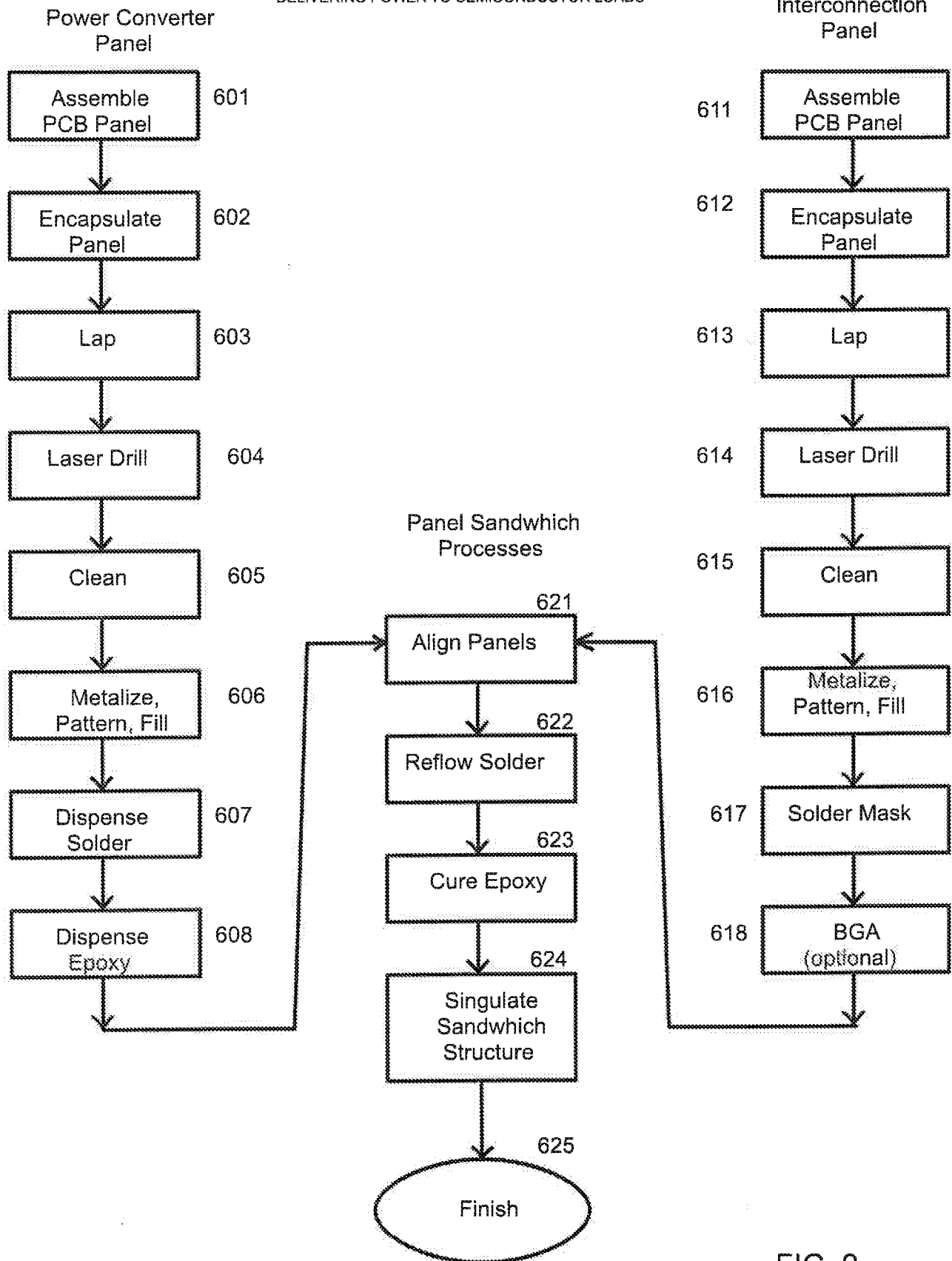


FIG. 9

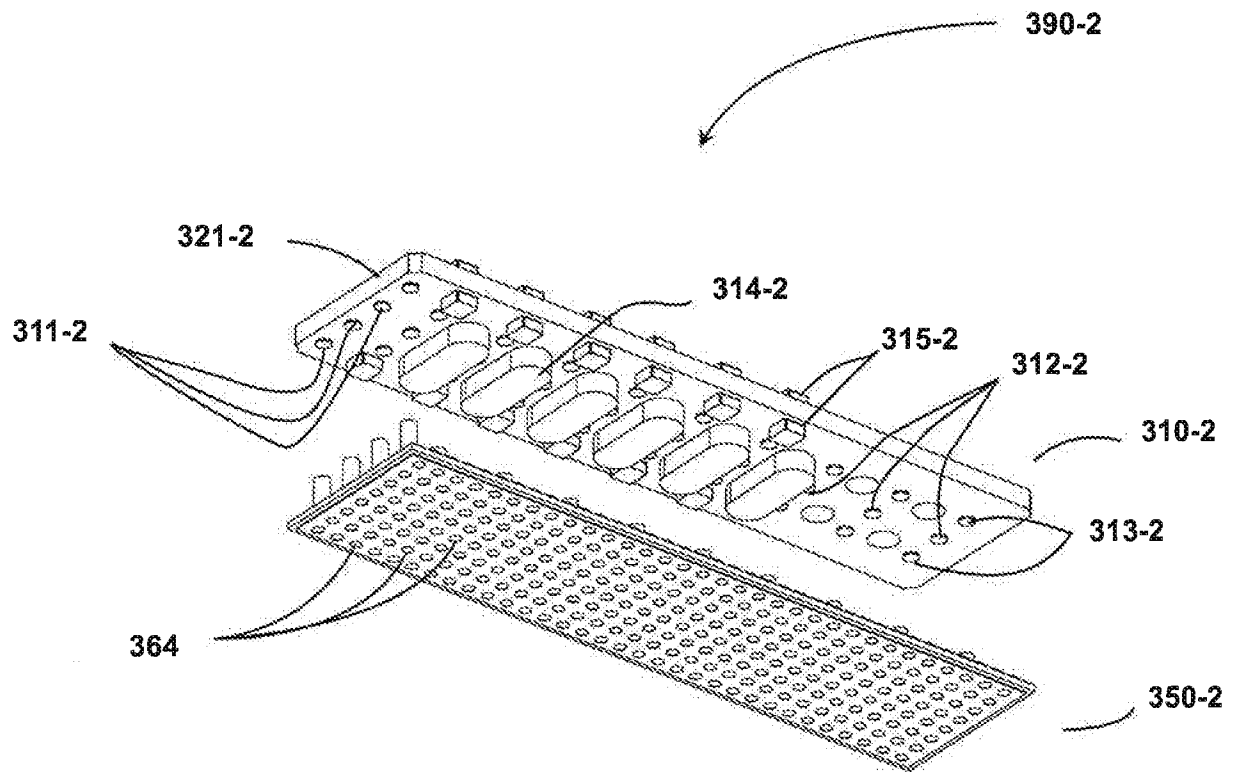


FIG. 10A

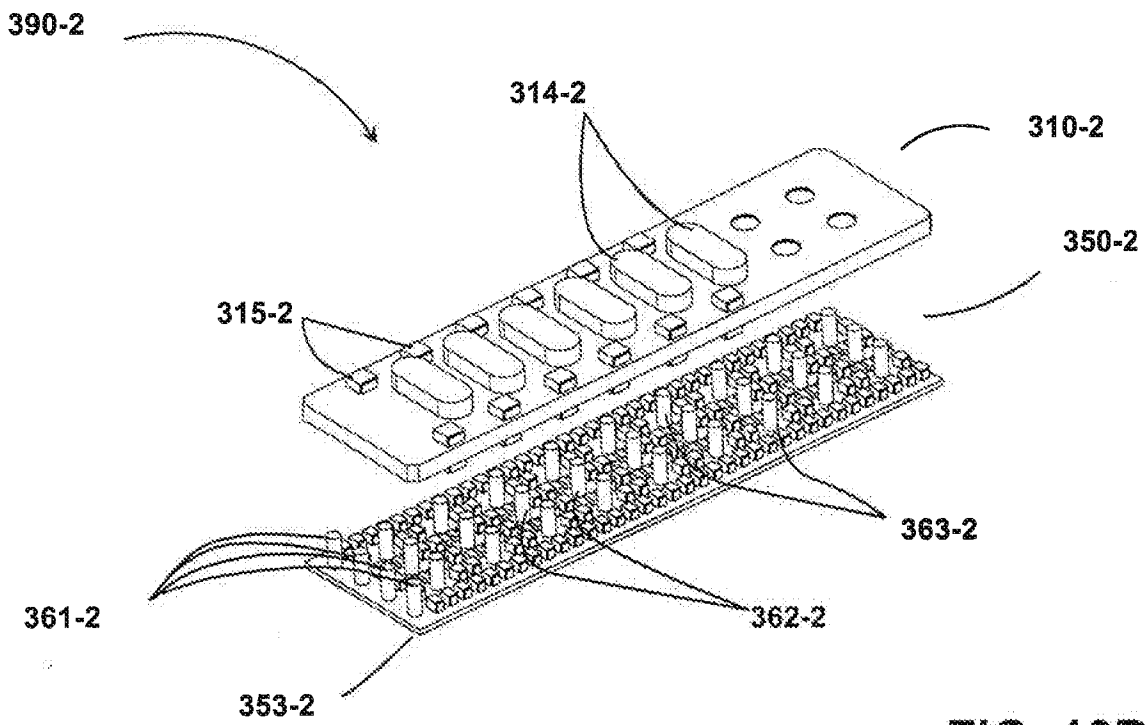


FIG. 10B

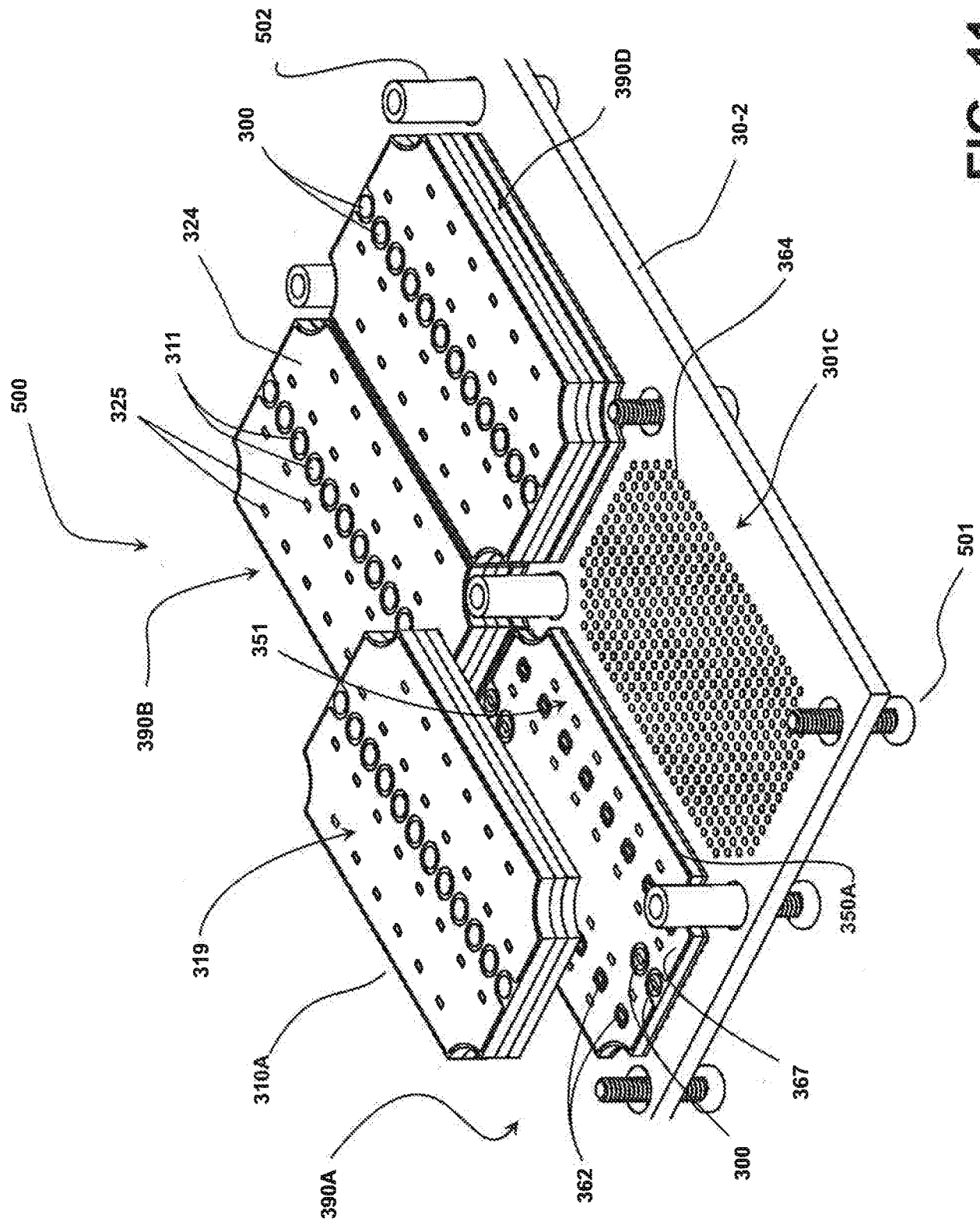


FIG. 11

ABSTRACT

Encapsulated electronic modules having complex contact structures may be formed by encapsulating panels containing a substrate comprising pluralities of electronic modules delineated by cut lines and having conductive interconnects buried within terminal holes and other holes drilled in the panel within the boundaries of the cut lines. Slots may be cut in the panel along the cut lines. The interior of the holes, as well as surfaces within the slots and on the surfaces of the panel may be metallized, e.g. by a series of processes including plating. Solder may be dispensed into the holes for surface mounting. Two or more panels may be stacked prior to singulation to form module stacks.

Multi-cell converters having a large cell pitch may be combined with an interconnection module to provide vertical power delivery to semiconductor devices through a semiconductor power grid having a small pitch. The converters and interconnection modules may be fabricated in panels and stacked prior to singulation.

WHAT IS CLAIMED IS:

1. A method of powering a semiconductor device having power input terminals, the method comprising:

providing a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells, each conversion cell including one or more converter output terminals, the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage, the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals;

providing an interconnection module having x-input terminals arranged on a first side of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a surface of a second opposite side of the interconnection module in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;

assembling the power conversion module to the first side of the interconnection module with each converter output terminal mated with and electrically connected to a respective x-input terminal;

wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern;

wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer.

2. The method of claim 1 wherein the interconnection module further comprises filter circuitry connected to the x-input terminals and the x-output terminals.

3. The method of claim 2 wherein the interconnection module further comprises a multilayer printed circuit board (“PCB”), and the filter circuitry comprises a plurality of capacitors mounted to a first surface of the PCB on the first side of the interconnection module.
4. The method of claim 2 wherein the second spacing is less than the first spacing.
5. The method of claim 2 wherein each conversion cell further comprises a first dimension and the first spacing is configured as a function of the first dimension.
6. The method of claim 5 wherein each conversion cell further comprises a second dimension, the first spacing further comprises a latitudinal spacing and a longitudinal spacing, and the latitudinal spacing is a function of the first dimension and the longitudinal spacing is a function of the second dimension.
7. The method of claim 1, further comprising configuring the conversion circuitry as a buck converter having a plurality of phases with each conversion cell configured to operate as a selected one of the phases.
8. The method of claim 1, further comprising configuring the conversion circuitry to convert power received from an input to the power conversion module at an input voltage, V_{IN} , and input current, I_{IN} , for delivery to the converter output terminals at an output voltage, V_{OUT} , and an output current, I_{OUT} , using a transformation ratio, K , where $K = V_{OUT} / V_{IN}$, that is essentially constant, subject to an equivalent output resistance.
9. The method of claim 8, further comprising encapsulating the conversion circuitry to form a first surface of the power conversion module; encapsulating the interconnection module to form a first surface on the first side of the interconnection module; affixing the first surface of the power conversion module to the first surface of the interconnection module with the converter output terminals and the x-input terminals aligned using an adhesive, and forming solder connections between respective one of the output terminals and the x-input terminals.

10. The method of claim 1 wherein the interconnection module further comprises a multilayer printed circuit board (“PCB”) and the method further comprises mounting at least one solder ball to the PCB for each x-output terminal.

11. The method of claim 1 wherein the interconnection module further comprises a multilayer printed circuit board (“PCB”) and the method further comprises mounting at least one conductive pin to the PCB for each x-output terminal.

12. The method of claim 1 wherein the interconnection module further comprises a multilayer printed circuit board (“PCB”) and the method further comprises providing the x-output terminals in a land grid array on the PCB.

13. The method of claim 1, further comprising attaching the semiconductor device to the interconnection module with the x-output terminals aligned with and electrically connected to respective ones of the power input terminals; and wherein the connections between the x-output terminals and the power input terminals form a fourth layer in the module sandwich.

14. The method of claim 1, further comprising
assembling a first panel comprising a plurality of unsingulated power conversion modules;

assembling a second panel comprising a plurality of unsingulated interconnection modules;

wherein the assembling of the power conversion module and the interconnection module is performed using the first and second panels prior to singulation to form a panel sandwich and further comprises dispensing epoxy between the first and second panels and forming solder connections between each converter output terminal and each respective x-input terminal; and

further comprising:

cutting the panel sandwich along predetermined cut lines to singulate a plurality of the module sandwiches.

15. The method of claim 14 wherein the dispensing epoxy further comprises dispensing epoxy on the surface of at least one of the panels along the cut lines.

16. The method of claim 15, further comprising reflowing solder and curing the epoxy.

17. A method of making power converters, the method comprising:
assembling an encapsulated panel including a printed circuit board (“PCB”) having electronic components mounted to the PCB and a plurality of conductive features enclosed within the panel and unexposed to an exterior surface of the panel, the panel comprising one or more power conversion modules having boundaries defined by one or more predetermined cut lines;
configuring each of the one or more power conversion modules in a multi-cell structure having a plurality of conversion cells, each conversion cell including one or more converter output nodes arranged in a first pattern having a first spacing between the output nodes, and wherein each output node is electrically connected to a respective conductive feature;
selectively forming a plurality of holes in the exterior surface of the panel, each hole being formed at a respective predetermined location aligned with a respective output node, spaced apart from the cut lines, and exposing within the hole the respective conductive feature;
selectively forming one or more conductive metal layers on the exterior surface of the panel including within the plurality of holes, the one or more conductive metal layers within each hole being in electrical contact with the respective conductive feature, to form a plurality of electrical contacts on the exterior surface of the panel;
patterning the one or more conductive metal layers on the exterior surface to electrically isolate at least one electrical contact from one or more of the plurality of electrical contacts;
cutting the panel along the one or more cut lines to singulate the one or more power conversion modules, each singulated power conversion module having a respective plurality of the electrical contacts formed on the exterior surface of the module.

18. The method of claim 17 further comprising selectively filling the plurality of holes with solder.

19. The method of claim 18, further comprising planarizing the solder to a predetermined height relative to the exterior surface.

20. The method of claim 17 wherein each conversion cell further comprises a first dimension and a second dimension, the first spacing further comprises a latitudinal spacing and a longitudinal spacing, and the latitudinal spacing is a function of the first dimension and the longitudinal spacing is a function of the second dimension.

21. The method of claim 17, further comprising:
providing an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern, the second pattern having a second spacing between the x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;

assembling the first surface of the interconnection module to the power conversion module with each converter output terminal mated with and electrically connected to a respective x-input terminal,

wherein the power conversion module and the interconnection module are stacked to form a sandwich structure including the power conversion module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer.

22. The method of claim 21, further comprising assembling the sandwich structure to a substrate with the x-output terminals aligned with and electrically connected to respective terminals on a surface of the substrate.

23. The method of claim 22 wherein the substrate includes a semiconductor device electrically connected to receive power from the power conversion module.

24. A method of making electronic assemblies, the method comprising:

- providing an electronic module having electronic circuitry including circuit terminals arranged in a first pattern having a first spacing between the circuit terminals;
- providing an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of circuit terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with external terminals of a third electronic device, the second pattern having a second spacing between the x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;
- assembling the electronic module to the first surface of the interconnection module with each circuit terminal mated with and electrically connected to a respective x-input terminal;
- wherein the interconnection module is configured to provide electrical and mechanical translation between the circuit terminals arranged in the first pattern and the external terminals of the third electronic device arranged in the second pattern;
- wherein the electronic module and the interconnection module are stacked to form a module sandwich including the electronic module in a first layer, electrical connections between the circuit terminals and the x-input terminals in a second layer, and the interconnection module in a third layer;
- assembling a first panel comprising a plurality of unsingulated electronic modules;
- assembling a second panel comprising a plurality of unsingulated interconnection modules;
- wherein the assembling of the electronic module and the interconnection module is performed using the first and second panels prior to singulation to form a panel sandwich and further comprises dispensing a curable adhesive compound between the first and second panels and forming solder connections between each circuit terminal and each respective x-input terminal; and
- cutting the panel sandwich along predetermined cut lines to singulate a plurality of the module sandwiches.

25. The method of claim 24 wherein the dispensing the curable adhesive compound further comprises dispensing the curable adhesive compound on the surface of at least one of the panels along the cut lines.

26. The method of claim 24, further comprising reflowing solder and curing the adhesive compound.

27. A method of forming modular circuit assemblies, the method comprising:
assembling first and second encapsulated panels each including a first multilayer printed circuit board (“PCB”) having a plurality of conductive layers, a first plurality of electronic components mounted to a first surface of the PCB, and a first layer of cured encapsulant covering the first plurality of electronic components and the surface of the PCB, the first layer of cured encapsulant forming a first exterior surface of the encapsulated panel, the encapsulated panel comprising a plurality of unsingulated electronic modules, each electronic module having perimeter boundaries defined by one or more predetermined cut lines;

selectively forming a plurality of terminal holes in the first exterior surface of each panel through the first layer at predetermined locations within the perimeter boundaries of each electronic module, each terminal hole being spaced apart from the cut lines and exposing within the terminal hole a respective conductive feature;

selectively forming one or more conductive metal layers on the first exterior surface of each panel including within the plurality of terminal holes, the one or more conductive metal layers within each terminal hole being in electrical contact with the respective conductive feature;

patterning the one or more conductive metal layers on the exterior surface, to form a plurality of electrical contacts on the exterior surface of each panel electrically isolated from at least one other electrical contact in the plurality of electrical contacts;

assembling the first and second encapsulated panels with electrical contacts on the first panel aligned and mated with respective electrical contacts on the second panel to form a panel stack, the panel stack comprising a plurality of un-singulated module stacks, each module stack including an unsingulated electronic module in the first panel and a respective unsingulated electronic module in the second panel; and

cutting the panel stack along the one or more cut lines to singulate the plurality of electronic module stacks.

28. The method of claim 27, further comprising dispensing a curable adhesive compound on the surface of at least one of the panels along the cut lines.

29. The method of claim 28, further comprising reflowing solder and curing the adhesive compound.

30. An apparatus comprising:

- a semiconductor device having power input terminals;
- a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells, each conversion cell including one or more converter output terminals, the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage, the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals; and
- an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;
- the converter output terminals of the power conversion module being mated with and electrically connected to respective x-input terminals of the interconnection module;
- wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern;
- wherein the power conversion module and the interconnection module are stacked with the power conversion module located at a first level, electrical connections between the converter

output terminals and the x-input terminals at a second level, and the interconnection module at a third level.

31. The method of claim 30 wherein the semiconductor device comprises a die.

32. The method of claim 30 wherein the semiconductor device comprises a wafer.

33. The method of claim 30 wherein the semiconductor device comprises a substrate.

34. The method of claim 30 wherein the semiconductor device comprises a semiconductor die or package mounted to a substrate.

35. A method of powering a semiconductor device having power input terminals, the method comprising:

providing a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells, each conversion cell including one or more converter output terminals, the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage, the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals;

providing an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;

providing a plurality of conductive pillars for electrically connecting the converter output terminals with respective x-input terminals;

assembling the power conversion module to the interconnection module with each converter output terminal aligned with and electrically connected to a respective x-input terminal by a respective one of the conductive pillars;

wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern;

wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, the conductive pillars connecting the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer; and

wherein the x-output terminals are arranged with alternating polarities to reduce interconnection inductance.

36. The method of claim 1 further comprising arranging the x-output terminals with alternating polarities.

37. The method of claim 1 further comprising arranging the x-output terminals with polarities that alternate between adjacent columns.

38. The method of claim 1 further comprising arranging the x-output terminals with polarities that alternate between adjacent rows.

39. The method of claim 1 further comprising arranging the x-output terminals with polarities that alternate between adjacent x-output terminals.

40. The method of claim 1 further comprising arranging the converter output terminals with alternating polarities.

41. The method of claim 1 further comprising arranging the converter output terminals with polarities that alternate between rows.

42. The method of claim 1 further comprising arranging the converter output terminals with polarities that alternate between columns.

43. The method of claim 1 further comprising arranging the converter output terminals with polarities that alternate between adjacent converter output terminals.

44. The method of claim 35 further comprising arranging the x-output terminals with alternating polarities.

45. The method of claim 35 further comprising arranging the converter output terminals with alternating polarities.

DELIVERING POWER TO SEMICONDUCTOR LOADS

CROSS-REFERENCE TO RELATED APPLICATION

[001] This is a continuation-in-part of U.S. Patent Application 15/616,288, filed on June 7, 2017 which is a continuation-in-part of U.S. Patent Application 15/091,346, filed on April 5, 2016, both of which are hereby incorporated in their entirety by reference.

FIELD OF THE INVENTION

[002] This invention relates to the field of powering semiconductor loads such as including CPUs, GPUs, ASICs, and wafer scale devices and more particularly to providing power encapsulated electronic assemblies, including encapsulated power converters.

BACKGROUND

[003] In contemporary electronic systems, space is at a premium on customer circuit boards, e.g. on a circuit board near a processor. Additionally, efficiency and thermal management considerations place limits on the power dissipation of power supplies at, or near, the point of load. Many very large scale integrated (“VLSI”) semiconductor dies such as central processing units (“CPU”), graphics processing units (“GPU”), and application specific integrated circuits (“ASIC”) are mounted to a multilayer ceramic substrate which translates the electrical connections from the die to larger connections suitable for interfacing with a customer motherboard. As feature sizes decrease and transistor counts increase, so too do the power supply current requirements for such large chips. Current requirements for a typical CPU can easily exceed 200 amps and for a typical GPU exceeding 500 amps creating challenges for the package and system designers to efficiently supply such high currents. For example, power connections between the component package (such as a chip carrier or substrate or other package in or on which the semiconductor die is mounted) and the printed circuit board (PCB) on which the package is mounted may demand a large number of connector pins, leads, solder bumps, etc., to carry very high currents challenging package designers to accommodate both power and signal requirements. In many cases the large number and high frequency demands of signals may limit

the maximum voltage, e.g. the interlayer breakdown voltage, to which the substrate or package may be subjected, in some cases as low as a few volts, further challenging power connections to and within the package or substrate.

SUMMARY

[004] In general, in one aspect, a method of powering a semiconductor device having power input terminals is provided. The method includes providing a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells, each conversion cell including one or more converter output terminals, the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage, the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals. The method includes providing an interconnection module having x-input terminals arranged on a first side of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a surface of a second opposite side of the interconnection module in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; and assembling the power conversion module to the first side of the interconnection module with each converter output terminal mated with and electrically connected to a respective x-input terminal. The interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern. The power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer.

[005] Implementations of the aspect can include one or more of the following features. The interconnection module can further include filter circuitry connected to the x-input terminals and the x-output terminals. The interconnection module can further include a multilayer printed circuit board ("PCB"), and the filter circuitry can include a plurality of capacitors mounted to a

first surface of the PCB on the first side of the interconnection module. The second spacing can be less than the first spacing. Each conversion cell can further include a first dimension and the first spacing can be configured as a function of the first dimension. Each conversion cell can further include a second dimension, the first spacing can further include a latitudinal spacing and a longitudinal spacing, and the latitudinal spacing can be a function of the first dimension and the longitudinal spacing can be a function of the second dimension. The method can further include configuring the conversion circuitry as a buck converter having a plurality of phases with each conversion cell configured to operate as a selected one of the phases. The method can further include configuring the conversion circuitry to convert power received from an input to the power conversion module at an input voltage, V_{IN} , and input current, I_{IN} , for delivery to the converter output terminals at an output voltage, V_{OUT} , and an output current, I_{OUT} , using a transformation ratio, K , where $K = V_{OUT} / V_{IN}$, that is essentially constant, subject to an equivalent output resistance.

[006] The method can further include encapsulating the conversion circuitry to form a first surface of the power conversion module, encapsulating the interconnection module to form a first surface on the first side of the interconnection module, affixing the first surface of the power conversion module to the first surface of the interconnection module with the converter output terminals and the x-input terminals aligned using an adhesive, and forming solder connections between respective one of the output terminals and the x-input terminals. The interconnection module can further include a multilayer printed circuit board ("PCB") and the method can further include mounting at least one solder ball to the PCB for each x-output terminal. The interconnection module can further include a multilayer printed circuit board ("PCB") and the method can further include mounting at least one conductive pin to the PCB for each x-output terminal. The interconnection module can further include a multilayer printed circuit board ("PCB") and the method can further include providing the x-output terminals in a land grid array on the PCB.

[007] The method can further include attaching the semiconductor device to the interconnection module with the x-output terminals aligned with and electrically connected to respective ones of the power input terminals; and wherein the connections between the x-output terminals and the power input terminals form a fourth layer in the module sandwich. The method

can further include assembling a first panel comprising a plurality of unsingulated power conversion modules; and assembling a second panel comprising a plurality of unsingulated interconnection modules; wherein the assembling of the power conversion module and the interconnection module is performed using the first and second panels prior to singulation to form a panel sandwich and further comprises dispensing epoxy between the first and second panels and forming solder connections between each converter output terminal and each respective x-input terminal. The method can further include cutting the panel sandwich along predetermined cut lines to singulate a plurality of the module sandwiches. The dispensing of epoxy can further include dispensing epoxy on the surface of at least one of the panels along the cut lines. The method can further include reflowing solder and curing the epoxy.

[008] The method can further include arranging the x-output terminals with alternating polarities. In some example, the method can further include arranging the x-output terminals with polarities that alternate between adjacent columns. In some examples, the method can further include arranging the x-output terminals with polarities that alternate between adjacent rows. In some examples, the method can further include arranging the x-output terminals with polarities that alternate between adjacent x-output terminals. The method can further include arranging the converter output terminals with alternating polarities. In some examples, the method can further include arranging the converter output terminals with polarities that alternate between rows. In some examples, the method can further include arranging the converter output terminals with polarities that alternate between columns. In some examples, the method can further include arranging the converter output terminals with polarities that alternate between adjacent converter output terminals.

[009] In general, in another aspect, a method of making power converters is provided. The method includes assembling an encapsulated panel including a printed circuit board ("PCB") having electronic components mounted to the PCB and a plurality of conductive features enclosed within the panel and unexposed to an exterior surface of the panel, the panel comprising one or more power conversion modules having boundaries defined by one or more predetermined cut lines; configuring each of the one or more power conversion modules in a multi-cell structure having a plurality of conversion cells, each conversion cell including one or more converter output nodes (it seems a conversion cell does not include a whole converter

output node, but rather shares portions of several output nodes, see comment in claim 1), wherein the output nodes are arranged in a first pattern having a first spacing between the output nodes, and wherein each output node is electrically connected to a respective conductive feature; selectively forming a plurality of holes in the exterior surface of the panel, each hole being formed at a respective predetermined location aligned with a respective output node, spaced apart from the cut lines, and exposing within the hole the respective conductive feature; selectively forming one or more conductive metal layers on the exterior surface of the panel including within the plurality of holes, the one or more conductive metal layers within each hole being in electrical contact with the respective conductive feature, to form a plurality of electrical contacts on the exterior surface of the panel; patterning the one or more conductive metal layers on the exterior surface to electrically isolate at least one electrical contact from one or more of the plurality of electrical contacts; and cutting the panel along the one or more cut lines to singulate the one or more power conversion modules, each singulated power conversion module having a respective plurality of the electrical contacts formed on the exterior surface of the module.

[010] Implementations of the aspect can include one or more of the following features. The method can further include selectively filling the plurality of holes with solder. The method can further include planarizing the solder to a predetermined height relative to the exterior surface. Each conversion cell can further include a first dimension and a second dimension, the first spacing can further include a latitudinal spacing and a longitudinal spacing, and the latitudinal spacing can be a function of the first dimension and the longitudinal spacing can be a function of the second dimension.

[011] The method can further include: providing an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern, the second pattern having a second spacing between the x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; and assembling the first surface of the interconnection module to the power conversion module with each converter output terminal mated with and electrically connected to a respective x-input terminal, wherein the power

conversion module and the interconnection module are stacked to form a sandwich structure including the power conversion module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer. The method can further include assembling the sandwich structure to a substrate with the x-output terminals aligned with and electrically connected to respective terminals on a surface of the substrate. The substrate can include a semiconductor device electrically connected to receive power from the power conversion module.

[012] In general, in another aspect, a method of making electronic assemblies is provided. The method can include: providing an electronic module having electronic circuitry including circuit terminals arranged in a first pattern having a first spacing between the circuit terminals; providing an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of circuit terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with external terminals of a third electronic device, the second pattern having a second spacing between the x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals; and assembling the electronic module to the first surface of the interconnection module with each circuit terminal mated with and electrically connected to a respective x-input terminal. The interconnection module is configured to provide electrical and mechanical translation between the circuit terminals arranged in the first pattern and the external terminals of the third electronic device arranged in the second pattern. The electronic module and the interconnection module are stacked to form a module sandwich including the electronic module in a first layer, electrical connections between the circuit terminals and the x-input terminals in a second layer, and the interconnection module in a third layer. The method includes assembling a first panel comprising a plurality of unsingulated electronic modules; and assembling a second panel comprising a plurality of unsingulated interconnection modules; wherein the assembling of the electronic module and the interconnection module is performed using the first and second panels prior to singulation to form a panel sandwich and further comprises dispensing a curable adhesive compound between the first and second panels and forming solder connections between each circuit terminal and each respective x-input terminal. The method includes cutting the panel sandwich along predetermined cut lines to singulate a plurality of the module sandwiches.

[013] Implementations of the aspect can include one or more of the following features. The dispensing of the curable adhesive compound can further include dispensing the curable adhesive compound on the surface of at least one of the panels along the cut lines. The method can further include reflowing solder and curing the adhesive compound.

[014] In another general aspect, a method of forming modular circuit assemblies is provided. The method includes: assembling first and second encapsulated panels each including a first multilayer printed circuit board (“PCB”) having a plurality of conductive layers, a first plurality of electronic components mounted to a first surface of the PCB, and a first layer of cured encapsulant covering the first plurality of electronic components and the surface of the PCB, the first layer of cured encapsulant forming a first exterior surface of the encapsulated panel, the encapsulated panel comprising a plurality of unsingulated electronic modules, each electronic module having perimeter boundaries defined by one or more predetermined cut lines; selectively forming a plurality of terminal holes in the first exterior surface of each panel through the first layer at predetermined locations within the perimeter boundaries of each electronic module, each terminal hole being spaced apart from the cut lines and exposing within the terminal hole a respective conductive feature; selectively forming one or more conductive metal layers on the first exterior surface of each panel including within the plurality of terminal holes, the one or more conductive metal layers within each terminal hole being in electrical contact with the respective conductive feature; patterning the one or more conductive metal layers on the exterior surface, to form a plurality of electrical contacts on the exterior surface of each panel electrically isolated from at least one other electrical contact in the plurality of electrical contacts; assembling the first and second encapsulated panels with electrical contacts on the first panel aligned and mated with respective electrical contacts on the second panel to form a panel stack, the panel stack comprising a plurality of un-singulated module stacks, each module stack including an unsingulated electronic module in the first panel and a respective unsingulated electronic module in the second panel; and cutting the panel stack along the one or more cut lines to singulate the plurality of electronic module stacks.

[015] Implementations of the aspect can include one or more of the following features. The method can further include dispensing a curable adhesive compound on the surface of at least

one of the panels along the cut lines. The method can further include reflowing solder and curing the adhesive compound.

[016] In another general aspect, an apparatus includes: a semiconductor device having power input terminals; a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells, each conversion cell including one or more converter output terminals (it seems a conversion cell does not include a whole converter output terminal, but rather shares portions of several output terminals, see comment in claim 1), the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage, the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals; and an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals. The converter output terminals of the power conversion module is mated with and electrically connected to respective x-input terminals of the interconnection module. The interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern. The power conversion module and the interconnection module are stacked with the power conversion module located at a first level, electrical connections between the converter output terminals and the x-input terminals at a second level, and the interconnection module at a third level.

[017] Implementations of the aspect can include one or more of the following features. The semiconductor device can include a die, a wafer, or a substrate. The semiconductor device can include a semiconductor die or package mounted to a substrate.

[018] In another general aspect, a method of powering a semiconductor device having power input terminals is provided. The method includes: providing a power conversion module having

conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells, each conversion cell including one or more converter output terminals, the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage, the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals; and providing an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals. The method further includes providing a plurality of conductive pillars for electrically connecting the converter output terminals with respective x-input terminals; and assembling the power conversion module to the interconnection module with each converter output terminal aligned with and electrically connected to a respective x-input terminal by a respective one of the conductive pillars. The interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern. The power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, the conductive pillars connecting the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer. The x-output terminals are arranged with alternating polarities to reduce interconnection inductance. The method can further include arranging the x-output terminals with alternating polarities. The method can further include arranging the converter output terminals with alternating polarities.

BRIEF DESCRIPTION OF DRAWINGS

[019] Fig. 1 shows a block diagram of a system for powering a semiconductor load such as a CPU, GPU, ASIC, or other high current demanding load.

[020] Fig. 2 shows a schematic diagram of a multi-cell converter.

[021] Fig. 3 shows a multi-cell converter module package.

[022] Fig. 4 shows an example of a power reticle for a semiconductor device such as a GPU.

[023] Fig. 5A and Fig. 5B show top and bottom isometric views of an exploded converter assembly including a multi-cell converter module and an interconnection module.

[024] Fig. 6 is a plan view of the un-encapsulated interconnection module.

[025] Fig. 7 shows a cross section of a portion of the converter assembly.

[026] Fig. 8A is an exploded view of a panel sandwich assembly including a panel of converter modules and a panel of the interconnection modules.

[027] Fig. 8B is side view of the panel sandwich assembly.

[028] Fig. 9 is a process flow for fabricating the converter assemblies.

[029] Figs. 10A and 10B show top and bottom isometric views of an exploded converter assembly including an open-frame multi-cell converter module and an open-frame interconnection module.

[030] Fig. 11 shows a vertical power delivery system.

[031] Like reference numbers and symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

[032] A schematic block diagram of a system for powering a semiconductor load, such as a high-power ASIC, is shown in Fig. 1. As shown, the power system 10 may include a first power converter 15 connected to convert power received from a source, e.g. a system power bus, (not shown) for delivery to a second stage converter 20 which converts power for delivery to the semiconductor load, e.g. ASIC 30. As shown the second stage converter 20 is preferably mounted in close proximity to the point of load (“PoL”), for example on or in the same package or assembly, e.g. assembly 31, as the ASIC 30. Preferably, power conversion stage 15 is a

switching regulator having a nominal output voltage at or near 48 Volts DC, and the second conversion stage is a fixed-ratio switching converter which converts the 48 VDC input to the low voltage required by the semiconductor, e.g. typically ranging from less than 2 VDC to less than 1 VDC.

[033] A Factorized Power Architecture well suited for supplying power to low voltage high current loads is described in Vinciarelli, *Factorized Power with Point of Load Sine Amplitude Converters*, U.S. Patent No. 6,975,098, issued Dec. 13, 2005 (the “Micro FPA Patent”) and U.S. Patent No. 6,984,965, issued Jan. 10, 2006 (the “FPA Patent”) (both assigned to VLT, Inc. of Andover, MA., and the entire disclosure of each patent is incorporated herein by reference). Power converters which function as DC-to-DC transformers called Voltage Transformation Modules (“VTM”) and Sine Amplitude Converters (“SAC”) which have a transfer function approximating $V_o = K_{VTM} * V_{in} - I_o * R_{VTM}$ are described in Vinciarelli, *Factorized Power with Point of Load Sine Amplitude Converters*, U.S. Patent No. 6,930,893, issued Aug. 16, 2005 (the “SAC Patent”) and in Vinciarelli, *Point of Load Sine Amplitude Converters and Methods*, U.S. Patent No. 7,145,786, issued Dec. 5, 2006 (the “POL SAC Patent”) (both assigned to VLT, Inc. of Andover, MA., the entire disclosure of each patent is incorporated herein by reference).

[034] In-package and on-package power conversion topologies are described in Vinciarelli et al., *Method and Apparatus for Delivering Power to Semiconductors*, U.S. Ser. No. 15/091,346 filed on April 5, 2016 (the “MCM disclosure”); and in U.S. Ser. No. 15/616,288 filed on June 7, 2017 (the multi-rail disclosure”); (collectively the “PoP” disclosures) both of which are assigned to VLT, Inc. of Andover, MA., the entire disclosure of each is incorporated herein by reference. As discussed in connection with Figs. 15A and 15B of the MCM disclosure, the fixed-ratio power converter may be mounted to the backside of the semiconductor package in the same or similar footprint of the semiconductor die to deliver power vertically to the semiconductor die.

[035] Referring to Fig. 2, a multi-output cell fixed-ratio converter adapted for use as the second stage converter 20 is shown schematically. As shown in Fig. 2, a single half-bridge input circuit drives the primary winding 214 of transformer T1 209 with power received at the input voltage V_{in} . A plurality of output cells 250-1, 250-2, 250-n are shown having respective secondary windings 215 coupled to the transformer core 225 to receive power from the primary

via the transformer. In the example shown, each secondary winding is center tapped comprising two windings 215a and 215b. It is assumed in the example shown that all of the secondary windings 215 (215a, 215b) have an equal number of turns providing a uniform turns ratio in each of the output cells 250. The outputs 227, 228 from each output cell may be connected in parallel to provide the required output current to the die. In each output cell two synchronous rectifier switches SR1 and SR2 selectively couple their respective winding 215a, 215b to complete the circuit providing full wave rectification. A switch controller (not shown) operates the primary switches S1 210 and S2 211 in the input circuit and the secondary switches SR1 216 and SR2 217 in each of the output cells in a series of converter operating cycles. Preferably the multi-cell converter uses the SAC topology described more fully in the SAC patent. Some or all of the input circuit may be located remotely from the transformer and output cells as described in Vinciarelli, *Driver and Output Circuit for Powering Semiconductor Loads*, U.S. Patent No. 10,014,798 issued July 3, 2018, assigned to VLT, Inc. of Andover, MA, and incorporated in its entirety here (the “Remote Driver Patent”) and in the MCM disclosure.

[036] Preferably, the transformer and output cells of the multi-cell power converter may be formed on a multi-layer printed circuit board (“PCB”) 321 and encapsulated to form a package amenable to mount directly to the semiconductor load. For example, as shown in Fig. 3, the multi-cell converter 310 is shown in an encapsulated package having a generally rectangular outline or footprint. Figs. 5A and 5B show edges of the internal PCB 321 of the converter 310 sandwiched between two cured encapsulant layers 322 and 323. One of the large surfaces 320 of the converter 310 is shown in Fig. 5A including a plurality of electrical terminations which will be discussed in more detail below in connection with Fig. 3. The other large surface 319 of the converter 310 is shown in Fig. 5B having a conductive layer 324 covering a majority of the surface 319 and which preferably may be electrically connected to a common terminal, e.g. using one or more electrical and thermal connections 325, to provide EMI shielding and also enhance thermal performance. For example, the surface 319 may be mated with a cold plate or heat sink. Encapsulated packages of the type shown in Figs. 3, 5A, and 5B, amenable for housing the multi-cell power converter in a format suitable for in-package and on-package applications is described in Vinciarelli et al., *Panel Molded Electronic Assemblies with Multi-Surface Contacts*, U.S. Patent Appln. Ser. No. 14/731,287 filed June 4, 2015 (the “Leadless Disclosure”) and in Vinciarelli et al., *Panel Molded Electronic Assemblies with Integral Terminals*, U.S. Patent

Appln. Ser. No. xxxxxxxx filed December-xx-2018, (the “Trench Disclosure”) both of which are assigned to VLT, Inc. of Andover, MA, and incorporated herein by reference in their entirety.

[037] The multi-layer PCB may include a plurality of conductive layers on which the transformer windings may be formed. Referring to the schematic diagram of Fig. 2, the transformer 209 includes a single primary winding 214 driving the transformer core 225 to which are coupled the plurality (one for each output cell 250) of center-tapped secondary windings 215, each including windings 215a and 215b. One example PCB layer stack-up may have a primary winding (P) on the top outer conductive layer, a first set of secondary windings (Sa) on the second layer, a second set of secondary windings (Sb) on the third layer, a primary winding (P) on the fourth layer, a first set of secondary windings (Sa) on the fifth layer, a second set of secondary windings (Sb) on a sixth layer, a primary winding (P) on the seventh layer, and so on. In this example, the primary windings may be connected in series to divide the input voltage and the secondary windings connected in parallel to multiply the output current and reduce output resistance. Preferable PCB transformer structures for use in the multi-cell converter are described in Vinciarelli, *Printed Circuit Transformer*, U.S. Patent No. 7,187,263 issued March 6, 2007 assigned to VLT, Inc. of Andover, MA, and incorporated by reference in its entirety here (the “Racetrack” patent).

[038] The multi-cell converter example shown in Fig. 3 includes sixteen output cells, e.g. cells 250-1, 250-2, 250-3, 250-4 ... 250-16, distributed in two horizontal rows and eight vertical columns in the orientation illustrated by the brackets shown in broken lines. A portion of the right side of the module 310 is shown cutaway to reveal portions of the output cell structure. Referring to output cell 250-1, a single turn winding 215-1 is shown surrounding a core leg 225-1. As shown, the footprint of semiconductor dice, e.g. die 315, lie on the border of the output cells 250, and referring to output cell 250-1 for example, overlap portions of the respective secondary windings, e.g. winding 215-1. In the layout of Fig. 3, each semiconductor die includes two devices, which may for example be MOSFET switches, used to implement the synchronous rectifiers (SR1 216 and SR2 217: Fig. 2). Accordingly, the die located between output cells 250-3 and 250-1 for example is labeled to show that switches 217-3 (on the left) and 216-1 (on the right), corresponding to switches SR2 of output cell 250-3 and SR1 of output cell 250-1 reside in that location. Note that the dice at the far ends of the multi-output cell converter 310 are not

shared between adjacent cells, allowing both switches to be connected in parallel, providing lower output resistance in the cells at the ends of the converter, which as described below may force better current distribution. Preferably, the dice are located on the top (shown) and bottom (not shown) PCB surfaces for improved electrical and thermal performance, e.g. as described in Vinciarelli et al., *Encapsulated Modular Power Converter with Symmetrical Heat Distribution*, U.S. Patent No. 9,516,761 issued December 6, 2016, assigned to VLT, Inc. of Andover, MA, and incorporated by reference in its entirety here (the “Symmetrical Patent”).

[039] Although only a single turn winding 215-1 is shown in cell 250-1 in Fig. 3, it should be understood that each output cell 250 may include a center-tapped winding (windings 215 as shown in Fig. 2) comprising two individual single turn windings (215a and 215b: Fig. 2), each of which, as described above in the exemplary PCB stack-up, may comprise a plurality of single turns connected in parallel to form lower resistance single turn windings. In the layout shown in Fig. 3, both halves of each center-tapped secondary, e.g. center-tapped secondary 215-1, may be coupled to a single core leg, e.g. core leg 225-1, allowing placement of the synchronous rectifier for each half winding (215a, 215b: Fig. 2) to be located at opposite edges of the output cell, e.g. 217-1, 216-1 in Fig. 3, respectively. In Fig. 3, the outline of output cells 250-1, 250-3, and 250-4 are shown with broken lines to generally indicate the boundaries of each cell, however, it should be understood that each cell may share features with adjacent cells.

[040] The power output terminations 312, e.g. positive, and 313, e.g. common, may be distributed on the generally planar top surface of the encapsulated converter for vertical power delivery. As shown in Fig. 3, nine positive terminations 312 and eighteen common terminations 313 provide a total of twenty-seven power connections distributed in an array of nine columns and three rows. (The last two columns on the right-hand side of the drawing are not shown in Fig. 3 to provide cut-away details.) Note that in the example of Fig. 3, all of the terminations in a row are connected together by metal on the surface of the encapsulant 323, e.g. metal strips 317 and 318 respectively connect all of the termination 313 in the bottom and top rows and metal 316 connects all of the terminations in the middle row.

[041] At the left-most side of the encapsulated module 310, an additional column of four electrical terminations, e.g. terminations 311, are shown to provide control and power input

connections to the converter. For example, two of the terminations 311 may be used as power input terminations, e.g. to provide connections to the primary winding (214 in Fig. 2) of the transformer 209; and two of the terminations 311 may be used as control terminations, e.g. to provide connections to the control terminals of the synchronous rectifiers (SR1 216 and SR2 217 in Fig. 2), which may e.g. be the gates of MOSFETs used to implement the synchronous rectifiers, where the switch controller is located separately from the encapsulated module 310.

[042] As shown in Fig. 3, magnetically permeable core pieces 314 (shown as broken oval lines) are disposed between each columnar set of power terminations. Beneath each core piece are circular core legs 225, e.g. 225-1, 225-4 and 225-16, arranged in pairs. The core legs extend through the PCB and are surrounded by the respective electrical windings as described above and in the Racetrack patent. In the layout shown, it will be appreciated that each output cell occupies a single core leg, that pairs of output cells share core pieces, and that output terminations 312, 313 are shared by adjacent output cells. For example, the windings of output cell 250-1 surround core leg 225-1; output cells 250-9 and 250-10 share a core piece 316 (shown on top) and another core piece (not shown) on the bottom of the module; cells 250-13, 250-14, 250-15, and 250-16 are all connected to the positive output termination 312 between them (as well as other positive output terminations), etc. It will be appreciated from the layout of Fig. 3, that there is a natural output termination pitch that results from the multi-cell output structure of the converter. For example, the lateral (or longitudinal) spacing between output termination columns is a function of the individual output cell geometry. Similarly, the vertical (or latitudinal) spacing between output termination rows is a function of, among other things, the individual cell geometry.

[043] For supplying the extremely large and fast load changes common for modern large format semiconductor devices, the interconnection impedance between the converter output terminals and the semiconductor device should be as low as possible. One way of reducing the impedance of the power connections between the power converter and the semiconductor device is to use interdigitated vertical power connections. Interdigitated vertical power connections are shown in Figs. 15A and 15B of the MCM Disclosure and described as preferably being spatially arranged to allow routing of signals laterally between the vertical power connections. However, to minimize the parasitic impedance of the vertical power connections to the semiconductor device, the number of connections should be large, the polarities should be alternating between

adjacent connections, i.e., interdigitated, the spacing between adjacent connections should be small, and the length of the connections should be minimized. Thus, the ideal cell size or pitch in a multi-cell converter for powering large semiconductor loads may be advantageously minimized. Using the SAC topology, a multi-cell fixed-ratio converter as shown in Figs. 2 and 3 operating at approximately 2 MHz may have a cell pitch of approximately 3.5 mm. As shown in Fig. 3, the polarities of the output terminations of the multi-cell converter alternate within each column, e.g. proceeding from top to bottom within each column, the output terminations are common (313), positive (312), common (313). While the interdigitation at the scale of the converter cell pitch is beneficial, it may not provide a sufficiently low impedance for the extremely large and fast load steps produced by many modern processors, GPUs, and ASICs.

[044] Referring to Fig. 4, a generalized example of a semiconductor die, semiconductor package, or intervening substrate, 30 is shown having a preferred array of contacts 304 through which power may be delivered vertically to circuitry on the die, package, or substrate 30. The array of contacts may be referred to as a vertical semiconductor power grid 301. Fig. 4 is representative of vertical power connections to a semiconductor device, which may be in the form of a wafer, a single die, a semiconductor package including one or more dice, or a substrate to which a wafer, one or more semiconductor die and/or semiconductor packages is/are mounted and reference to the die 30 should be understood to refer to any such configuration. As shown, a large multiplicity of closely-spaced and interdigitated grid contacts 304 are provided in an array of eight rows, e.g. rows 303, and thirty two columns, e.g. columns 302, to both satisfy the current requirements of, and reduce the parasitic impedance of vertical power connections to, the semiconductor die 30. The vertical connections in the grid 301 are large in number, closely-spaced, e.g. on a 1mm or less pitch, and interdigitated, i.e. the polarity of each individual power connection 304 is preferably opposite to the polarity of its adjacent power connections to provide a very low interconnection impedance. The contacts in the grid 301 as shown include a vertical (in plan view) spacing, e.g. D_v 305, between rows 303 and a horizontal (in plan view) spacing, e.g. D_H 306, between adjacent columns 302. Although shown as a simple grid of contacts in uniform rows and columns, it will be appreciated that the contacts may be distributed in any manner necessary or desired for distributing power to the semiconductor device. To illustrate the difference between the power grid 301 of the semiconductor and the natural contact structure of the multi-cell power converter, the power grid in Fig. 4 is shown having the same footprint as the

multi-cell converter package of Fig. 3. The actual size of the semiconductor die, package, or intervening substrate, 30, the area in which the power grid is located, and the density of the contacts within the grid 301 may be a function of the semiconductor, package, or intervening substrate design. The configuration, size, and number of multi-cell converters 310 used to power the die 30 may be a function of the die current or power requirements, heat density, and available cooling. Generally, the size and spacing, e.g. D_V 305 and D_H 306 (preferably 1 mm or less), of the contacts, e.g. 304, in the power grid 301 should be much smaller than the spacing available for the power terminations 312, 313 of the multi-cell converter 310, e.g. 3.5 mm in the example described above.

[045] Figs. 5A and 5B respectively show exploded isometric top and bottom views of a converter assembly 390, which, as shown, includes the multi-cell converter 310 mated with an interconnection module 350. Fig. 5A shows the converter terminations 311, 312, 313, on surface 320 of the multi-cell converter 310 (discussed above in connection with Fig. 3). The interconnection module 350 may be used to establish the low impedance vertical connections to the semiconductor die 30 using the interdigitated contacts 304 in the power grid 301 to which the semiconductor die, package, or intervening substrate, 30, may preferably be adapted to mate, provide energy storage to the die 30 via the low impedance vertical connections, and translation to the power terminations of the multi-cell converter 310, which are fewer in number, larger, and spaced further apart than the interdigitated contacts 304 in the power grid. Thus the interconnection module 350 may provide electrical and mechanical translation between the converter output terminals arranged in a first pattern and the power input terminals of the semiconductor device arranged in a second pattern, while enabling very low impedance vertical connections. The interconnection module may further provide a mechanical platform allowing the die, package, or intervening substrate, 30 and the converter 310 to be assembled together providing signal connectivity and mechanical integrity.

[046] The interconnection module 350 as shown may include a PCB 353 and an optional layer of cured encapsulant 354. Electrically-conductive pads, e.g. pads 361, 362, and 363, may be formed on the surface 351 (Fig. 5B) of the cured encapsulant in locations configured as shown to align and mate with corresponding terminations 311, 312, 313 on surface 320 (Fig. 5A) of the multi-cell module 310. On the surface 352 of the interconnection module 350, which may

as shown be a surface of the PCB 353, conductive pads 364 (Fig. 5A) may be formed in a pattern configured to mate and align with corresponding contacts 304 in the power grid 301 (Fig. 4) of semiconductor die 30. It is important to note that the conductive pads are formed on a surface of the PCB 353 in a pattern that mates with similar contacts provided on the semiconductor die, wafer, package, or intervening substrate, 30 to provide the shortest possible connections between the two. Although the pads 364 are shown as a land grid area, solder balls, e.g. in a ball grid array, or other contact structures (e.g., conductive pins) may be used. Preferably, the contacts 364 are arranged to alternate between positive and common polarities (interdigitated) to reduce the parasitic inductance of the interconnections. Although shown as individual pads, a copper plane, e.g. connected to the negative or common terminals may be provided on PCB surface 353 with solder mask defining each individual pad and with isolation around the pads connected to the positive terminals.

[047] Internally, the PCB 353 may include a plurality of conductive layers separated by intervening dielectric layers. For example, the PCB 353 may include a conductive layer for each power plane (positive and common), a surface conductive layer for components and a second surface conductive layer on the opposite surface for contact pads 364. It may be advantageous to include one or more additional conductive layers for each of the power planes, e.g. to reduce interconnection resistance, in alternating layers to increase the capacitance between the power planes and reduce parasitic inductances. Referring to Fig. 6, a plan view of the component side of the PCB 353 is shown having conductive pads 371, 372, and 373 which form the foundation for the pads 361, 362, and 363 on the finished interconnection module; a multiplicity of capacitors 365 arranged between and among the pads 371, 372, and 373; and a temperature sensor 370. The capacitors 365, which provide energy storage for the die, can be part of filter circuitry electrically connected to the pads, e.g. pads 372 and 373 (Fig. 6) and pads 362 and 363 (Fig. 5B). The capacitors may be preferably mounted as shown on the opposite surface of the interconnection module PCB to avoid extending the vertical distance between the interconnection PCB and the semiconductor die 30. The components and pads as shown in Fig. 6 may be covered by encapsulant to form the surface 351 of the interconnection module. The electrical contacts 361, 362, and 363 may be formed on the surface 351 of the interconnection module 350 (cured encapsulant 354) by forming holes in the encapsulant 354 at the desired locations exposing the respective conductive pads 371, 372, and 373 on the PCB surface;

forming one or more metal layers on the surface 351 including in the holes to contact the conductive pads 371, 372, and 373 at the bottom of the holes; patterning the metal layers as desired, e.g. into strips 366, 367, and 368 (Fig. 5B) which may match the strips 316, 317, 318 on the converter 310 as shown; and optionally filling the metalized holes; as described in the Leadless and Trench disclosures. As shown in Fig. 5B, the strips 366, 367, and 368 are electrically isolated from one another. Preferably, the pads 371, 372, and 373 are formed on the external surface 351 of the PCB 353 and connected to the respective internal power planes using conductive vias to avoid the need for large clearance holes in the power planes.

[048] Some semiconductor devices may tend to concentrate current draw in specific regions of the die, e.g. in the centrally located contacts in the power grid. The interconnection module or multi-cell converter or both may be configured to force a desired current distribution on the semiconductor die, e.g. by providing a series resistance that is slightly lower in areas in which current flow is to be encouraged and slightly higher in areas where the current flow is to be reduced. In the interconnection module, for example, connections between the power planes and selected contact pads 364 may be configured to provide a series resistance that is slightly lesser, than the nominal average series resistance of the contacts 364, e.g. using additional vias near the selected contacts, or conversely provide a slightly greater resistance, using fewer vias or other PCB features, for contacts through which lower current flow is desired. By tailoring the series resistance of the pads 364 in selected areas of the grid compared to other areas, the interconnection module may compensate for the inequalities in current distribution on the die. It may however be preferable to modulate the equivalent output resistance of individual output cells directly to achieve a similar effect more efficiently and dynamically. Methods of controlling the output resistance of the converter are described in Vinciarelli, *Output Resistance Modulation in Power Converters*, U.S. Patent No. 6,934,166 issued August 23, 2005, assigned to VLT, Inc. of Andover, MA, and incorporated in its entirety here by reference (the “Modulation” patent); in Vinciarelli, *Adaptive Control of Resonant Power Converters*, U.S. Patent No. 10,020,752 issued July 10, 2018, assigned to VLT, Inc. of Andover, MA, and incorporated in its entirety here by reference (the “ASAC” patent); and in Vinciarelli, *Digital Control of Resonant Power Converters*, U.S. Patent No. 9,166,481 issued October 20, 2015, assigned to VLT, Inc. of Andover, MA, and incorporated in its entirety here by reference (the “Digital Controller” patent). For example, gate drive signals to the rectifier switches, SR1, SR2 (Fig. 2) of each output cell

(250: Fig. 2) or group of cells may be modulated to affect output resistance changes in each cell. For example, separate gate drive inputs may be provided for each cell or group of cells, or circuitry that attenuates the gate drive signal for specific output cells or groups of output cells may be provided. Alternatively, gate drive circuitry may be provided in the same package with the multi-cell converter. The gate drive modulation approach is preferable for the ability to dynamically reconfigure the output resistance of the cells or groups of cells during operation, e.g. in response to signals from the die. Alternatively, the circuitry may provide one-time gate drive programmability as a configuration option for more static compensation.

[049] A preferred method of manufacturing the converter assembly 390 will be described in connection with Figs. 7, 8A, 8B, and 9. Figure 7 shows a cross section of a portion of the converter assembly 390 revealing solder connections 391 between converter terminations 311 and 313 and interconnection pads 361 and 363, respectively. Holes 326 formed in the encapsulant 323 of the multi-cell converter 310 to form the contact terminations 311 and 313 are shown in broken lines extending from the surface 320 to the surface of PCB 321. Similarly, holes 356 formed in the encapsulant 354 of the interconnection module 350 to form the contact terminations 361 and 363 are shown in broken lines extending from the surface 351 to the surface of the PCB 353. As shown in Figs. 8A and 8B, a panel 340 comprising a plurality of the multi-cell converters 310 may be assembled with a panel 380 comprising a plurality of the interconnection modules 350 to form a panel-sandwich structure 400 comprising a plurality of the converter assemblies 390 prior to singulation. A side view of the panel-sandwich structure 400 is shown in Fig. 8B having solder balls attached (optional process step 618: Fig. 9) to contacts 364 as an alternative to the land grid array shown in Figs. 5A and 7. An adhesive, e.g. epoxy (not shown), may be dispensed between the solder connections 391 and preferably along the boundary lines between each individual module prior to forming the solder connections 391 between the two panels 340 and 380, e.g., on the surface of at least one of the panels 340 and 380 along the cut lines between the modules. The cured epoxy may provide mechanical strength and serve as a seal between the panels during singulation.

[050] A preferred manufacturing process flow will be described with reference to Fig. 9 which shows two parallel paths for fabrication of the power converter panel 340 and the interconnection panel 380 prior to step 621. The power converter panels may be prepared as

described in the Panel Mold, Leadless, and Trench disclosures as shown in steps 601 through 608 of Fig. 9. For example, the PCB panel may be assembled (601) and encapsulated (602) then lapped (603), holes drilled (604), e.g. holes 326 (Fig. 7), cleaned (605) and then metalized (606) to form the contacts, e.g. contacts 311, 312, 313 (Figs. 3, 5A, and 7). The metalization step 606 may include patterning the metal layer(s) on the surfaces, e.g. to form the conductive strips 316, 317, and 318 (Fig. 3) described above. Solder may then be dispensed onto the contacts, e.g. 311, 312, 313 (Fig. 5A) and epoxy dispensed in preselected locations between the contacts and preferably along the perimeter of each individual module, e.g. along the cut lines.

[051] The interconnection panels may be prepared in a similar manner as shown in steps 611 through 618 of Fig. 9. For example, the PCB panel may be assembled (611) and encapsulated (612), preferably on one side, and then lapped (613), holes drilled (604), e.g. holes 356 (Fig. 7), cleaned (615) and then metalized (616) to form the contacts, e.g. contacts 361, 362, 363 (Figs. 5B and 7) with pads, e.g. pads 371 and 373 on the surface of the PCB 353. The metalization step (616) may include patterning the metal layer(s) on the surface 351, e.g. to form the above-described conductive strips 366, 367, and 368 (Fig. 5B).

[052] The converter panels 340 and interconnection panels 380 may then be assembled together as shown in steps 621-625. The panels 340 and 380 may be aligned (621) with each other, e.g. using a fixture to maintain coplanarity of the panels during subsequent solder reflow (622) and epoxy cure (623). Afterward, the panel sandwich may be cut along predetermined cut lines to singulate the converter assemblies 390. Preferably, the cuts are made in a manner that leaves an epoxy connection between the converter module 310 and the interconnection module 350 along the entire periphery of the assembly 390 providing a seal during singulation and subsequent manufacturing steps.

[053] Although the above example of the multi-cell converter is based upon a fixed-ratio sine amplitude converter topology, other converter topologies may be used. For example, the converter assembly 390 may be implemented using a multiphase buck converter or switched capacitor converter, e.g. with each phase defining an individual output cell, together with the interconnection module 350 providing interconnection translation and optional filtering. Fixed ratio, regulating, or quasi-regulating topologies may be used.

[054] Similarly, although the above examples are based upon encapsulated multi-cell converter and interconnection modules, it will be appreciated that either or both may be open-frame assemblies. Referring to Figs. 10A and 10B, open frame examples of a multi-cell converter module 310-2 and interconnection module 350-2 are shown forming a converter assembly 390-2. As shown, vertical connections between the converter PCB 321-2 and the interconnection module PCB 353-2 may be made using conductive pillars or pins, e.g. electrically conductive pillars 361-2, 362-2, and 363-2 extending from interconnection module PCB 353-2 to mating pads, e.g. conductive pads 311-2, 312-2, and 313-2 on the surface of the multi-cell converter PCB 321-2. Although the pillars and pads are shown respectively as part of the interconnection module 350-2 and multi-cell converter module 310-2, it will be understood that they may be reversed. The pillars 361-2, 362-2, and 363-2 may provide vertical spacing and mechanical support between the two modules. The conductive pads 364 may be arranged on the surface of PCB 353-2 as described above in connection with Fig. 5A in a pattern configured to mate and align with corresponding contacts 304 in the power grid 301 (Fig. 4) of the semiconductor die, wafer, package, or intervening substrate 30 as described above. The capacitors should be mounted as shown on the surface of the interconnection module PCB opposite the conductive pads 364 to avoid extending the vertical distance between the interconnection PCB and the semiconductor die 30.

[055] The converter assembly 390 is shown comprising separate multi-cell converter 310 and interconnection module 350 assemblies which accommodates the three-dimensional nature of the SAC topology shown (components and cores mounted to both sides of the converter PCB 321). However, with other converter topologies that may be amenable to more planar designs, e.g. analogous to a system in a package that occupies only one side of the PCB, the converter and interconnection module may use the same PCB, i.e. occupy a single PCB, or may have their respective PCBs in contact or assembled together, e.g. laminated, to form a composite PCB structure, optionally with internal spaces for the energy storage capacitors.

[056] Referring to Fig. 11, a system 500 is shown comprising a plurality of multi-cell converter assemblies, e.g. 390A, 390B, and 390D, each comprising a multi-cell converter, e.g. 310A, and an interconnection module, e.g. 350A, mounted to a portion of a very large scale semiconductor package 30-2, which may carry a plurality of semiconductor dice or a

semiconductor wafer. Mounting hardware including screws, e.g. screw 501, and standoffs or pressure cylinders, e.g. 502, as shown may be included to provide mechanically robust connections in the assembly. Although the package 30-2 is shown schematically having a plurality of contacts 364 arranged in a plurality of power grids, e.g. grid 301C, it will be appreciated that the size, number, and configurations of the grids and the converter assemblies may be adapted to a large variety of configurations. In the example 500 of Fig. 11, a single almost continuous conductive shield is provided on the surfaces of the interconnection module 350A and multi-cell converter 310A. Additionally, input power and control signals may be applied to input terminals 311 on the top surface of the converter assemblies 390 as shown in Fig. 11. Some signals may pass through the converter assembly 390 from the top to the semiconductor package, e.g. using dedicated terminals 300 as shown. Using the manufacturing processes described above and in the panel mold disclosures, the converter assembly package may be arranged as shown to provide clearance for mounting hardware while still maximizing utilization of the available area for active circuitry, e.g. conversion cells and energy storage.

[057] A number of embodiments of the invention have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the invention. For example, the converter assembly 390 may be implemented using other converter topologies for the multi-cell converter 310, e.g. multiphase buck converter topologies or switched capacitor converter topologies may be used, e.g. with each phase defining an individual output cell. Fixed ratio, regulating, or quasi-regulating topologies may be used. The multi-cell converter described above is an example in which the driver and switch control circuitry are packaged separately, however the control and drive electronics also may be incorporated into the multi-cell converter. The interconnection module 350 may be fabricated using other methods to provide interconnection translation and optional filtering. Although a single assembly 390 is shown for powering the ASIC 30, it will be appreciated a plurality of modules may be used. Similarly, a single interconnection module may be used to mount a plurality of converter modules 310 to the ASIC 30 or substrate to which the ASIC is mounted. The assembly 390 may be mounted directly to a wafer, a die, a substrate on which the substrate or die is mounted, or a package on or in which the wafer or die is mounted. Although the preferred method for manufacturing the converter assembly forms the panel sandwich prior to

singulation, the interconnection module may be fabricated and singulated prior to attachment to one or more converter modules.

[058] In this document, the term “x-input terminal” may be used to refer to the input terminal of the interconnection module 350, so as to differentiate from the input terminal of the converter 310 and the input terminal of the ASIC 30. The ‘x’ in the “x-input terminal” does not represent a number. Two input terminals of the interconnection module 350 may be referred to as two x-input terminals. Similarly, the term “x-output terminal” may be used to refer to the output terminal of the interconnection module 350, so as to differentiate from the output terminal of the converter 310 and the output terminal of the ASIC 30. The ‘x’ in the “x-output terminal” does not represent a number. Two output terminals of the interconnection module 350 may be referred to as two x-output terminals.

[059] Accordingly, other embodiments are within the scope of the following claims.

Electronic Patent Application Fee Transmittal				
Application Number:				
Filing Date:				
Title of Invention:		DELIVERING POWER TO SEMICONDUCTOR LOADS		
First Named Inventor/Applicant Name:		PATRIZIO VINCIARELLI		
Filer:		Rex I. Huang/anne roy		
Attorney Docket Number:		00614-0183001		
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
UTILITY APPLICATION FILING	1011	1	300	300
UTILITY SEARCH FEE	1111	1	660	660
UTILITY EXAMINATION FEE	1311	1	760	760
Pages:				
Claims:				
CLAIMS IN EXCESS OF 20	1202	25	100	2500
INDEPENDENT CLAIMS IN EXCESS OF 3	1201	3	460	1380
Miscellaneous-Filing:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				5600

Electronic Acknowledgement Receipt

EFS ID:	34570136
Application Number:	16218418
International Application Number:	
Confirmation Number:	1092
Title of Invention:	DELIVERING POWER TO SEMICONDUCTOR LOADS
First Named Inventor/Applicant Name:	PATRIZIO VINCIARELLI
Customer Number:	26214
Filer:	Rex I. Huang/anne roy
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0183001
Receipt Date:	12-DEC-2018
Filing Date:	
Time Stamp:	21:33:10
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$5600
RAM confirmation Number	121318INTEFSW00007035061050
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal Letter	0183trans.pdf	115575	no	2
			da377e1bc7fd0fe324a0f0146b380da4f50bb7bb		
Warnings:					
Information:					
2	Nonpublication request from applicant	nonpub5.pdf	141446	no	1
			ceed56743522dad6aa39b1937b671c5fdcc6a3e		
Warnings:					
Information:					
3	Application Data Sheet	0184001ads.pdf	1822764	no	8
			5792ec98e0e50c7d1ab417c34fda810f06450fb8		
Warnings:					
Information:					
4	Drawings-only black and white line drawings	12202536_Document_001.pdf	641818	no	13
			31fa7782bd299664d27ca7e04050a853d061049f		
Warnings:					
Information:					
5		0183app.pdf	266486	yes	36
			be2ced5fee2dea63a612c751f7ebccda114d16a5		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Abstract		36	36	
	Claims		25	35	
	Specification		1	24	

Warnings:					
Information:					
6	Fee Worksheet (SB06)	fee-info.pdf	38000	no	2
			467eb2b98e7e8757c1c8ba0080bd38e0d7750435		
Warnings:					
Information:					
Total Files Size (in bytes):			3026089		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	Unknown
Application No. :	16/218,418	Examiner :	Unknown
Filed :	December 12, 2018	Conf. No. :	1092
Title :	DELIVERING POWER TO SEMICONDUCTOR LOADS		

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

PRELIMINARY AMENDMENT

Prior to examination, please amend the application as indicated on the following pages.

Electronic Patent Application Fee Transmittal				
Application Number:		16218418		
Filing Date:				
Title of Invention:		Delivering Power To Semiconductor Loads		
First Named Inventor/Applicant Name:		Patrizio Vinciarelli		
Filer:		Rex I. Huang/Lisa Gray		
Attorney Docket Number:		00614-0183001		
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
CLAIMS IN EXCESS OF 20	1202	2	100	200
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				200

Electronic Acknowledgement Receipt

EFS ID:	34610936
Application Number:	16218418
International Application Number:	
Confirmation Number:	1092
Title of Invention:	Delivering Power To Semiconductor Loads
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Lisa Gray
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0183001
Receipt Date:	17-DEC-2018
Filing Date:	
Time Stamp:	17:15:18
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$200
RAM confirmation Number	121818INTEFSW00004800061050
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		00614-0183001prelimamend.pdf	132656	yes	14
			c028105fd3af9cfe3db6d12283c661209c37ea68		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Applicant Arguments/Remarks Made in an Amendment		14	14	
	Claims		2	13	
	Preliminary Amendment		1	1	
Warnings:					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	30262	no	2
			14cfafc899fd57a5bd9c340918fb194a01e6d7f		
Warnings:					
Information:					
Total Files Size (in bytes):			162918		

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

First Named Inventor : Patrizio Vinciarelli
Application No. : 16/218,418
Filed : December 12, 2018
Page : 14 of 14

Attorney Docket: 00614-0183001 / 0269-US-1

REMARKS

Applicant asks that all claims be examined in view of the amendments to the claims.

The fee in the amount of \$200 for excess claims is being paid with this reply on the Electronic Filing System. In addition, please apply any other necessary charges or credits to Deposit Account 06-1050, referencing the above attorney docket number.

Respectfully submitted,

Date: December 17, 2018_____

/Rex I. Huang/_____

Rex I. Huang

Reg. No. 57,661

Customer Number 26214
Fish & Richardson P.C.
Telephone: (617) 542-5070
Facsimile: (877) 769-7945

Amendments to the Claims:

This listing of claims replaces all prior versions and listings of claims in the application:

Listing of Claims:

1. (Original) A method of powering a semiconductor device having power input terminals, the method comprising:

providing a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells, each conversion cell including one or more converter output terminals, the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage, the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals;

providing an interconnection module having x-input terminals arranged on a first side of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a surface of a second opposite side of the interconnection module in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;

assembling the power conversion module to the first side of the interconnection module with each converter output terminal mated with and electrically connected to a respective x-input terminal;

wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern;

wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer.

2. (Original) The method of claim 1 wherein the interconnection module further comprises filter circuitry connected to the x-input terminals and the x-output terminals.

3. (Original) The method of claim 2 wherein the interconnection module further comprises a multilayer printed circuit board ("PCB"), and the filter circuitry comprises a plurality of capacitors mounted to a first surface of the PCB on the first side of the interconnection module.

4. (Original) The method of claim 2 wherein the second spacing is less than the first spacing.

5. (Original) The method of claim 2 wherein each conversion cell further comprises a first dimension and the first spacing is configured as a function of the first dimension.

6. (Original) The method of claim 5 wherein each conversion cell further comprises a second dimension, the first spacing further comprises a latitudinal spacing and a longitudinal spacing, and the latitudinal spacing is a function of the first dimension and the longitudinal spacing is a function of the second dimension.

7. (Original) The method of claim 1, further comprising configuring the conversion circuitry as a buck converter having a plurality of phases with each conversion cell configured to operate as a selected one of the phases.

8. (Original) The method of claim 1, further comprising configuring the conversion circuitry to convert power received from an input to the power conversion module at an input

voltage, V_{IN} , and input current, I_{IN} , for delivery to the converter output terminals at an output voltage, V_{OUT} , and an output current, I_{OUT} , using a transformation ratio, K , where $K = V_{OUT} / V_{IN}$, that is essentially constant, subject to an equivalent output resistance.

9. (Original) The method of claim 8, further comprising encapsulating the conversion circuitry to form a first surface of the power conversion module; encapsulating the interconnection module to form a first surface on the first side of the interconnection module; affixing the first surface of the power conversion module to the first surface of the interconnection module with the converter output terminals and the x-input terminals aligned using an adhesive, and forming solder connections between respective one of the output terminals and the x-input terminals.

10. (Original) The method of claim 1 wherein the interconnection module further comprises a multilayer printed circuit board ("PCB") and the method further comprises mounting at least one solder ball to the PCB for each x-output terminal.

11. (Original) The method of claim 1 wherein the interconnection module further comprises a multilayer printed circuit board ("PCB") and the method further comprises mounting at least one conductive pin to the PCB for each x-output terminal.

12. (Original) The method of claim 1 wherein the interconnection module further comprises a multilayer printed circuit board ("PCB") and the method further comprises providing the x-output terminals in a land grid array on the PCB.

13. (Original) The method of claim 1, further comprising attaching the semiconductor device to the interconnection module with the x-output terminals aligned with and electrically connected to respective ones of the power input terminals; and wherein the connections between the x-output terminals and the power input terminals form a fourth layer in the module sandwich.

14. (Original) The method of claim 1, further comprising

assembling a first panel comprising a plurality of unsingulated power conversion modules;

assembling a second panel comprising a plurality of unsingulated interconnection modules;

wherein the assembling of the power conversion module and the interconnection module is performed using the first and second panels prior to singulation to form a panel sandwich and further comprises dispensing epoxy between the first and second panels and forming solder connections between each converter output terminal and each respective x-input terminal; and further comprising:

cutting the panel sandwich along predetermined cut lines to singulate a plurality of the module sandwiches.

15. (Original) The method of claim 14 wherein the dispensing epoxy further comprises dispensing epoxy on the surface of at least one of the panels along the cut lines.

16. (Original) The method of claim 15, further comprising reflowing solder and curing the epoxy.

17. (Original) A method of making power converters, the method comprising:
assembling an encapsulated panel including a printed circuit board ("PCB") having electronic components mounted to the PCB and a plurality of conductive features enclosed within the panel and unexposed to an exterior surface of the panel, the panel comprising one or more power conversion modules having boundaries defined by one or more predetermined cut lines;

configuring each of the one or more power conversion modules in a multi-cell structure having a plurality of conversion cells, each conversion cell including one or more converter output nodes arranged in a first pattern having a first spacing between the output nodes, and wherein each output node is electrically connected to a respective conductive feature;

selectively forming a plurality of holes in the exterior surface of the panel, each hole being formed at a respective predetermined location aligned with a respective output node, spaced apart from the cut lines, and exposing within the hole the respective conductive feature;

selectively forming one or more conductive metal layers on the exterior surface of the panel including within the plurality of holes, the one or more conductive metal layers within each hole being in electrical contact with the respective conductive feature, to form a plurality of electrical contacts on the exterior surface of the panel;

patterning the one or more conductive metal layers on the exterior surface to electrically isolate at least one electrical contact from one or more of the plurality of electrical contacts;

cutting the panel along the one or more cut lines to singulate the one or more power conversion modules, each singulated power conversion module having a respective plurality of the electrical contacts formed on the exterior surface of the module.

18. (Original) The method of claim 17 further comprising selectively filling the plurality of holes with solder.

19. (Original) The method of claim 18, further comprising planarizing the solder to a predetermined height relative to the exterior surface.

20. (Original) The method of claim 17 wherein each conversion cell further comprises a first dimension and a second dimension, the first spacing further comprises a latitudinal spacing and a longitudinal spacing, and the latitudinal spacing is a function of the first dimension and the longitudinal spacing is a function of the second dimension.

21. (Original) The method of claim 17, further comprising:
providing an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern, the second pattern having a second spacing between the x-output terminals, the second spacing being

different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;

assembling the first surface of the interconnection module to the power conversion module with each converter output terminal mated with and electrically connected to a respective x-input terminal,

wherein the power conversion module and the interconnection module are stacked to form a sandwich structure including the power conversion module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer.

22. (Original) The method of claim 21, further comprising assembling the sandwich structure to a substrate with the x-output terminals aligned with and electrically connected to respective terminals on a surface of the substrate.

23. (Original) The method of claim 22 wherein the substrate includes a semiconductor device electrically connected to receive power from the power conversion module.

24. (Original) A method of making electronic assemblies, the method comprising:
providing an electronic module having electronic circuitry including circuit terminals arranged in a first pattern having a first spacing between the circuit terminals;

providing an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of circuit terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with external terminals of a third electronic device, the second pattern having a second spacing between the x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;

assembling the electronic module to the first surface of the interconnection module with each circuit terminal mated with and electrically connected to a respective x-input terminal;

wherein the interconnection module is configured to provide electrical and mechanical translation between the circuit terminals arranged in the first pattern and the external terminals of the third electronic device arranged in the second pattern;

wherein the electronic module and the interconnection module are stacked to form a module sandwich including the electronic module in a first layer, electrical connections between the circuit terminals and the x-input terminals in a second layer, and the interconnection module in a third layer;

assembling a first panel comprising a plurality of unsingulated electronic modules;

assembling a second panel comprising a plurality of unsingulated interconnection modules;

wherein the assembling of the electronic module and the interconnection module is performed using the first and second panels prior to singulation to form a panel sandwich and further comprises dispensing a curable adhesive compound between the first and second panels and forming solder connections between each circuit terminal and each respective x-input terminal; and

cutting the panel sandwich along predetermined cut lines to singulate a plurality of the module sandwiches.

25. (Original) The method of claim 24 wherein the dispensing the curable adhesive compound further comprises dispensing the curable adhesive compound on the surface of at least one of the panels along the cut lines.

26. (Original) The method of claim 24, further comprising reflowing solder and curing the adhesive compound.

27. (Original) A method of forming modular circuit assemblies, the method comprising:

assembling first and second encapsulated panels each including a first multilayer printed circuit board ("PCB") having a plurality of conductive layers, a first plurality of electronic components mounted to a first surface of the PCB, and a first layer of cured encapsulant

covering the first plurality of electronic components and the surface of the PCB, the first layer of cured encapsulant forming a first exterior surface of the encapsulated panel, the encapsulated panel comprising a plurality of unsingulated electronic modules, each electronic module having perimeter boundaries defined by one or more predetermined cut lines;

selectively forming a plurality of terminal holes in the first exterior surface of each panel through the first layer at predetermined locations within the perimeter boundaries of each electronic module, each terminal hole being spaced apart from the cut lines and exposing within the terminal hole a respective conductive feature;

selectively forming one or more conductive metal layers on the first exterior surface of each panel including within the plurality of terminal holes, the one or more conductive metal layers within each terminal hole being in electrical contact with the respective conductive feature;

patterning the one or more conductive metal layers on the exterior surface, to form a plurality of electrical contacts on the exterior surface of each panel electrically isolated from at least one other electrical contact in the plurality of electrical contacts;

assembling the first and second encapsulated panels with electrical contacts on the first panel aligned and mated with respective electrical contacts on the second panel to form a panel stack, the panel stack comprising a plurality of un-singulated module stacks, each module stack including an unsingulated electronic module in the first panel and a respective unsingulated electronic module in the second panel; and

cutting the panel stack along the one or more cut lines to singulate the plurality of electronic module stacks.

28. (Original) The method of claim 27, further comprising dispensing a curable adhesive compound on the surface of at least one of the panels along the cut lines.

29. (Original) The method of claim 28, further comprising reflowing solder and curing the adhesive compound.

30. (Original) An apparatus comprising:

a semiconductor device having power input terminals;

a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells, each conversion cell including one or more converter output terminals, the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage, the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals; and

an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;

the converter output terminals of the power conversion module being mated with and electrically connected to respective x-input terminals of the interconnection module;

wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern;

wherein the power conversion module and the interconnection module are stacked with the power conversion module located at a first level, electrical connections between the converter output terminals and the x-input terminals at a second level, and the interconnection module at a third level.

31. (Original) The method of claim 30 wherein the semiconductor device comprises a die.

32. (Original) The method of claim 30 wherein the semiconductor device comprises a wafer.

33. (Original) The method of claim 30 wherein the semiconductor device comprises a substrate.

34. (Original) The method of claim 30 wherein the semiconductor device comprises a semiconductor die or package mounted to a substrate.

35. (Original) A method of powering a semiconductor device having power input terminals, the method comprising:

providing a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells, each conversion cell including one or more converter output terminals, the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage, the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals;

providing an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals, the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;

providing a plurality of conductive pillars for electrically connecting the converter output terminals with respective x-input terminals;

assembling the power conversion module to the interconnection module with each converter output terminal aligned with and electrically connected to a respective x-input terminal by a respective one of the conductive pillars;

wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern;

wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, the conductive

pillars connecting the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer; and

wherein the x-output terminals are arranged with alternating polarities to reduce interconnection inductance.

36. (Original) The method of claim 1 further comprising arranging the x-output terminals with alternating polarities.

37. (Original) The method of claim 1 further comprising arranging the x-output terminals with polarities that alternate between adjacent columns.

38. (Original) The method of claim 1 further comprising arranging the x-output terminals with polarities that alternate between adjacent rows.

39. (Original) The method of claim 1 further comprising arranging the x-output terminals with polarities that alternate between adjacent x-output terminals.

40. (Original) The method of claim 1 further comprising arranging the converter output terminals with alternating polarities.

41. (Original) The method of claim 1 further comprising arranging the converter output terminals with polarities that alternate between rows.

42. (Original) The method of claim 1 further comprising arranging the converter output terminals with polarities that alternate between columns.

43. (Original) The method of claim 1 further comprising arranging the converter output terminals with polarities that alternate between adjacent converter output terminals.

44. (Original) The method of claim 35 further comprising arranging the x-output terminals with alternating polarities.

45. (Original) The method of claim 35 further comprising arranging the converter output terminals with alternating polarities.

46. (New) The method of claim 39, further comprising assembling the sandwich structure to a substrate with the x-output terminals aligned with and electrically connected to respective terminals on a surface of the substrate, the substrate including a semiconductor device electrically connected to receive power from the power conversion module; and

further comprising conducting heat vertically from the converter module through the sandwich structure including through the interconnection module and substrate.

47. (New) The method of claim 35, further comprising assembling the sandwich structure to a substrate with the x-output terminals aligned with and electrically connected to respective terminals on a surface of the substrate, the substrate including a semiconductor device electrically connected to receive power from the power conversion module; and

further comprising conducting heat vertically from the converter module through the sandwich structure and out the top of the substrate.



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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001

CONFIRMATION NO. 1092

26214
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

INFORMAL NOTICE



Date Mailed: 01/11/2019

INFORMATIONAL NOTICE TO APPLICANT

Applicant is notified that the above-identified application contains the deficiencies noted below. No period for reply is set forth in this notice for correction of these deficiencies. However, if a deficiency relates to the inventor's oath or declaration, the applicant must file an oath or declaration in compliance with 37 CFR 1.63, or a substitute statement in compliance with 37 CFR 1.64, executed by or with respect to each actual inventor no later than the expiration of the time period set in the "Notice of Allowability" to avoid abandonment. See 37 CFR 1.53(f).

The item(s) indicated below are also required and should be submitted with any reply to this notice to avoid further processing delays.

- A properly executed inventor's oath or declaration has not been received for the following inventor(s):
Patrizio Vinciarelli

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/fasrat/

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
16/218,418

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	47 minus 20 =	* 27
INDEPENDENT CLAIMS (37 CFR 1.16(h))	6 minus 3 =	* 3
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

OR

RATE(\$)	FEE(\$)
N/A	300
N/A	660
N/A	760
x 100 =	2700
x 460 =	1380
	0.00
	0.00
TOTAL	5800

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
16/218,418	12/12/2018	2833	5960	00614-0183001	47	6

CONFIRMATION NO. 1092

FILING RECEIPT

26214
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022



Date Mailed: 01/11/2019

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Patrizio Vinciarelli, Boston, MA;

Applicant(s)

VLT, Inc., Andover, MA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CIP of 15/616,288 06/07/2017
which is a CIP of 15/091,346 04/05/2016 PAT 10158357

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 01/04/2019

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 16/218,418**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

Title

Delivering Power To Semiconductor Loads

Preliminary Class

439

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

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Title 37, Code of Federal Regulations, 5.11 & 5.15

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The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

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No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

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To: PATDOCTC@fr.com,,
From: PAIR_eOfficeAction@uspto.gov
Cc: PAIR_eOfficeAction@uspto.gov
Subject: Private PAIR Correspondence Notification for Customer Number 26214

Jan 11, 2019 04:05:53 AM

Dear PAIR Customer:

FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022
UNITED STATES

The following USPTO patent application(s) associated with your Customer Number, 26214 , have new outgoing correspondence. This correspondence is now available for viewing in Private PAIR.

The official date of notification of the outgoing correspondence will be indicated on the form PTOL-90 accompanying the correspondence.

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Application	Document	Mailroom Date	Attorney Docket No.
16218418	M327	01/11/2019	00614-0183001
	APP.FILE.REC	01/11/2019	00614-0183001

To view your correspondence online or update your email addresses, please visit us anytime at <https://sportal.uspto.gov/secure/myportal/privatepair>.

If you have any questions, please email the Electronic Business Center (EBC) at EBC@uspto.gov with 'e-Office Action' on the subject line or call 1-866-217-9197 during the following hours:

Monday - Friday 6:00 a.m. to 12:00 a.m.

Thank you for prompt attention to this notice,

UNITED STATES PATENT AND TRADEMARK OFFICE
PATENT APPLICATION INFORMATION RETRIEVAL SYSTEM

POWER OF ATTORNEY BY APPLICANT

I hereby revoke all previous powers of attorney given in the application identified in either the attached transmittal letter or the boxes below.

Application Number	Filing Date
16/218,418	12-12-2018

(Note: The boxes above may be left blank if information is provided on the Transmittal.)

- ☒ I hereby appoint the Patent Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter, which may be an application filed after the date of this power of attorney, or identified above:

26214

OR

- ☐ I hereby appoint Practitioner(s) named in the attached list as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the patent application referenced in the attached transmittal letter, which may be an application filed after the date of this power of attorney, or identified above.

Please recognize or change the correspondence address for the application identified in the attached transmittal letter to:

- ☐ The address associated with the above-mentioned Customer Number

OR

- ☒ The address associated with Customer Number: **26214**

OR

Firm or Individual Name					
Address					
City		State		Zip	
Country					
Telephone		Email			

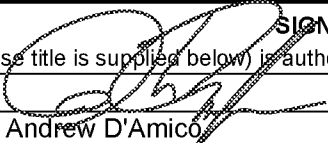
I am the Applicant (if the Applicant is a juristic entity, list the Applicant name in the box):

VLT, Inc.

- ☐ Inventor or Joint Inventor (title not required below)
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)
- ☒ Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)
- ☐ Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document) (provide signer's title if applicant is a juristic entity)

SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).

Signature		Date (Optional)	
Name	Andrew D'Amico		
Title	V.P., Intellectual Property		

NOTE: Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. If more than one applicant, use multiple forms.

☐ *Total of _____ forms are submitted.

Electronic Acknowledgement Receipt

EFS ID:	34855260
Application Number:	16218418
International Application Number:	
Confirmation Number:	1092
Title of Invention:	Delivering Power To Semiconductor Loads
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Rita Liston
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0183001
Receipt Date:	15-JAN-2019
Filing Date:	12-DEC-2018
Time Stamp:	11:52:14
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		00614_0183001_Trans_POA.pdf	151772	yes	2
			817084a317e36cbc7b672174d73399b9cae64f58		

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Transmittal Letter	1	1
	Power of Attorney	2	2
Warnings:			
Information:			
Total Files Size (in bytes):		151772	
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.			

TRANSMITTAL FOR POWER OF ATTORNEY TO ONE OR MORE REGISTERED PRACTITIONERS

Application Number	16/218,418
Filing Date	December 12, 2018
First Named Inventor	Patrizio Vinciarelli
Title	DELIVERING POWER TO SEMICONDUCTOR LOADS
Art Unit	
Examiner Name	
Attorney Docket Number	00614-0183001

SIGNATURE of Applicant or Patent Practitioner

Signature	/Rex I. Huang/	Date (Optional)	
Name	Rex I. Huang	Registration Number	57,661
Title (if Applicant is a juristic entity)			
Applicant Name (if Applicant is a juristic entity)			

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. If more than one applicant, use multiple forms.

☐ *Total of _____ forms are submitted.



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UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
16/218,418	12/12/2018	Patrizio Vinciarelli	00614-0183001

26214
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

CONFIRMATION NO. 1092
POA ACCEPTANCE LETTER



Date Mailed: 01/22/2019

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 01/15/2019.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/qnghuyen/

To: PATDOCTC@fr.com,,
From: PAIR_eOfficeAction@uspto.gov
Cc: PAIR_eOfficeAction@uspto.gov
Subject: Private PAIR Correspondence Notification for Customer Number 26214

Jan 22, 2019 03:24:38 AM

Dear PAIR Customer:

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P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022
UNITED STATES

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Application	Document	Mailroom Date	Attorney Docket No.
16218418	N570	01/22/2019	00614-0183001

To view your correspondence online or update your email addresses, please visit us anytime at <https://portal.uspto.gov/secure/myportal/privatepair>.

If you have any questions, please email the Electronic Business Center (EBC) at EBC@uspto.gov with 'e-Office Action' on the subject line or call 1-866-217-9197 during the following hours:

Monday - Friday 6:00 a.m. to 12:00 a.m.

Thank you for prompt attention to this notice,

UNITED STATES PATENT AND TRADEMARK OFFICE
PATENT APPLICATION INFORMATION RETRIEVAL SYSTEM

DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR 1.76)

Title of Invention	DELIVERING POWER TO SEMICONDUCTOR LOADS
--------------------	---

As the below named inventor, I hereby declare that:

This declaration is directed to: ☐ The attached application, or ☒ United States application or PCT international application number 16/218,418

filed on December 12, 2018.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

WARNING:

Petitioner/applicant is cautioned to avoid submitting personal information in documents filed in a patent application that may contribute to identity theft. Personal information such as social security numbers, bank account numbers, or credit card numbers (other than a check or credit card authorization form PTO-2038 submitted for payment purposes) is never required by the USPTO to support a petition or an application. If this type of personal information is included in documents submitted to the USPTO, petitioners/applicants should consider redacting such personal information from the documents before submitting them to the USPTO. Petitioner/applicant is advised that the record of a patent application is available to the public after publication of the application (unless a non-publication request in compliance with 37 CFR 1.213(a) is made in the application) or issuance of a patent. Furthermore, the record from an abandoned application may also be available to the public if the application is referenced in a published application or an issued patent (see 37 CFR 1.14). Checks and credit card authorization forms PTO-2038 submitted for payment purposes are not retained in the application file and therefore are not publicly available.

LEGAL NAME OF INVENTOR

Inventor: Patrizio Vinciarelli Date (Optional): Jan. 22, 2019

Signature: 

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 1 minute to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Electronic Patent Application Fee Transmittal

Application Number:	16218418			
Filing Date:	12-Dec-2018			
Title of Invention:	Delivering Power To Semiconductor Loads			
First Named Inventor/Applicant Name:	Patrizio Vinciarelli			
Filer:	Rex I. Huang/Rita Liston			
Attorney Docket Number:	00614-0183001			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
LATE FILING FEE FOR OATH OR DECLARATION	1051	1	160	160
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				160

Electronic Acknowledgement Receipt

EFS ID:	35279266
Application Number:	16218418
International Application Number:	
Confirmation Number:	1092
Title of Invention:	Delivering Power To Semiconductor Loads
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Rita Liston
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0183001
Receipt Date:	28-FEB-2019
Filing Date:	12-DEC-2018
Time Stamp:	11:01:23
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 160
RAM confirmation Number	022819INTEFSW00012710061050
Deposit Account	061050
Authorized User	Rita Liston

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:
37 CFR 1.21 (Miscellaneous fees and charges)

--

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Oath or Declaration filed	00614-0183001_Signed_Dec.pdf	736746	no	1
			10f51e072bf08bd2f8b6657ac4f853f435f737d4		

Warnings:

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	30311	no	2
			04e571cfd548e5782e4ec728bcacced71ae70b		

Warnings:

Information:

Total Files Size (in bytes):			767057
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	Unknown
Application No. :	16/218,418	Examiner :	Unknown
Filed :	December 12, 2018	Conf. No. :	1092

Title : DELIVERING POWER TO SEMICONDUCTOR LOADS

MAIL STOP PGPUB-DRAWINGS

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

TRANSMITTAL OF FORMAL DRAWINGS

Please substitute the enclosed eight (8) sheets of formal drawings for the corresponding drawings presently in the application. No new matter has been added.

Please apply any necessary charges or credits to Deposit Account 06-1050, referencing the above attorney docket number.

Respectfully submitted,

Date: April 2, 2019_____

/Rex I. Huang/_____

Rex I. Huang
Reg. No. 57,661

Customer Number 26214
Fish & Richardson P.C.
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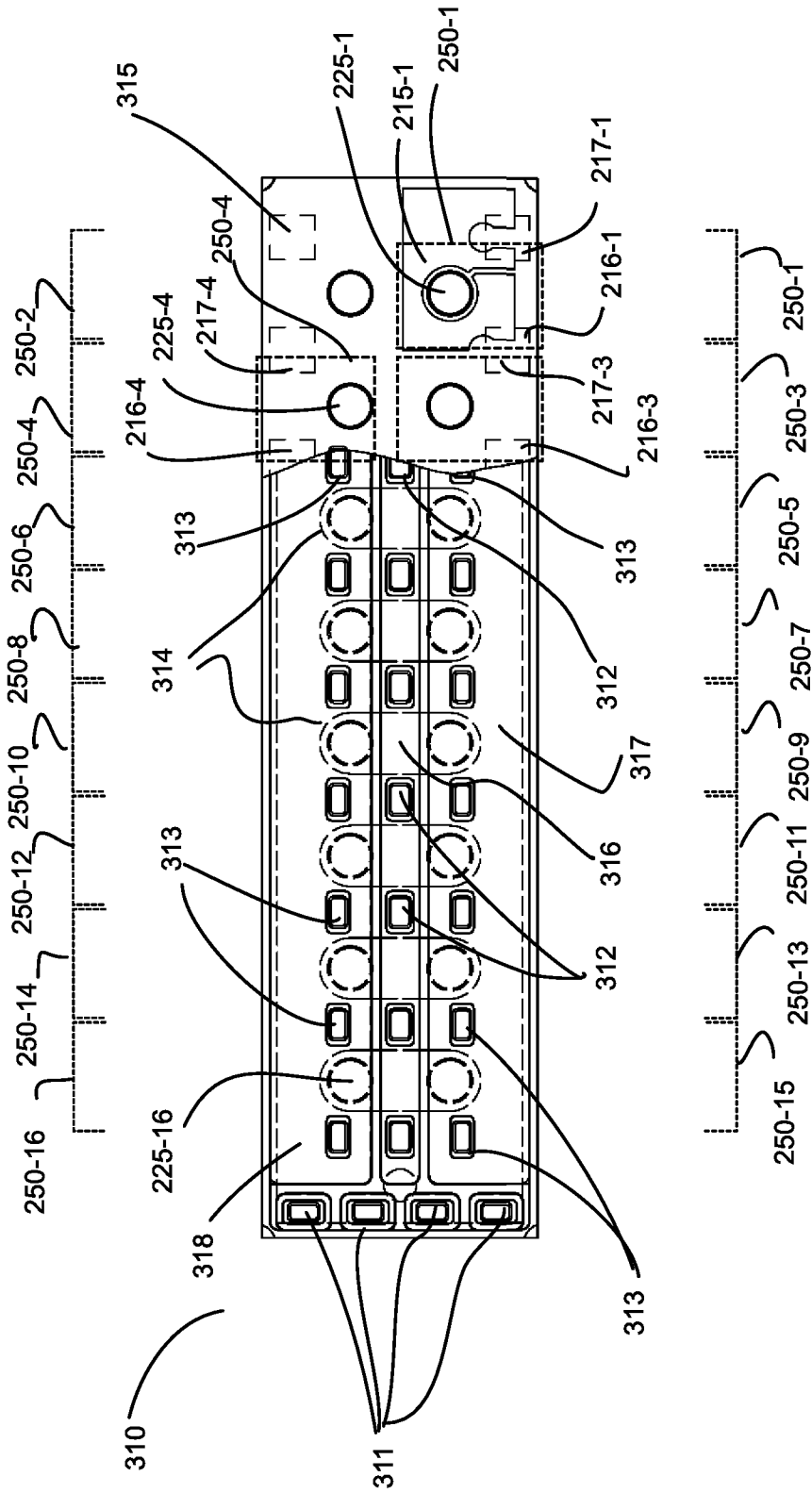


FIG. 3

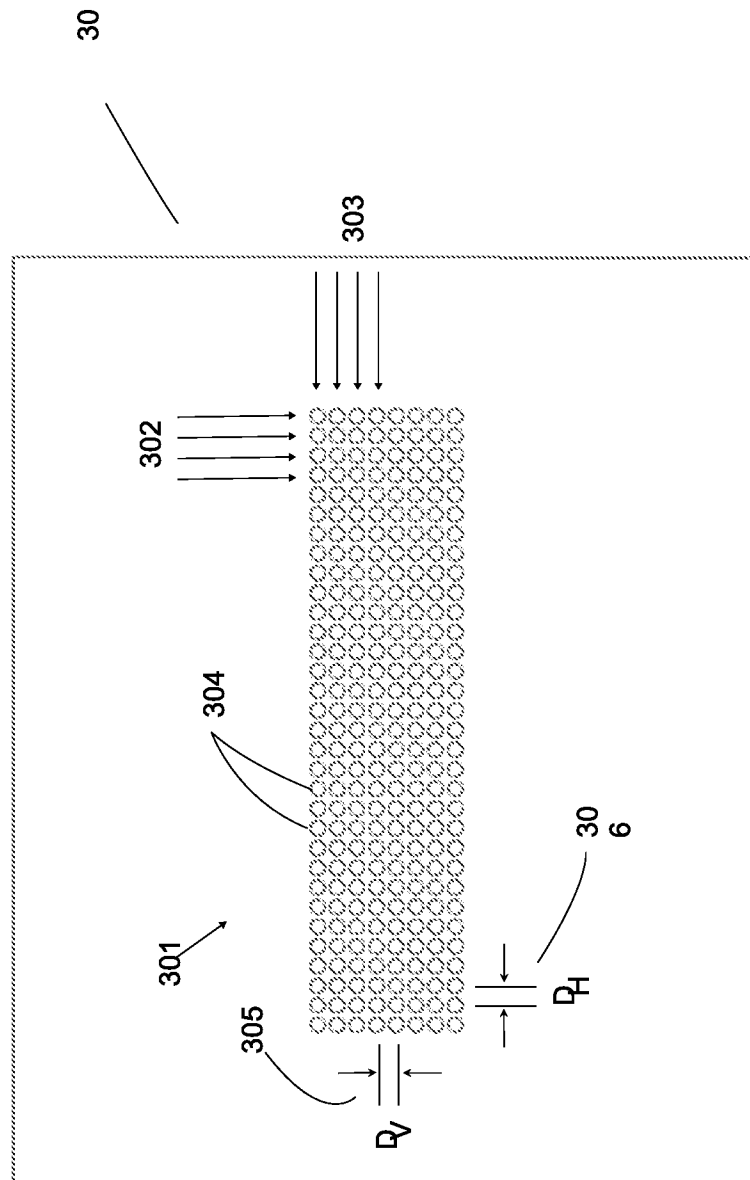
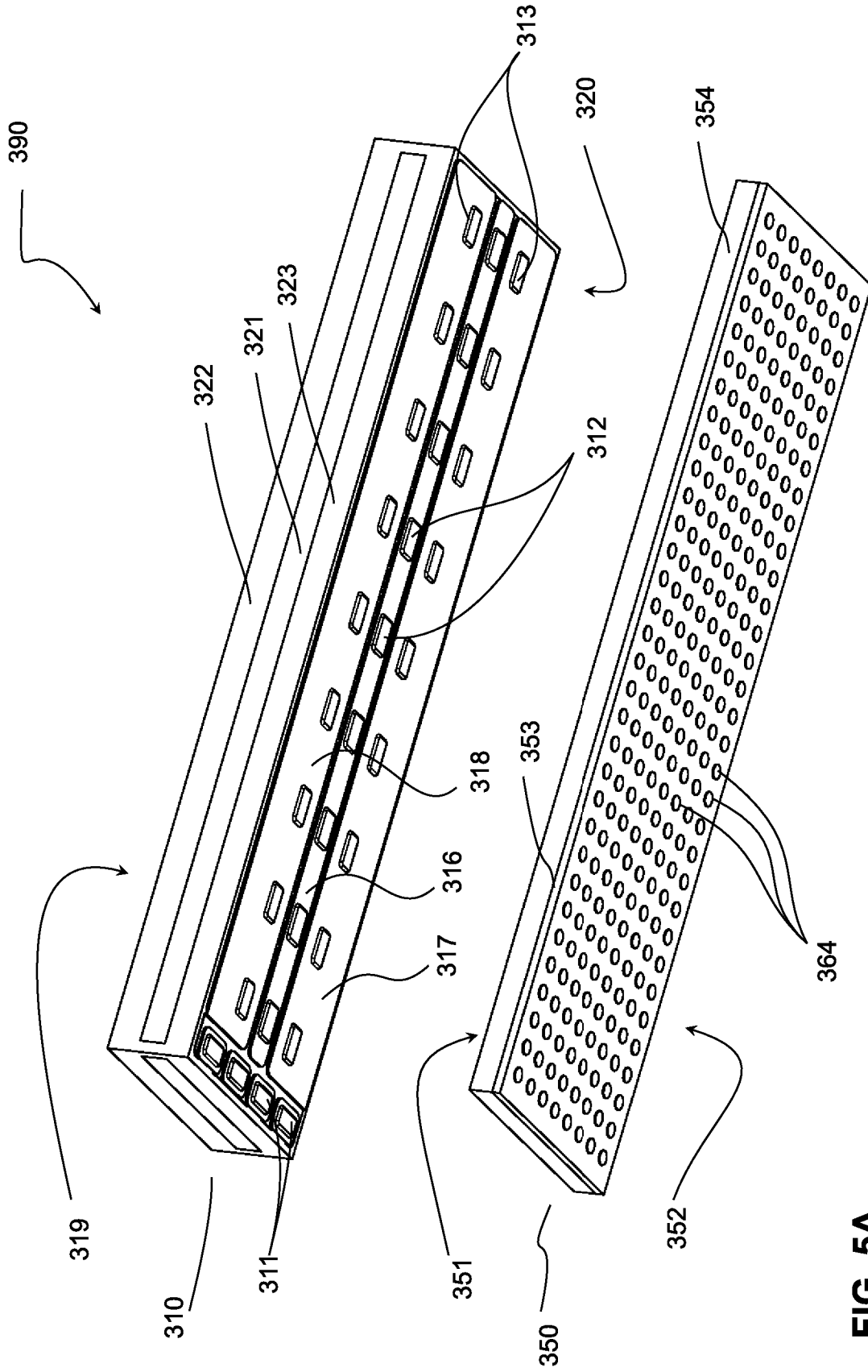


FIG. 4



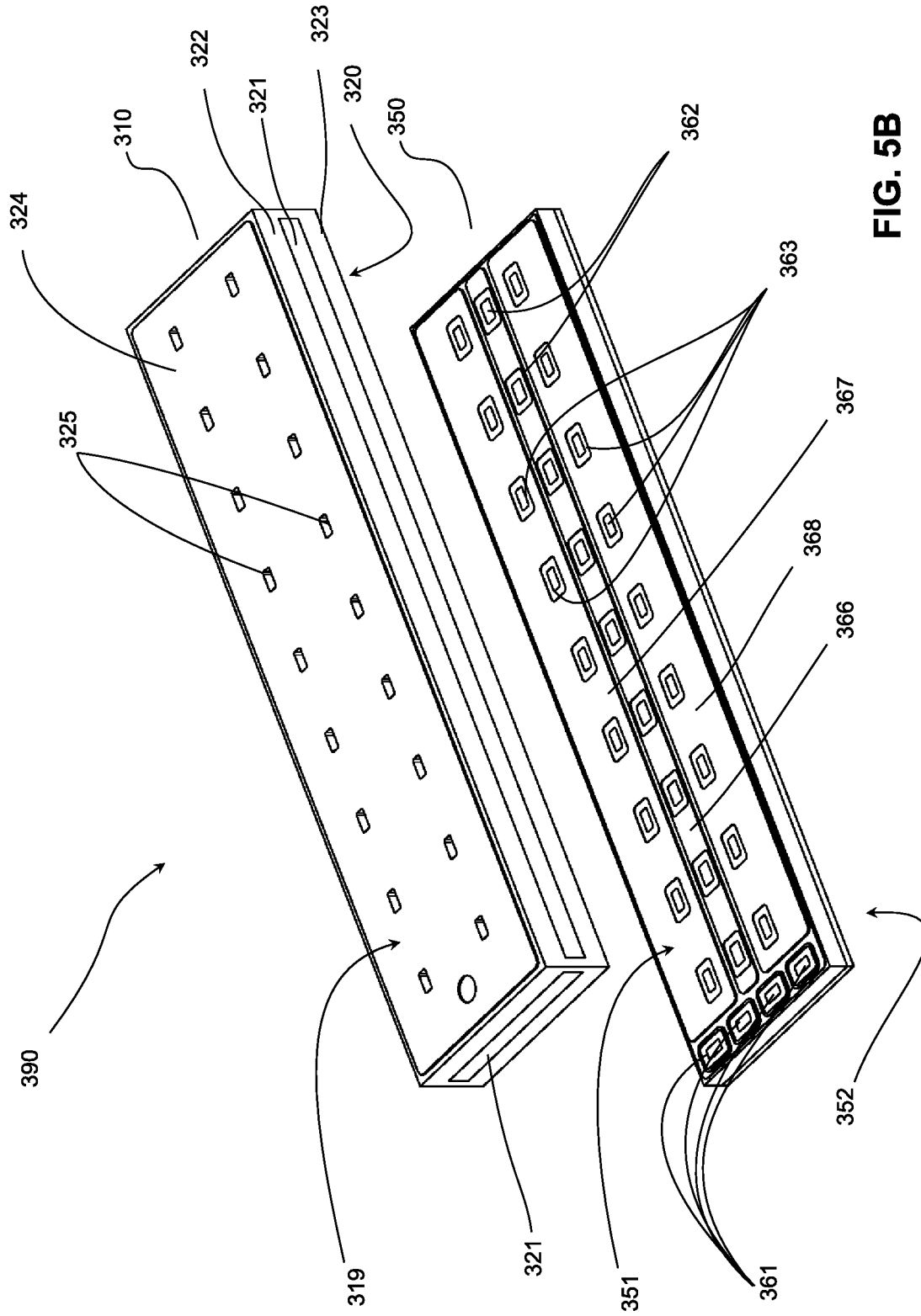
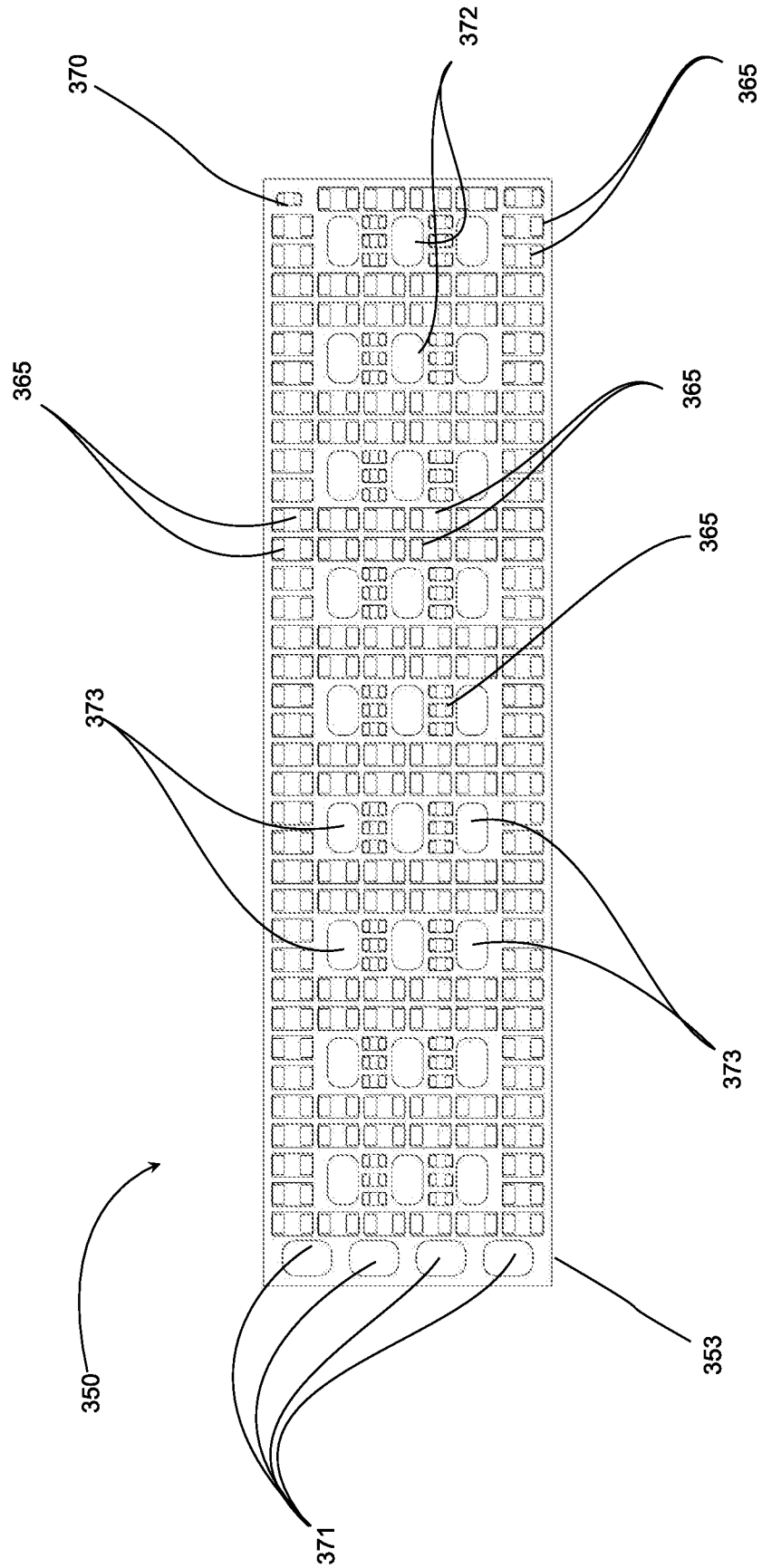


FIG. 5B



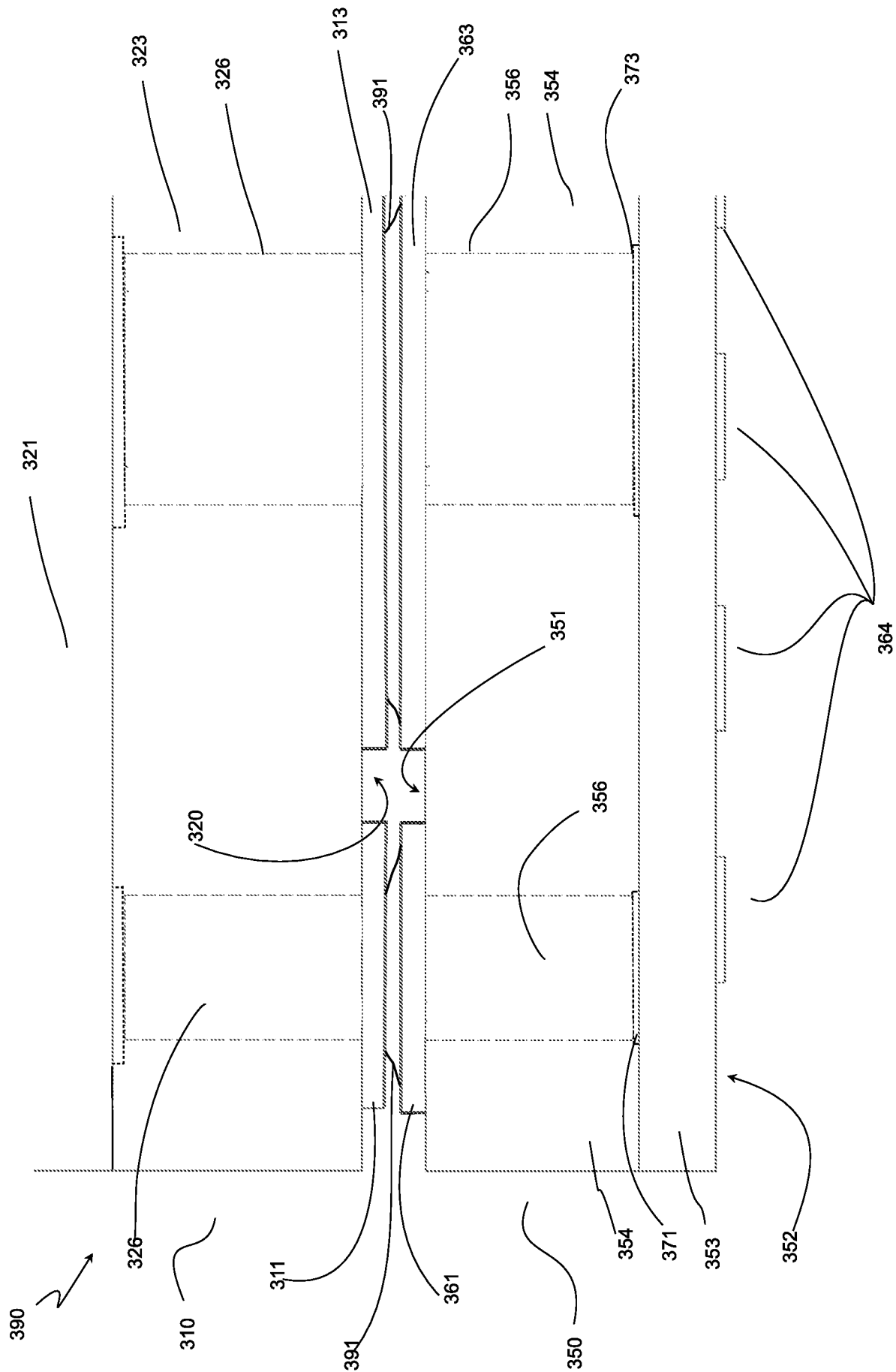
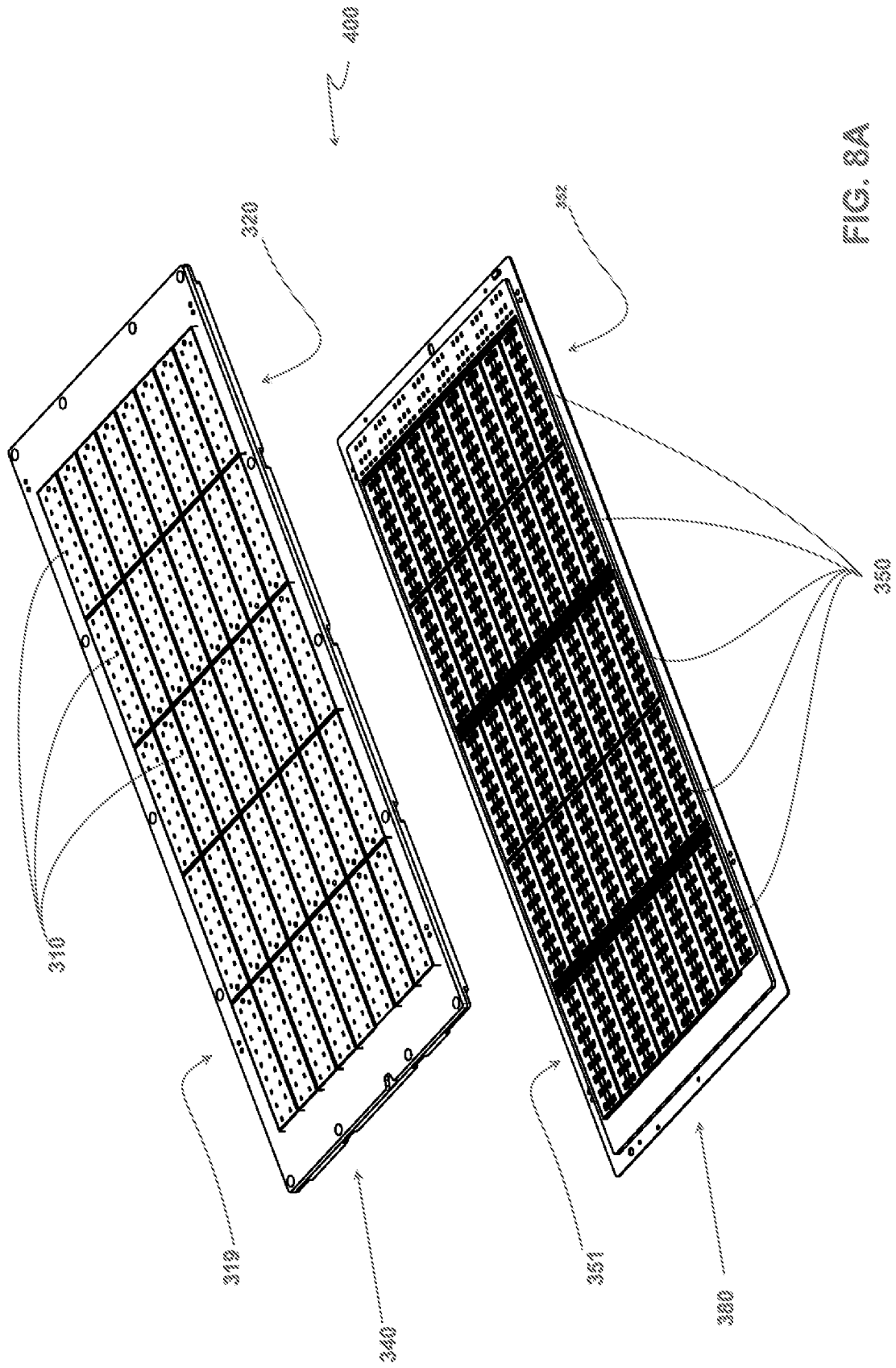


FIG. 7



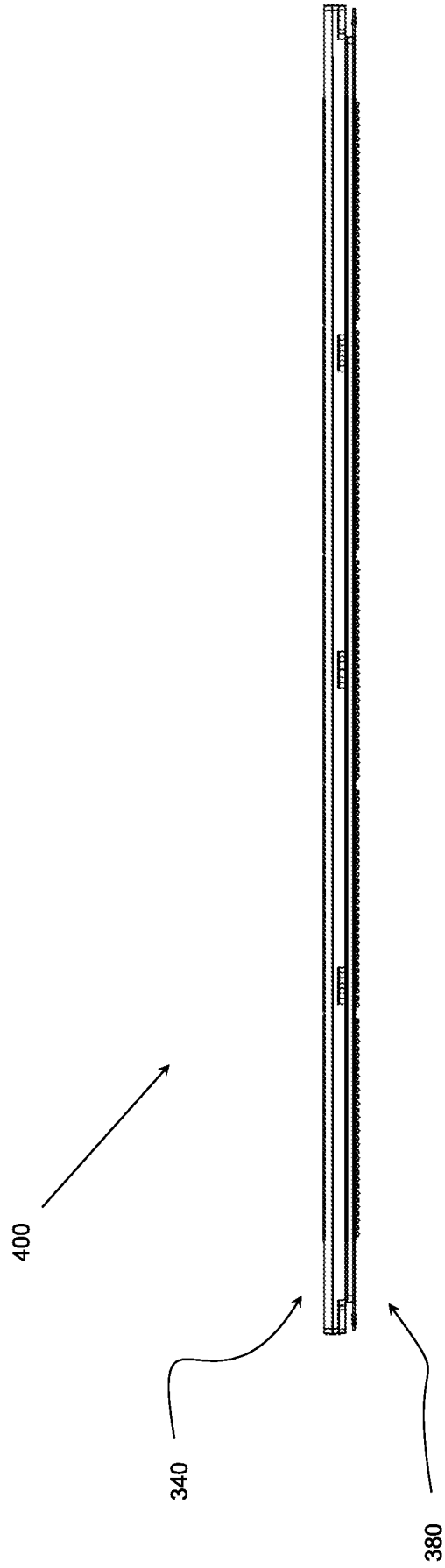


FIG. 8B

Electronic Acknowledgement Receipt

EFS ID:	35603373
Application Number:	16218418
International Application Number:	
Confirmation Number:	1092
Title of Invention:	Delivering Power To Semiconductor Loads
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Rita Liston
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0183001
Receipt Date:	02-APR-2019
Filing Date:	12-DEC-2018
Time Stamp:	14:37:56
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		00614_0183001_Trans_Replace ment_Dwgs.pdf	1206153	yes	9
			b8ae06056fcd223337f25310005c111be99 84c08		

	Multipart Description/PDF files in .zip description		
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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	Unknown
Application No. :	16/218,418	Examiner :	Unknown
Filed :	December 12, 2018	Conf. No. :	1092

Title : DELIVERING POWER TO SEMICONDUCTOR LOADS

MAIL STOP PGPUB-DRAWINGS

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

TRANSMITTAL OF FORMAL DRAWINGS

Please substitute the enclosed one (1) sheet of formal drawing for the corresponding drawing presently in the application. No new matter has been added.

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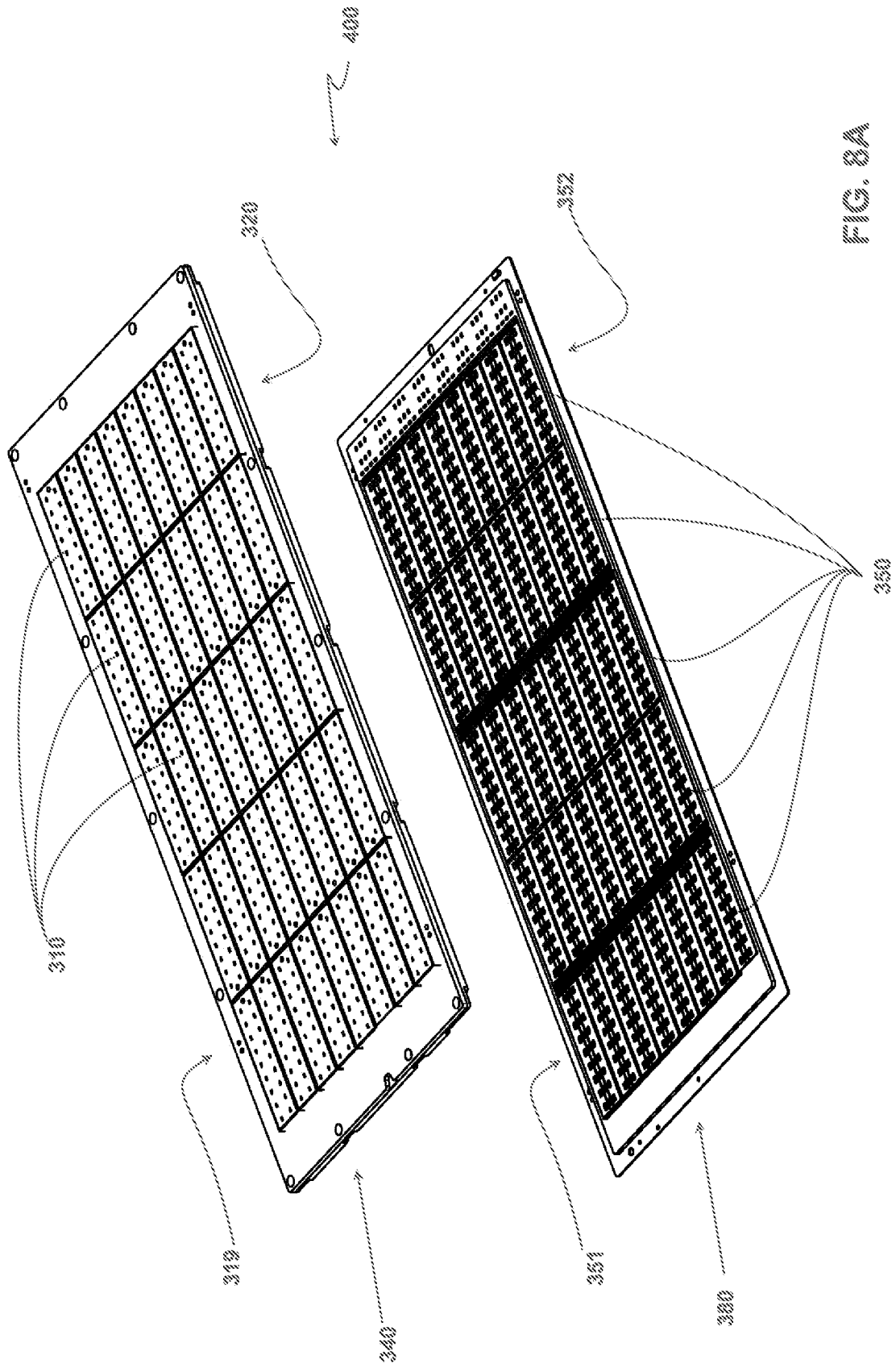
Respectfully submitted,

Date: April 23, 2019_____

/Rex I. Huang/_____

Rex I. Huang
Reg. No. 57,661

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Electronic Acknowledgement Receipt

EFS ID:	35801671
Application Number:	16218418
International Application Number:	
Confirmation Number:	1092
Title of Invention:	Delivering Power To Semiconductor Loads
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Rita Liston
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0183001
Receipt Date:	23-APR-2019
Filing Date:	12-DEC-2018
Time Stamp:	12:43:04
Application Type:	Utility under 35 USC 111(a)

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Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		00614_0183001_Transmittal. pdf	1203502	yes	2
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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor : Patrizio Vinciarelli Art Unit : Unknown
 Application No. : 16/218,418 Examiner : Unknown
 Filed : December 12, 2018 Conf. No. : 1092
 Title : DELIVERING POWER TO SEMICONDUCTOR LOADS

MAIL STOP AMENDMENT

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INFORMATION DISCLOSURE STATEMENT

Please consider the documents listed on the disclosure form submitted with this statement. Copies of the documents are being submitted with this statement to the extent required under 37 C.F.R. § 1.98(a)(2).

The applicant wishes to make the examiner aware of the following related application. The prosecution histories for this application are available on PAIR, and are not provided with this statement.

Application No.	Filing Date	First Named Inventor	Attorney Docket No.
16/218,395	December 12, 2018	Vinciarelli	00614-0182001

This statement is being filed before the receipt of a first action on the merits. Apply any necessary charges or credits applicable under 37 C.F.R. § 1.17(p) to deposit account 06-1050, referencing the above attorney docket number.

Please contact the undersigned if there are any questions regarding this statement.

Respectfully submitted,

Date: August 7, 2019 _____

/Rex I. Huang/ _____

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First Named Inventor : Patrizio Vinciarelli
Application No. : 16/218,418
Filed : December 12, 2018
Page : 2 of 2

Attorney Docket: 00614-0183001 / 0269-US-1
Gearbox

Substitute Disclosure Form	U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0183001	Application No. 16/218,418
Information Disclosure Statement by Applicant (Use several sheets if necessary)		Applicant VLT, Inc.	
		Filing Date December 12, 2018	Group Art Unit
(37 CFR §1.98(b))			

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1.	5438294	8/1/1995	Smith			
	2.	5728600	3/17/1998	Vinciarelli et al.			
	3.	6031743	02/29/2000	Carpenter et al.			
	4.	6035261	03/07/2000	Carpenter et al.			
	5.	6118186	09/12/2000	Scott et al.			
	6.	6275958	08/14/2001	Carpenter et al.			
	7.	6300749	10/01/2001	Castelli et al.			
	8.	6403009	6/11/2002	Vinciarelli et al.			
	9.	6421262	07/16/2002	Saxelby et al.			
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	11.	6603292	08/01/2003	Schouten et al.			
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	13.	6788033	09/07/2004	Vinciarelli			
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	15.	6930893	8/16/2005	Vinciarelli et al.			
	16.	6934166	8/23/2005	Vinciarelli et al.			
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	26.	7521909	04/01/2009	Dow et al.			

Examiner Signature	Date Considered
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

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	Applicant VLT, Inc.	
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U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	27.	7561446	07/14/2009	Vinciarelli			
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	29.	8232783	07/01/2012	Yanagawa, Kenji			
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	39.	9765750	09/19/2017	Ferguson et al.			
	40.	9936580	4/3/2018	Vinciarelli et al.			
	41.	10014798	7/3/2018	Vinciarelli et al.			
	42.	10020752	7/10/2018	Vinciarelli et al.			
	43.	10158357	12/18/2018	Vinciarelli et al.			
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	51.	2007/0297198	12/27/2017	Chang			
	52.	2009/0016083	1/15/2009	Soldano et al.			

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	Filing Date December 12, 2018	Group Art Unit

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	53.	2013/0030534	1/31/2013	DeLurio et al.			
	54.	2013/0069608	03/01/2013	Gakhar et al.			
	55.	2015/0188407	07/02/2015	Golder et al.			
	56.	RE40,072	02/19/2008	Prager, et al.			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No
	57.	EP2863531	4/22/2015	EP				
	58.	JPH02280666	11/16/1990	JP			English Abstract	

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	59.	Amendment after Allowance Pursuant to 37 C.F.R. §1.312 in U.S. Application No. 13/044,063, dated March 3, 2015, 17 pages
	60.	AN2738, Application Note AN2738, "L6390 half-bridge gate driver," STMicroelectronics, August 2009
	61.	AND8311/D (Understanding the LLC Structure in Resonant Applications," Christophe Basso, ON Semiconductor, Jan., 2008) (Year: 2008)
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	64.	Cao et al., "A Family of Zero Current Switching Switched-Capacitor DC-DC converters," Applied Power Electronics Conference and Exposition (APEC), 2010 Twenty-Fifth Annual IEEE, Mar. 18, 2010
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	66.	DiBene II et al., "A 400 Amp Fully Integrated Silicon Voltage Regulator with in-die Magnetically Coupled Embedded Inductors," retrieved from http://www.psma.com/sites/default/files/uploads/tech-forums-nanotechnology/resources/400a-fully-integrated-silicon-voltage-regulator.pdf , retrieved on August 3, 2016, 25 pages.
	67.	European Office Action in European Application No. 16173089.0, dated April 10, 2017, 2 pages

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	Applicant VLT, Inc.	
	Filing Date December 12, 2018	Group Art Unit

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	68.	Extended Search Report in European Application No. 16173089.0, dated February 14, 2017, 8 pages
	69.	ISL6210, Datasheet ISL6210, "Dual Synchronous Rectified MOSFET Drivers," Intersil Americas Inc., December 2008
	70.	Miller, "12V PowerStage in Embedded Die System-in-Package," APEC 2015, 33 pages
	71.	Notice of Allowance in U.S. Application No. 13/044,063, Examiner Nguyen, dated February 20, 2015, 15 pages
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	73.	Notice of Allowance in U.S. Application No. 14/682,187, dated January 17, 2017, 22 pages
	74.	Notice of Allowance in U.S. Application No. 15/091,346, dated August 13, 2018, 8 pages
	75.	Notice of Allowance in U.S. Application No. 15/091,346, dated May 21, 2018, 9 pages
	76.	Notice of Allowance in U.S. Application No. 15/616,288, dated December 11, 2018, 9 pages
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	83.	TND359 ("High-Efficiency 255W ATX Power Supply Reference Design Documentation Package," ON Semiconductor, Jan. 2009) (Year: 2009)
	84.	Watson, "New Techniques in the Design of Distributed Power System," Dissertation, 1998, 32 pages
	85.	Yeung et al., "Generalised analysis of switched-capacitor step-down quasi-resonant converter," Electronics Letters, vol. 38, pp. 263-264, 2002
	86.	Yeung et al., "Zerocurrent switching switched-capacitor quasiresonant step-down converter," Electric Power Applications, IEE Proceedings -, vol. 149, pp. 111-121, 2002

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(19)



(11)

EP 2 863 531 A1

(12)

EUROPEAN PATENT APPLICATION

published in accordance with Art. 153(4) EPC

(43) Date of publication:

22.04.2015 Bulletin 2015/17

(51) Int Cl.:

H02M 3/335 (2006.01)**H02M 3/28 (2006.01)**(21) Application number: **13804652.9**

(86) International application number:

PCT/JP2013/003211(22) Date of filing: **21.05.2013**

(87) International publication number:

WO 2013/186991 (19.12.2013 Gazette 2013/51)

(84) Designated Contracting States:

**AL AT BE BG CH CY CZ DE DK EE ES FI FR GB
GR HR HU IE IS IT LI LT LU LV MC MK MT NL NO
PL PT RO RS SE SI SK SM TR**

Designated Extension States:

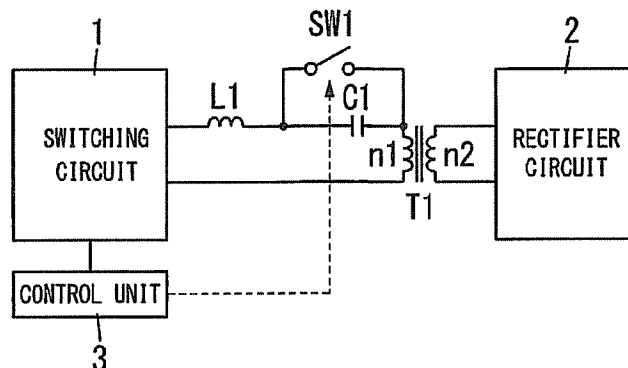
BA ME• **MASUDA, Takuya****Chuo-ku, Osaka 540-6207 (JP)**• **MUTO, Keisuke****Chuo-ku, Osaka 540-6207 (JP)**• **KOBAYASHI, Susumu****Chuo-ku, Osaka 540-6207 (JP)**• **ADACHI, Masakazu****Chuo-ku, Osaka 540-6207 (JP)**(30) Priority: **15.06.2012 JP 2012136174**(71) Applicant: **Panasonic Intellectual Property
Management Co., Ltd.****Osaka-shi, Osaka 540-6207 (JP)**(74) Representative: **Appelt, Christian W.****Boehmert & Boehmert****Anwaltpartnerschaft mbB****Patentanwälte Rechtsanwälte****Pettenkoferstrasse 20-22****80336 München (DE)**

(72) Inventors:

• **SANUKI, Maiya****Chuo-ku, Osaka 540-6207 (JP)**(54) **ELECTRIC POWER CONVERSION DEVICE**

(57) A switching circuit is configured to switch on/off a current that flows through a resonance circuit constituted by a primary winding of a transformer, a capacitance element, and an inductor that are connected in series. Electric power that is induced in a secondary winding of the transformer is rectified by a rectifier circuit. A switch is connected in parallel with the capacitance element. A control unit is configured to control the switching

circuit and the switch. The control unit is configured to select a first operation in which an operation frequency of the switching circuit is controlled by switching off the switch, and select a second operation in which a period during which a current flows from the switching circuit to the primary winding of the transformer is controlled by switching on the switch.

FIG. 1

Description

TECHNICAL FIELD

[0001] The present invention relates to an electric power conversion device that performs electric power conversion using a switching element and transmits electric power between an input and an output via a transformer.

BACKGROUND ART

[0002] Conventionally, electric power conversion devices are known that perform electric power conversion using a switching element and transmit electric power between an input and an output via a transformer. For example, a bidirectional DC/DC converter that transmits DC power bi-directionally is disclosed in JP 2011-234541 A (hereinafter referred to as "Document 1") as this type of electric power conversion device. The electric power conversion device described in Document 1 is provided with a primary circuit and a secondary circuit that sandwich a transformer.

[0003] The primary circuit and the secondary circuit each include a bridge circuit configured by a reverse conducting type semiconductor switch (switching element). The electric power conversion device can transmit desired DC power bi-directionally by controlling duty cycles (on duties) of the switching element of the primary circuit and the switching element of the secondary circuit.

[0004] Incidentally, although the electric power conversion device described in Document 1 can transmit desired DC power between the primary circuit and the secondary circuit by controlling the duty cycles of the switching elements, a voltage ratio between an input voltage and an output voltage depends on the winding ratio of the transformer. That is, the electric power conversion device described in Document 1 cannot output voltage by boosting the input voltage and only performs current adjustment, since bridge circuits are used in the primary circuit and the secondary circuit, and only the duty cycles are controlled.

[0005] On the other hand, an electric power conversion device that is used for utilizing electric power of a storage battery that is mounted in an automobile having an electric motor serving as a power source, a storage battery used by being attached to a building, or the like is required to have a function of adjusting a relationship between an input voltage and an output voltage over a wide range.

DISCLOSURE OF INVENTION

[0006] An object of the present invention is to provide an electric power conversion device configured to perform electric power conversion by switching and transmit electric power between an input and an output via a transformer, and is capable of adjusting a relationship between an input voltage and an output voltage over a wide range according to the use.

[0007] An electric power conversion device according to one aspect of the present invention includes a transformer, a resonance capacitance element, a switching circuit, a rectifier circuit, a switch, and a control unit. The transformer is configured to transmit electric power between an input and an output. The capacitance element is connected in series with a primary winding of the transformer. The switching circuit is configured to switch on/off current that flows through a series circuit constituted by the primary winding of the transformer and the capacitance element. The rectifier circuit is configured to rectify and output electric power induced in a secondary winding of the transformer. The switch is connected in parallel with the capacitance element. The control unit is configured to control a period during which current flows through the series circuit constituted by the primary winding of the transformer and the capacitance element and on/off of the switch. The control unit is configured to switch off the switch so as to configure, as an input-side circuit of the rectifier circuit, a first conversion circuit configured to change a relationship between an input voltage and an output voltage based on an operation frequency of the switching circuit. The control unit is configured to switch on the switch so as to configure, as the input-side circuit of the rectifier circuit, a second conversion circuit configured to change a relationship between an input voltage and an output voltage based on a period during which current flows from the switching circuit to the primary winding of the transformer. The control unit is configured to select an operation of the switching circuit from a first operation for adjusting the relationship in the first conversion circuit by controlling the operation frequency and a second operation for adjusting the relationship in the second conversion circuit by controlling the period during which the current flows.

[0008] In the electric power conversion device, the control unit is preferably configured to match a timing at which the switch is switched on with a zero-crossing point of the current that flows from the switching circuit to the series circuit constituted by the primary winding of the transformer and the capacitance element, when transitioning from the first operation of the first conversion circuit to the second operation of the second conversion circuit.

[0009] Preferably, the electric power conversion device further includes a series circuit constituted by a second switch and a resistor. The series circuit is preferably connected in parallel with the capacitance element. The control unit is preferably configured to switch on the switch after switching on the second switch when transitioning from the first operation of the first conversion circuit to the second operation of the second conversion circuit.

[0010] In the electric power conversion device, the control unit is preferably configured to provide a transition period during which an operation of the switching circuit is an operation different from the first operation of the first conversion circuit and the second operation of the

second conversion circuit, when switching between the first operation of the first conversion circuit and the second operation of the second conversion circuit.

[0011] In the electric power conversion device, the control unit is preferably configured to store the operation frequency of the switching circuit in the first operation of the first conversion circuit in association with the relationship in the first conversion circuit, and store the period during which current flows from the switching circuit to the primary winding of the transformer in the second operation of the second conversion circuit in association with the relationship in the second conversion circuit. The control unit is preferably configured to control the switching circuit in accordance with the stored operation frequency and the stored period during which current flows, such that the relationship between the input voltage and the output voltage after switching coincides with the relationship between the input voltage and the output voltage before switching, when switching between the first operation of the first conversion circuit and the second operation of the second conversion circuit.

[0012] In the electric power conversion device, the control unit is preferably configured to store a conversion efficiency in the first operation of the first conversion circuit in association with the relationship in the first conversion circuit, and store a conversion efficiency in the second operation of the second conversion circuit in association with the relationship in the second conversion circuit. The control unit is preferably configured to select whichever of the first operation and the second operation has a higher stored conversion efficiency, when both the first operation of the first conversion circuit and the second operation of the second conversion circuit are operable, in an overlapping region between the relationship in the first conversion circuit and the relationship in the second conversion circuit.

[0013] Preferably, the electric power conversion device further includes a resonance inductor. The inductor is preferably connected between the primary winding of the transformer and the switching circuit.

[0014] Preferably, the electric power conversion device further includes a plurality of resonance inductors and a short-circuit switch. The plurality of inductors are preferably connected between the primary winding of the transformer and the switching circuit. The short-circuit switch is preferably configured to short-circuit at least one of the plurality of inductors.

[0015] In the electric power conversion device, the capacitance element preferably includes a plurality of resonance capacitors. The plurality of capacitors are preferably connected between the primary winding of the transformer and the switching circuit. The switch is preferably configured to short-circuit at least one of the plurality of capacitors.

[0016] In the electric power conversion device, the capacitance element preferably includes a resonance capacitor. The capacitor is preferably connected between the primary winding of the transformer and the switching

circuit. The switch is preferably configured to short-circuit the capacitor.

[0017] In the electric power conversion device, both the relationship in the first conversion circuit and the relationship in the second conversion circuit are preferably a voltage ratio between the input voltage and the output voltage.

[0018] In the electric power conversion device, the first conversion circuit is preferably a resonance type conversion circuit. The resonance type conversion circuit is preferably configured to change the relationship in the first conversion circuit using a resonance circuit constituted by the capacitance element and the transformer by adjusting the operation frequency of the switching circuit. The second conversion circuit is preferably a phase shift type conversion circuit. The phase shift type conversion circuit is preferably configured to change the relationship in the second conversion circuit by adjusting the period during which current flows from the switching circuit to the primary winding of the transformer through changing a phase of switching on a switching element of the switching circuit.

[0019] According to the one aspect of the present invention, the electric power conversion is performed by switching, and the electric power is transmitted between the input and the output via the transformer. It is possible to adjust the relationship between the input voltage and the output voltage over a wide range according to the use by selecting one of the operation of the first conversion circuit and the operation of the second conversion circuit.

BRIEF DESCRIPTION OF DRAWINGS

[0020] Preferable embodiments according to the present invention will be described in more detail. Other features and advantages of the present invention will be better understood with reference to the following detailed description and the attached drawings:

FIG. 1 is a circuit diagram illustrating an electric power conversion device according to Embodiment 1; FIG. 2 is a specific circuit diagram illustrating an example of the electric power conversion device according to Embodiment 1;

FIG. 3 is an explanatory diagram illustrating an operation of the electric power conversion device according to Embodiment 1 as an LLC resonance type conversion circuit;

"A" of FIG. 4 is an explanatory diagram illustrating an operation of a switching element Q11 in a phase shift type conversion circuit of the electric power conversion device according to Embodiment 1; "B" of FIG. 4 is an explanatory diagram illustrating an operation of a switching element Q12 in the phase shift type conversion circuit of the electric power conversion device according to Embodiment 1; "C" of FIG. 4 is an explanatory diagram illustrating an operation

of a switching element Q13 in the phase shift type conversion circuit of the electric power conversion device according to Embodiment 1; "D" of FIG. 4 is an explanatory diagram illustrating an operation of a switching element Q14 in the phase shift type conversion circuit of the electric power conversion device according to Embodiment 1; and "E" of FIG. 4 is a diagram illustrating a conduction state of a primary winding of a transformer in the phase shift type conversion circuit of the electric power conversion device according to Embodiment 1;

FIG. 5 is an explanatory diagram illustrating an operation of the electric power conversion device according to Embodiment 1 as the phase shift type conversion circuit;

"A" of FIG. 6 is a diagram illustrating a waveform of current flowing through a resonance circuit of the electric power conversion device according to Embodiment 1; and "B" of FIG. 6 is an explanatory operation diagram illustrating a switching timing of a switching element used in the electric power conversion device according to Embodiment 1;

"A" of FIG. 7 is an explanatory diagram illustrating an operation of a switching element Q11 of a switching circuit in the electric power conversion device according to Embodiment 1; "B" of FIG. 7 is an explanatory diagram illustrating an operation of a switching element Q12 of the switching circuit in the electric power conversion device according to Embodiment 1; "C" of FIG. 7 is an explanatory diagram illustrating an operation of a switching element Q13 of the switching circuit in the electric power conversion device according to Embodiment 1; and "D" of FIG. 7 is an explanatory diagram illustrating an operation of a switching element Q14 of the switching circuit in the electric power conversion device according to Embodiment 1;

FIG. 8 is a circuit diagram illustrating an electric power conversion device according to Embodiment 2;

FIG. 9 is a circuit diagram illustrating an electric power conversion device according to Embodiment 3;

FIG. 10 is a circuit diagram illustrating another configuration example of the electric power conversion device according to Embodiment 3;

FIG. 11 is a circuit diagram illustrating an electric power conversion device according to Embodiment 4; and

FIG. 12 is a circuit diagram illustrating another configuration example of the electric power conversion device according to Embodiment 4.

BEST MODE FOR CARRYING OUT INVENTION

Embodiment 1

[0021] Although, in an embodiment described below, a case where an electric power conversion device performs DC/DC conversion uni-directionally will be de-

scribed as an example, the technical idea described below can also be applicable to an electric power conversion device that performs DC/AC conversion or an electric power conversion device that performs electric power conversion bi-directionally. Also, although the electric power conversion device described below is assumed to be used for utilizing electric power of a storage battery mounted in an automobile having an electric motor as a power source, a storage battery used by being attached to a building, or the like, the application is not limited thereto. For example, the technique of the electric power conversion device described below may be adopted in a distributed power source such as a solar power generation device or a fuel cell.

[0022] The electric power conversion device according to the present embodiment includes a switching circuit 1 and a rectifier circuit 2 with a transformer T1 interposed therebetween, as shown in FIG. 1. The switching circuit 1 includes four switching elements Q11 to Q14 (refer to FIG. 2) described later, and a control unit 3 is attached to the switching circuit 1 so as to control the switching elements Q11 to Q14. A primary winding n1 of the transformer T1 is connected to the switching circuit 1 via a series circuit constituted by a capacitor C1 serving as a capacitance element and an inductor L1. The rectifier circuit 2 is connected to a secondary winding n2 of the transformer T1. Furthermore, a switch SW1 is connected in parallel with the capacitor C1. The control unit 3 is also configured to control on/off of the switch SW1.

[0023] The switching circuit 1 and the rectifier circuit 2 includes a configuration shown in FIG. 2, for example. The configuration shown in FIG. 2 is an example of the switching circuit 1 and the rectifier circuit 2, and the configuration of the switching circuit 1 and the rectifier circuit 2 is not limited thereto. Also, a DC power supply serving as an input power supply is connected to the switching circuit 1. The DC power supply may be a DC power supply obtained by rectifying a commercial AC power supply, other than being a storage battery, a fuel cell, a solar cell, or the like. Also, the DC power outputted from the rectifier circuit 2 may be used for charging a storage battery, in addition to being supplied to a load.

[0024] The switching circuit 1 shown in the diagram includes a bridge circuit constituted by the four switching elements Q11 to Q14. A series circuit of the switching elements Q11 and Q12 and a series circuit of the switching elements Q13 and Q14 are connected in parallel with a smoothing capacitor C11.

[0025] Also, the rectifier circuit 2 shown in the diagram uses a circuit having the same configuration as the switching circuit 1. That is, the rectifier circuit 2 includes a bridge circuit constituted by four switching elements Q21 to Q24. A series circuit of the switching elements Q21 and Q22 and a series circuit of the switching elements Q23 and Q24 are connected in parallel with a smoothing capacitor C21. By controlling on/off of the switching elements Q21 to Q24 in the rectifier circuit 2, electric power can be bi-directionally transmitted via the

transformer T1. A description of operations to perform an electric power conversion bi-directionally will, however, be omitted in the present embodiment.

[0026] Each of the switching elements Q11 to Q14 and Q21 to Q24 is assumed to be a MOSFET. Note that a configuration in which a diode is connected anti-parallelly between an emitter and a collector of a bipolar transistor, an IGBT (Insulated Gate Bipolar Transistor), or the like may be used as each of the switching elements Q11 to Q14 and Q21 to Q24. These switching elements Q11 to Q14 and Q21 to Q24 can pass current in one direction or bi-directionally in an on-state, and can pass current in the other direction in an off-state. For example, in the case where the switching elements Q11 to Q14 and Q21 to Q24 are MOSFETs, the switching elements Q11 to Q14 and Q21 to Q24 can pass current via a path passing through a body diode (parasitic diode) in the off-state.

[0027] The rectifier circuit 2 shown in the diagram performs full-wave rectification using a function in the off-state of the switching elements Q21 to Q24. Note that synchronous rectification can also be performed if on/off of the switching elements Q21 to Q24 is controlled.

[0028] The control unit 3 includes a processor (selected from a microcontroller, a DSP (Digital Signal Processor), an FPGA (Field-Programmable Gate Array), or the like) that operates according to a program. The control unit 3 has a function of monitoring an output current of the rectifier circuit 2 using an output from a current detector (not shown) provided at an appropriate location, and includes an AD converter that converts analog information outputted from the current detector to digital information so as to be handled by the processor. On/off of each of the switching elements Q11 to Q14 is controlled by a pulse signal given by the control unit 3, and the frequency and the phase of the pulse signal are adjustable. That is, the control unit 3 includes a pulse generator that generates pulse signals, and performs control on the switching circuit 1, selecting from PFM (Pulse Frequency Modulation) control and PPM (Pulse Phase shift Modulation) control, by giving instructions to the pulse generator from the processor. Furthermore, the control unit 3 includes a memory that stores a program for causing the processor to operate, data for determining an operation condition of the processor, data that is generated during an operation of the processor, and the like.

[0029] A frequency for switching on/off the switching elements Q11 to Q14 is selected appropriately from a range of several tens of Hz to several hundreds of kHz. Here, since the switch SW1 is controlled based on the on/off timings of the switching elements Q11 to Q14, the switch SW1 is required to respond in a time period of approximately several μ s to several tens of ms from the time when an instruction is given. Accordingly, it is desirable that a semiconductor switch having a short response time is used for the switch SW1, instead of using an electromagnetic relay including a mechanical contact.

[0030] Incidentally, when the switch SW1 is off, the capacitor C1 is electrically connected between the inductor

L1 and the primary winding n1 of the transformer T1, and a resonance circuit is formed by the capacitor C1, the inductor L1, and the transformer T1. Accordingly, if the frequency of the switching circuit 1 is adjusted appropriately with respect to the resonance frequency of the resonance circuit, this resonance circuit and the switching circuit 1 operate as a first conversion circuit that changes the voltage that is outputted from the secondary winding n2 of the transformer T1. That is, a circuit that is on the input side of the rectifier circuit 2 operates as the first conversion circuit. The first conversion circuit is configured to change the relationship (voltage ratio) between an input voltage and an output voltage based on the operation frequency of the switching circuit 1. In this way, a configuration, in which the resonance circuit constituted by the inductor L1 and the transformer T1 with one capacitor C1 is used to change the output voltage of the rectifier circuit 2 by adjusting the operation frequency of the switching circuit 1, will be hereinafter referred to as "LLC resonance type conversion circuit". Note that the inductor L1 is not an essential constituent element in the first conversion circuit.

[0031] Hereinafter, in the configuration shown in FIG. 2, the two switching elements Q11 and Q14 and the two switching elements Q12 and Q13 that each have a positional relationship of being connected in series with the primary winding n1 of the transformer T1 interposed therebetween are each referred to as "opposing switching elements". When operating as the LLC resonance type conversion circuit, on/off of the opposing switching elements Q11 and Q14 are switched at the same time, and on/off of the opposing switching elements Q12 and Q13 are switched at the same time. Also, the two switching elements Q11 and Q12 that are connected in series are not made to be in on-states at the same time, and the two switching elements Q13 and Q14 that are connected in series are not made to be in on-states at the same time.

[0032] FIG. 3 shows a relationship between the operation frequency of the switching circuit 1 in the LLC resonance type conversion circuit and a voltage gain of the output voltage of the rectifier circuit 2. Also, the curves denoted by reference signs A1 to A6 in FIG. 3 show results that are taken on different load sizes. The size of the load increases in the order from the curve A1 to the curve A6. That is, the load corresponding to the curve A1 is the smallest and the load corresponding to the curve A6 is the largest, among the curves A1 to A6.

[0033] In the illustrated example, when the load is light, the voltage gain reaches a peak when an operation frequency ratio is approximately 0.4, and the voltage gain decreases as the operation frequency ratio moves away from 0.4. The operation frequency ratio is a ratio of the operation frequency of the switching circuit 1 with respect to the resonance frequency determined by the capacitor C1 and the inductor L1. The lighter the load is, the stronger the resonance is that occurs in the LLC resonance type conversion circuit, and thus, in a case of a light load, the output voltage changes largely by the operation fre-

quency of the switching circuit 1 merely changing slightly in the vicinity of the resonance point, possibly resulting in the output voltage becoming unstable. Furthermore, in the case of a light load, many noise components are included in the output of the rectifier circuit 2.

[0034] On the other hand, in an example shown in FIG. 3, when the load increases, the change of the output voltage with respect to the change of the operation frequency of the switching circuit 1 decreases, and an adjustable range of the voltage gain also decreases. That is, in the LLC resonance type conversion circuit, the resonance weakens in a region in which the load is large, and a change of the output voltage with respect to the change in the operation frequency of the switching circuit 1 decreases. At this time, the noise component included in the output of the rectifier circuit 2 decreases compared with the case of a light load.

[0035] As described above, in the LLC resonance type conversion circuit, the voltage ratio (voltage gain) of the output voltage to the input voltage can, depending on the size of the load, be changed by changing the operation frequency ratio. Furthermore, a voltage boost is possible, and adjustment of the voltage boost ratio becomes possible. However, since the operation becomes unstable when the load is light, the adjustable range of the voltage gain is relatively narrow in the load range where a stable operation can be achieved.

[0036] On the other hand, when the switch SW1 is in an on-state, the capacitor C1 is short-circuited, and thus only the inductor L1 and the transformer T1 are electrically interposed between the switching circuit 1 and the rectifier circuit 2. In this case, due to a change in the current flowing through the inductor L1 and the primary winding n1 of the transformer T1, the output voltage of the rectifier circuit 2 changes. In this operation, electromagnetic energy accumulated in the transformer T1 due to the current flowing through the primary winding n1 of the transformer T1 is transferred to the secondary winding n2. Accordingly, by adjusting the electromagnetic energy accumulated in the transformer T1 per unit time period, the circuit constituted by the inductor L1, the transformer T1, and the switching circuit 1 functions as a second conversion circuit that adjusts the output voltage of the rectifier circuit 2. That is, the input side circuit of the rectifier circuit 2 functions as the second conversion circuit. The second conversion circuit changes a relationship (voltage ratio) between the input voltage and the output voltage based on the period during which current flows from the switching circuit 1 to the primary winding n1 of the transformer T1. Note that the inductor L1 is not an essential constituent element in the second conversion circuit.

[0037] In order to adjust the period during which current flows through the primary winding n1 of the transformer T1, the phases of switching on the switching elements Q11 to Q14, as shown in "A" to "D" of FIG. 4, are adjusted, for example. In this way, a configuration in which the phases of switching on the switching elements Q11 to

Q14 of the switching circuit 1 are changed in order to adjust the current that flows through the primary winding n1 of the transformer T1 per unit time period (that is, second conversion circuit) will hereinafter be referred to as a "phase shift type conversion circuit".

[0038] In the phase shift type conversion circuit, without changing the duty cycles of the four switching elements Q11 to Q14 that constitute the switching circuit 1, the timings at which the switching elements are turned on in each pair of switching elements that need to be on to allow current to flow through the transformer T1, out of the switching elements Q11 to Q14, are changed such that the period in which the opposing switching elements are on at the same time is changed. That is, the output electric power is adjusted by adjusting the timings at which the opposing switching elements Q11 and Q14 are turned on such that the period of being on at the same time is adjusted, and by adjusting the timings at which the opposing switching elements Q12 and Q13 are turned on such that the period of being on at the same time is adjusted.

[0039] In the circuit shown in FIG. 2, current flows through the primary winding n1 of the transformer T1 in the period during which the opposing switching elements Q11 and Q14 are on at the same time, and the period during which the opposing switching elements Q12 and Q13 are on at the same time (refer to "E" of FIG. 4). Accordingly, by adjusting the timings at which the opposing switching elements Q11 and Q14 are turned on, or adjusting the timings at which the opposing switching elements Q12 and Q13 are turned on, the period during which current flows through the primary winding n1 is adjusted, and as a result the output voltage of the rectifier circuit 2 is adjusted. Note that the switching elements Q11 and Q12 that are connected in series or the switching elements Q13 and Q14 that are connected in series are prohibited from being on at the same time, and the switching on/off of the switching elements Q11 and Q12 or the switching elements Q13 and Q14 are controlled such that the phases thereof being on/off are inverted. That is, the on/off of the switching element Q11 is inverted from the on/off of the switching element Q12, and the on/off of the switching element Q13 is inverted from the on/off of the switching element Q14.

[0040] Referring to operations shown in "A" to "E" of FIG. 4, duty cycles of the switching elements Q11 to Q14 are set to 50% and the on period of the switching element Q11 differs from the on period of the switching element Q13 by a phase difference T_s , as shown in "A" to "D" of FIG. 4. Accordingly, the on periods of the opposing switching elements Q11 and Q14 overlap during a period corresponding to the phase difference T_s , and the on periods of the opposing switching elements Q12 and Q13 overlap during a period corresponding to the phase difference T_s . The phase difference T_s is assumed to be 0 degrees when the on periods of the switching element Q11 and the switching element Q13 do not overlap, and to be 180 degrees when the on periods of the switching

element Q11 and the switching element Q13 coincide. That is, the phase difference T_s corresponds to a value represented by an angle the period in which the opposing switching elements Q11 and Q14 are on at the same time and the period in which the opposing switching elements Q12 and Q13 are on at the same time, with the angle being 180 degrees when the on periods of the switching element Q11 and the switching element Q13 coincide.

[0041] When the switching circuit 1 performs the operations described above, a current flows through the primary winding n1 of the transformer T1 only in the period corresponding to the phase difference T_s among the on periods of the switching elements Q11 to Q14, as shown in "E" of FIG. 4. The direction of the current flowing through the primary winding n1 of the transformer T1 in the period during which the opposing switching elements Q11 and Q14 are on at the same time is inverted from the direction of the current in the period during which the opposing switching elements Q12 and Q13 are on at the same time. According to these operations, an alternating voltage is induced in the secondary winding n2 of the transformer T1.

[0042] FIG. 5 shows the relationship between the phase difference T_s and the voltage gain of the output voltage of the rectifier circuit 2 in the phase shift type conversion circuit. As shown in FIG. 5, the voltage gain has an approximately linear relationship with the phase difference T_s in the phase shift type conversion circuit. That is, in the phase shift type conversion circuit, the output voltage from the rectifier circuit 2 can be changed by adjusting the phase difference T_s relating to the period during which the opposing switching elements Q11 and Q14 are on at the same time and the period during which the opposing switching elements Q12 and Q13 are on at the same time.

[0043] Also, the phase shift type conversion circuit is characterized in that the adjustable range of the voltage ratio (voltage gain) between the input voltage and the output voltage is relatively wide. When the configuration shown in FIG. 2 is operated as the phase shift type conversion circuit, noise is reduced compared with forward type converters or flyback type converters, since a period during which current does not flow is unlikely to occur. Note that the voltage adjustment can only be performed in a voltage range not exceeding an upper limit value, with the upper limit value being a voltage determined by the winding ratio (number of winding of secondary winding n2/number of winding of primary winding n1) of the transformer T1.

[0044] As described above, the control unit 3 is configured to select the operation of the switching circuit 1 from a first operation and a second operation. The first operation is an operation in which the relationship (voltage ratio) between the input voltage and the output voltage is adjusted by controlling the operation frequency of the switching circuit 1. The second operation is an operation in which the relationship (voltage ratio) between the input voltage and the output voltage is adjusted by con-

trolling the period during which a current flows from the switching circuit 1 to the primary winding n1 of the transformer T1. Operations of both the LLC resonance type conversion circuit and the phase shift type conversion circuit are possible by only changing the operation of the switching circuit 1 by switching on/off the switch SW1. Accordingly, both features of the LLC resonance type conversion circuit and the phase shift type conversion circuit can be utilized so as to complement each other according to the state of the load.

[0045] Also, the configuration according to the present embodiment is realized by adding the switch SW1 to the LLC resonance type conversion circuit and adding, to the control unit 3, a program for enabling an operation as the phase shift type conversion circuit. Accordingly, in a case where the LLC resonance type conversion circuit has already been designed, an electric power conversion device having the functions of both the LLC resonance type conversion circuit and the phase shift type conversion circuit can be provided without a large-scale design change.

[0046] For example, the case is assumed where the input voltage is 400V and the output voltage is changed in a range from 100 to 500V. In this case, in a range of the output voltage from 100 to 400V, it is possible to adjust the output voltage in a wide range by causing the conversion circuit to operate as the phase shift type conversion circuit by switching on the switch SW1. On the other hand, in a range of the output voltage from 400 to 500V, it is possible to obtain an output voltage in a voltage range higher than the input voltage by causing the conversion circuit to operate as the LLC resonance type conversion circuit by switching off the switch SW1.

[0047] Also, in a case where the current detector is provided as described above, the on/off of the switch SW1 may be determined based on the output current of the rectifier circuit 2 (electric power conversion device). The input side circuit of the rectifier circuit 2 may be caused to operate as the phase shift type conversion circuit if the output current is a prescribed value (1A, for example) or less in a case where the load is light, and the input side circuit of the rectifier circuit 2 may be caused to operate as the LLC resonance type conversion circuit if the output current exceeds the prescribed value. In this operation, a stable operation with a small noise component is possible by causing the input side circuit of the rectifier circuit 2 to operate as the phase shift type conversion circuit in a light load region in which the operation of the LLC resonance type conversion circuit becomes unstable and a noise component increases in the output of the rectifier circuit 2.

[0048] There are cases where an electric power conversion efficiency is considered as a condition to switch on/off the switch SW1. For example, in a case where both the LLC resonance type conversion circuit and the phase shift type conversion circuit can be used such that a ratio of the input voltage and the output voltage becomes a desired voltage ratio, it is possible to determine

the on/off of the switch SW1 based on the conversion efficiency. For example, assuming that the conversion efficiency of the LLC resonance type conversion circuit is E1, the conversion efficiency of the phase shift type conversion circuit is E2, and E1 and E2 satisfy a condition of $E1 > E2$ when the input voltage and the output voltage are desired values, it is desirable to use the LLC resonance type conversion circuit, since the conversion efficiency is higher compared with a case where the phase shift type conversion circuit is used.

[0049] That is to say, the control unit 3 is configured to select whichever of the first operation and the second operation has a higher stored conversion efficiency, when both the first operation of the first conversion circuit and the second operation of the second conversion circuit are operable, in an overlapping region between the relationship (voltage ratio) in the first conversion circuit and the relationship (voltage ratio) in the second conversion circuit.

[0050] Note that, in a case where the on/off of the switch SW1 is determined while taking the conversion efficiency into consideration as described above, the relationship between the efficiency and the input voltage and the output voltage needs to be stored in the control unit 3. Also, a function of notifying the input voltage and the output voltage of the control unit 3 is required.

[0051] As described above, an operation as the LLC resonance type conversion circuit and an operation as the phase shift type conversion circuit can be switched by selecting the on/off of the switch SW1 so as to change the operation of the switching circuit 1. Accordingly, usability is improved by switching on/off the switch SW1 while focusing on the output voltage range, the size of the load, the conversion efficiency, or the like.

[0052] Incidentally, when the switch SW1 is switched from off to on in order to switch from operating as the LLC resonance type conversion circuit to operating as the phase shift type conversion circuit, there are cases where charges are accumulated in the capacitor C1. When the switch SW1 is switched on in a state in which charges are accumulated in the capacitor C1, it is possible that a rush current flows through the switch SW1. The occurrence of such a rush current puts much stress on the switch SW1. Also, since the voltage gain differs in the operation of the LLC resonance type conversion circuit and in the operation of the phase shift type conversion circuit, the output voltage changes when the on/off of the switch SW1 is switched if the voltage gain is not considered when the on/off of the switch SW1 is switched.

[0053] In order to prevent the rush current, the control unit 3 should switch on the switch SW1 at the time of a zero-cross point of the current flowing through the resonance circuit constituted by the inductor L1, the capacitor C1, and the transformer T1 in a period during which the conversion circuit is operated as the LLC resonance type conversion circuit. The current Iq flowing through the resonance circuit has a waveform as shown in "A" of FIG.

6. The waveform of the current Iq is aligned with the on/off timings of the switching elements Q11 and Q14 shown in "B" of FIG. 6, and the current Iq becomes zero at the timings of switching on/off the switching elements Q11 and Q14, and the polarity of the current Iq inverts.

[0054] Accordingly, switching on the switch SW1 at the timing of switching on/off the switching element Q11 and Q14 prevents the rush current caused by the charges in the capacitor C1 from flowing through the switch SW1. In the illustrated example, the timing at which the switch SW1 is switched on is designated by an arrow, and this timing is matched with the timing at which the switching elements Q11 and Q14 are switched on. Although the timing shown in the diagram (timing at which switch SW1 is switched on) is matched with the timing at which the switching element Q11 and Q14 are switched on, the timing may be matched with the timing at which the switching element Q11 and Q14 are switched off.

[0055] Also, since the above operations are aligned with the on/off timing of the switching elements Q11 and Q14, the control unit 3 can easily determine the timing at which the switch SW1 is switched on. Note that it is possible, by detecting the voltage between two terminals of the capacitor C1 or by detecting the resonance current that flows through the resonance circuit, to determine the timing at which the switch SW1 is switched on according to the detected state.

[0056] Due to the above operations, the switch SW1 is switched on in a state in which charges are not accumulated in the capacitor C1 when transitioning from the LLC resonance type conversion circuit to the phase shift type conversion circuit. Accordingly, a rush current is prevented from flowing from the capacitor C1 to the switch SW1 when the switch SW1 is switched on. That is, the transition from the LLC resonance type conversion circuit to the phase shift type conversion circuit can be performed seamlessly.

[0057] Incidentally, the on/off timings of the switching elements Q11 to Q14 differs between a period during which the conversion circuit operates as the LLC resonance type conversion circuit and a period during which the conversion circuit operates as the phase shift type conversion circuit, as described above. Accordingly, the control unit 3 needs to also change the on/off timings of the switching elements Q11 to Q14 when switching on/off the switch SW1. However, it is difficult to adjust the phases of on/off of the switching elements Q11 to Q14 instantly when shifting between the LLC resonance type conversion circuit and the phase shift type conversion circuit. Therefore, it is desirable that a transition period during which the on/off timings of the switching elements Q11 to Q14 are changed is provided before and after the timing at which the switch SW1 is switched on/off. In the transition period, the on/off of the switching elements Q11 to Q14 is temporarily suspended or the on/off timings of the switching elements Q11 to Q14 are forcedly changed.

[0058] An example of on/off timings of the switching elements Q11 to Q14 is shown in "A" to "D" of FIG. 7 in

the case of transitioning from operating as the LLC resonance type conversion circuit to operating as the phase shift type conversion circuit. When operating as the LLC resonance type conversion circuit, the on/off timings are the same for the opposing switching elements Q11 and Q14, and the on/off timings are the same for the opposing switching elements Q12 and Q13. Also, when operating as the phase shift type conversion circuit, an appropriate phase difference is given between on timings of the opposing switching elements Q11 and Q14, and an appropriate phase difference is given between on timings of the opposing switching elements Q12 and Q13.

[0059] When transitioning from the operation as the LLC resonance type conversion circuit to the operation as the phase shift type conversion circuit, a transition period P_s during which all the switching elements Q11 to Q14 are switched off is provided. The control unit 3 is configured to switch on the switch SW1 in this transition period P_s .

[0060] Incidentally, when transitioning between the LLC resonance type conversion circuit and the phase shift type conversion circuit, the output voltage of the rectifier circuit 2 preferably does not vary. Accordingly, when the operation is transitioned, the control unit 3 needs to adjust the on/off timings of the switching elements Q11 to Q14 such that the output voltage of the rectifier circuit 2 remains the same. In order to realize this operation, the relationship between the operation frequency and the voltage gain in the LLC resonance type conversion circuit and the relationship between the phase difference T_s and the voltage gain in the phase shift type conversion circuit are stored in a storage unit provided in the control unit 3. Also, the control unit 3 is configured to monitor the input voltage, the output voltage, and the output current, and determine the on/off timings of the switching elements Q11 to Q14 according to the voltage gain immediately before transitioning between the LLC resonance type conversion circuit and the phase shift type conversion circuit by referring to the storage unit. Variation in the output voltage of the rectifier circuit 2 is suppressed by this operation when switching between operations of the LLC resonance type conversion circuit and the phase shift type conversion circuit.

[0061] The relationship between the voltage gain and the conversion efficiency of the LLC resonance type conversion circuit and the relationship between the voltage gain and the conversion efficiency of the phase shift type conversion circuit may be stored in the storage unit provided in the control unit 3. That is, if the relationships between the conversion efficiency and the voltage ratio (voltage gain), which is a ratio of the input voltage and the output voltage, are known for both the LLC resonance type conversion circuit and the phase shift type conversion circuit, an operation having higher conversion efficiency can be selected out of the operation of the LLC resonance type conversion circuit and the operation of the phase shift type conversion circuit according to the relationship between the input voltage and the output

voltage. That is, when both the LLC resonance type conversion circuit and the phase shift type conversion circuit are operable in order to obtain a desired voltage ratio between the input voltage and the output voltage, the control unit 3 determines the on/off of the switch SW1 based on the conversion efficiency, as described above.

[0062] In the embodiment described above, although the capacitor C1 is inserted between the inductor L1 and the transformer T1, a similar operation can be obtained in a configuration in which an inductor L1 is inserted between the capacitor C1 and the transformer T1.

[0063] Note that although, in the present embodiment, only a configuration in which electric power is supplied from the switching circuit 1 to the transformer T1 is described, it is possible to supply electric power from the rectifier circuit 2 to the transformer T1 by controlling the on/off of the switching elements Q21 to Q24 provided in the rectifier circuit 2. That is, electric power can be transmitted bi-directionally. Also, the switching circuit 1 is not limited to the configuration shown in FIG. 2. The switching circuit 1 may include a configuration in which a period during which current flows through the series circuit constituted by the primary winding n_1 of the transformer T1 and the capacitor C1 is controlled by switching.

Embodiment 2

[0064] As shown in FIG. 8, an electric power conversion device according to the present embodiment includes a configuration in which a series circuit constituted by a switch SW2 and a resistor R2 is connected in parallel with the switch SW1 in the configuration of Embodiment 1.

[0065] When transitioning from an LLC resonance type conversion circuit to a phase shift type conversion circuit in the configuration according to the present embodiment, a control unit 3 is configured to switch on the switch SW2 before switching on the switch SW1. That is, the control unit 3 is configured to switch on the switch SW1 after switching on the switch SW2 when transitioning from a first operation of the LLC resonance type conversion circuit to a second operation of the phase shift type conversion circuit. A timing at which the switch SW2 is switched on need not be matched with the on/off timings of switching elements Q11 to Q14. When the switch SW2 is switched on, charges accumulated in the capacitor C1 are discharged via the resistor R2, and the current flowing through the switch SW2 is limited by the resistance value of the resistor R2.

[0066] Furthermore, after the switch SW2 is switched on and the charge amount in the capacitor C1 decreases, the control unit 3 is configured to switch on the switch SW1. The charge amount over time in the capacitor C1 after the switch SW2 is switched on is determined by a voltage between the two terminals and the capacitance of the capacitor C1 and a resistance value of the resistor R2. Accordingly, the time period from switching on the switch SW2 to switching on the switch SW1 is dependent

on the voltage between the two terminals of the capacitor C1. When simplified control is desired, the time period may be set to a fixed time period using the capacitance of the capacitor C1 and the resistance value of the resistor R2.

[0067] In this configuration, since the timing at which the switch SW1 is switched on does not depend on the waveform of the current flowing through the resonance circuit, the response time of the switch SW1 need not be taken into consideration, and it is possible to use an electromagnetic relay having a longer response time than the semiconductor switch as the switch SW1. The other configurations and operations of the present embodiment are similar to those of Embodiment 1.

Embodiment 3

[0068] An electric power conversion device according to the present embodiment is a modification of Embodiment 1, and uses a plurality of (three each in diagram) inductors L31 to L33 and switches (short-circuit switches) SW31 to SW33, as shown in FIG. 9, instead of using the inductor L1 shown in FIG. 1. The switches SW31 to SW33 are respectively connected in parallel with the inductors L31 to L33. That is, the switch SW31 is connected in parallel with the inductor L31, the switch SW32 is connected in parallel with the inductor L32, and the switch SW33 is connected in parallel with the inductor L33. Accordingly, at least one of the inductors L31 to L33 is used in a resonance circuit, depending on the combination of on/off of the switches SW31 to SW33. In other words, the inductance used in the resonance circuit is adjusted by combining the on/off of the switches SW31 to SW33, and as a result the resonance frequency can be selected from a plurality of frequencies.

[0069] For example, in the configuration shown in the diagram, if the switch SW31 is switched off and the switches SW32 and SW33 are switched on, only the inductor L31 effectively functions in the resonance circuit. Also, if all the switches SW31 to SW33 are switched off, a series circuit of the three inductors L31 to L33 effectively functions in the resonance circuit. Accordingly, it is possible to change the resonance frequency of the resonance circuit in a wide range.

[0070] Instead of the configuration shown in FIG. 9, the electric power conversion device may use a plurality of (three each in diagram) inductors L41 to L43 and switches SW41 to SW43 so as to have the connection relationship shown in FIG. 10, instead of using the inductor L1 shown in FIG. 1. In this configuration, the three inductors L41 to L43 are connected in series, and one terminal (first terminal) of each of the three switches SW41 to SW43 is connected to one terminal of the series circuit of the inductors L41 to L43. Also, the other terminal (second terminal) of each of the three switches SW41 to SW43 is respectively connected to one terminal of the three inductors L41 to L43. That is, the switch SW41 is connected in parallel with the inductor L41, the switch

SW42 is connected in parallel with a series circuit of the inductors L41 and L42, and the switch SW43 is connected in parallel with the series circuit of the inductors L41 to L43.

[0071] In the configuration shown in FIG. 9, the number of types of selectable inductances is determined by the number of combinations of on/off of the switches SW31 to SW33. In the configuration shown in FIG. 10, the number of types of selectable inductances is one more than the number of switches SW41 to SW43. That is, in the configuration shown in FIG. 10, only four states are selectable, namely a state in which the switch SW43 is on, a state in which the switch SW43 is off and the switch SW42 is on, a state in which the switches SW42 and SW43 are off and the switch SW41 is on, and a state in which the switches SW41 to SW43 are off. However, whereas the number of switches SW41 to SW43 included in the resonance circuit is one or less in the configuration shown in FIG. 10, the number of switches SW31 to SW33 included in the resonance circuit changes in a range from zero to three in the configuration shown in FIG. 9. Accordingly, an average loss of the resonance circuit caused by the switch in the configuration shown in FIG. 10 is smaller than that in the configuration shown in FIG. 9.

[0072] Although the electric power conversion device according to the present embodiment includes three each of the inductors and the switches, the numbers of the inductors and the switches are selected arbitrarily according to the number of frequencies to which the resonance frequency is changed. The other configurations and operations of the present embodiment are similar to those of Embodiment 1, and also, the configuration of the present embodiment can be combined with a configuration in which the switch SW2 and the resistor R2 are provided similarly to Embodiment 2.

Embodiment 4

[0073] The present embodiment is a modification of Embodiment 1, and uses a plurality of (three each in diagram) capacitors C51 to C53 and switches SW51 to SW53 as shown in FIG. 11 as a capacitance element, instead of using the capacitor C1 shown in FIG. 1. The switches SW51 to SW53 are respectively connected in parallel with the capacitors C51 to C53. That is, the switch SW51 is connected in parallel with the capacitor C51, the switch SW52 is connected in parallel with the capacitor C52, and the switch SW53 is connected in parallel with the capacitor C53. Accordingly, at least one of the capacitors C51 to C53 is used in the resonance circuit depending on the combination of on/off of the switches SW51 to SW53. In other words, the capacitance used in the resonance circuit is adjusted by combining the on/off of the switches SW51 to SW53, and as a result the resonance frequency can be selected from a plurality of frequencies.

[0074] Also, in the diagram shown, an inductor L3 is

used instead of the inductor L1, and a switch SW3 is connected in parallel with the inductor L3. Accordingly, whether or not the inductor L3 is included in the resonance circuit can be selected by the on/off of the switch SW3.

[0075] For example, in the configuration shown in the diagram, if the switch SW51 is switched off and the switches SW52 and SW53 are switched on, only the capacitor C51 effectively functions in the resonance circuit. Also, if all the switches SW51 to SW53 are switched off, a series circuit of the three capacitors C51 to C53 effectively functions in the resonance circuit. Accordingly, it is possible to change the resonance frequency of the resonance circuit in a wide range.

[0076] Instead of the configuration shown in FIG. 11, the electric power conversion device may use a plurality of (three each in diagram) capacitors C61 to C63 and the switches SW61 to SW63 so as to have the connection relationship shown in FIG. 12, instead of using the capacitor C1 shown in FIG. 1. In this configuration, the three capacitors C61 to C63 are connected in series, and one terminal (first terminal) of each of the three switches SW61 to SW63 is connected to one terminal of the series circuit of the capacitors C61 to C63. Also, the other terminal (second terminal) of each of the three switches SW61 to SW63 is respectively connected to one terminal of the three capacitors C61 to C63. That is, the switch SW61 is connected in parallel with the capacitor C61, the switch S62 is connected in parallel with the series circuit of capacitors C61 and C62, and the switch SW63 is connected in parallel with the series circuit of the capacitors C61 to C63.

[0077] In the configuration shown in FIG. 11, the number of types of selectable capacitances is determined by the number of combinations of on/off of the switches SW51 to SW53. In the configuration shown in FIG. 12, the number of types of selectable capacitances is one more than the number of switches SW61 to SW63. That is, in the configuration shown in FIG. 12, only four states are selectable, namely a state in which the switch SW63 is on, a state in which the switch SW63 is off and the switch SW62 is on, a state in which the switches SW62 and SW63 are off and the switch SW61 is on, and a state in which the switches SW61 to SW63 are off. However, whereas the number of switches SW61 to SW63 included in the resonance circuit is one or less in the configuration shown in FIG. 12, the number of switches SW51 to SW53 included in the resonance circuit changes in a range from zero to three in the configuration shown in FIG. 11. Accordingly, an average loss of the resonance circuit due to the switch in the configuration shown in FIG. 12 is smaller than that in the configuration shown in FIG. 11.

[0078] Also, as described in Embodiment 1, the timings at which the switches SW51 to SW53 or SW61 to SW63 are switched on are controlled so as to be matched with the on/off timings of the switching elements Q11 to Q14 provided in the switching circuit 1. Therefore, a rush cur-

rent caused by the charges in the capacitors C51 to C53 or C61 to C63 is prevented from flowing through the switches SW51 to SW53 or SW61 to SW63.

[0079] Although the electric power conversion device according to the present embodiment includes three each of the capacitors and the switches, the numbers of the capacitors and the switches are selected arbitrarily according to the number of frequencies to which the resonance frequency is changed. The other configurations and operations of the present embodiment are similar to those of Embodiment 1, and also, the configuration of the present embodiment may be combined with a configuration in which the switch SW2 and the resistor R2 are provided similarly to Embodiment 2. Furthermore, the configuration of the present embodiment and the configuration shown in Embodiment 3 may be combined.

[0080] Although the present invention has been described in a preferred embodiment, various modifications and variations are possible by those skilled in the art without departing from the spirit or scope of this invention, that is, the claims.

Claims

1. An electric power conversion device comprising:

a transformer configured to transmit electric power between an input and an output;
a resonance capacitance element connected in series with a primary winding of the transformer;
a switching circuit configured to switch on/off current that flows through a series circuit constituted by the primary winding of the transformer and the capacitance element;
a rectifier circuit configured to rectify and output electric power induced in a secondary winding of the transformer;
a switch connected in parallel with the capacitance element; and
a control unit configured to control a period during which current flows through the series circuit constituted by the primary winding of the transformer and the capacitance element and on/off of the switch,
wherein the control unit is configured to switch off the switch so as to configure, as an input-side circuit of the rectifier circuit, a first conversion circuit configured to change a relationship between an input voltage and an output voltage based on an operation frequency of the switching circuit,
the control unit being configured to switch on the switch so as to configure, as the input-side circuit of the rectifier circuit, a second conversion circuit configured to change a relationship between an input voltage and an output voltage based on a period during which current flows from the

- switching circuit to the primary winding of the transformer,
the control unit being configured to select an operation of the switching circuit from a first operation for adjusting the relationship in the first conversion circuit by controlling the operation frequency and a second operation for adjusting the relationship in the second conversion circuit by controlling the period during which the current flows.
2. The electric power conversion device according to claim 1, wherein the control unit is configured to match a timing at which the switch is switched on with a zero-crossing point of current that flows from the switching circuit to the series circuit constituted by the primary winding of the transformer and the capacitance element, when transitioning from the first operation of the first conversion circuit to the second operation of the second conversion circuit.
 3. The electric power conversion device according to claim 1, further comprising a series circuit constituted by a second switch and a resistor and connected in parallel with the capacitance element, wherein the control unit is configured to switch on the switch after switching on the second switch when transitioning from the first operation of the first conversion circuit to the second operation of the second conversion circuit.
 4. The electric power conversion device according to any one of claims 1 to 3, wherein the control unit is configured to provide a transition period during which an operation of the switching circuit is an operation different from the first operation of the first conversion circuit and the second operation of the second conversion circuit, when switching between the first operation of the first conversion circuit and the second operation of the second conversion circuit.
 5. The electric power conversion device according to any one of claims 1 to 4, wherein the control unit is configured to store the operation frequency of the switching circuit in the first operation of the first conversion circuit in association with the relationship in the first conversion circuit, and store the period during which current flows from the switching circuit to the primary winding of the transformer in the second operation of the second conversion circuit in association with the relationship in the second conversion circuit, the control unit being configured to control the switching circuit in accordance with the stored operation frequency and the stored period during which current flows, such that the relationship between the input voltage and the output voltage after switching coincides with the relationship between the input voltage and the output voltage before switching, when switching between the first operation of the first conversion circuit and the second operation of the second conversion circuit.
 6. The electric power conversion device according to any one of claims 1 to 5, wherein the control unit is configured to store a conversion efficiency in the first operation of the first conversion circuit in association with the relationship in the first conversion circuit, and store a conversion efficiency in the second operation of the second conversion circuit in association with the relationship in the second conversion circuit, the control unit being configured to select whichever of the first operation and the second operation has a higher stored conversion efficiency, when both the first operation of the first conversion circuit and the second operation of the second conversion circuit are operable, in an overlapping region between the relationship in the first conversion circuit and the relationship in the second conversion circuit.
 7. The electric power conversion device according to any one of claims 1 to 6, further comprising a resonance inductor that is connected between the primary winding of the transformer and the switching circuit.
 8. The electric power conversion device according to any one of claims 1 to 6, further comprising:
 - a plurality of resonance inductors that are connected between the primary winding of the transformer and the switching circuit; and
 - a short-circuit switch configured to short-circuit at least one of the plurality of inductors.
 9. The electric power conversion device according to any one of claims 1 to 8, wherein the capacitance element comprises a plurality of resonance capacitors that are connected between the primary winding of the transformer and the switching circuit, and wherein the switch is configured to short-circuit at least one of the plurality of capacitors.
 10. The electric power conversion device according to any one of claims 1 to 8, wherein the capacitance element comprises a resonance capacitor that is connected between the primary winding of the transformer and the switching circuit, and wherein the switch is configured to short-circuit the capacitor.
 11. The electric power conversion device according to any one of claims 1 to 10,

wherein both the relationship in the first conversion circuit and the relationship in the second conversion circuit are a voltage ratio between the input voltage and the output voltage.

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12. The electric power conversion device according to any one of claims 1 to 11,

wherein the first conversion circuit is a resonance type conversion circuit configured to change the relationship in the first conversion circuit using a resonance circuit constituted by the capacitance element and the transformer by adjusting the operation frequency of the switching circuit, and

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wherein the second conversion circuit is a phase shift type conversion circuit configured to change the relationship in the second conversion circuit by adjusting the period during which current flows from the switching circuit to the primary winding of the transformer through changing a phase of switching on a switching element of the switching circuit.

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FIG. 1

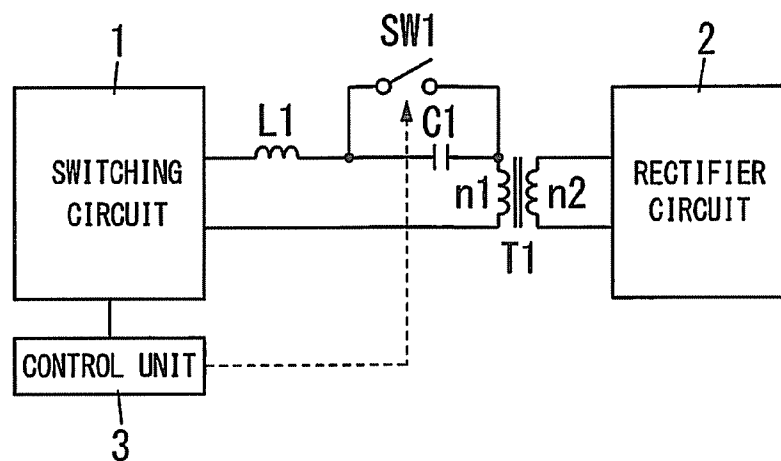


FIG. 2

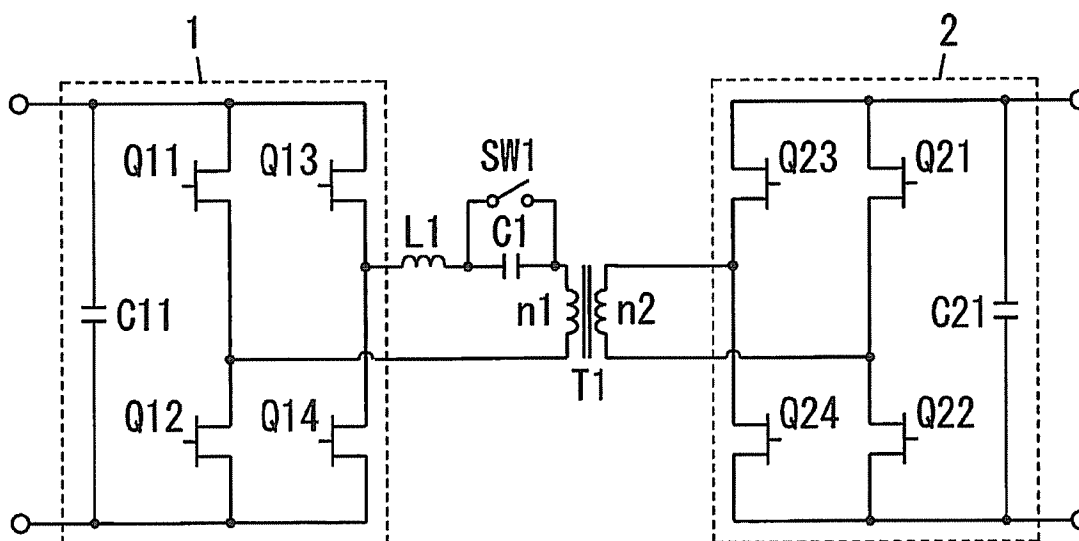


FIG. 3

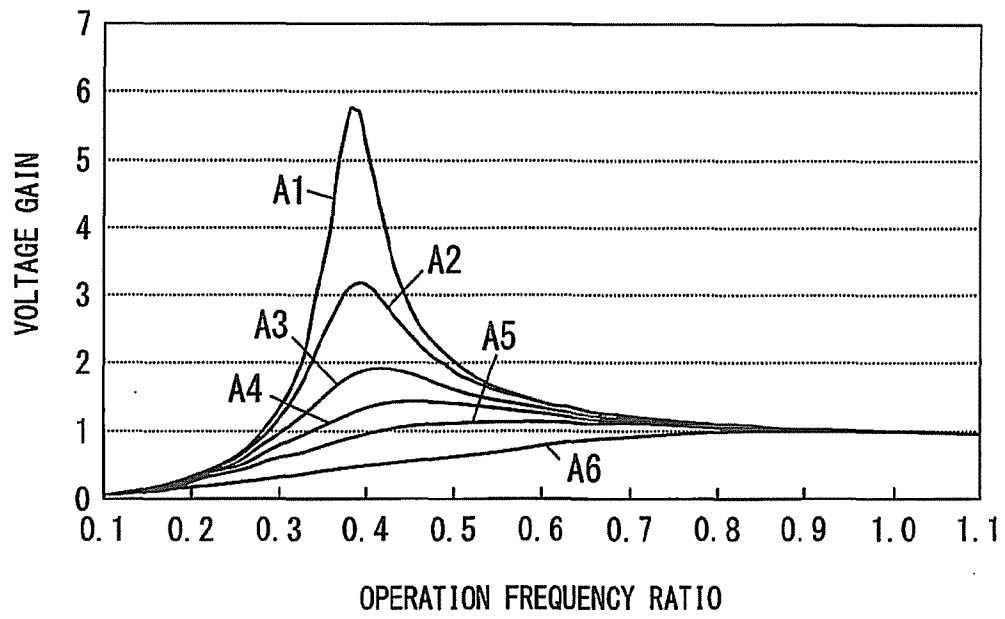


FIG. 4

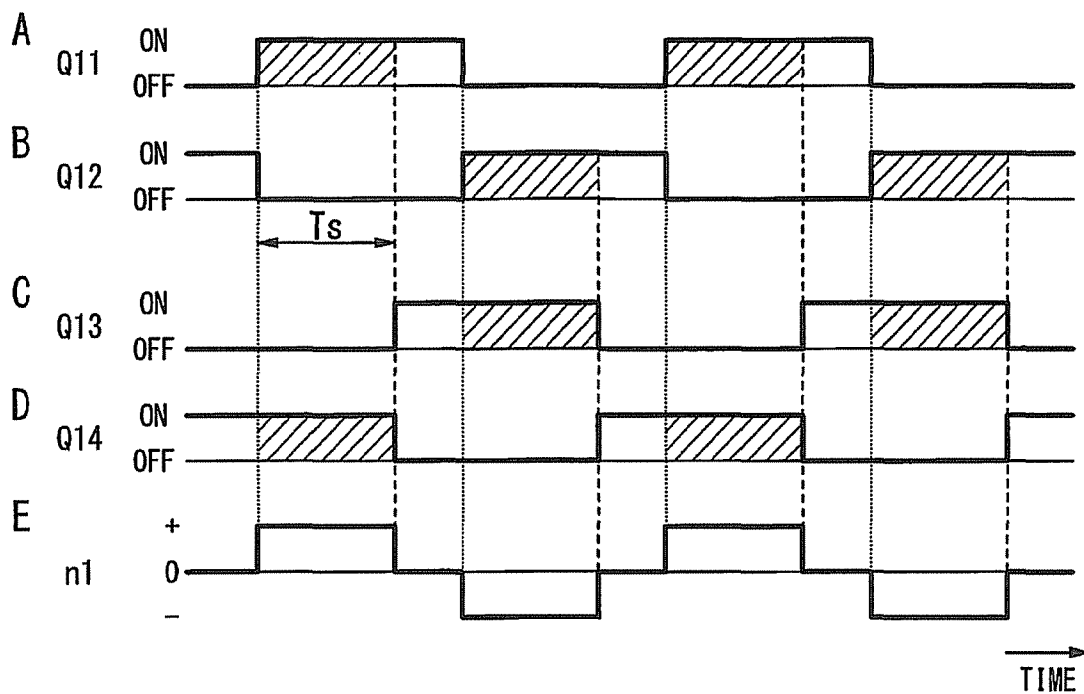


FIG. 5

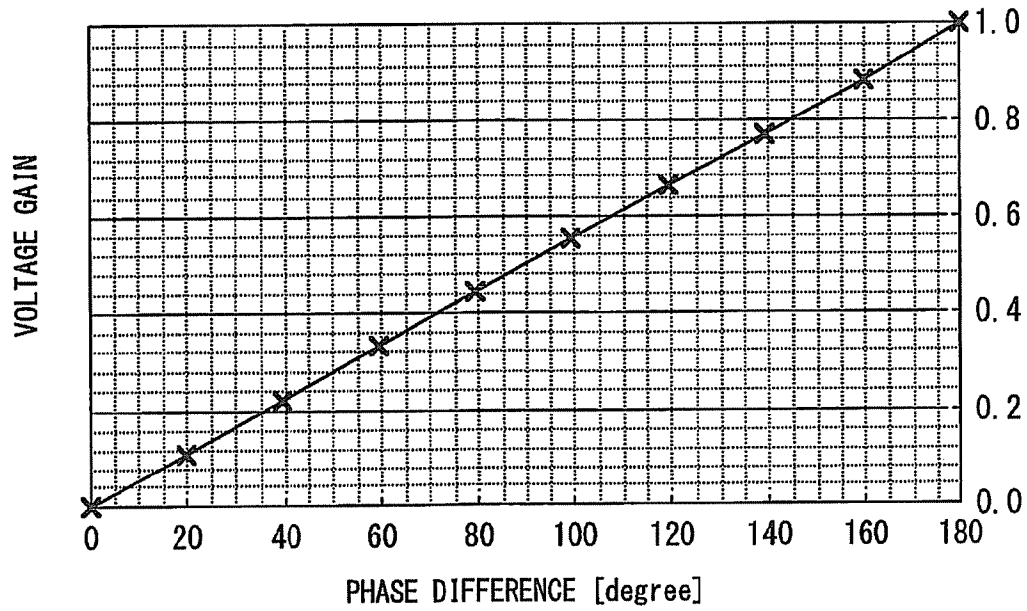


FIG. 6

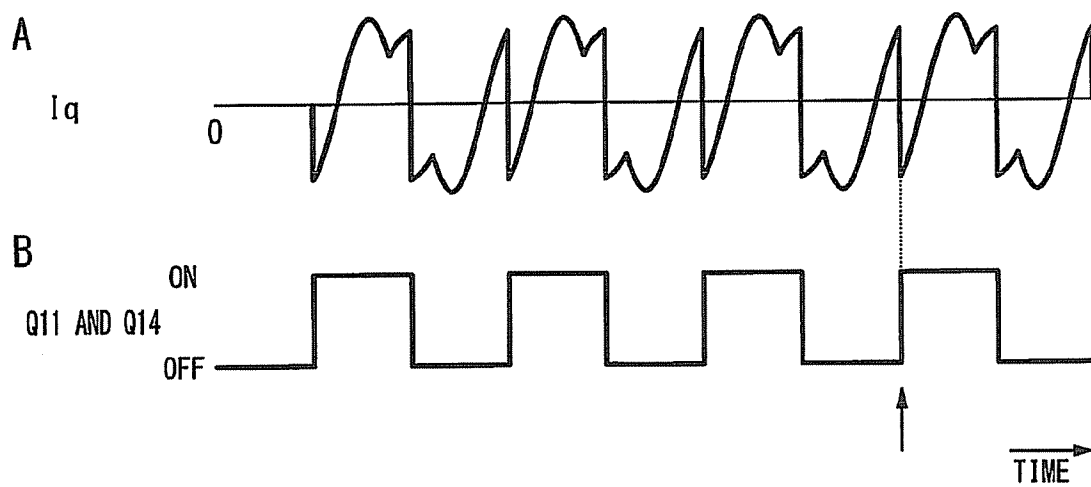


FIG. 7

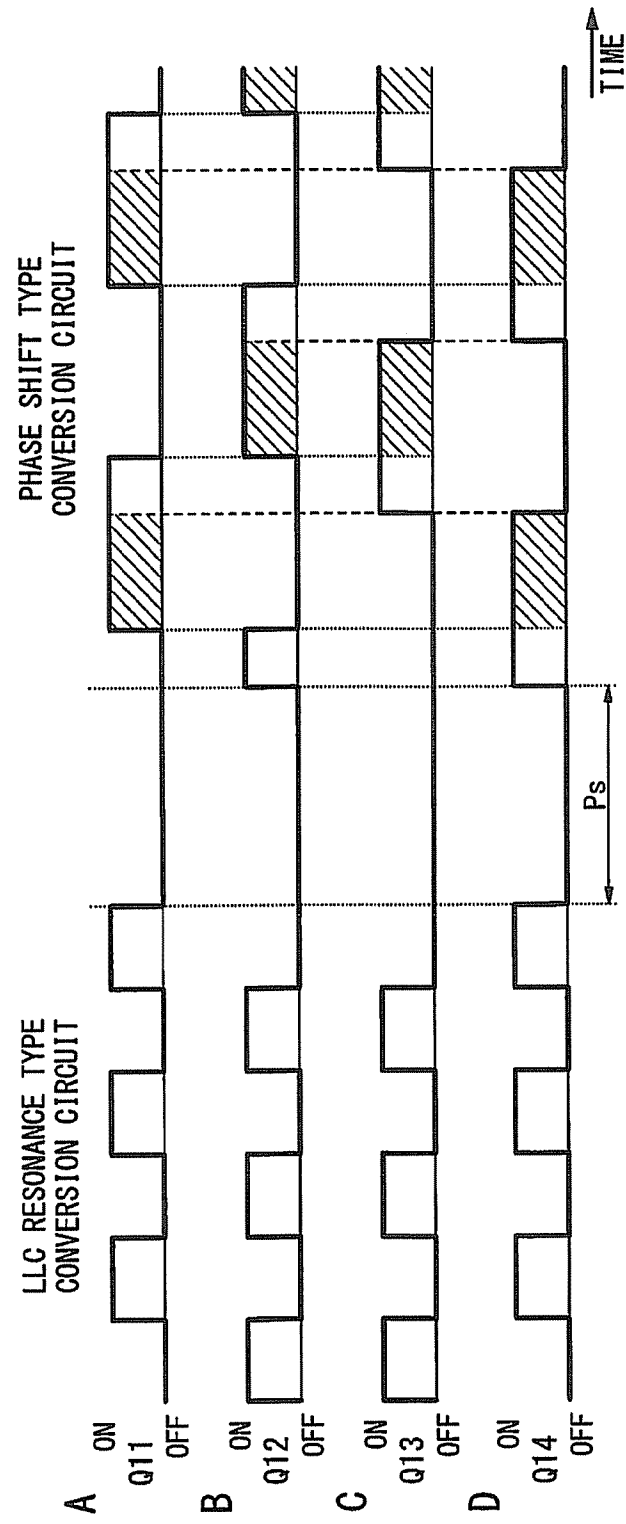


FIG. 8

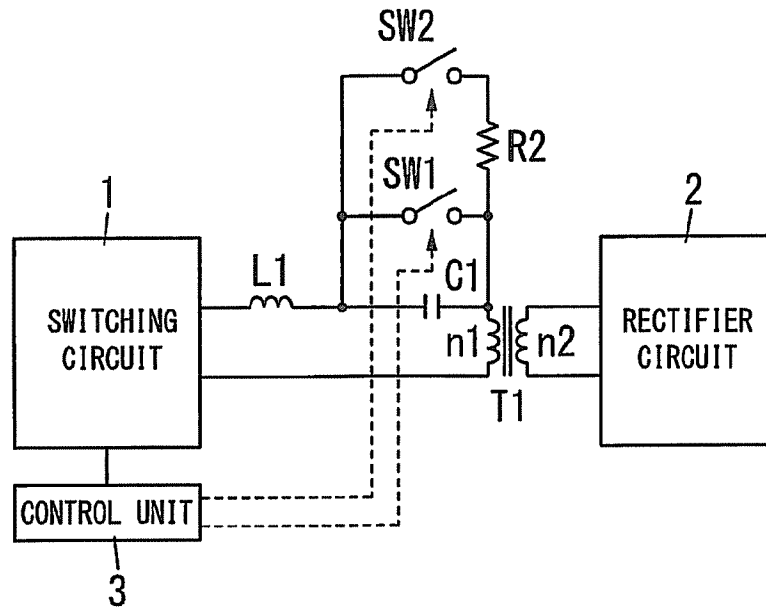


FIG. 9

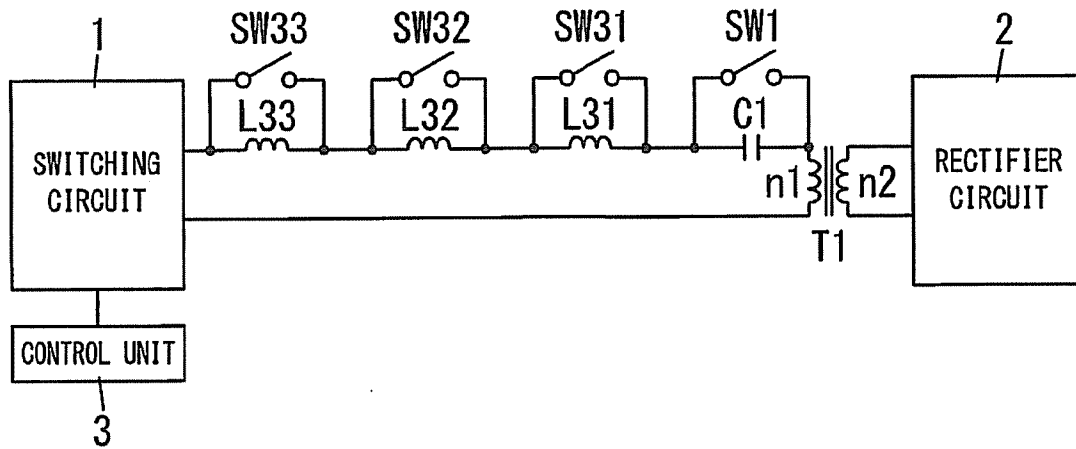


FIG. 10

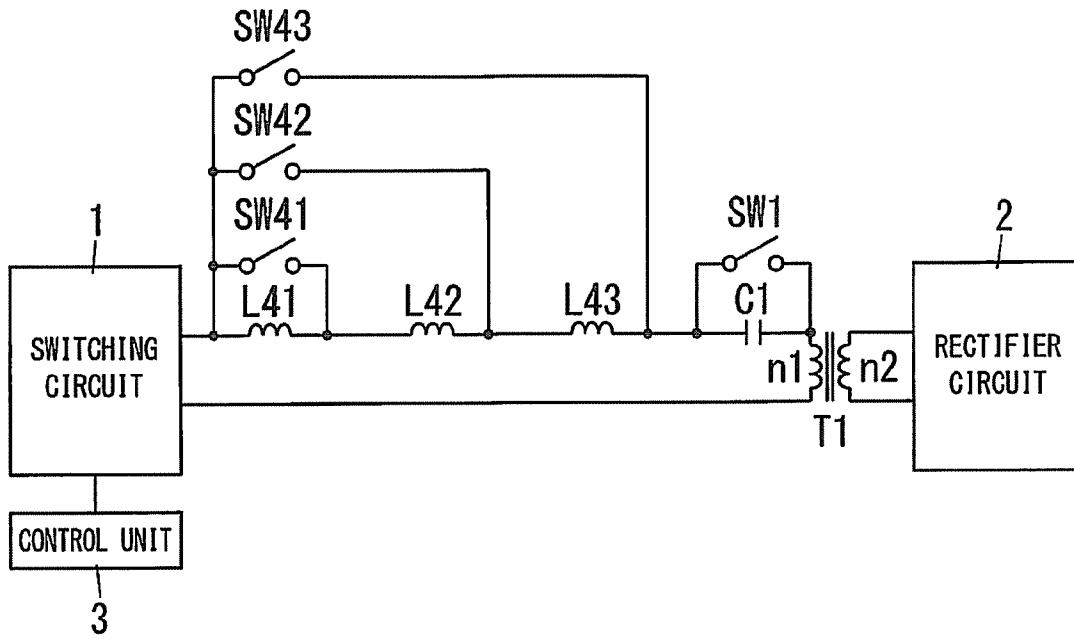


FIG. 11

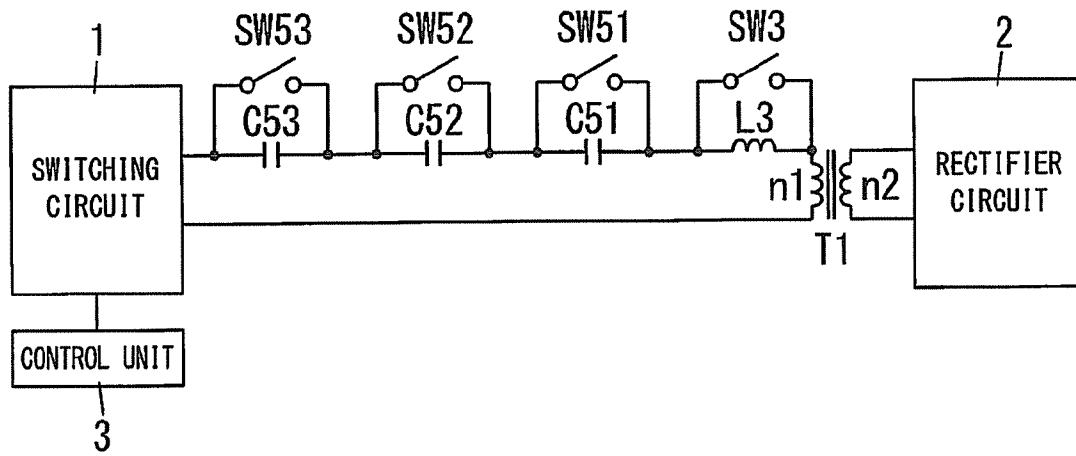
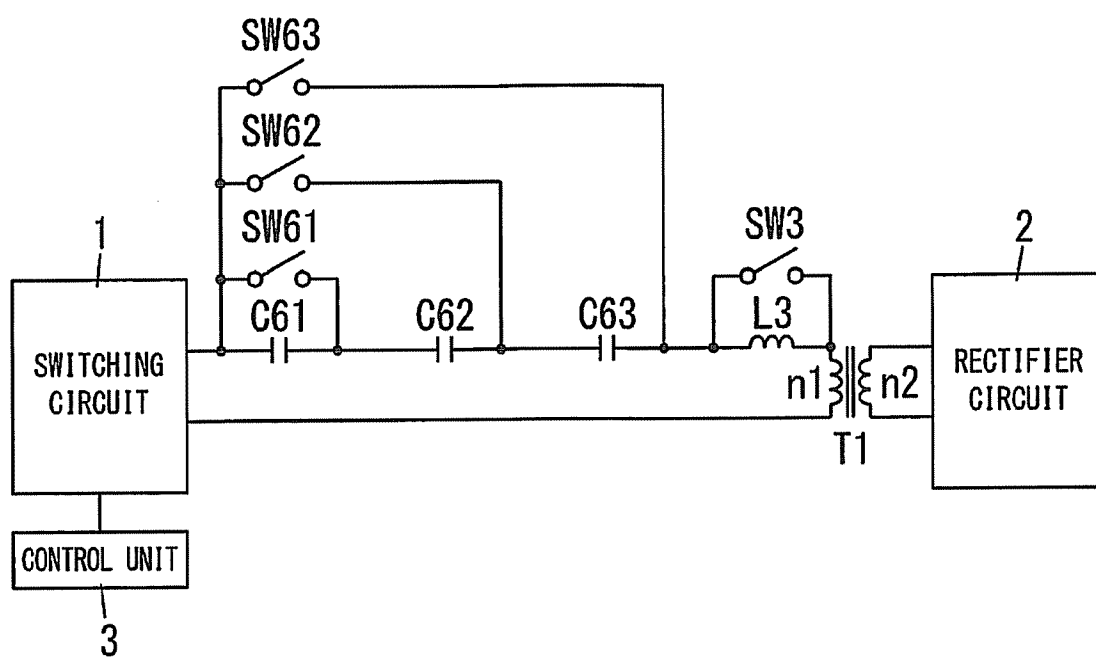


FIG. 12



INTERNATIONAL SEARCH REPORT

International application No.

PCT/JP2013/003211

A. CLASSIFICATION OF SUBJECT MATTER

H02M3/335(2006.01) i, H02M3/28(2006.01) i

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
H02M3/00-3/44

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Jitsuyo Shinan Koho	1922-1996	Jitsuyo Shinan Toroku Koho	1996-2013
Kokai Jitsuyo Shinan Koho	1971-2013	Toroku Jitsuyo Shinan Koho	1994-2013

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	JP 2006-158137 A (TDK Corp.), 15 June 2006 (15.06.2006), abstract; fig. 1 (Family: none)	1-12
A	JP 4-125062 A (Toyoda Automatic Loom Works, Ltd.), 24 April 1992 (24.04.1992), claims; drawings (Family: none)	1-12
A	US 2009/0046482 A1 (SMITH, Stephen William), 19 February 2009 (19.02.2009), fig. 1 (Family: none)	1-12

☒ Further documents are listed in the continuation of Box C.
 ☐ See patent family annex.

* Special categories of cited documents:

"A" document defining the general state of the art which is not considered to be of particular relevance

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"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search
22 July, 2013 (22.07.13)Date of mailing of the international search report
30 July, 2013 (30.07.13)Name and mailing address of the ISA/
Japanese Patent Office

Authorized officer

Facsimile No.

Telephone No.

INTERNATIONAL SEARCH REPORT

International application No.

PCT/JP2013/003211

C (Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	JP 2010-207014 A (Fujitsu Telecom Networks Ltd.), 16 September 2010 (16.09.2010), abstract; paragraphs [0017], [0019] to [0024]; fig. 1 (Family: none)	1-12
A	JP 2004-260928 A (Omron Corp.), 16 September 2004 (16.09.2004), abstract; drawings (Family: none)	1-12
A	JP 3-128672 A (General Electric Co.), 31 May 1991 (31.05.1991), entire text & US 4951185 A & EP 0408253 A2 & CA 2012811 A1	1-12
A	JP 3-251079 A (Origin Electric Co., Ltd.), 08 November 1991 (08.11.1991), claims; drawings (Family: none)	1-12
A	JP 2011-234541 A (Panasonic Electric Works Co., Ltd.), 17 November 2011 (17.11.2011), abstract & WO 2011/135415 A1 & US 2013/0044519 A1 & EP 2566027 A1 & CN 102986125 A & KR 10-2013-0036226 A	1-12

REFERENCES CITED IN THE DESCRIPTION

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Patent documents cited in the description

- JP 2011234541 A [0002]

**Espacenet****Bibliographic data: JPH02280666 (A) — 1990-11-16**

RESONANT SWITCHING POWER SOURCE

Inventor(s): TANIYAMA KOREYASU ± (TANIYAMA KOREYASU)

Applicant(s): SANKEN ELECTRIC CO LTD ± (SANKEN ELECTRIC CO LTD)

Classification: - **international:** *H02M3/28; H02M3/335*; (IPC1-7): H02M3/28; H02M3/335
- **cooperative:**

Application number: JP19890100847 19890420

Priority number(s): JP19890100847 19890420

Also published as: JPH0746903 (B2)

Abstract of JPH02280666 (A)

PURPOSE: To reduce power loss by connecting a switching element in parallel with a resonant capacitor. **CONSTITUTION:** A resonant DC-DC converter comprising a DC power source 1, first and second transistor switching elements Q1, Q2, a series resonant inductance element L1, a transformer T, and the like, feeds power through an output smoothing/rectifying circuit 7 to a load 8. First and second switching elements S1, S2 for controlling charge/discharge operation are connected in parallel with the resonant capacitor C1, and a PWM pulse forming circuit 25 is provided. By such arrangement, power to be fed to the resonant circuit can be controlled through control of ON time width of the switching elements S1, S2.

⑫ 公開特許公報(A) 平2-280666

⑤ Int. Cl.³H 02 M 3/335
3/28

識別記号

E 7829-5H
Q 7829-5H
H 7829-5H

庁内整理番号

④ 公開 平成2年(1990)11月16日

審査請求 未請求 請求項の数 7 (全11頁)

⑬ 発明の名称 共振型スイッチング電源回路

⑰ 特 願 平1-100847

⑱ 出 願 平1(1989)4月20日

⑲ 発 明 者 谷 山 之 康 埼玉県新座市北野3丁目6番3号 サンケン電気株式会社
内

⑳ 出 願 人 サンケン電気株式会社 埼玉県新座市北野3丁目6番3号

㉑ 代 理 人 弁理士 高野 則次

明 細 書

1. 発明の名称

共振型スイッチング電源回路

2. 特許請求の範囲

〔1〕 第1の端子と第2の端子と前記第1及び第2の端子の midpoint 電位を与える第3の端子とを有する直流電源回路と、

前記第1の端子と前記第2の端子との間に互いに直列になるように接続された第1及び第2のスイッチング素子と、

前記第1及び第2のスイッチング素子を交互にオン・オフ制御するための制御回路と、

一端が前記第3の端子に接続されている共振用コンデンサと、

前記共振用コンデンサの他端と前記第1及び第2のスイッチング素子の接続 midpoint との間に接続された共振用インダクタンス素子と、

前記インダクタンス素子に直列に接続されているか又は前記インダクタンス素子に電磁結合されている負荷回路と

から成る共振型スイッチング電源回路において、

前記コンデンサに並列に前記コンデンサの充電及び／又は放電を制御するための素子を接続したことを特徴とする共振型スイッチング電源回路。

〔2〕 更に、前記コンデンサの他端と前記電源回路の第1及び第2の端子との間に前記コンデンサの電圧をクランプするためのクランプ用素子をそれぞれ接続したことを特徴とする請求項1記載の共振型スイッチング電源回路。

〔3〕 第1及び第2の端子を有する直流電源回路と、

前記第1及び第2の端子の間に互いに直列になるように接続された第1及び第2のスイッチング素子と、

前記第1及び第2の端子の間に互いに直列になるように接続された第3及び第4のスイッチング素子と、

前記第1及び第4のスイッチング素子と前記第2及び第3のスイッチング素子とを交互にオン・オフ制御するための制御回路と、

一端が前記第3及び第4のスイッチング素子の接続中点に接続された共振用コンデンサと、

前記共振用コンデンサの他端と前記第1及び第2のスイッチング素子の接続中点との間に接続された共振用インダクタンス素子と、

前記インダクタンス素子に直列に接続されているか、又は前記インダクタンス素子に電磁結合されている負荷回路と

から成る共振型スイッチング電源回路において、

前記コンデンサに並列に前記コンデンサの充電及び／又は放電を制御するための素子を接続したことを特徴とする共振型スイッチング電源回路。

〔4〕 更に、前記コンデンサの他端と前記電源回路の第1及び第2の端子との間に前記コンデンサの電圧をクランプするためのクランプ用素子をそれぞれ接続したことを特徴とする請求項3記載の共振型スイッチング電源回路。

〔5〕 第1及び第2の端子を有する直流電源回路と、

前記第1及び第2の端子の間に互いに直列にな

路と、

前記第1及び第2の端子の間に互いに直列になるように接続された第1及び第2のスイッチング素子と、

前記第1及び第2の端子の間に互いに直列になるように接続された第1及び第2のダイオードと、

前記第1のスイッチング素子と前記第2のスイッチング素子とを交互にオン・オフ制御するための制御回路と、

前記第1及び第2のダイオードに並列に接続されている第1及び第2のコンデンサと、

前記第1及び第2のスイッチング素子の接続中点と前記第1及び第2のダイオードの接続中点との間に接続された共振用インダクタンス素子と、

前記インダクタンス素子に直列に接続されているか又は前記インダクタンス素子に電磁結合されている負荷回路と、

前記第1及び第2のコンデンサに並列に接続された前記コンデンサの充電及び／又は放電を制御するための第1及び第2のコンデンサ制御素子と

るように接続された第1及び第2のスイッチング素子と、

前記第1及び第2のスイッチング素子を交互にオン・オフするための制御回路と、

前記第1及び第2の端子の間に互いに直列になるように接続された第1及び第2のダイオードと、

前記第2のダイオードに並列に接続された共振用コンデンサと、

前記第1及び第2のスイッチング素子の接続中点と前記第1及び第2のダイオードの接続中点との間に接続された共振用インダクタンス素子と、

前記インダクタンス素子に直列に接続されているか、又は前記インダクタンス素子に電磁結合されている負荷回路と

から成る共振型スイッチング電源回路において、

前記共振用コンデンサに並列に前記共振用コンデンサの充電及び／又は放電を制御するための素子を接続したことを特徴とする共振型スイッチング電源回路。

〔6〕 第1及び第2の端子を有する直流電源回

を具備した共振型スイッチング電源回路。

〔7〕 第1の端子と第2の端子と前記第1及び第2の端子の midpoint 電位を与える第3の端子とを有する直流電源回路と、

センタタップを有する1次巻線を備えているトランスと、

前記電源回路の前記第3の端子と前記1次巻線のセンタタップとの間に接続されたインダクタンス素子とコンデンサとから成る直列共振回路と、

前記第1の端子と前記1次巻線の一端との間に接続された第1のスイッチング素子と、

前記第2の端子と前記1次巻線の他端との間に接続された第2のスイッチング素子と、

前記コンデンサに並列に接続された前記コンデンサの充電及び／又は放電を制御するための素子と

から成る共振型スイッチング電源回路。

3. 発明の詳細な説明

〔産業上の利用分野〕

本発明は、直流安定化電源等に使用される共振

振型スイッチング電源回路に関するものである。

〔従来の技術〕

現在、電子計算機、通信機器等に使用される電子通信用電源は、小型、軽量、高効率の利点を持つPWM方式のスイッチングレギュレータが多く使用されている。しかし、スイッチング周波数が20kHz以下の場合には、大きな問題が生じないが、100kHz以上になると、電力伝送用トランスの洩れインダクタンスや、スイッチング素子の浮遊容量の影響により、スイッチング素子のオン・オフ転換時におけるスイッチング素子の両端の電圧波形と電流波形とが重なり合う割合が大きくなり、この電圧、電流波形が互いに重なり合う部分がスイッチング損失となり、発振周波数が高くなるに従って損失が増加する。スイッチング損失が増加すると、電力変換効率が低下すると共に、発熱する。このため大きな放熱器を必要とし、電源装置の小型化の妨げとなる。

上述の如き問題点を解決するための方式として、第11図に示す共振型スイッチング電源回路があ

タタツプ構成の2次巻線 N_2 、 N_3 には、ダイオード D_3 、 D_4 と平滑用コンデンサ C_2 とから成る出力整流平滑回路7を介して負荷8が接続されている。

第1及び第2のスイッチング素子 Q_1 、 Q_2 をオン・オフ制御するための制御回路9は、出力平滑整流回路7の出力電圧を検出する電圧検出回路10と、この電圧検出回路10から得られる検出電圧と基準電圧源11の基準電圧との差に対応する出力を得るための誤差増幅器12と、この誤差増幅器12の出力電圧に対応した周波数信号を出力するVCO(電圧制御発振器)13と、このVCO13の出力周波数でスイッチング素子 Q_1 、 Q_2 を交互にオン・オフ制御するためのスイッチ制御信号形成回路14とから成る。スイッチ制御信号形成回路14はVCOの出力に基づいて第1のスイッチング素子 Q_1 の制御信号を形成し、又この反転信号で第2のスイッチング素子 Q_2 の制御を形成し、各スイッチング素子 Q_1 、 Q_2 即ちトランジスタのベースに供給するものである。

スイッチング素子 Q_1 、 Q_2 は、直列共振回路の共

振。この共振型スイッチング電源回路の直流電源1は第1及び第2の電源2a、2bの直列回路から成り、第1、第2及び第3の端子3、4、5を有するセンタタップ型に形成されている。第1の端子3と第2の端子4との間にはトランジスタから成る第1及び第2のスイッチング素子 Q_1 、 Q_2 の直列回路が接続されている。交互にオン・オフ制御される第1及び第2のスイッチング素子 Q_1 、 Q_2 にはバイパス用ダイオード D_1 、 D_2 がそれぞれ逆並列に接続されている。

電源1の第3の端子5と第1及び第2のスイッチング素子 Q_1 、 Q_2 の接続中点6との間には共振用コンデンサ C_1 と直列共振用インダクタンス素子(リアクトル) L_1 と負荷回路を構成するためのトランスTの1次巻線 N_1 との直列回路が接続されている。即ち、コンデンサ C_1 の一端が第3の端子5に接続され、コンデンサ C_1 の他端がインダクタンス素子 L_1 と1次巻線 N_1 とを介してスイッチング素子 Q_1 、 Q_2 の接続中点6に接続されている。

トランスTは2次巻線 N_2 、 N_3 を有し、このセン

振周波数よりも低い周波数で第12図(A)(B)に示すように交互にオン・オフ制御される。第1のスイッチング素子 Q_1 がオン、第2のスイッチング素子 Q_2 がオフの期間(第7図の $t_1 \sim t_2$)では、まず、直列共振に基づく電流Iが、コンデンサ C_1 、電源2a、第1のスイッチング素子 Q_1 、1次巻線 N_1 、インダクタンス素子 L_1 から成る回路で流れる。コンデンサ C_1 の電圧は、第12図(C)に示すように t_1 時点では $-E$ であるが、零に向つて減少し、しかる後は $+E$ になる。コンデンサ C_1 及びインダクタンス素子 L_1 を通つて流れる電流Iは、第12図(D)に示すようにコンデンサ電圧 V_C が零になる時点で正の最大値になり、その後減少する。 $t_2 \sim t_3$ で示すコンデンサ C_1 の放電期間においては、コンデンサ C_1 、インダクタンス素子 L_1 、1次巻線 N_1 、ダイオード D_1 、電源2aから成る回路で逆方向電流が流れる。第2のスイッチング素子 Q_2 がオンになる $t_3 \sim t_4$ 期間においても $t_1 \sim t_2$ 期間と同様な動作が生じる。

〔発明が解決しようとする課題〕

ところで、共振型スイッチング電源回路の出力電圧又は電力の制御は、周波数制御方式で行われる。即ち、第1及び第2のスイッチング素子 Q_1 、 Q_2 のオン・オフ制御の周波数を変えることによつて負荷8の電圧を変える。従つて、最低周波数で回路定数を決めなければならず、損失を大幅に低減することが困難になつた。また、ダイオード D_1 が $t_2 \sim t_3$ でオンしている状態において、第2のスイッチング素子 Q_2 がオンになると、ダイオード D_1 の蓄積キャリアに基づくリカバリー電流（短絡電流）がスイッチング素子 Q_2 に流れ込む。このリカバリー電流（短絡電流）が第2のスイッチング素子 Q_2 のコレクタ・エミッタ間電圧がEボルトから零ボルトに変化する期間に流れるので、損失となり、効率が低下する。ダイオード D_1 、 D_2 を省くとリカバリー電流（短絡電流）による損失が生じなくなるが、コンデンサ C_1 がEよりも高くなる。即ち、電源投入時、負荷急変時等における第1及び第2のスイッチング素子 Q_1 、 Q_2 に流れる共振電流のアンバランスによつてコンデンサ C_1 の電圧がE

続中点との間に接続された共振用インダクタンス素子と、前記インダクタンス素子に直列に接続されているか又は前記インダクタンス素子に電磁結合されている負荷回路とから成る共振型スイッチング電源回路において、前記コンデンサに並列に前記コンデンサの充電及び／又は放電を制御するための素子を接続したことを特徴とする共振型スイッチング電源回路に係わるものである。

請求項2に示すようにクランプ用ダイオードを接続することが望ましい。

請求項3に示すように第1～第4のスイッチング素子をブリッジ型に接続する場合にも請求項1の技術思想を適用することができる。

請求項4に示すように、請求項3の回路にクランプ用ダイオードを付加することが望ましい。

請求項5に示すように1つのダイオードに並列にコンデンサを接続する変形ハーフブリッジ型の回路にも請求項1の技術思想を適用することができる。

請求項6に示すように2つのダイオードにコン

よりも高くなり、スイッチング素子 Q_1 、 Q_2 及びダイオード D_1 、 D_2 に耐圧の高いものを使用することが必要になる。高耐圧のスイッチング素子 Q_1 、 Q_2 及びダイオード D_1 、 D_2 を使用すると、オン時における抵抗（電圧降下）が大きくなり、損失も多くなるので、共振型に基づく損失低減効果が期待出来なくなる。

そこで、本発明の目的は効率の良い共振型スイッチング電源回路を提供することにある。

〔課題を解決するための手段〕

上記目的を達成するための本発明は、第1の端子と第2の端子と前記第1及び第2の端子の中点電位を与える第3の端子とを有する直流電源回路と、前記第1の端子と前記第2の端子との間に互いに直列になるように接続された第1及び第2のスイッチング素子と、前記第1及び第2のスイッチング素子を交互にオン・オフ制御するための制御回路と、一端が前記第3の端子に接続されている共振用コンデンサと、前記共振用コンデンサの他端と前記第1及び第2のスイッチング素子の接

デンサをそれぞれ並列接続する回路にも請求項1の技術思想を適用することができる。

請求項7に示すように並列型のスイッチング回路にも請求項1の技術思想を適用することができる。

〔作用〕

各請求項に従う発明において、コンデンサを短絡させるためのスイッチング素子のオン時間幅を制御することによつて1周期中に電源から共振回路に供給する電力量が制御される。これにより、出力電力、又は電圧又は電流の制御をスイッチング周波数固定状態で行うことが可能になる。

請求項2、4、5、6に従う発明ではコンデンサの電圧の上昇をダイオードで抑制することができる。

〔第1の実施例〕

次に、第1図及び第2図を参照して本発明の第1の実施例に係わる共振型スイッチング電源回路即ち共振型DC-DCコンバータを説明する。但し、第1図において第11図と実質的に同一の部分には同一の符号を付してその説明を省略する。

第1図の本実施例の共振型スイッチング電源回路は、コンデンサ C_1 に並列接続された第1及び第2のコンデンサ制御用スイッチング素子 S_1 、 S_2 を有し、更に、コンデンサ C_1 の右端15と電源1の第1の端子3との間及びコンデンサ C_1 の右端15と電源1の第2の端子4との間にそれぞれ接続された第1及び第2のクランプ用ダイオード16、17を有する。第1図の第1及び第2のスイッチング素子 Q_1 、 Q_2 は出力電圧の変化に無関係に一定の周波数で交互にオン・オフ制御される。

18はのこぎり波(三角波)発生器であつて、この実施例では第1及び第2のスイッチング素子 Q_1 、 Q_2 のオン・オフ繰返し周波数の2倍の周波数を有するのこぎり波を発生する。19はスイッチ制御信号形成回路(スイッチ制御回路)であつて、のこぎり波発生回路18から供給されたのこぎり波に同期した第2図(A)(B)に示す固定の周波数の方形波の制御信号を形成し、第1及び第2のスイッチング素子 Q_1 、 Q_2 に供給する。なお、第2図(A)(B)の制御信号の周波数は C_1L_1 共振回路の共振周波数

は第1及び第2のコンデンサ制御スイッチング素子 S_1 、 S_2 に供給される。なお、のこぎり波発生回路18の周波数が第1及び第2のスイッチング素子 Q_1 、 Q_2 のスイッチング周波数と同一になるように構成することも可能である。

第1図の回路の動作を第2図を参照して説明すると、第2図(A)に示すように t_0 時点で第1のスイッチング素子 Q_1 がオンになつた時には従来の回路と同様に共振による電流 I が第2図(B)に示すように流れ始める。これにより、コンデンサ C_1 の電圧が第2図(C)に示すように $-E/2$ から0に向つて変化する。なお、コンデンサ C_1 の充電電圧の変化範囲は、クランプ用ダイオード16、17の働きで $-E/2$ から $+E/2$ までとなる。 t_0 時点で第1のコンデンサ制御用スイッチング素子 S_1 が第2図(C)のパルスでオン制御されているが、コンデンサ C_1 の左端が正になるように充電されているために、第1のコンデンサ制御用スイッチング素子 S_1 が逆バイアス状態にあり、 $t_0 \sim t_1$ 期間はオフに保たれる。電源回路1の第1の端子3、第1のスイッ

よりも低い。本実施例では出力電圧の制御を第1及び第2のスイッチング素子 Q_1 、 Q_2 のオン・オフ繰返し周波数の制御によつて行わずに、共振コンデンサ C_1 の充放電制御によつて行う。

出力整流平滑回路7は4つのダイオード20、21、22、23と、平滑用コンデンサ24とから成り、トランスTの2次巻線 N_2 と負荷8との間に接続されている。出力整流平滑回路7の出力電圧は検出回路10で検出され、誤差増幅器12の入力となり、基準電圧源11の基準電圧と比較される。PWM(パルス幅変調)パルス形成回路25は、のこぎり波発生回路18と誤差増幅器12とに接続され、のこぎり波と誤差出力とを比較してPWMパルスを出力する電圧比較器を含んでおり、第2図(C)(D)に示すPWMパルスを発生する。なお、のこぎり波は第2図(A)～(D)に示すパルスの繰返し周波数の2倍の周波数を有するので、比較器の出力を交互に分配することによつて第2図(C)(D)に示す180度位相差の2つのPWMパルス列を得る。PWMパルス形成回路25の出力ライン

ング素子 Q_1 、トランスT、インダクタンス素子 L_1 、コンデンサ C_1 の回路でコンデンサ C_1 の逆充電が進み、 t_2 時点で充電電圧が零になると、第1のコンデンサ制御用スイッチング素子 S_1 の逆バイアスが解除され、これがオン状態になる。この結果、スイッチング素子 Q_1 、 Q_2 、 S_1 、 S_2 の電圧降下を無視して考えると、コンデンサ C_1 の充電電圧は第2図(C)のパルスが終了する t_2 時点まで零に保たれる。 $t_1 \sim t_2$ 期間にはインダクタンス素子 L_1 に制限された電流が第2図(B)に示すように流れ続ける。 t_2 時点で第1のコンデンサ制御用スイッチング素子 S_1 がオフに転換すると、 C_1L_1 共振回路に基づく電流が流れ始め、コンデンサ C_1 の電圧は第2図(D)に示すように正方向に上昇し、電流 I は徐々に減少する。 t_3 時点でコンデンサ C_1 の電圧が $E/2$ よりも僅かに高くなるとクランプ用ダイオード16がオンになり、コンデンサ C_1 の電圧はほぼ $E/2$ にクランプされる。 $t_3 \sim t_4$ 期間ではインダクタンス素子 L_1 のエネルギーの放出で電流が流れるが、第2図(D)に示すように徐々に減少し、 t_4 で零になる。第

1 図では第 1 及び第 2 のスイッチング素子 Q_1 、 Q_2 にダイオードが逆並列接続されていないが、第 1 図に示すようにダイオード D_1 、 D_2 が接続されている場合又は FET のためにダイオードが内蔵されている場合であっても、第 1 図ではコンデンサ C_1 の電圧が $E/2$ にクランプされているために、 $t_3 \sim t_5$ の期間に第 1 及び第 2 のスイッチング素子 Q_1 、 Q_2 に並列のダイオードがオンにならない。従つて、第 2 図の t_5 時点での第 1 及び第 2 のスイッチング素子 Q_1 、 Q_2 の切換時にダイオードの蓄積キャリアに基づく過大電流が流れない。

第 2 図の $t_5 \sim t_6$ 期間には第 2 のスイッチング素子 Q_2 がオン制御され、 $t_6 \sim t_5$ 期間と同様な動作が生じる。勿論この期間の電流 I 、電圧 V_C の向きは $t_6 \sim t_5$ 期間と逆になる。

出力電圧が変動すると、誤差増幅器 12 の出力が変化し、第 2 図(C)(D)のパルスの変化し、第 2 図の $t_1 \sim t_2$ 期間も変化する。この結果、第 1 のスイッチング素子 Q_1 の固定されたオン期間 $t_0 \sim t_5$ 内における電源回路 1 からのエネルギー供給時

質的に同一部分には同一の符号を付してその説明を省略する。この実施例では第 1 ～第 4 のスイッチング素子 $Q_1 \sim Q_4$ でスイッチング回路がフルブリッジ型に構成されている。第 3 及び第 4 のスイッチング素子 Q_3 、 Q_4 は第 1 図のコンデンサ 2 a、2 b の位置に接続されている。第 1 ～第 4 のスイッチング素子 $Q_1 \sim Q_4$ は第 4 図(A)～(D)に示すように制御され、第 1 及び第 2 のコンデンサ制御用スイッチング素子 S_1 、 S_2 は第 4 図(E)(F)に示すように制御される。

第 4 図の $t_0 \sim t_1$ 期間中における共振回路の電流 I は、 Q_1 、 N_1 、 L_1 、 C_1 、 Q_4 から成る回路で流れ、 $t_1 \sim t_2$ 期間中における共振回路の電流 I は、 Q_3 、 C_1 、 L_1 、 N_1 、 Q_2 から成る回路で流れる。共振による電流 I 及び電圧 V_C の変化の原理は第 1 図と同一である。また、出力電圧調整方式の原理も第 1 図と同一であり、第 4 図(E)(F)に示す第 1 及び第 2 のコンデンサ制御用スイッチング素子 S_1 、 S_2 のオン時間幅を変えることによつてコンデンサ C_1 の電圧が零になる時間幅が変化し、出力電圧が変化

間及び量が変化し、出力整流平滑回路 7 に与えられる電力量も変化し、出力電圧が変化する。

上述のように、本方式は周波数一定制御方式であるので、トランス T 等を最適設計することが可能になり、効率の高い共振型スイッチング電源回路を提供することができる。

また、クランプ用ダイオード 16、17 の働きにより、コンデンサ C_1 の電圧が $-E/2 \sim +E/2$ に制限されるため、第 1 及び第 2 のスイッチング素子 Q_1 、 Q_2 の耐圧を下げるのが可能になり、これに基づく電力損失低減効果も生じる。また、第 1 及び第 2 のスイッチング素子 Q_1 、 Q_2 に逆並列にダイオードを接続しなくても、コンデンサ C_1 の電圧上昇を抑えることが可能になり、且つダイオードの蓄積キャリアによる過大電流の問題も前述したように解決される。

〔第 2 の実施例〕

次に、第 3 図及び第 4 図を参照して本発明の第 2 の実施例に係わる共振型スイッチング電源回路を説明する。但し、第 3 図において、第 1 図と実

する。従つて、本実施例によつても、第 1 の実施例と同一の作用効果を得ることができる。

〔第 3 の実施例〕

次に、第 5 図及び第 6 図を参照して本発明の第 3 の実施例の共振型スイッチング電源回路を説明する。但し、第 5 図において第 1 図及び第 3 図と実質的に同一の部分には同一の符号を付してその説明を省略する。

この実施例では第 1 及び第 2 のスイッチング素子 Q_1 、 Q_2 の接続中点 6 と第 1 及び第 2 のダイオード 16、17 の接続中点 15 との間にトランス T とインダクタンス素子 L_1 とが接続され、第 2 のダイオード 17 に並列にコンデンサ C_1 とこの制御用スイッチング素子 S_1 とが接続されている。

第 1 及び第 2 のスイッチング素子 Q_1 、 Q_2 は第 6 図(A)(B)に示すように第 1 図の回路と同様に一定の周波数で交互にオン・オフ制御される。コンデンサ制御用スイッチング素子 S_1 は第 6 図(C)に示す PWM パルスで制御される。第 6 図の t_0 時点で $I = 0$ 、 $V_C = 0$ であるとすれば、 $t_0 \sim t_1$ 期間では Q_1 、

Q_3 がオン、 Q_2 がオフであるので、電流 I は零から直線上に上昇する。なお、この電流 I は Q_1 、 N_1 、 L_1 、 S_1 の回路で流れる。

$t_1 \sim t_2$ 期間では、 Q_1 がオン、 Q_2 、 Q_3 がオフであり、 Q_1 、 N_1 、 L_1 、 C_1 の回路で電流 I が流れる。

$t_2 \sim t_3$ 期間では、コンデンサ C_1 の充電電圧 V_C が電源電圧 E よりも僅かに高くなっているために、第1のダイオード16がオンになり、コンデンサ C_1 の電圧 V_C がほぼ電源電圧 E にクランプされる。インダクタンス素子 L_1 の蓄積エネルギーの放出は t_3 時点よりも前に終了し、電流 I は t_3 で零である。従つて、第2のスイッチング素子 Q_2 が電流零の状態でもオンに転換する。

$t_3 \sim t_4$ 期間では、 Q_2 がオン、 Q_1 、 Q_3 がオフであるので、 C_1 、 L_1 、 N_1 、 Q_2 の閉回路が形成され、コンデンサ C_1 の放電が起きる。 t_4 時点になつてコンデンサ C_1 の電圧が零になり、その後逆方向に充電されるとダイオード17がオンになり、コンデンサ C_1 の電圧 V_C はほぼ零になる。コンデンサ制御用スイッチング素子 S_1 を駆動するための第6図

のコンデンサ制御用スイッチング素子 S_1 、 S_2 は第8図(C)(D)のPWMパルスで制御されるため、出力電圧が第1、第2及び第3の実施例と同様に制御され、同様な作用効果が得られる。

〔変形例〕

本発明は上述の実施例に限定されるものでなく、例えば次の変形が可能なものである。

(1) 第9図に示すようにトランス T の1次巻線にセンタタップを設け、このセンタタップと第3の端子5との間にインダクタンス素子 L_1 とコンデンサ C_1 との共振回路を接続し、第1の端子3とセンタタップとの間に第1のスイッチング素子 Q_1 を介して1次巻線の半分 N_{1a} を接続し、第2の端子4とセンタタップとの間に第2のスイッチング素子 Q_2 を介して1次巻線の残りの半分 N_{1b} を接続し、第1及び第2のスイッチング素子 Q_1 、 Q_2 と第1及び第2のコンデンサ制御用スイッチング素子 S_1 、 S_2 とを第1図と同様に制御してもよい。

(2) コンデンサ C_1 を制御するためのスイッチング素子 S_1 、 S_2 を第10図に示すように2つのFE

(C) のパルスの後縁を制御すると、1サイクル ($t_0 \sim t_5$) 中に電源回路1から供給される電力量が変化し、出力電圧も変化する。従つて、第1及び第2の実施例と同一の作用効果を得ることができる。
〔第4の実施例〕

次に、第7図及び第8図を参照して第4の実施例の共振型スイッチング電源回路を説明する。但し、第7図において、第1図、第3図、第5図と実質的に同一の部分には同一の符号を付してその説明を省略する。第7図の回路は第5図のダイオード16に対して並列にコンデンサ C_2 とコンデンサ制御用スイッチング素子 S_2 とを接続した回路と同一である。

第8図の $t_0 \sim t_3$ 期間の動作は第6図の $t_0 \sim t_3$ 期間と同一である。 $t_3 \sim t_6$ 期間は、 $t_0 \sim t_3$ 期間の反対極性の動作であり、 $t_3 \sim t_4$ で Q_1 、 S_1 がオフ、 Q_2 及び S_2 がオンになり、 $t_4 \sim t_6$ 期間で Q_1 、 S_1 、 S_2 がオフ、 Q_2 がオンになり、電流 I は第8図(D)に示すように対称的に変化し、コンデンサ C_1 の電圧は第8図(E)に示すように変化する。第1及び第2

T31、32と2つのダイオード33、34とで構成してもよい。

(3) 第1～第4のスイッチング素子 $Q_1 \sim Q_4$ をFETとしてもよい。

(4) 第1図、及び第3図の第1及び第2のスイッチング素子 Q_1 、 Q_2 に逆並列にダイオードを接続してもよい。

(5) 第1図の電源回路1を2個の $E/2$ の電源によつて構成することができる。

(6) S_1 、 S_2 を制御するためのPWMパルスを例えば第2図の $t_1 \sim t_2$ 期間に対応するパルス幅を有するものにしてもよい。

(7) トランス T が大きなインダクタンスを有する場合にはこれを共振用のインダクタンスに兼用し、インダクタンス素子 L_1 を省いてもよい。

(8) トランス T の位置に負荷8を直接に接続し、負荷8に交流を供給する場合にも適用可能である。

〔発明の効果〕

上述のようにいずれの請求項の発明によつてもスイッチング周波数を固定した状態で電力又は電

圧又は電流を調整することが可能になる。また、請求項2、4、5、6によれば、コンデンサの電圧の上昇を抑えることができると共に、スイッチング素子のオン・オフ切換時の電力損失の低減を図ることができる。

4. 図面の簡単な説明

第1図は本発明の第1の実施例の共振型スイッチング電源回路を示す回路図、

第2図は第1図の各部の状態を原理的に示す波形図、

第3図は第2の実施例の共振型スイッチング電源回路を示す回路図、

第4図は第3図の各部の状態を原理的に示す波形図、

第5図は第3の実施例の共振型スイッチング電源回路を示す回路図、

第6図は第5図の各部の状態を原理的に示す波形図、

第7図は第4の実施例の共振型スイッチング電源回路を示す波形図、

第8図は第7図の各部の状態を原理的に示す波形図、

第9図は変形例の共振型スイッチング電源回路を示す回路図、

第10図は変形例のコンデンサ制御用スイッチ回路を示す回路図、

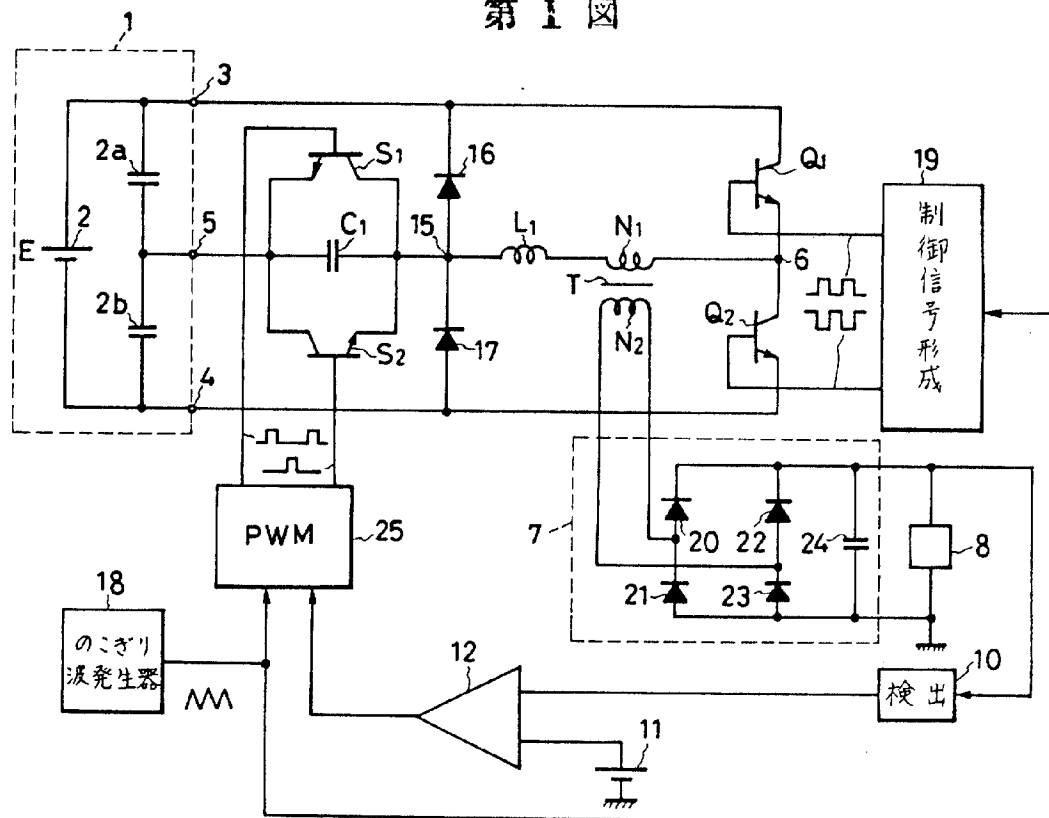
第11図は従来の共振型スイッチング電源回路を示す回路図、

第12図は第11図の各部の状態を原理的に示す波形図である。

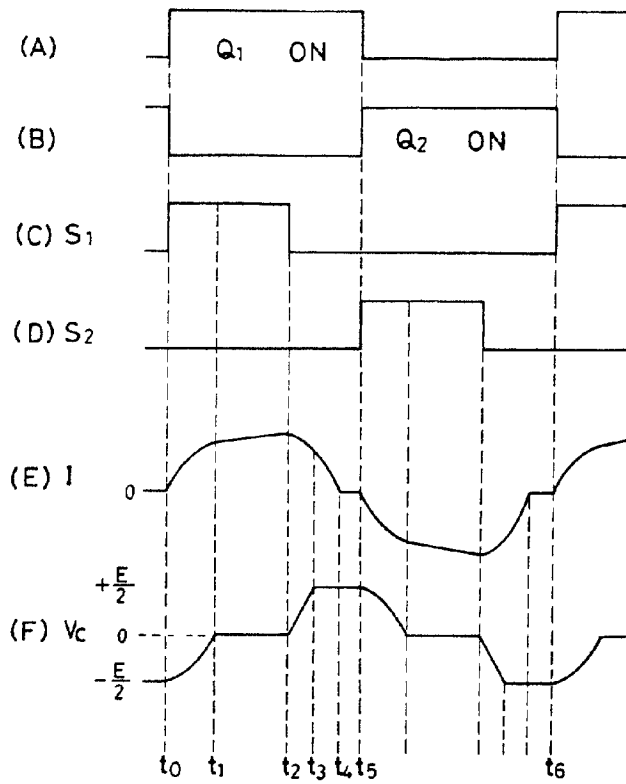
1…直流電源回路、3…第1の端子、4…第2の端子、 Q_1 …第1のスイッチング素子、 Q_2 …第2のスイッチング素子、 C_1 …共振用コンデンサ、 L_1 …直列共振用インダクタンス素子、 S_1 、 S_2 …コンデンサ制御用スイッチング素子。

代理人 高野 則 次

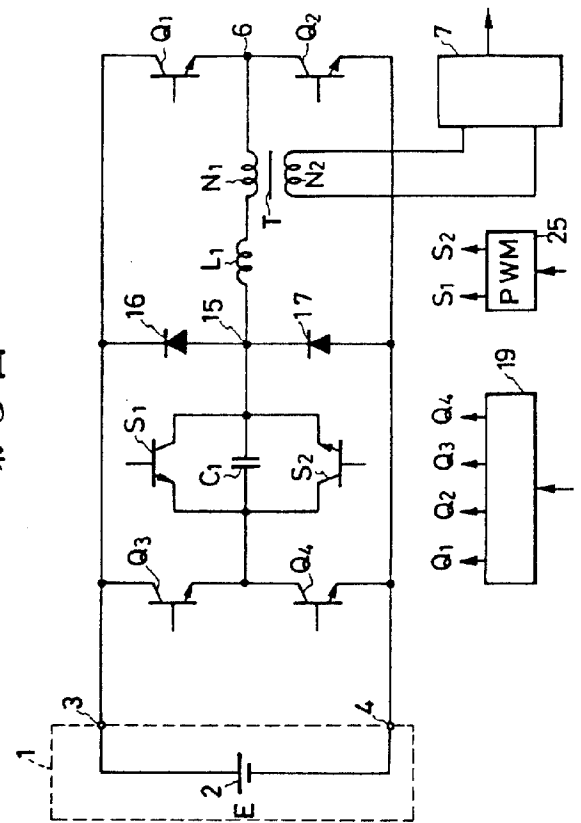
第1図



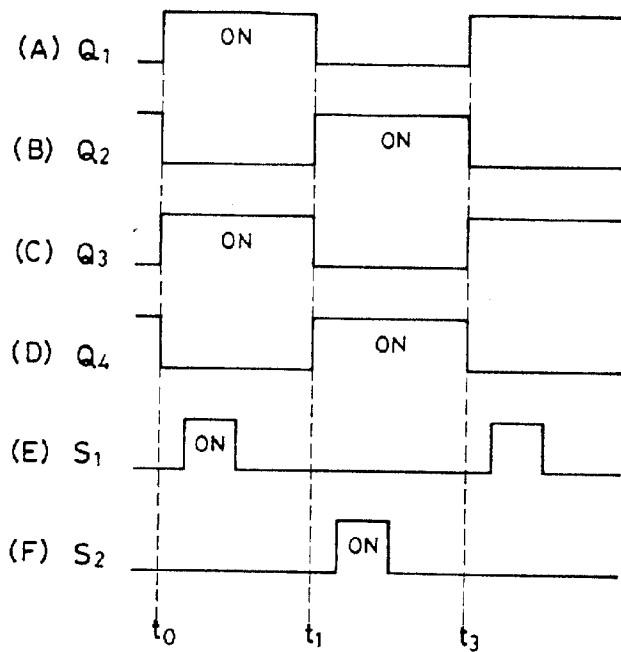
第 2 図



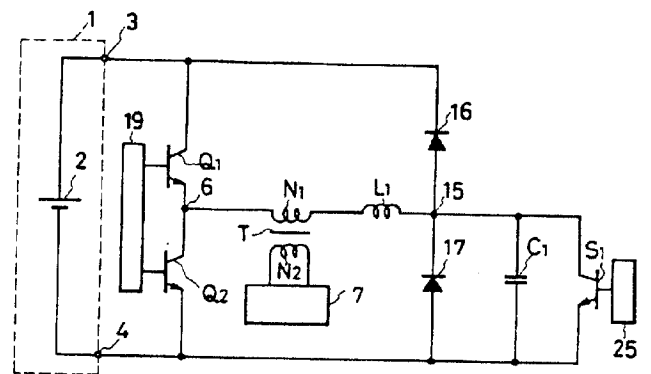
第 3 図



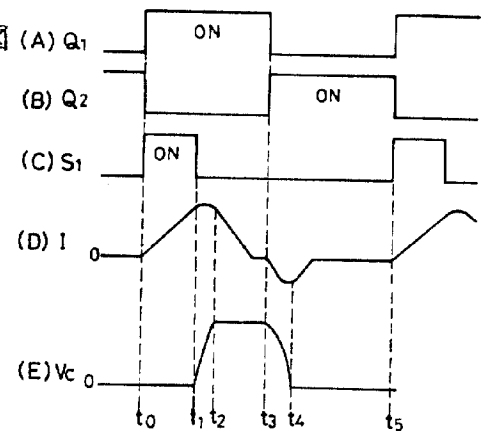
第 4 図



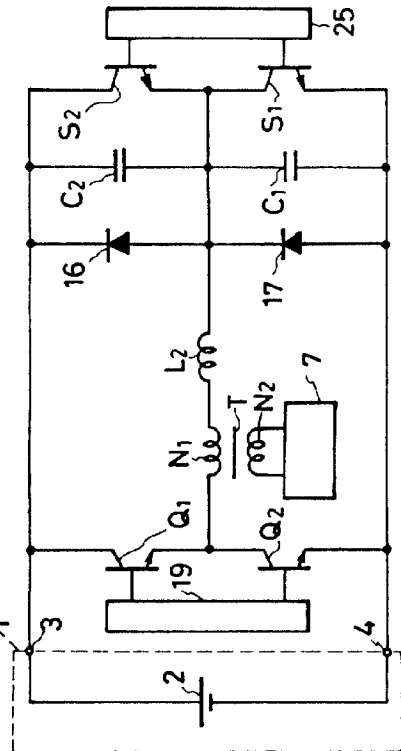
第 5 図



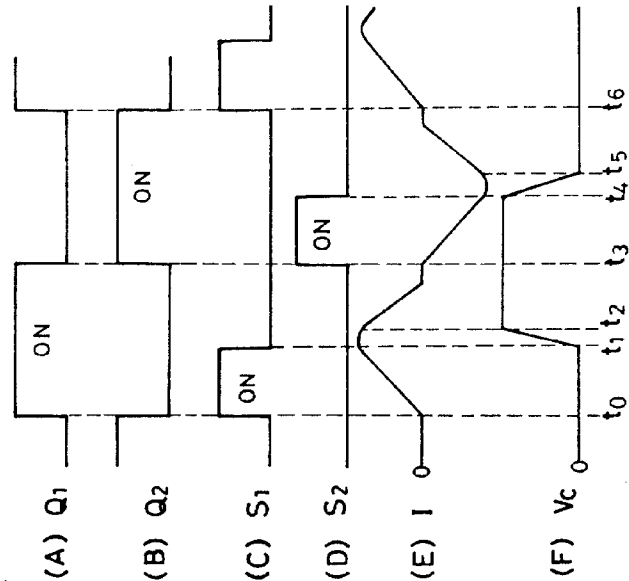
第 6 図



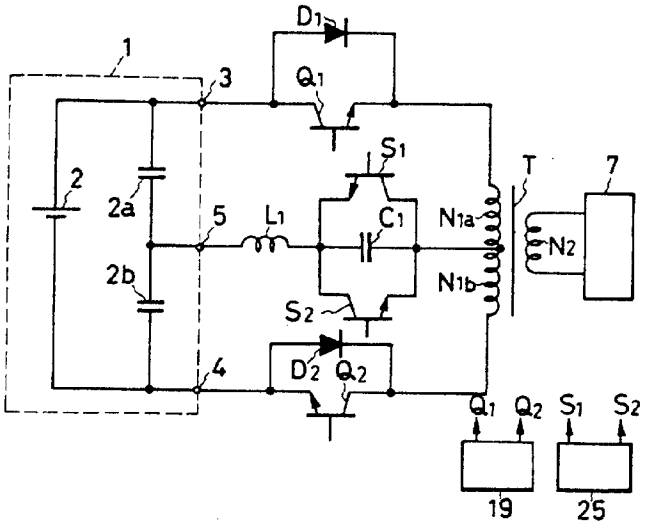
第7図



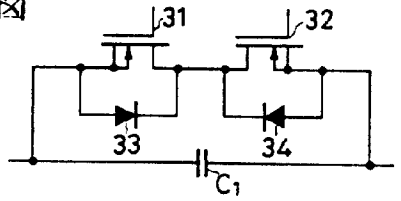
第8図



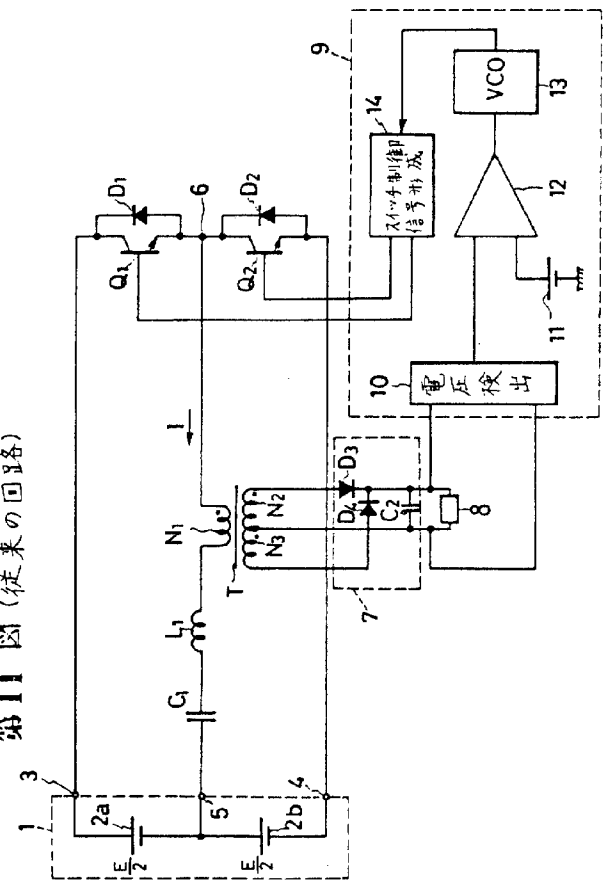
第9図



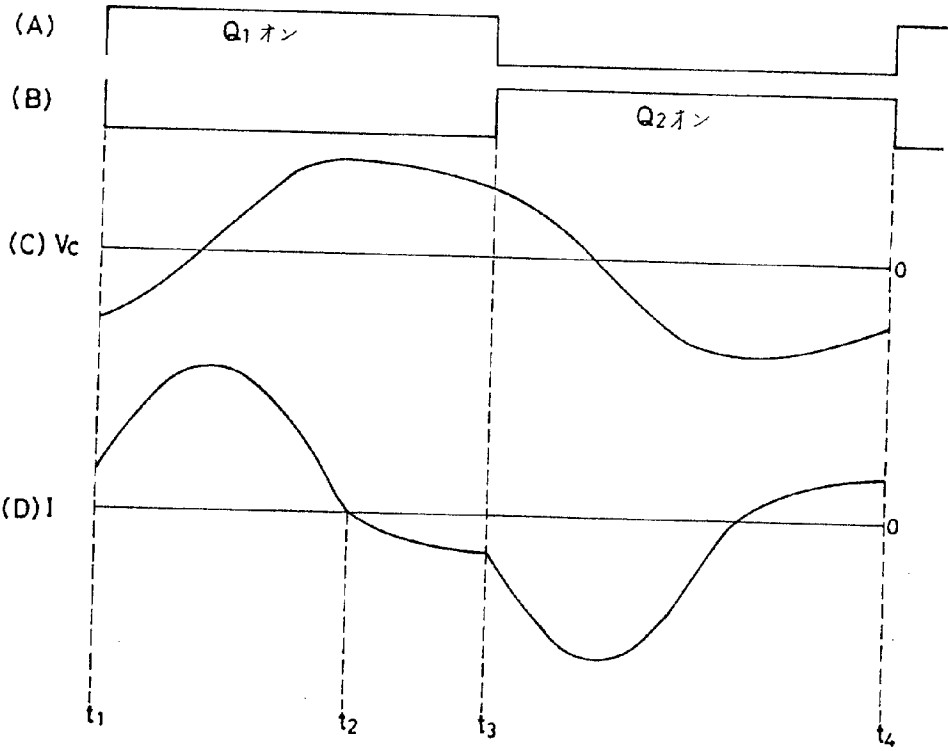
第10図



第11図 (従来の回路)



第12図



IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor : Patrizio Vinciarelli Art Unit : 2838
Serial No. : 13/044,063 Examiner : Jye-June Lee
Filed : March 9, 2011 Confirmation No. : 1081
Notice of Allowance Date: 2/20/2015
Title : FAULT TOLERANT POWER CONVERTER

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Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT AFTER ALLOWANCE
PURSUANT TO 37 C.F.R. §1.312

Please amend the application as indicated on the following pages. This amendment is being filed prior to payment of the issue fee.

Amendments to the Claims:

This listing of claims replaces all prior versions and listings of claims in the application:

Listing of Claims:

1. (Currently Amended) A method of converting power via a transformer for supplying a low voltage output comprising:

providing a secondary winding having a first terminal and a second terminal;

providing a first switch and a second switch each adapted to block current flow in an OFF state in at least one direction;

forming a first series circuit with the first switch connected to the second switch at a first junction, the first series circuit having two end terminals;

connecting the first terminal of the secondary winding to the first junction;

connecting the first series circuit across an output with each of the two end terminals connected to a respective terminal of the output and with each of the first and second switches being arranged in the OFF state to block current flow through the first series circuit in a direction that would short the output, wherein a short across one of the first or second switches does not short the output;

sensing an output of the second switch and preventing the first switch from turning ON if the sensing indicates that the second switch is not open; and

sensing an output of the ~~second~~ first switch and preventing the second switch from turning ON if the sensing indicates that the first switch is not open.

2. (Previously Presented) The method of claim 1 further comprising:

sensing across each of the first and second switches, a respective voltage polarity; and

turning each of the first and second switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite of the first polarity.

3. (Previously Presented) The method of claim 1 further comprising:
providing a third switch and a fourth switch each adapted to block current in an OFF state in at least one direction;

forming a second series circuit with the third switch connected to the fourth switch at a second junction, the second series circuit having two end terminals;

connecting the second terminal of the secondary winding to the second junction; and

connecting the second series circuit across the output with each of the two end terminals connected to a respective terminal of the output and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output, wherein a short across one of the third or fourth switches does not short the output.

4. (Previously Presented) The method of claim 3 further comprising:

preventing the third switch from turning ON if the fourth switch fails to open;

preventing the fourth switch from turning ON if the third switch fails to open.

5. (Previously Presented) The method of claim 4 further comprising:

sensing across each of the third and fourth switches, a respective voltage polarity; and

turning each of the third and fourth switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite of the first polarity.

6. (Previously Presented) The method of claim 5 further comprising:

providing a first control circuit to operate the first switch;

providing power to operate the first control circuit and first switch from across the third switch;

providing a third control circuit to operate the third switch; and

providing power to operate the third control circuit and third switch from across the first switch.

7. – 10. (Canceled)

11. (Previously Presented) The method of claim 6 further comprising:
providing a second control circuit to operate the second switch;
providing power to operate the second control circuit and second switch from across the fourth switch;
providing a fourth control circuit to operate the fourth switch; and
providing power to operate the fourth control circuit and fourth switch from across the second switch.

12. – 15. (Cancelled).

16. (Previously Presented) The method of claim 1 further comprising:
forming a second series circuit with a first capacitor connected to a second capacitor at a second junction, the second series circuit having two end terminals;
connecting the second terminal of the secondary winding to the second junction; and
connecting the second series circuit across the output with each of the two end terminals connected to a respective terminal of the output.

17. (Previously Presented) The method of claim 16 further comprising:
sensing across each of the switches, a respective voltage polarity; and
turning each of the switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

18. (Previously Presented) A method of providing fault tolerance in a power converter comprising:

supplying an alternating current from a secondary winding to an input of a full-bridge rectifier circuit having switches,

operating the switches as rectifiers to provide full-wave rectification of the alternating current at an output of the full-bridge rectifier circuit,

connecting the output of the full-bridge rectifier circuit to an output of the power converter,

monitoring selected circuit conditions;

determining from the selected circuit conditions whether a predetermined fault is present, disabling the switches in response to a determination that the predetermined fault is present, and

providing a first one of the switches connected in series with a second one of the switches across the output of the full-bridge rectifier circuit with the first switch connected to a first terminal of the output of the full-bridge rectifier circuit and the second switch connected to a second terminal of the output of the full-bridge rectifier circuit; and

sensing an output of the second switch and preventing the first switch from turning ON if the sensing indicates that the second switch is not open; and

sensing an output of the second switch and preventing the second switch from turning ON if the sensing indicates that the first switch is not open.

19. (Previously Presented) The method of claim 18 further comprising:
sensing across each of the first and second switches, a respective voltage polarity; and
turning each of the first and second switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

20. (Previously Presented) The method of claim 18 further comprising:
providing a third one of the switches and a fourth one of the switches each adapted to block current in an OFF state in at least one direction;

forming a second series circuit with the third switch connected to the fourth switch at a second junction, the second series circuit having two end terminals;

connecting the second terminal of the secondary winding to the second junction; and

connecting the second series circuit across the output of the full-bridge rectifier circuit with each of the two end terminals connected to a respective terminal of the output of the full-bridge rectifier circuit and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output of the full-bridge rectifier circuit, wherein a short across one of the third or fourth switches does not short the output of the full-bridge rectifier circuit.

21. (Previously Presented) The method of claim 20 further comprising:

preventing the third switch from turning ON if the fourth switch fails to open; and

preventing the fourth switch from turning ON if the third switch fails to open.

22. (Previously Presented) The method of claim 20 further comprising:

sensing across each of the third and fourth switches, a respective voltage polarity; and

turning each of the third and fourth switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

23. (Previously Presented) The method of claim 20 further comprising:

providing a first control circuit to operate the first switch;

providing power to operate the first control circuit and first switch from across the third switch;

providing a third control circuit to operate the third switch; and

providing power to operate the third control circuit and third switch from across the first switch.

24. – 27. (Cancelled)

28. (Previously Presented) The method of claim 20 further comprising:
providing a second control circuit to operate the second switch;
providing power to operate the second control circuit and second switch from across the fourth switch;
providing a fourth control circuit to operate the fourth switch; and
providing power to operate the fourth control circuit and fourth switch from across the second switch.

29. – 43. (Cancelled).

44. (Previously Presented) A fault-tolerant converter comprising:
a transformer having a primary winding and a secondary winding, the secondary winding having a first terminal and a second terminal;
a first series circuit having a first switch and a second switch each adapted to block current flow in an OFF state in at least one direction, the first switch being connected to the second switch at a first junction, the first series circuit having two end terminals, the first junction being connected to the first terminal of the secondary winding, the first series circuit being connected across an output of the converter with each of the two end terminals connected to a respective terminal of the output and with each of the first and second switches being arranged in the OFF state to block current flow through the first series circuit in a direction that would short the output, wherein a short across one of the first or second switches does not short the output; and
a switch control circuit configured to sense an output of the second switch and prevent the first switch from turning ON if the sensing indicates that the second switch is not open, and to sense an output of the first switch and prevent the second switch from turning ON if the sensing indicates that the first switch is not open.

45. (Previously Presented) The apparatus of claim 44 wherein the switch control circuit is configured to sense across each of the first and second switches, a respective voltage polarity,

and to turn each of the first and second switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

46. (Previously Presented) The apparatus of claim 44, comprising a second series circuit that comprises a third switch and a fourth switch each adapted to block current in an OFF state in at least one direction, the third switch being connected to the fourth switch at a second junction, the second series circuit having two end terminals, the second terminal of the secondary winding being connected to the second junction, the second series circuit being connected across the output with each of the two end terminals connected to a respective terminal of the output and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output, wherein a short across one of the third or fourth switches does not short the output.

47. (Previously Presented) The apparatus of claim 46 wherein the switch control circuit comprises a first control circuit and a third control circuit, the first control circuit operates the first switch, the third control circuit operates the third switch, the first control circuit and the first switch are powered using a voltage across the third switch, and the third control circuit and third switch are powered using a voltage across the first switch.

48. – 51. (Cancelled).

52. (Previously Presented) The apparatus of claim 47 wherein the switch control circuit comprises a second control circuit and a fourth control circuit, the second control circuit operates the second switch, the fourth control circuit operates the fourth switch, the second control circuit and the second switch are powered using a voltage across the fourth switch, and the fourth control circuit and fourth switch are powered using a voltage across the second switch.

53. – 56. (Cancelled).

57. (Previously Presented) The apparatus of claim 44, comprising a second series circuit having a first capacitor connected to a second capacitor at a second junction that is connected to the second terminal of the secondary winding, the second series circuit having two end terminals connected across the output of the converter with each of the two end terminals connected to a respective terminal of the output of the converter.

58. (Previously Presented) The apparatus of claim 57 wherein the switch control circuit is configured to sense across each of the first and second switches, a respective voltage polarity, and to turn each of the first and second switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

59. – 88. (Cancelled).

89. (Previously Presented) A method of delivering power to an output, comprising:
providing a transformer having a secondary winding, the secondary winding having a first terminal and a second terminal;

providing a first switch, a second switch, a third switch and a fourth switch, each adapted to block current flow in an OFF state in at least one direction;

forming a first series circuit with the first switch connected to the second switch at a first junction, the first series circuit having two end terminals;

forming a second series circuit with the third switch connected to the fourth switch at a second junction, the second series circuit having two end terminals;

connecting the first terminal of the secondary winding to the first junction and the second terminal of the secondary winding to the second junction;

connecting the first series circuit across the output with each of the two end terminals connected to a respective terminal of the output, sensing outputs of the first and second switches, arranging the first switch in the OFF state if the sensing indicates that the second switch is not

open, and arranging the second switch in the OFF state if the sensing indicates that the first switch is not open to block current flow through the first series circuit in a direction that would short the output, wherein a short across one of the first or second switch does not short the output;

connecting the second series circuit across the output with each of the two end terminals connected to a respective terminal of the output, sensing outputs of the third and fourth switches, arranging the third switch in the OFF state if the sensing indicates that the fourth switch is not open, and arranging the fourth switch in the OFF state if the sensing indicates that the third switch is not open to block current flow through the second series circuit in a direction that would short the output, wherein a short across one of the third or fourth switch does not short the output;

converting power via the transformer to deliver a unipolar voltage of five Volts or less to the output.

90. (Previously Presented) The method of claim 89 wherein the unipolar output is approximately 1 Volt.

91. (New) The method of claim 6 further comprising:
providing a common-source dual MOSFET semiconductor device including a first MOSFET and a third MOSFET integrated on a semiconductor die;
providing a common source terminal for connecting to the first and third MOSFETs in the common-source dual MOSFET semiconductor device; and
using the first MOSFET and the third MOSFET of the common-source dual MOSFET semiconductor device for the first and third switches.

92. (New) The method of claim 91 further comprising integrating the first and third control circuits in the common-source dual MOSFET semiconductor device.

93. (New) The method of claim 92 further comprising:

providing a plurality of individual first and third MOSFET elements on the semiconductor die;

arranging the individual first and third MOSFET elements in an alternating pattern on the semiconductor die;

connecting the first MOSFET elements together to form the first MOSFET;

connecting the third MOSFET elements together to form the third MOSFET; and

using a common source metallization to interconnect the source terminals of the first and third MOSFET elements.

94. (New) The method of claim 93 wherein the common-source dual MOSFET semiconductor device has exactly three terminals.

95. (New) The method of claim 11 further comprising:

providing a first common-source dual MOSFET semiconductor device including a first MOSFET and a third MOSFET integrated on a first semiconductor die;

providing a common source terminal for connecting to the first and third MOSFETs in the first common-source dual MOSFET semiconductor device;

using the first MOSFET and the third MOSFET of the first common-source dual MOSFET semiconductor device for the first and third switches;

providing a second common-source dual MOSFET semiconductor device including a second MOSFET and a fourth MOSFET integrated on a second semiconductor die;

providing a common source terminal for connecting to the second and fourth MOSFETs in the second common-source dual MOSFET semiconductor device;

using the second MOSFET and the fourth MOSFET of the second common-source dual MOSFET semiconductor device for the second and fourth switches.

96. (New) The method of claim 95 further comprising:

integrating the first and third control circuits in the first common-source dual MOSFET semiconductor device; and

integrating the second and fourth control circuits in the second common-source dual MOSFET semiconductor device.

97. (New) The method of claim 96 further comprising:
providing a plurality of individual first and third MOSFET elements on the first semiconductor die;
arranging the individual first and third MOSFET elements in an alternating pattern on the first semiconductor die;
connecting the first MOSFET elements together to form the first MOSFET;
connecting the third MOSFET elements together to form the third MOSFET;
using a common source metallization to interconnect the source terminals of the first and third MOSFET elements;
providing a plurality of individual second and fourth MOSFET elements on the second semiconductor die;
arranging the individual second and fourth MOSFET elements in an alternating pattern on the second semiconductor die;
connecting the second MOSFET elements together to form the second MOSFET;
connecting the fourth MOSFET elements together to form the fourth MOSFET; and
using a common source metallization to interconnect the source terminals of the second and fourth MOSFET elements.

98. (New) The method of claim 97 wherein the first and second common-source dual MOSFET semiconductor devices each have exactly three terminals.

99. (New) The method of claim 20 further comprising:
providing a common-source dual MOSFET semiconductor device including a first MOSFET and a third MOSFET integrated on a semiconductor die;
providing a common source terminal for connecting to the first and third MOSFETs in the common-source dual MOSFET semiconductor device; and

using the first MOSFET and the third MOSFET of the common-source dual MOSFET semiconductor device for the first and third switches.

100. (New) The method of claim 99 further comprising integrating the first and third control circuits in the common-source dual MOSFET semiconductor device.

101. (New) The method of claim 99 further comprising:
providing a plurality of individual first and third MOSFET elements on the semiconductor die;
arranging the individual first and third MOSFET elements in an alternating pattern on the semiconductor die;
connecting the first MOSFET elements together to form the first MOSFET;
connecting the third MOSFET elements together to form the third MOSFET; and
using a common source metallization to interconnect the source terminals of the first and third MOSFET elements.

102. (New) The method of claim 99 wherein the common-source dual MOSFET semiconductor device has exactly three terminals.

103. (New) The method of claim 20 further comprising:
providing a first common-source dual MOSFET semiconductor device including a first MOSFET and a third MOSFET integrated on a first semiconductor die;
providing a common source terminal for connecting to the first and third MOSFETs in the first common-source dual MOSFET semiconductor device;
using the first MOSFET and the third MOSFET of the first common-source dual MOSFET semiconductor device for the first and third switches;
providing a second common-source dual MOSFET semiconductor device including a second MOSFET and a fourth MOSFET integrated on a second semiconductor die;
providing a common source terminal for connecting to the second and fourth MOSFETs in the second common-source dual MOSFET semiconductor device; and

using the second MOSFET and the fourth MOSFET of the second common-source dual MOSFET semiconductor device for the second and fourth switches.

104. (New) The method of claim 103 further comprising:
integrating the first and third control circuits in the first common-source dual MOSFET semiconductor device; and
integrating the second and fourth control circuits in the second common-source dual MOSFET semiconductor device.

105. (New) The method of claim 103 further comprising:
providing a plurality of individual first and third MOSFET elements on the first semiconductor die;
arranging the individual first and third MOSFET elements in an alternating pattern on the first semiconductor die;
connecting the first MOSFET elements together to form the first MOSFET;
connecting the third MOSFET elements together to form the third MOSFET;
using a common source metallization to interconnect the source terminals of the first and third MOSFET elements;
providing a plurality of individual second and fourth MOSFET elements on the second semiconductor die;
arranging the individual second and fourth MOSFET elements in an alternating pattern on the second semiconductor die;
connecting the second MOSFET elements together to form the second MOSFET;
connecting the fourth MOSFET elements together to form the fourth MOSFET; and
using a common source metallization to interconnect the source terminals of the second and fourth MOSFET elements.

106. (New) The method of claim 103 wherein the first and second common-source dual MOSFET semiconductor devices each have exactly three terminals.

107. (New) The apparatus of claim 46, comprising a common-source dual MOSFET semiconductor device having a first MOSFET and a third MOSFET that are integrated on a semiconductor die, the first MOSFET is used for the first switch, the third MOSFET is used for the third switch, and the common-source dual MOSFET semiconductor device has a source terminal connected to the source of the first MOSFET and the source of the third MOSFET.

108. (New) The apparatus of claim 107 wherein the switch control circuit comprises a first control circuit and a third control circuit, the first control circuit operates the first switch, the third control circuit operates the third switch, the first control circuit and the first switch are powered using a voltage across the third switch, the third control circuit and the third switch are powered using a voltage across the first switch, and the first and third control circuits are integrated in the semiconductor die of the common-source dual MOSFET semiconductor device.

109. (New) The apparatus of claim 108 wherein the first MOSFET comprises a plurality of individual first MOSFET elements on the semiconductor die, the third MOSFET comprises a plurality of individual third MOSFET elements on the semiconductor die, the individual first and third MOSFET elements are arranged in an alternating pattern on the semiconductor die, and a common source metallization interconnects the source terminals of the first and third MOSFET elements.

110. (New) The apparatus of claim 109 wherein the common-source dual MOSFET semiconductor device has exactly three terminals that includes the source terminal, a terminal connected to the drain of the first MOSFET, and a terminal connected to the drain of the third MOSFET.

111. (New) The apparatus of claim 52, comprising:
a first common-source dual MOSFET semiconductor device that comprises a first MOSFET and a third MOSFET integrated on a first semiconductor die, the first common-source dual MOSFET semiconductor device comprising a common source terminal for connecting to

the first and third MOSFETs, wherein the first MOSFET and the third MOSFET are used for the first and third switches; and

a second common-source dual MOSFET semiconductor device that comprises a second MOSFET and a fourth MOSFET integrated on a second semiconductor die, the second common-source dual MOSFET semiconductor device comprising a common source terminal for connecting to the second and fourth MOSFETs, wherein the second MOSFET and the fourth MOSFET are used for the second and fourth switches.

112. (New) The apparatus of claim 111 wherein the first and third control circuits are integrated on the first semiconductor die, and the second and fourth control circuits are integrated on the second semiconductor die.

113. (New) The apparatus of claim 112 wherein the first MOSFET comprises a plurality of individual first MOSFET elements on the first semiconductor die, the third MOSFET comprises a plurality of individual third MOSFET elements on the first semiconductor die, the individual first and third MOSFET elements are arranged in an alternating pattern on the first semiconductor die, a common source metallization interconnects the source terminals of the first and third MOSFET elements, the second MOSFET comprises a plurality of individual second MOSFET elements on the second semiconductor die, the individual second and fourth MOSFET elements are arranged in an alternating pattern on the second semiconductor die, and a common source metallization interconnects the source terminals of the second and fourth MOSFET elements.

114. (New) The apparatus of claim 113 wherein the first common-source dual MOSFET semiconductor device has exactly three terminals that includes the common source terminal, a terminal connected to the drain of the first MOSFET, and a terminal connected to the drain of the third MOSFET, and

the second common-source dual MOSFET semiconductor device has exactly three terminals that include the common source terminal, a terminal connected to the drain of the second MOSFET, and a terminal connected to the drain of the fourth MOSFET.

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Serial No. : 13/044,063
Filed : March 9, 2011
Page : 17

Attorney Docket: 00614-0166001

REMARKS

Claim 1 has been amended. New claims 91-144 have been added.

Claims 91-94 correspond to original claims 7-10, which depend on allowable claim 1.

Claims 95-98 correspond to original claims 12-15, which depend on allowable claim 1.

Claims 99-102 correspond to original claims 24-27, which depend on allowable claim 18.

Claims 103-106 correspond to original claims 29-32, which depend on allowable claim 18.

Claims 107-110 correspond to original claims 48-51, which depend on allowable claim 44.

Claims 111-114 correspond to original claims 53-56, which depend on allowable claim 44.

Applicant asks that all claims be allowed in view of the amendment to the claims. Please apply any necessary charges or credits to Deposit Account 06-1050, referencing the above attorney docket number.

Respectfully submitted,

Date: March 3, 2015 _____

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L6390 half-bridge gate driver

Introduction

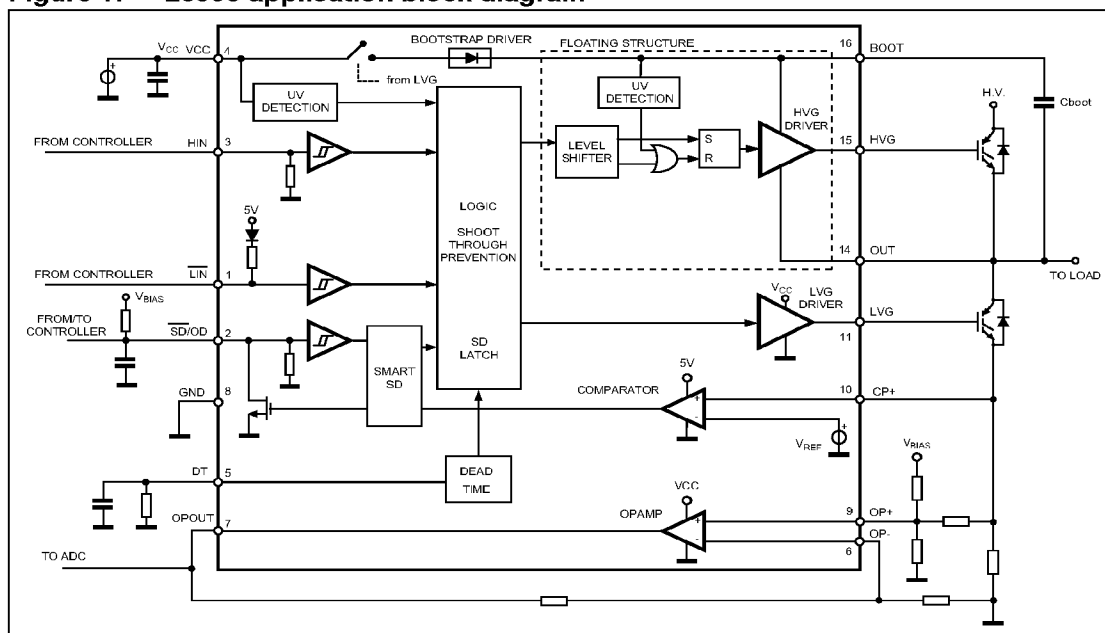
The L6390 is a versatile high voltage gate driver IC which is particularly suited for field oriented control (FOC) motor driving applications. It simplifies the design of control systems for a wide range of motor applications such as home appliances, industrial drives, DC motors and fans.

Designed using BCD off-line technology, this device is capable of operating with voltage rails up to 600 V. The gate driver provides all the functions and current capability necessary for high side and low side power MOSFET and IGBT driving.

The L639x series are high voltage half-bridge gate drivers. These devices can be used in all applications where high voltage shifted control is necessary. The devices have a driver current capability best suited for home appliance motor driving ratings, and they are also equipped with patented internal circuitry which replaces the external bootstrap diode. This feature is achieved by means of a high voltage DMOS synchronously driven with the low side gate driver.

The L6390 is a half-bridge driver with several functions such as externally adjustable dead-time, interlocking, smart shutdown (patented), fault comparator and a dedicated high performance op-amp for advanced current sensing. The outputs can be driven by two dedicated logic signals or, alternatively, only one logic signal by connecting the two inputs together. The device is available in the DIP16 or SO16 packages.

Figure 1. L6390 application block diagram



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1 Pin description

Table 1. Pin description

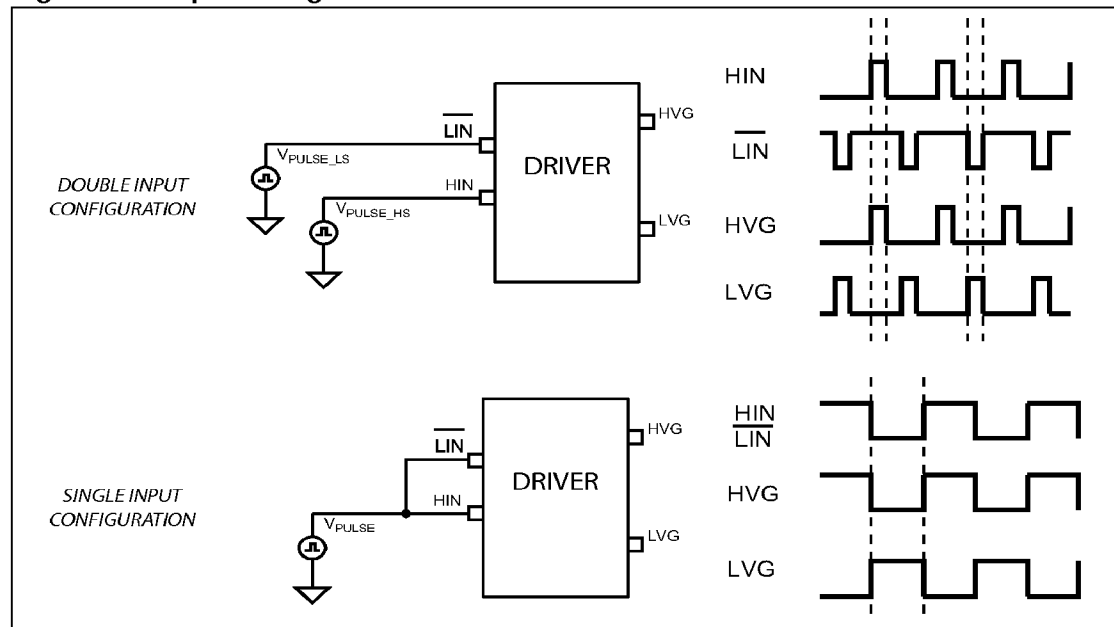
Pin n°	Pin name	Type	Function
1	$\overline{\text{LIN}}$	I	Low side driver logic input (active low)
2	$\overline{\text{SD}}/\text{OD}^{(1)}$	I/O	Shutdown logic input (active low)/open drain (comparator output)
3	HIN	I	High side driver logic input (active high)
4	VCC	P	Lower section supply voltage
5	DT	I	Dead time setting
6	OP-	I	Op-amp inverting input
7	OPOUT	O	Op-amp output
8	GND	P	Ground
9	OP+	I	Op-amp non inverting input
10	CP+	I	Comparator input
11	LVG ⁽¹⁾	O	Low side driver output
12,13	NC		Not connected
14	OUT	P	High side (floating) common voltage
15	HVG ⁽¹⁾	O	High side driver output
16	BOOT	P	Floating section (bootstrap) supply voltage

1. The circuit provides less than 1 V on the LVG and HVG pins (@ $I_{\text{sink}} = 10 \text{ mA}$), with $V_{\text{CC}} > 3 \text{ V}$. This allows the omission of the “bleeder” resistor connected between the gate and the source of the external MOSFET normally used to hold the pin low. The gate driver ensures low impedance in SD conditions also. See *Section 4*.

2 Logic inputs

The L6390 has two logic inputs, HIN and $\overline{\text{LIN}}$, to separately control the high side and low side outputs, HVG and LVG. HIN is in phase with HVG, while $\overline{\text{LIN}}$ is out of phase with LVG. The signal inversion on the low side input allows control of the half-bridge output with only one control signal (see *Figure 2*).

Figure 2. Input configuration



Note that by connecting the two logic input signals together, the resulting dead time is defined by the resistor connected between pin 5 and ground. The dead time can be set to a wide range of values from hundreds of nanoseconds to a few microseconds (see *Figure 5* or the L6390 datasheet). All the logic inputs are provided with hysteresis (~ 1 V) for low noise sensitivity and are TTL/CMOS 3.3 V compatible. Thanks to this low voltage interface logic compatibility, the L6390 can be used with any kind of high performance controller, such as microcontrollers, DSPs or FPGAs.

As shown in the block diagram in *Figure 1*, the logic inputs have internal pull-down (or pull-up) resistors. The purpose of these resistors is to set a proper logic level in case, for example, there is an interruption in the logic lines or the controller outputs are in tri-state conditions. If logic inputs are left floating, the gate driver outputs LVG and HVG are set to low level. The internal resistors are:

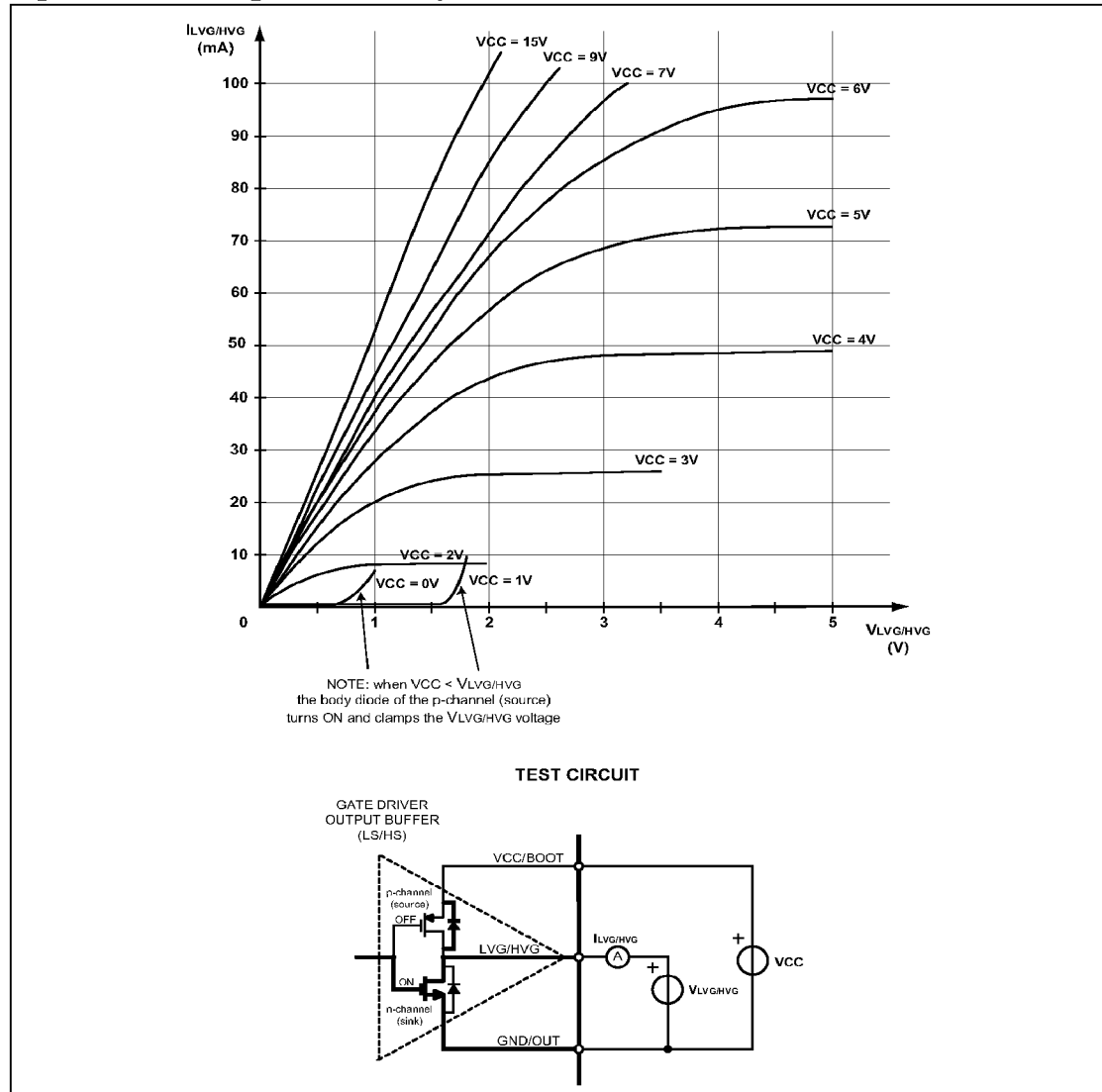
- HIN logic input: 85 k Ω (typ.) pull-down
- $\overline{\text{LIN}}$ logic input: 720 k Ω (typ.) pull-up connected to an internal 5 V regulator through a diode
- $\overline{\text{SD}}$ logic input: 375 k Ω (typ.) pull-down

If the logic inputs are connected together as in the single input configuration (*Figure 2*) and they are left floating, the internal pull-down and pull-up resistors form a resistive divider providing a voltage value (about 460 mV) which keeps the HVG off and LVG on, thus turning on the low side power switch.

3 UVLO function

The L6390 supply voltage V_{CC} is continuously monitored by an under-voltage lockout (UVLO) circuitry which turns off the IC outputs when the supply voltage goes below the V_{CC_thOFF} threshold (see L6390 datasheet for values) and turns on the device when the supply voltage goes above the V_{CC_thON} voltage. A hysteresis of about 1.5 V is provided for noise rejection purpose. The high voltage floating supply V_{BOOT} is provided with a similar under-voltage lockout circuitry also. When the L6390 is in UVLO condition, both gate driver outputs are set to low level, setting the half-bridge power stage output to high impedance. *Figure 3* below shows the I-V characteristics of the output buffers at different V_{CC} values.

Figure 3. L6390 gate driver outputs in UVLO condition



4 Dead time and interlocking function management

In order to avoid any possible cross-conduction between the power MOSFETs/IGBTs of the half-bridge, the L6390 provides both the dead time and the interlocking functions. The interlocking function is a logic operation which sets both the outputs to low level when the inputs are simultaneously active (HIN to high level and LIN to low level). The dead time function is a safety time introduced by the device between the falling edge transition of one driver output and the rising edge of the other output. If the rising edge set externally by the user occurs before the end of this dead time, it is ignored and results delayed until the end of the dead time. The dead time can be adjusted externally through the value of the DT resistor connected between pin 5 and pin 8 (see *Figure 5*). A 100 nF ceramic capacitor in parallel with this resistor is recommended for noise immunity. In *Figure 4* the details of dead time and interlocking function management are described.

Table 2. L6390 truth table

Inputs			Outputs	
SD	LIN	HIN	LVG	HVG
L	X ⁽¹⁾	X ⁽¹⁾	L	L
H	H	L	L	L
H	L	H	L	L
H	L	L	H	L
H	H	H	L	H

1. Don't care



Figure 4. Timing waveforms

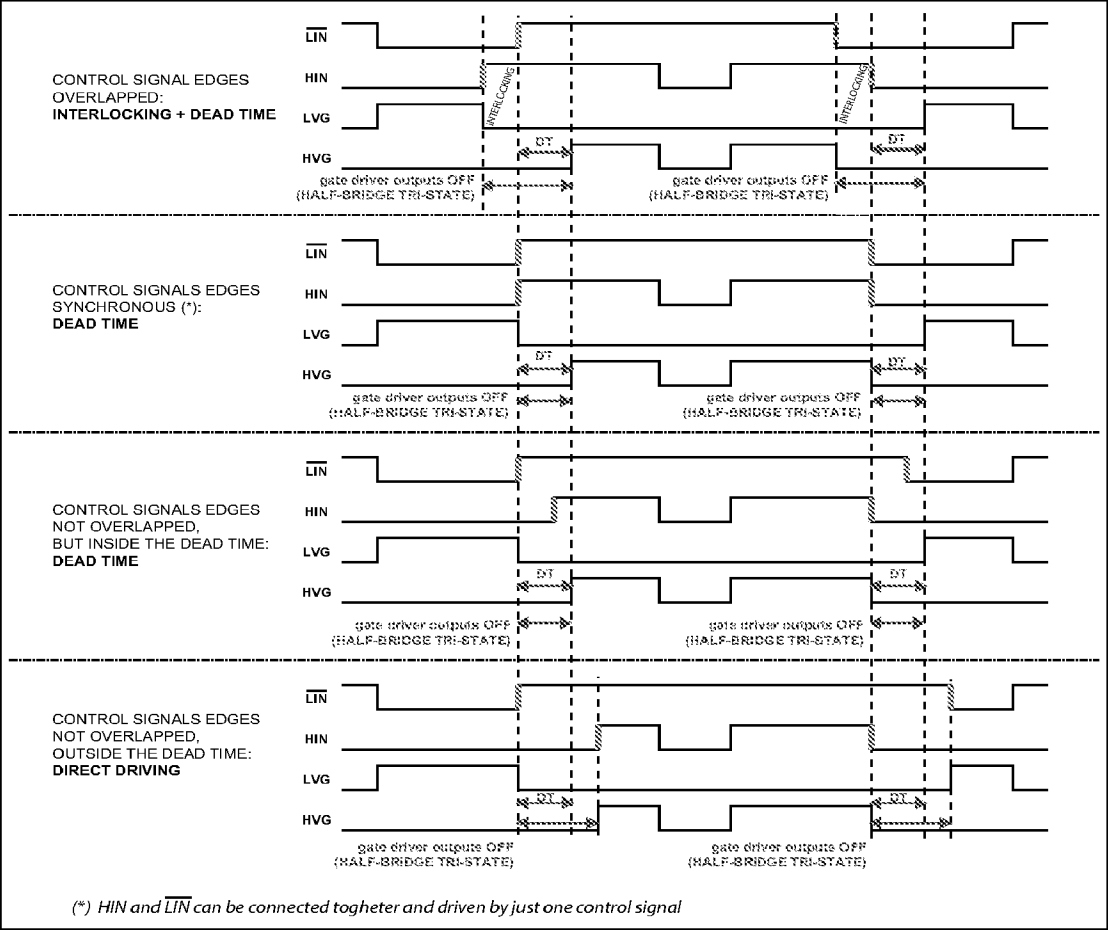
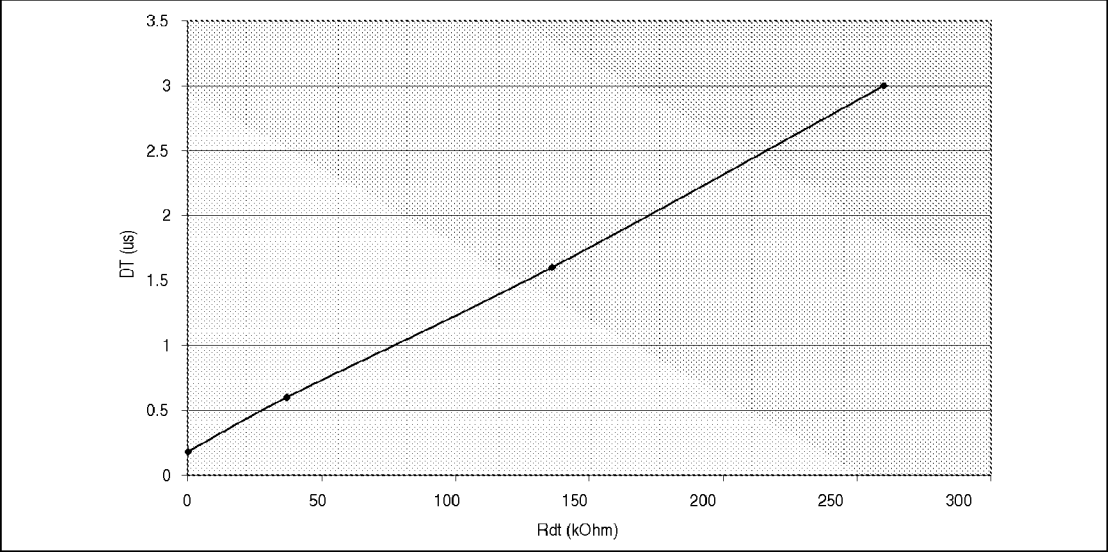


Figure 5. Typical dead time vs. DT resistor value

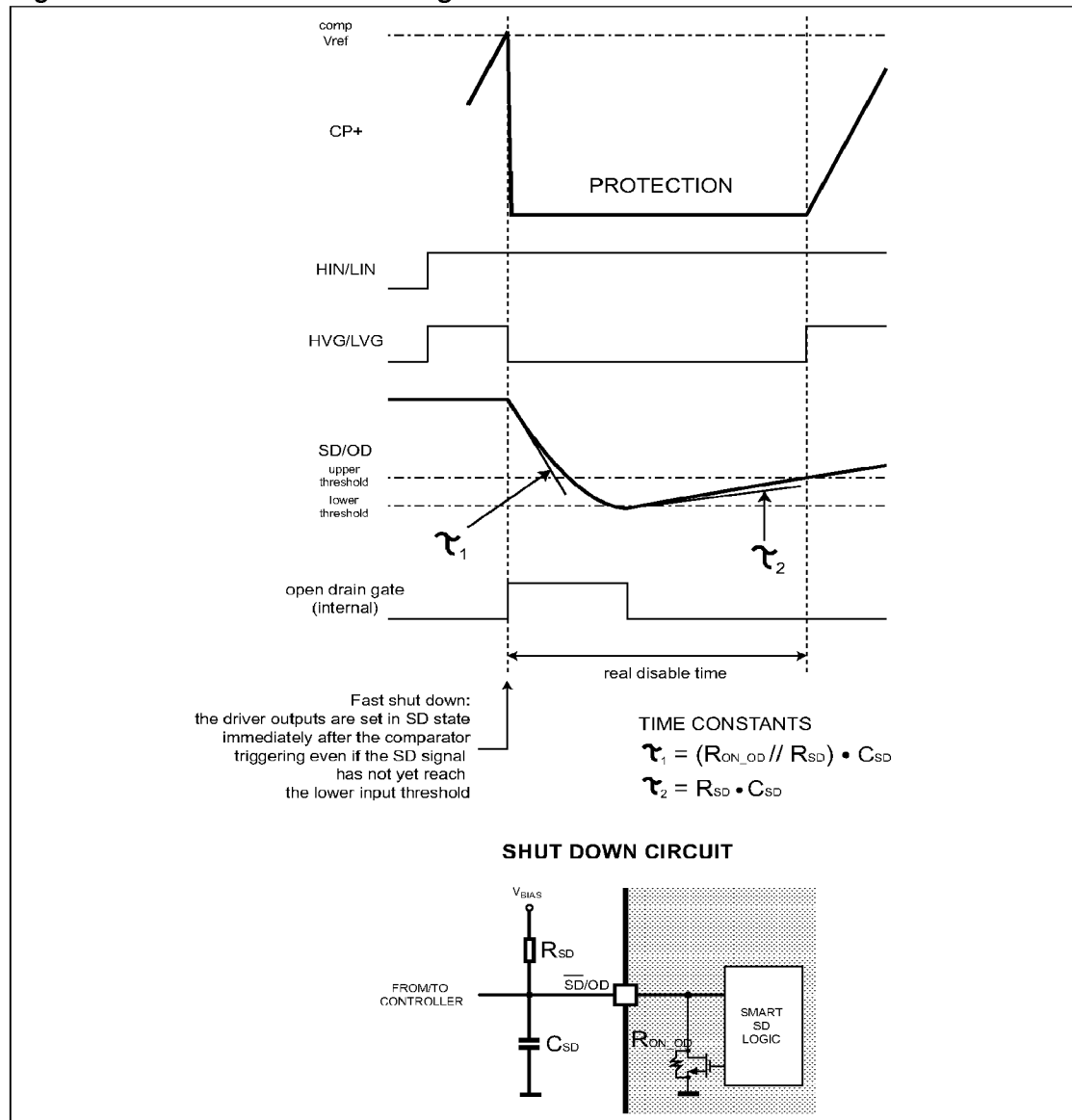


5 Smart shutdown function

The L6390 integrates a comparator for fault sensing purposes. The comparator has an internal reference voltage V_{ref} on its inverting input (see L6390 datasheet), while the non-inverting input is available on pin 10. The comparator input can be connected to an external shunt resistor in order to implement a simple over-current detection function. The output signal of the comparator is fed to an integrated MOSFET with the open drain available on pin 2, shared with the $\overline{SD}$ input.

When the comparator triggers, the device is set in shutdown state and both its outputs are set to low level leading the half-bridge in tri-state.

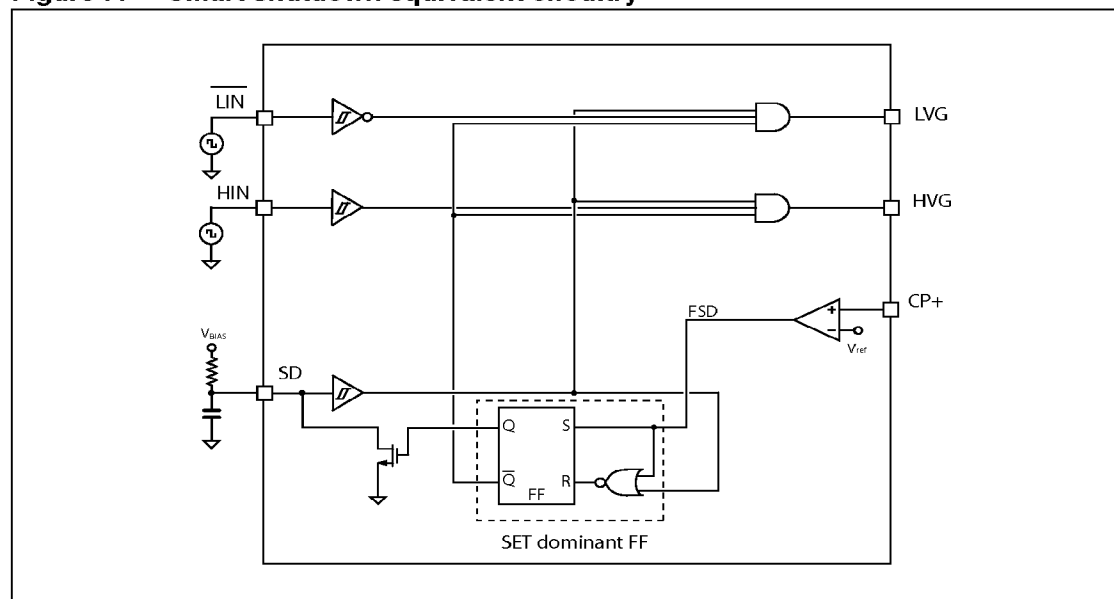
Figure 6. Smart shutdown timing waveforms



In common over-current protection architectures the comparator output is usually connected to the $\overline{SD}$ input and an RC network is connected to this $\overline{SD}/OD$ line in order to provide a mono-stable circuit, which implements a protection time that follows the fault condition. Unlike common fault detection systems, the L6390 smart shutdown architecture allows to immediately turn-off the output gate driver in case of fault, by minimizing the propagation delay between the fault detection event and the actual outputs switch-off. In fact the time delay between the fault and the outputs turn-off is no more dependent on the RC value of the external network connected to the pin. In the smart shutdown circuitry, the fault signal has a preferential path which directly switches off the outputs after the comparator triggering. At the same time the internal logic turns on the open drain output and holds it on until the SD voltage goes below the SD logic input lower threshold. The smart SD system provides the possibility to increase the time constant of the external RC network (that is the disable time after the fault event) without increasing the delay time of the protection. Any external signal provided to the SD pin is not latched and can be used as control signal in order to perform, for instance, PWM chopping through this pin. In fact when a PWM signal is applied to the SD input and the logic inputs of the gate driver are stable, the outputs switch from the low level to the state defined by the logic inputs and vice-versa.

A block diagram of the smart shutdown architecture is depicted in *Figure 7*.

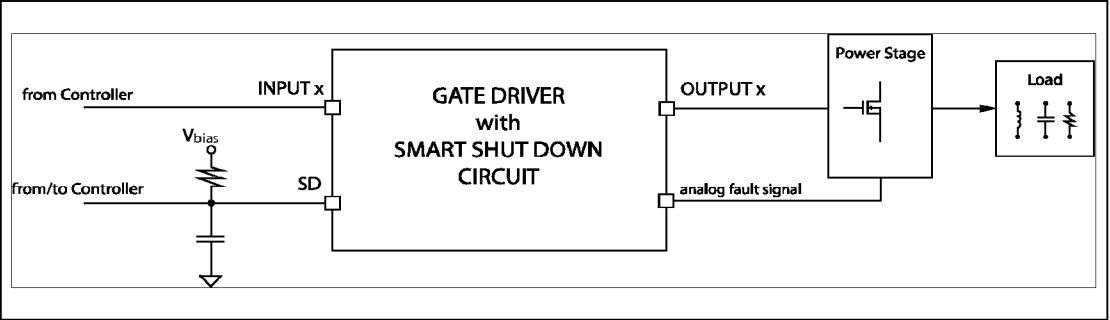
Figure 7. Smart shutdown equivalent circuitry



In normal operation the outputs follow the commands received from respective input signals. When a fault detection event occurs the fault signal (FSD) is set to high by the fault detection circuit output (LVG, HVG) and the FF receives a SET input signal. Consequently the FF outputs set output signals to low level and, at the same time, turn-on the open drain MOSFET which works as active pull-down for the SD signal. Note that the gate driver outputs stay at low level until the $\overline{SD}$ pin has experienced both a falling edge and a rising edge, although the fault signal could be returned to low level immediately after the fault sensing. In fact even if the FF is reset by the falling edge of the $\overline{SD}$ input, the SD signal also works as enable for the outputs, thanks to the two AND ports. Moreover once the internal open-drain transistor has been activated, due to the latch, it cannot be turned-off until the $\overline{SD}$ pin voltage reaches the low logic level. Note that, since the FF is SET dominant,

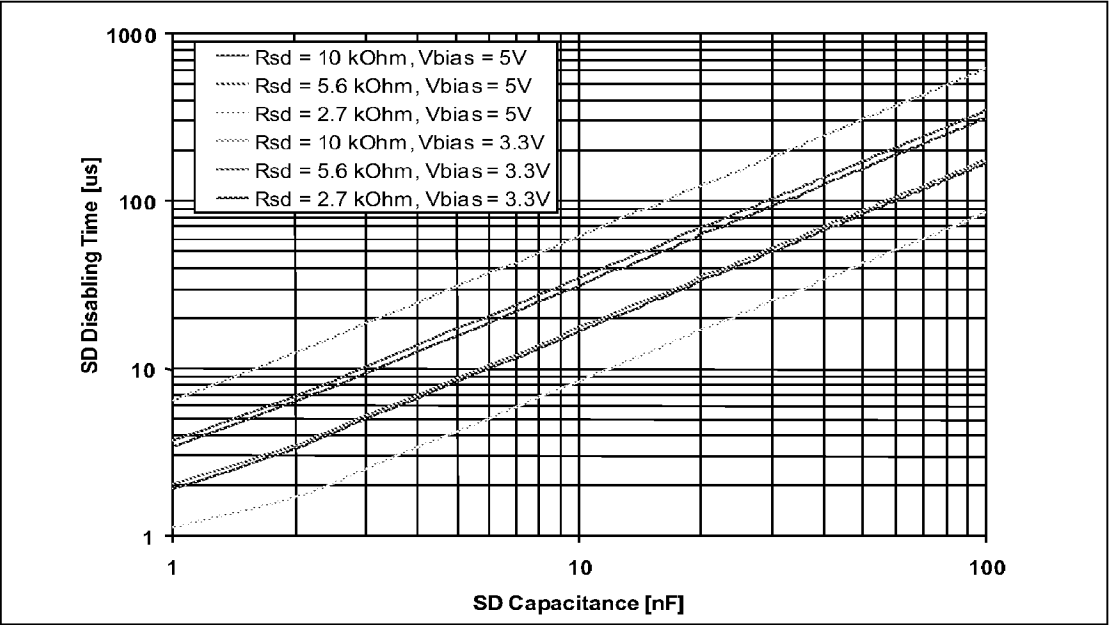
oscillations of the $\overline{\text{SD}}$ pin are avoided if the fault signal remains steady at high level. The block diagram of a power system using the gate driver with the smart shutdown architecture is shown in *Figure 8*. An RC network is used to implement the disable time after a fault detection event.

Figure 8. Protection scheme



In *Figure 9* the typical duration of the disable time vs. the SD capacitance with different R_{sd} values and V_{bias} values is shown.

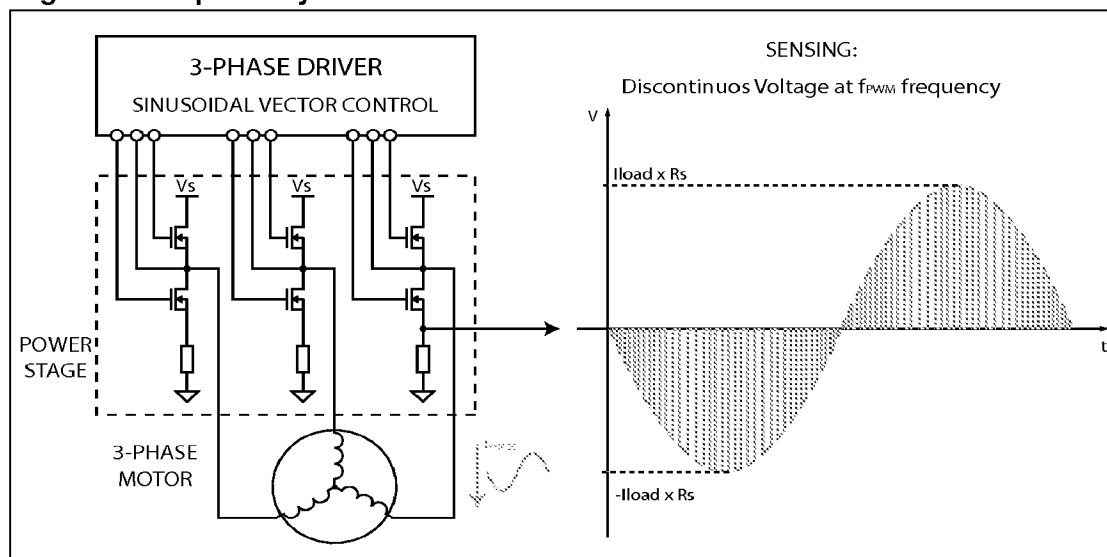
Figure 9. Disable time vs. SD capacitance (typical values)



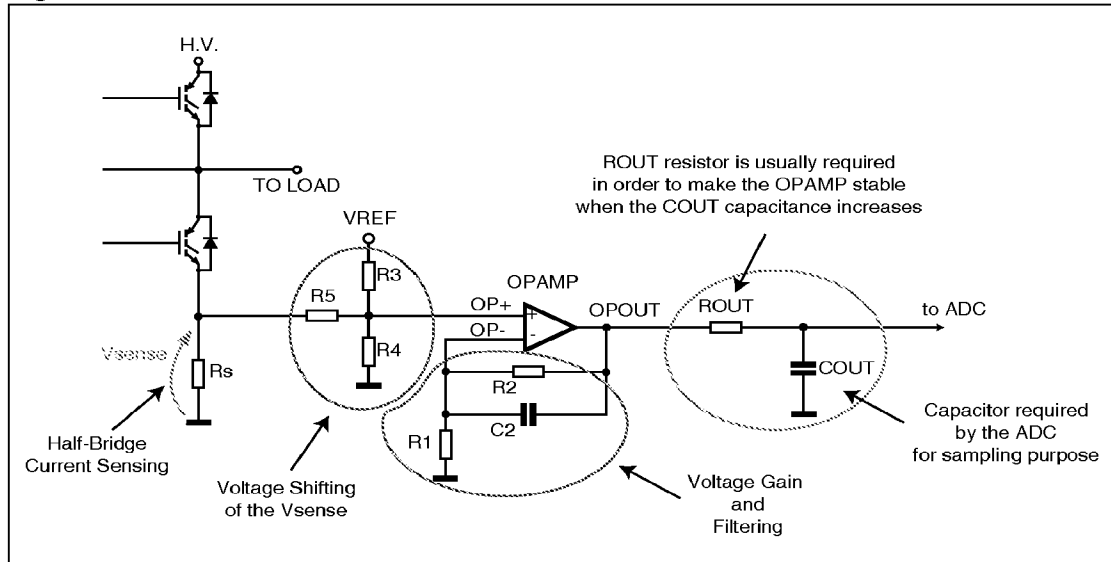
6 L6390 op-amp

The L6390 integrates an operational amplifier optimized for field oriented control (FOC) applications. In a typical FOC application a tri-phase power bridge is used and the currents in the three half-bridges are sensed using a shunt resistor on the source of each low side power switch. The analog current information is transformed in a discontinuous sense voltage signal, available only when the current is flowing in the low side path and having the same frequency of the PWM signal driving the bridge. The sense voltage is a bipolar analog signal, which sign depends on the direction of the current (see *Figure 10*):

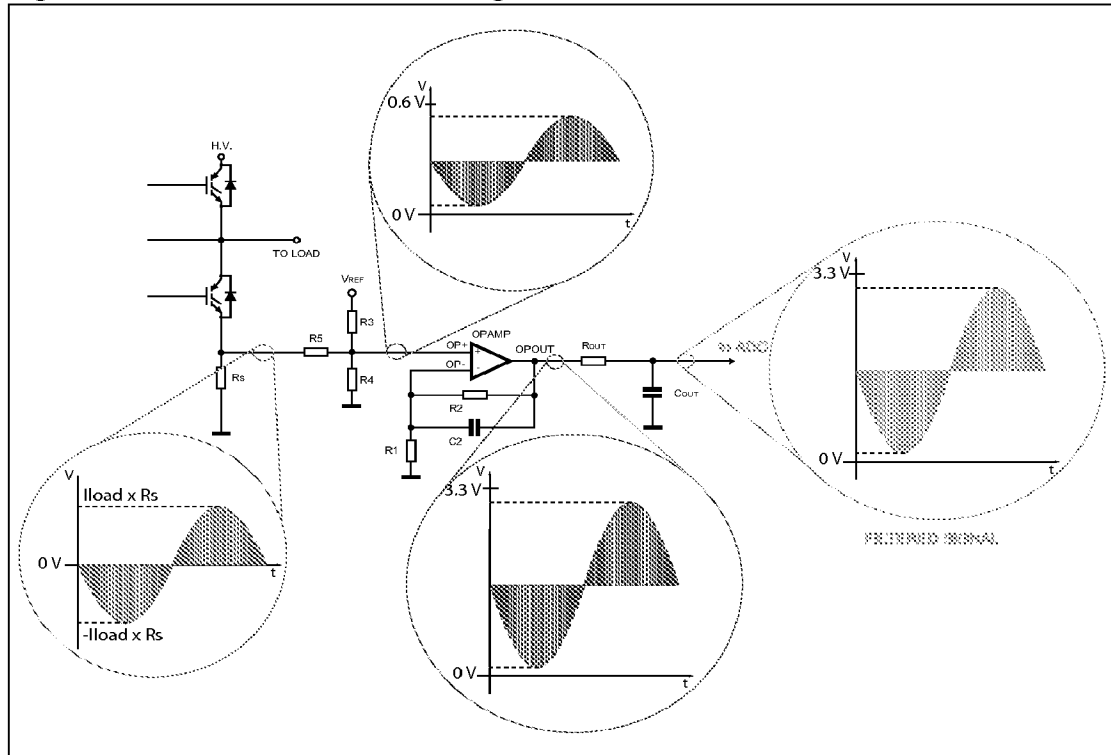
Figure 10. 3-phase system



PWM frequency in a typical home appliance motor driver application is in the range of 5-20 kHz. The sense voltage signals must be provided to an A/D converter in order to perform the matrix calculation related to a certain control technique. Those sense signals are usually shifted and amplified by dedicated op-amps in order to exploit the full range of the A/D converter. The typical scheme shown in *Figure 11* is used.

Figure 11. General advanced current sense scheme

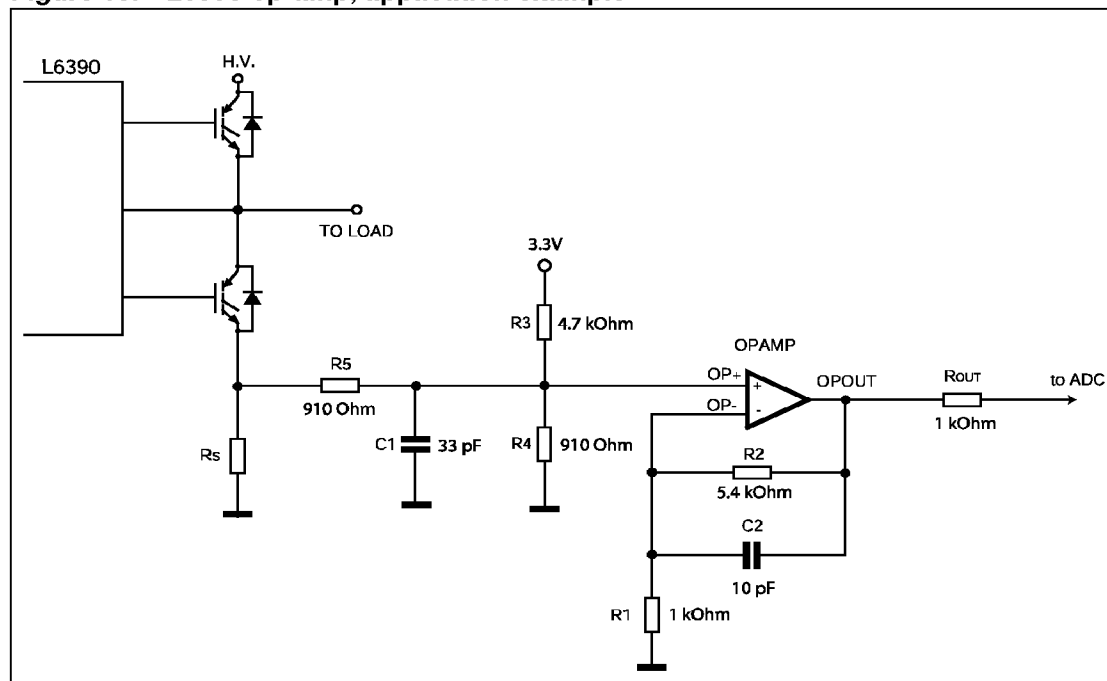
Principle waveforms corresponding to the above scheme are provided in *Figure 12*:

Figure 12. Advanced current sensing waveforms

ADCs used in vector control applications have a typical FSR of about 3.3 V. The sense signals have to be shifted and centered on FSR/2 voltage (about 1.65 V) and amplified with a gain which provides the matching between the maximum value of the sensed signal and

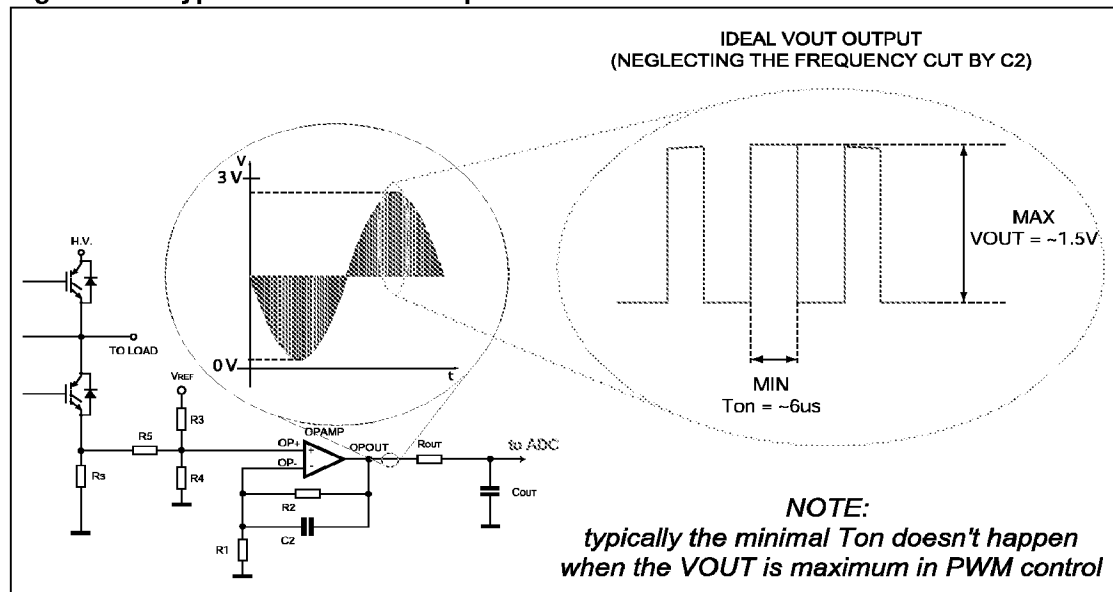
the FSR of the ADC. In *Figure 13* an application circuit of the op-amp with typical passive components values is shown.

Figure 13. L6390 op-amp, application example

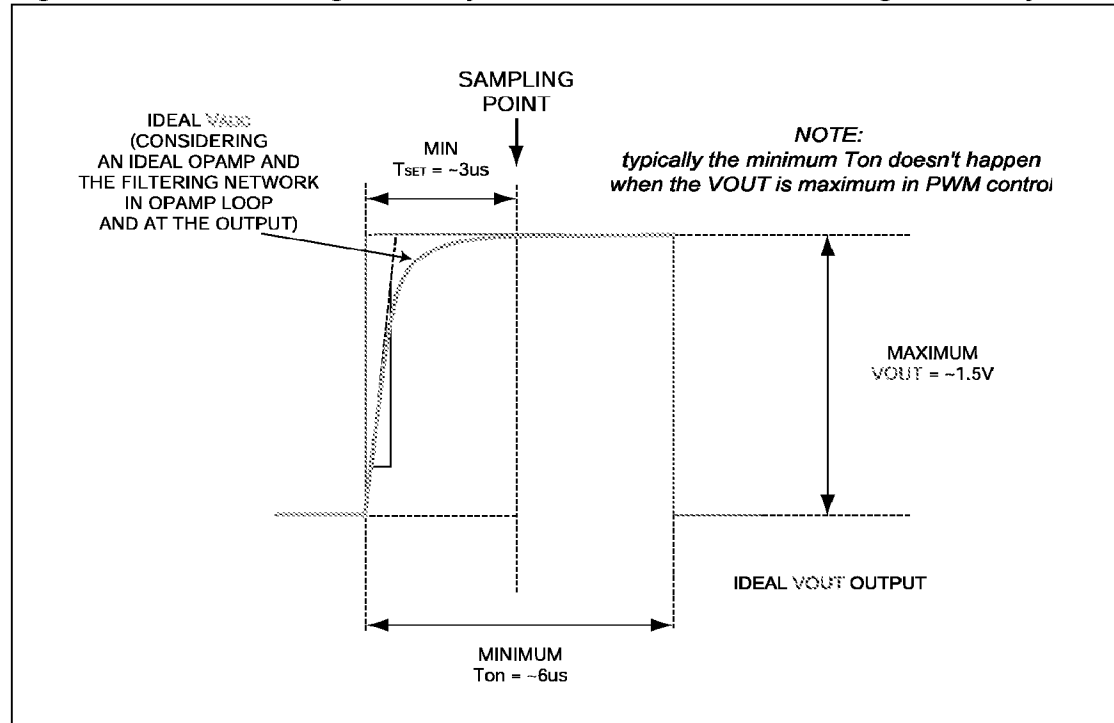


Following *Figure 14* shows the output voltage waveform of the L6390 op-amp considering the voltage level shifting and the gain amplification provided by the amplifier configuration of the op-amp, neglecting the low-pass filtering action of the feedback network. In particular the waveform is zoomed on the maximum amplitude of PWM sinusoidal voltage signal which represents the extreme condition for the external slew rate of the op-amp. The required external slew rate increases proportionally to the voltage excursion and reverse proportionally to the ON time of PWM output.

Figure 14. Typical L6390 ideal output



Typically the maximum voltage variation on op-amp output is $FSR/2$ (about 1.65 V) and the ON time usually does not show its minimum value where the PWM voltage amplitude is higher. In fact the duty cycle of each half-bridge of the 3-phase power stage is proportional to the average voltage applied to each 3-phase terminal which normally is not in phase with the output current of the same bridge, due to the load angle and the BEMF of the running motor. The ADC front-end usually samples the op-amp output voltage in the middle of the ON time, so the output must approach its final value as much as possible within one half of the ON-time. Considering a minimum ON time of about 6 μs , the op-amp output settling time must be lower than 3 μs . If the maximum output voltage variation is 1.5 V and the maximum time needed to reach the desired value is 3 μs , the minimum SR requirement for proper operation would be $1.5 V / 3 \mu s = 0.5 V/\mu s$ (see Figure 15).

Figure 15. Detail on single PWM cycle in advanced current sensing for FOC systems

7 Bootstrap driver

A bootstrap circuitry is needed to supply the high voltage section. This function is normally accomplished by a high voltage fast recovery diode (*Figure 16 a*). In the L6390 a patented integrated structure replaces the external diode. It is realized with a high voltage DMOS driven synchronously with the low side driver (LVG), with a diode in series, as shown in *Figure 16 b*. An internal charge pump (*Figure 16.b*) provides the DMOS driving voltage.

7.1 C_{BOOT} selection and charging

To choose the proper C_{BOOT} value the external MOSFET can be seen as an equivalent capacitor. This capacitor C_{EXT} is related to the MOSFET total gate charge:

Equation 1

$$C_{EXT} = \frac{Q_{GATE}}{V_{GATE}}$$

The ratio between the capacitors C_{EXT} and C_{BOOT} is proportional to the cyclical voltage loss. It has to be:

Equation 2

$$C_{BOOT} \gg C_{EXT}$$

For example: if Q_{GATE} is 30 nC and V_{GATE} is 10 V, C_{EXT} is 3 nF. With C_{BOOT} = 100 nF the drop would be 300 mV.

If HVG has to be supplied for a long time, the C_{BOOT} selection has to take into account also the leakage and quiescent losses.

For example: since the HVG steady state consumption is lower than 200 μA, if HVG T_{ON} is 5 ms then C_{BOOT} has to supply 1 μC to C_{EXT}. This charge on a 1 μF capacitor means a voltage drop of 1 V.

The internal bootstrap driver gives a great advantage: the external fast recovery high voltage diode can be avoided (it usually has great leakage current). This structure works if V_{OUT} is close to GND (or lower) and in the meanwhile the LVG is on. The charging time (T_{charge}) of C_{BOOT} is the time in which both conditions are fulfilled and it has to be long enough to charge the capacitor.

The bootstrap driver introduces a voltage drop due to the DMOS R_{DS(on)}. At low frequency of operation this drop can be neglected, but if the frequency is increased the drop must be taken into account.

The following equation is useful to compute the drop on the bootstrap DMOS:

Equation 3

$$V_{DROP} = I_{CHARGE} \cdot R_{DS(on)} \rightarrow V_{DROP} = \frac{Q_{GATE}}{T_{CHARGE}} \cdot R_{DS(on)}$$

where Q_{GATE} is the gate charge of the external power MOSFET, $R_{DS(on)}$ is the on resistance of the bootstrap DMOS and T_{CHARGE} is the charging time of the bootstrap capacitor.

For example: given the typical value of $120\ \Omega$ for $R_{DS(on)}$ and using a power MOS with a total gate charge of $30\ nC$, the drop on the bootstrap DMOS is about $1\ V$ if the T_{CHARGE} is $5\ \mu s$. In fact:

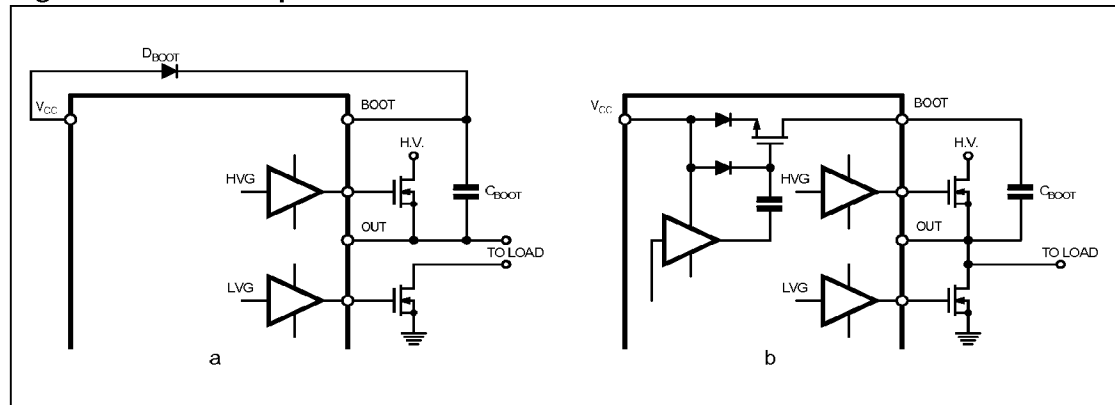
Equation 4

$$V_{DROP} = \frac{30nC}{5\mu s} \cdot 120\Omega \approx 0.7V$$

V_{DROP} has to be considered when the voltage drop on C_{BOOT} is calculated: if this drop is too high, or the circuit topology doesn't allow a sufficient charging time, an external diode can be used.

Working at very low frequencies the high side driver on-time can be very long. So C_{BOOT} voltage can drop because of HVG steady state consumption. To avoid extremely large capacitor ($> 1\text{-}2\ \mu F$) an external charge pump can be added (see *Figure 17* as example). It is mandatory for the diodes to have a low parasitic capacitance, because C1 and C2 should be greater than diodes capacitance. The oscillator has to work in order to balance the high voltage side consumption, and the minimum frequency is fixed by C1 and C2 values (with $C1 = C2 = 33\ pF$ then $f > 250\text{-}300\ kHz$).

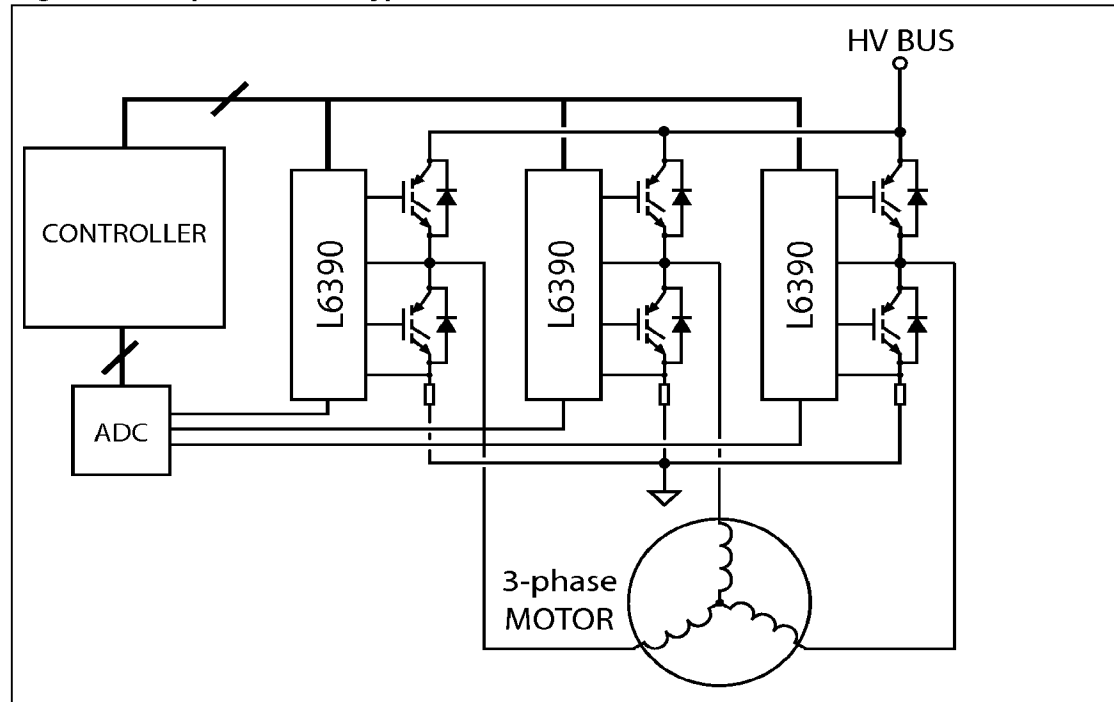
Figure 16. Bootstrap driver



8 Application example

A typical scheme for a 3-phase inverter application is shown in *Figure 18*. In order to drive a 3-phase load as for example a BLDC or Induction AC motor, three L6390 ICs can be used.

Figure 18. 3-phase drive- typical scheme

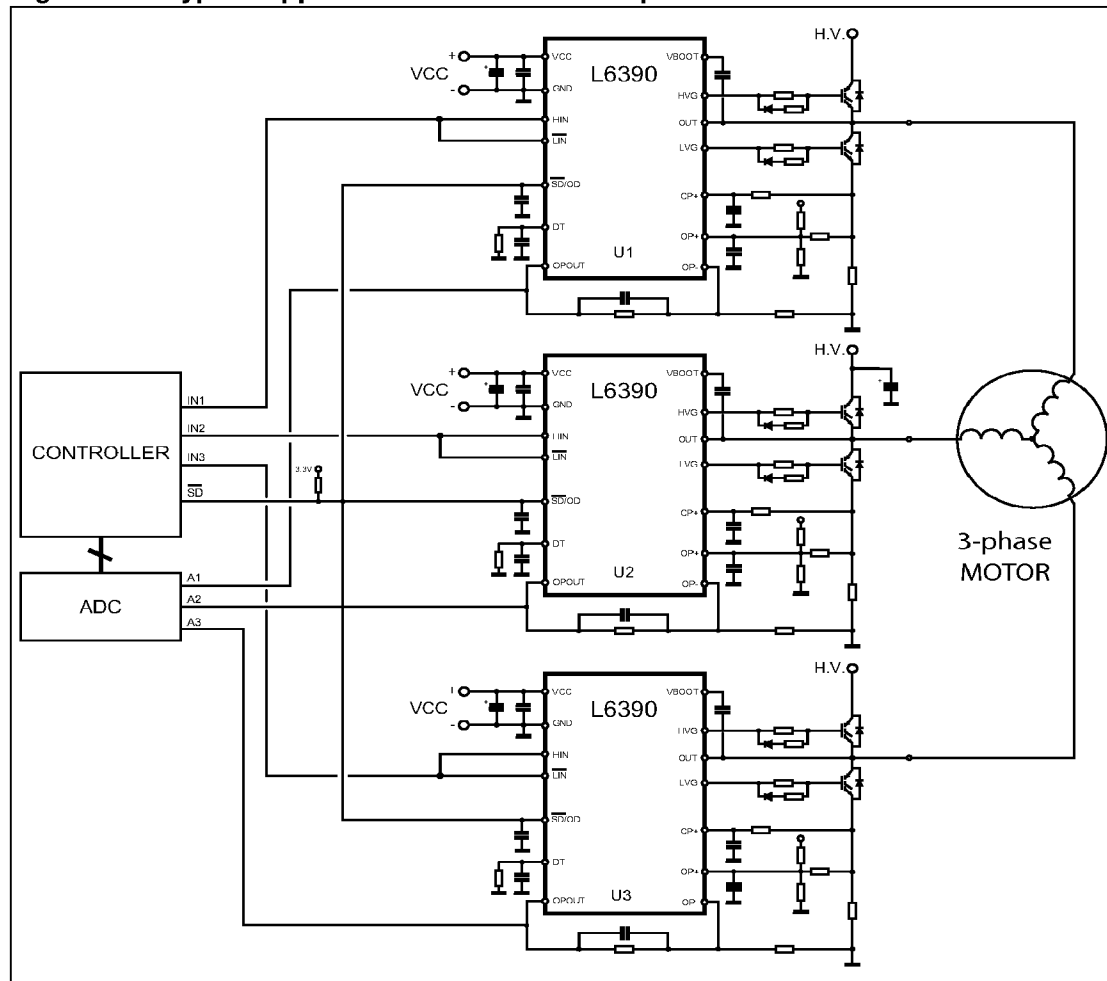


Each L6390 IC drives one half-bridge of the 3-phase power stage. The three gate drivers are driven by the controller and the sensing signals coming from the power stage are managed directly by the L6390 analog blocks (comparators and op-amps) which provide the proper feedback signals to the ADC and the system controller. Each half-bridge comprises two power switches such as IGBTs (or MOSFETs), one high side and one low side. When the high side switch is ON, it brings the output voltage of the half-bridge to the HV bus voltage, which can be a high DC voltage power supply with large power availability, while the low side switch shorts the output to power ground voltage when it is ON. Thanks to the internal floating structure for the high side switch driving, n-type IGBTs (or N-channel MOSFETs) can be used. The gate drivers work exactly as digital/analog-power interface between the controller and the power stage. A shunt resistor can be placed between each low side switch and the power ground, to sense the current on each low side. The information contained in each current sense signal can be conditioned by the L6390 op-amp as described in *Section 6*.

In *Figure 19* a more detailed schematic of a 3-phase power stage topology is shown. Note that usually is recommended (not mandatory) to have some gate resistances between each power switch gate and the correspondent gate driver output, in order to limit the current during the gate charge. Then some filtering and/or level shifting RC networks should be added between the shunt resistors and the correspondent comparator and op-amp input pins of each gate driver. A unique RC network can be used for the $\overline{SD}$ pin even if it is strongly recommended to split the capacitor in three equal components in parallel in order to

place each one very close to the $\overline{\text{SD}}$ input pin of each IC. This capacitance and, in general, each capacitance indicated in the schematic must be placed as close as possible to the IC in order to guarantee a good noise filtering action.

Figure 19. Typical application schematic of a 3-phase FOC



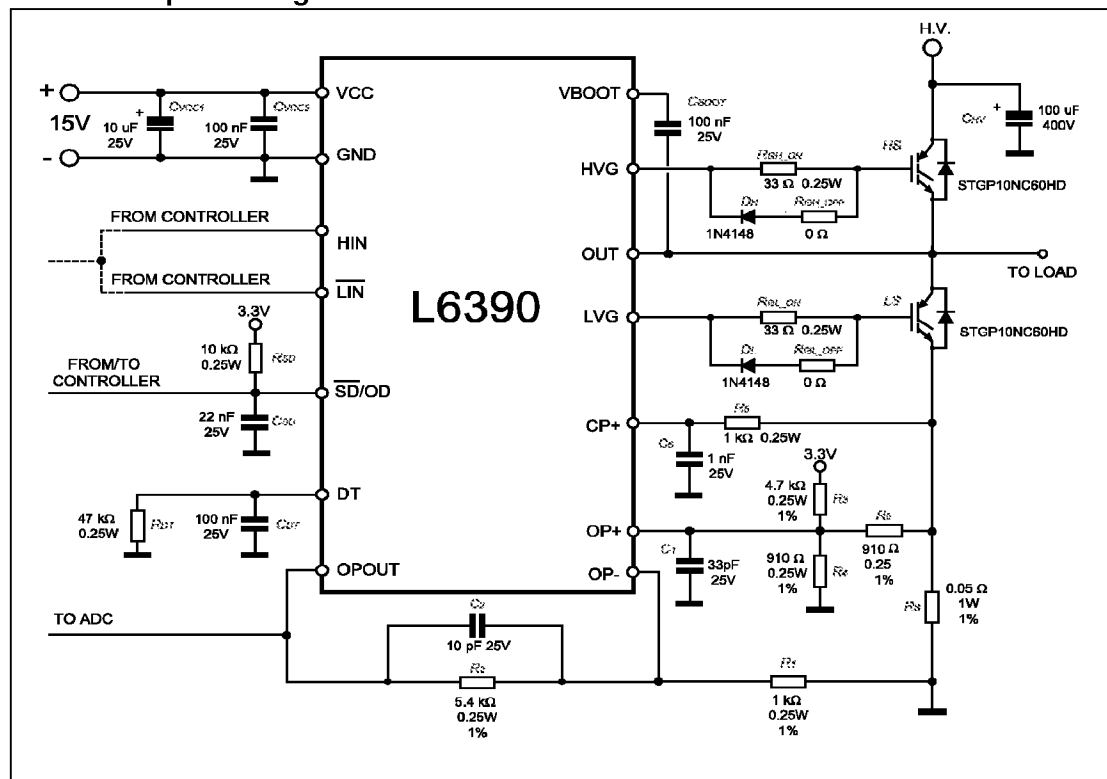
In the example of *Figure 19* the logic inputs HIN and $\overline{\text{LIN}}$ are connected together, using just one single signal to drive each L6390.

In Figure 20 a special focus on a single gate driver circuit of the 3-phase L6390 scheme is provided.

8.1 VCC supply pin

Regarding the VCC pin, a local filtering of the supply voltage very close to the L6390 IC is recommended. Generally the suggestion is to use two capacitors, one electrolytic with greater value ($C_{VCC1} = 10\ \mu\text{F}$, for example) which has a great energy capability but also a not negligible ESR (so is quite slow in providing the current) and a second smaller ceramic capacitor ($C_{VCC2} = 100\ \text{nF}$) which has a lower ESR value but a lower energy capability. The first capacitor works mainly as bulk energy storage while the second one is able to supply the dynamic current spikes required by the commutations of the device, so is better for high frequency decoupling of the IC supply voltage. On the other hand, the selection of the proper value for the CBOOT capacitor is already described in *Section 7*.

Figure 20. Ex. of an application circuit for one of the three half-bridges of a 3-phase power stage



In the example a VCC supply voltage of 15 V is used. This voltage is approximately the same voltage provided to the gate of the power switches.

8.2 BOOT (floating) supply pin

The bootstrap capacitor to supply the high-side floating section of the gate driver must be placed between the BOOT pin 16 and the OUT pin 14. For a deep description of the proper dimensioning of the bootstrap capacitor please refer to *Section 7 on page 18*. The capacitor must be placed as close as possible to the related IC pins. The bootstrap diode required for the charge of the bootstrap capacitor is integrated inside the L6390 device.

8.3 Logic input pins

The logic input pins can be connected directly to the controller with the suggestions provided in *Section 2 on page 6*. If the application environment is very noisy and the logic input voltage is low (e.g. 3.3 V), it can be useful to place some small RC network (not showed in *Figure 20*) in series with the logic input lines, in order to avoid false input triggering due external noise.

8.4 Shutdown pin

Dimensioning of the SD network (R_{SD} and C_{SD}) is provided in *Section 5*.

8.5 Dead time pin

The resistance value on the DT pin must be selected following the indications in *Section 4* and in *Figure 5*. It is recommended to connect between DT and GND pins a capacitor with a value of at least 100 nF, as close as possible to the IC and with short PCB tracks.

8.6 Op-amp

As explained in the previous paragraphs, the L6390 op-amp is completely uncommitted so a large amount of amplifier configurations can be implemented by the application designer. An example of amplifying network is provided in *Section 6*. Typically low tolerance resistors are used, if high accuracy is required in the conditioning of analog signals.

8.7 Comparator input

No particular external circuits are required apart some noise filtering network useful to avoid false triggering of the comparator due to voltage spikes on sense resistor. In the example of *Figure 20* a common RC network is implemented (time constant $\sim 1 \mu s$).

8.8 Sense resistor

The value of the sense resistor must be chosen considering mainly the current rating of the application and the amplitude of the sense voltage desired. The power rating of the sense resistor must be large enough to withstand the maximum current of the application. More resistors in parallel are often used in order to withstand higher power requirements and to obtain low resistance values with low parasitic inductance, which must be as low as possible in order to reduce the dynamic below-ground voltage on the OUT pin of the gate driver.

8.9 Gate driver outputs: gate lines

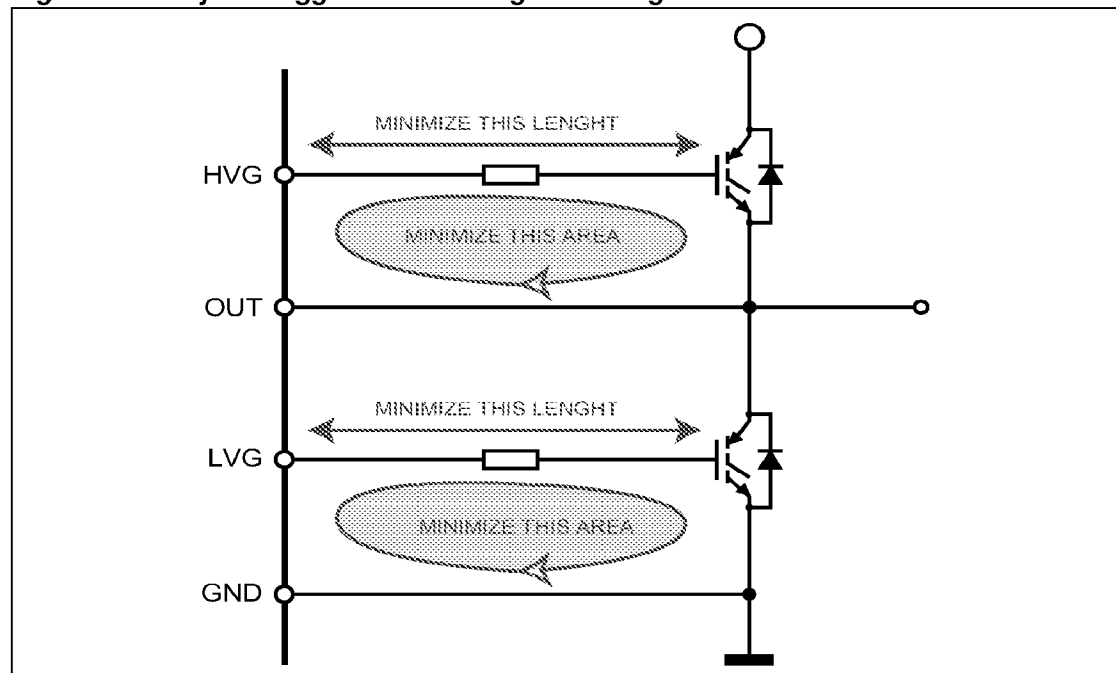
The gates of the power switches and the gate driver outputs can be connected directly, but usually some gate resistors are placed in series on the gate lines in order to limit the gate current during commutations. The final target is to control the dV_{OUT}/dt of each half-bridge output and then reducing the EMI. A more detailed explanation of the mechanisms behind the dV_{OUT}/dt control through the gate resistors is provided in the next sub-paragraph.

Consider that all the following considerations should be considered as approximated analyses of the gate charge phenomenon, so for a proper sizing of the gate resistor is always strongly recommended to evaluate the resulting power bridge transitions in bench analyses.

As shown in *Figure 20*, the gate line is split into separated paths, one for the turn-on (with, in the example, a gate resistor of $33\ \Omega$) and the other one for the turn-off, using a small signal diode as path selector the equivalent turn OFF resistance is in a first approximation the parallel of the turn OFF and the turn ON resistances (neglecting the diode drop). In the example the turn-off resistance is set to $0\ \Omega$ to provide the lowest resistance path for the turn-off of the IGBT. In fact, as explained in the two following paragraphs, low impedance on the gate driver turn-off helps in reducing the induced turn-on phenomena.

Regarding the layout of such gate lines, it is always strongly recommended to place the power IGBT (or MOSFETs) very close to the gate driver. It is important to reduce as much as possible the lengths of such line paths as well the areas included in the gate circuits, because these can act as weak antennas and could catch noise from the surrounding environment. The larger these areas, the higher the gain of such undesired antenna circuits.

Figure 21. Layout suggestion for the gate driving circuits



8.10 Gate driving: principle of working with inductive load

In this paragraph a deeper description of the power IGBT (or MOSFET) gate driving, in case of inductive load, will be provided. The following explanations and calculations are just intended to provide a general understanding of physical principles lying behind the phenomenon of the dV_{OUT}/dt control through the gate current limitation performed by the gate resistors. The target is to help the application designer which uses L6390 ICs to be aware of the various contributes that different parameters of the gate driver system have on the power transitions and also to distinguish the main effects on which he should focus on.

Calculations and formulae should be considered as qualitative indications and not for an accurate quantitative use since, as explained in the above paragraph, they are the result of a first approximation study. The experimental verification of the design choices is always recommended.

The description distinguishes and explores the two main actors of the gate driver system: by one side the inner structure of the gate driver output buffers is described and on the other side the switching mechanism of a generic IGBT (or MOSFET) is analyzed in detail. In *Figure 22* a simplified view of the L6390 gate driver output buffers is provided. Each one can be considered as a CMOS push-pull stage where a P-channel MOSFET works as source driver while an N-channel MOSFET works as sink driver. The structure is similar both for the low side and for the high side and the behavior can be considered the same. In fact the high side driver can be thought as a floating buffer having as supply the VBOOT voltage and as reference the OUT pin. The C_{BOOT} capacitor represents the floating supply voltage source of the high side driver. During the charge of the power switch gate, each source/sink MOSFET can be considered (in first approximation, for simplicity) as if it would be in the ohmic region, so it can be represented as an equivalent resistor with a value equal to its $R_{DS(on)}$. Thanks to this approximation it is possible to use a simplified equivalent circuit for the turn-on and the turn-off commutation (see *Figure 22* and *23*). Regarding the turn-on, this gate charge circuit has two resistors in series (R_{GATE_ON} and $R_{DS(on)_SOURCE}$) and a supply voltage which is VCC for the low side and VBOOT-OUT for the high side. Regarding the turn-off the equivalent circuit is composed just by two resistors (R_{GATE_OFF} and $R_{DS(on)_SINK}$) connected to the source of the power switch.

Gate driver output buffers: equivalent circuit

Figure 22. Gate driver output: equivalent circuit for turn-on

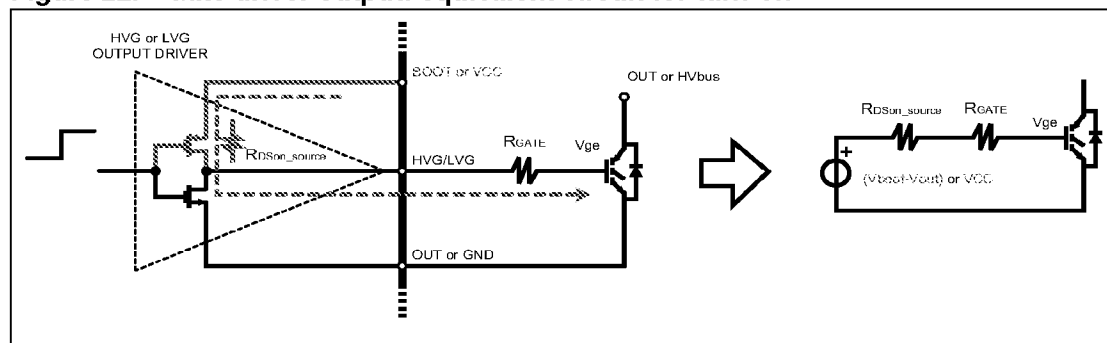
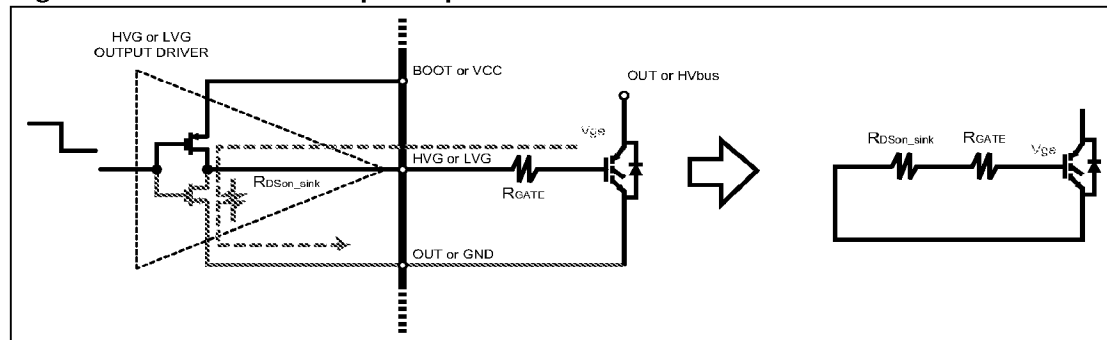


Figure 23. Gate driver output: equivalent circuit for turn-off



This first simplification is related to the inner structure of the gate driver circuitry. Let now focus for a while on the side of the power bridge and its commutations. In a common half-bridge composed by a high side and a low side switch the transitions of the power IGBTs (or MOSFETs) are not all similar, but they can be distinguished in two main types: soft switching and hard switching (see *Figure 24* and *25*). The main element playing a key role in the dynamic of transitions is the direction of the current flowing in the power IGBT (or MOSFET) under evaluation which depends on the direction of the load current respect to the half-bridge. Mainly if the current has the same direction of the power switch the commutation is hard. If it is in opposite direction the commutation is soft.

Figure 24. Hard switching

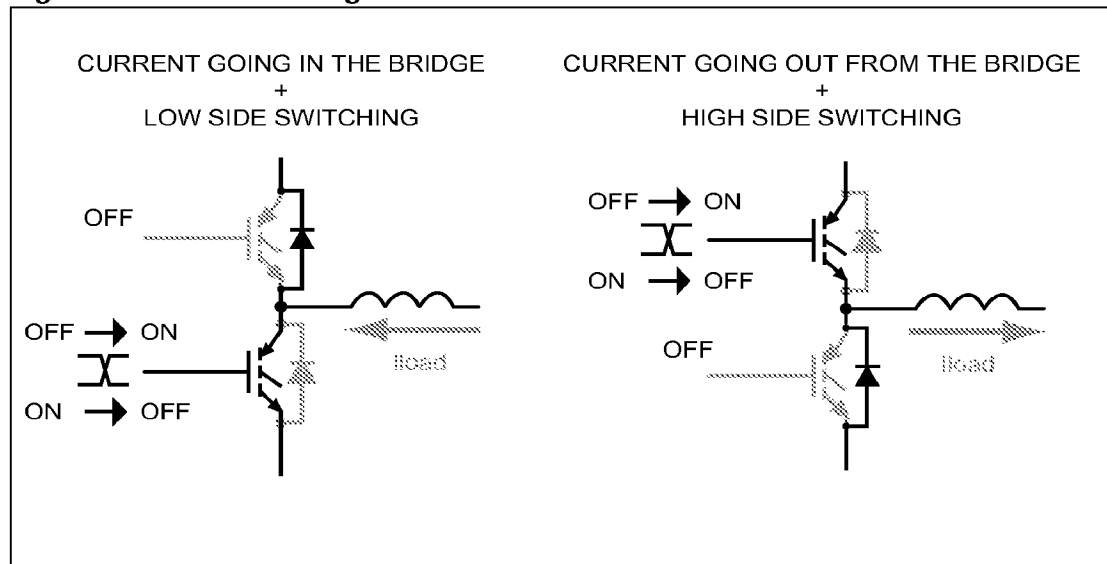
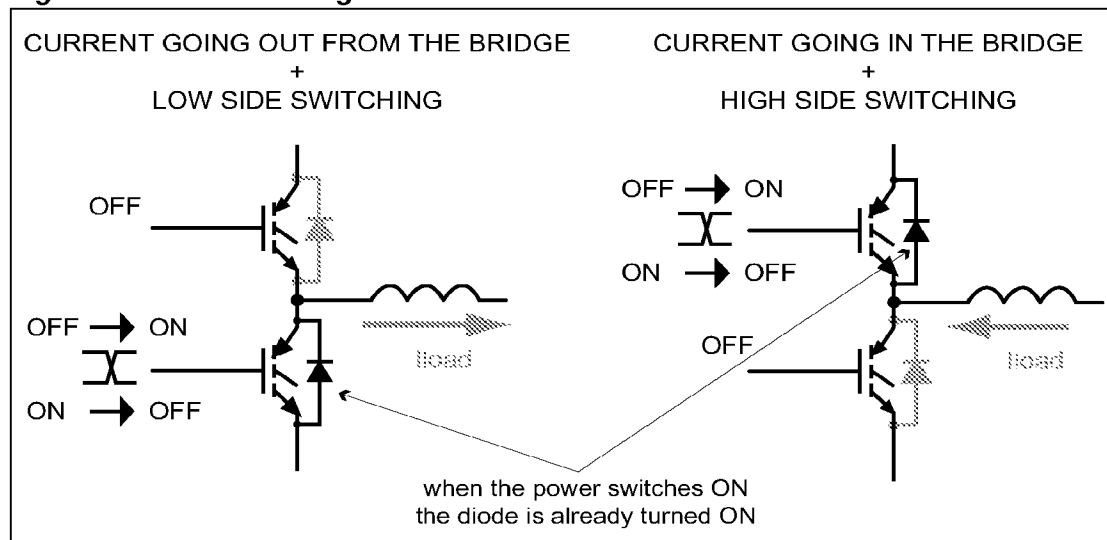


Figure 25. Soft switching



As shown in *Figure 24* and *25*, there is hard switching when the high side is commutating and the load current is exiting the bridge or when the low side is commutating and the load

current is entering the bridge. This type of transition is called “hard” because the power switch turns on when the related V_{CE} (V_{DS}) is at maximum; V_{CE} (V_{DS}) goes back to maximum during turn-off (V_{CE} is close to zero while the switch is steadily ON). The transition is completely managed by the switch that, during the commutation, dissipates some energy. On the other hand, if the switch turns on when its V_{CE} is already closed to zero or V_{CE} remains close to zero after turn-off, the transition is soft and, during the commutation, the switch dissipates almost no energy. This last condition happens when the free-wheeling diode of the switch is bringing the current because the diode is in the same direction of the load current. As a general rule, consider that the dynamic of the transition (dV_{OUT}/dt , V_{CE} rise/fall time etc.) is always managed and controlled by the power switch in hard switching, while its companion switch is necessarily in soft switching because it is in the opposite direction of the load current. Considering this, in order to understand the power IGBT (MOSFET) transition, the hard switching transition will be further investigated. Note that all of the descriptions that follow consider an inductive load connected to the half-bridge stage output.

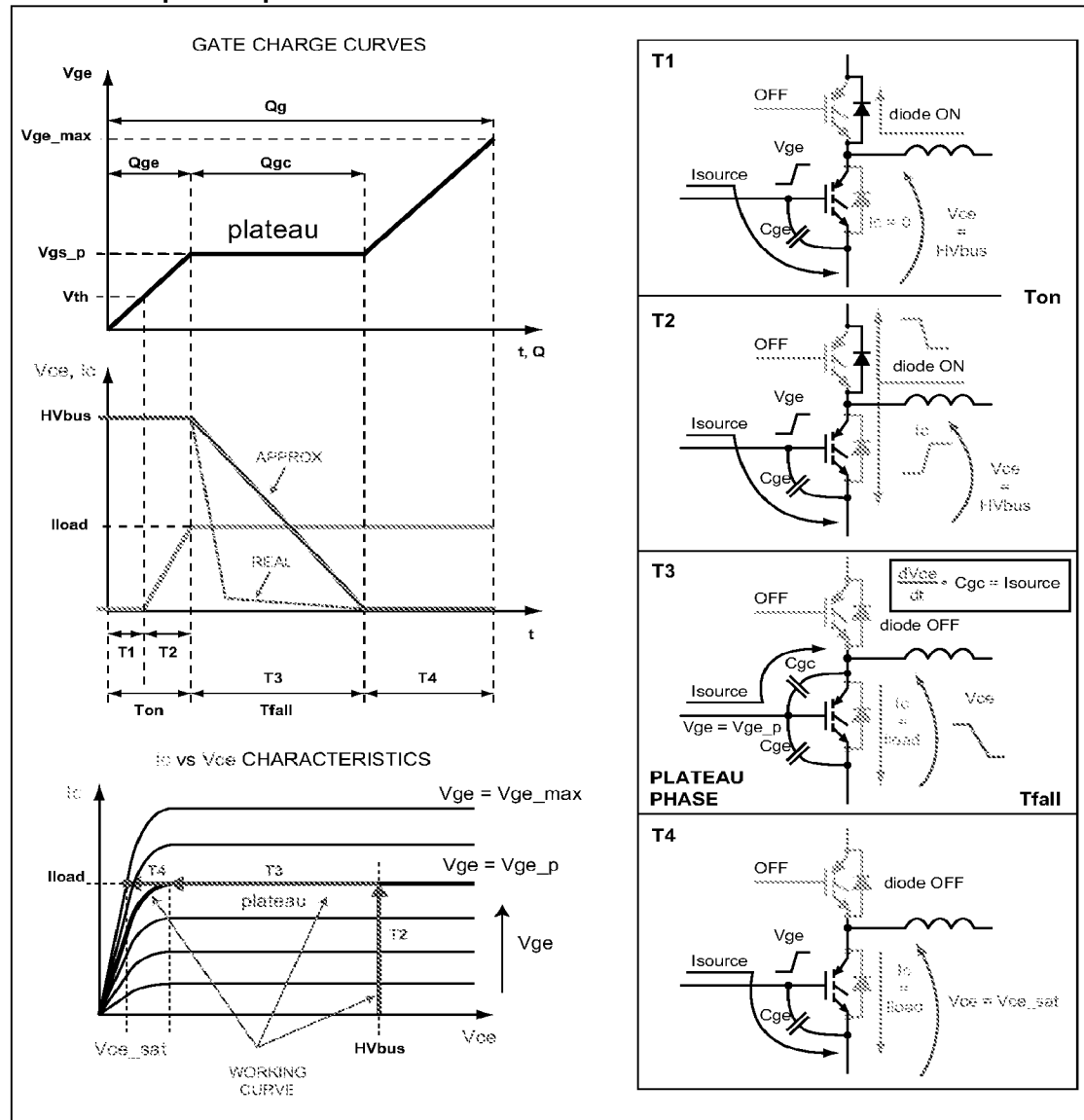
In *Figure 26*, the dynamics of the hard switching for the turn-on transition are described. The first graph is related to the gate charge curves of the power IGBTs (or MOSFETs). This is a typical graph which is also available in the datasheets for the IGBT or MOSFET and describes the dependence of the amount of charge required by the IGBT gate on the voltage drop between the gate and the emitter (or source). In the second graph the collector current (I_C) and V_{CE} voltage versus time are shown. The third graph shows the working point of the IGBT traced on the various I_C vs. V_{CE} characteristic curves for different V_{GE} voltages. For a single turn-on transition, four main phases related to the power IGBT (MOSFET) commutation can be distinguished. On the left side of *Figure 26*, the IGBT conditions for each phase are described. During phase T1 the gate begins to be charged, but the power IGBT still is not conducting because V_{GE} is below the threshold of the IGBT. In this state, the IGBT current is zero and its V_{CE} is at maximum, while V_{GE} is gradually increasing. In phase T2 the V_{GE} voltage goes above to the IGBT threshold and the IGBT starts to bring part of the load current. V_{CE} remains fixed at maximum level because the free-wheeling diode of the companion IGBT is still bringing the rest of the load current and is still conducting, thereby clamping the V_{CE} voltage. Both in T1 and T2 phases, the gate current contributes to charge the equivalent V_{GE} parasitic capacitance of the IGBT. In the I_C vs. V_{CE} characteristic plot the T2 phase is the vertical section of the working curve trace, because the IGBT is moving on its own characteristic with a constant V_{CE} and an increasing current, while its V_{GE} is increasing also. When the amount of current flowing in the IGBT is equal to the load current, the diode turns off and the V_{CE} voltage starts to decrease because it is no longer clamped by the free-wheeling diode of the other IGBT. The working point on I_C vs. V_{CE} curve reaches the load value and starts to move horizontally on the I_C constant curve in the direction of decreasing V_{CE} voltages. This is the T3 phase, usually called the “plateau” phase. This name derives from the fact that the gate charge curve is horizontal for the entire T3 phase, until V_{CE} reaches the V_{CE_sat} value corresponding to the load current. Note that the V_{GE} voltage is constant, although the gate current flowing in the IGBT gate is not zero. The reason for this is that the whole gate current is used to charge the C_{GC} (Miller) parasitic capacitance which then experiences a dV/dt on its terminals because the V_{CE} voltage is decreasing after the diode turn-off. As a first approximation, if C_{GC} was constant (currently it is not), the dV_{OUT}/dt could be calculated as follows:

Equation 5

$$\left(\frac{dV_{OUT}}{dt}\right)_{FALL} = \frac{I_{SOURCE}}{C_{GC}}$$

The formula above shows how the source current coming from the gate driver can directly control the dV_{OUT}/dt of the power half-bridge. This calculation is just a first approximation, since typically the C_{GC} Miller capacitance value is not constant but depends on the V_{CE} voltage, C_{GC} is not linear. The main result of the abrupt variation of the C_{GC} is that the actual slope of the OUT transition is composed of two different dV_{OUT}/dt slopes, one faster and the other slower compared to the value obtained by the above formula (see *Figure 26*). In this document only the first approximation approach will be used.

Figure 26. Turn-on hard switching details with induction load: gate charge and plateau phase



Due to C_{GC} non-linearity, the IGBT (or MOSFET) datasheet also reports the equivalent total amount of charge required during the different gate charge phases: the total gate charge Q_g required for turning on the IGBT (or MOSFET) completely, the Q_{GE} required for increasing

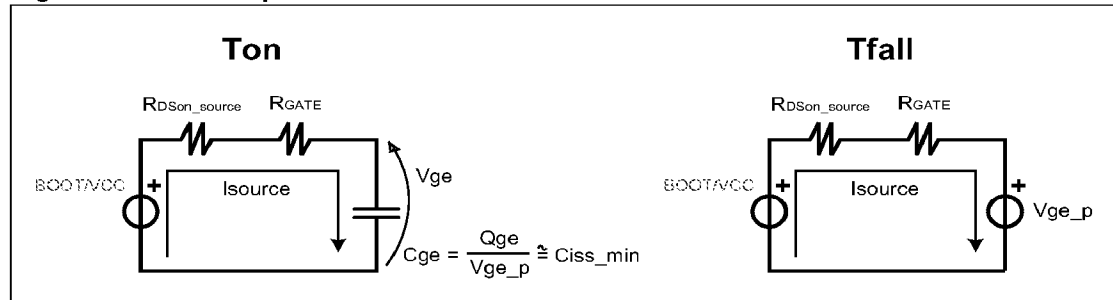
the V_{GE} up to the plateau voltage and the Q_{GC} required during the whole plateau phase. This last amount of charge is also called the plateau charge. The time required to provide the complete plateau charge is the T_{fall} time, and this is the same time necessary for completing the V_{CE} transition. For T_{on} time is intended the time delay between the beginning of the gate charge and the full conduction of the IGBT (or MOSFET), when the power switch current equals the full load current. So T_{fall} can be calculated as follows:

Equation 6

$$T_{fall} = \frac{Q_{GC}}{I_{SOURCE}}$$

It is now possible to merge the equivalent gate driver output circuit reported in figures 24 and 25 and the considerations regarding the different gate charge phases in the total equivalent circuit reported in Figure 27.

Figure 27. Total equivalent circuit for the turn-on



From the above circuit the value of the transition times could be calculated (the following formulae are related to the low side transition, but the same are suitable also for the high side by exchanging "VCC" with "VBOOT-VOUT"):

Equation 7

$$T_{ON} = (R_{DS(on)} + R_{GATE}) \cdot C_{ISSmin} \cdot \ln\left(\frac{VCC}{VCC - V_{GEP}}\right)$$

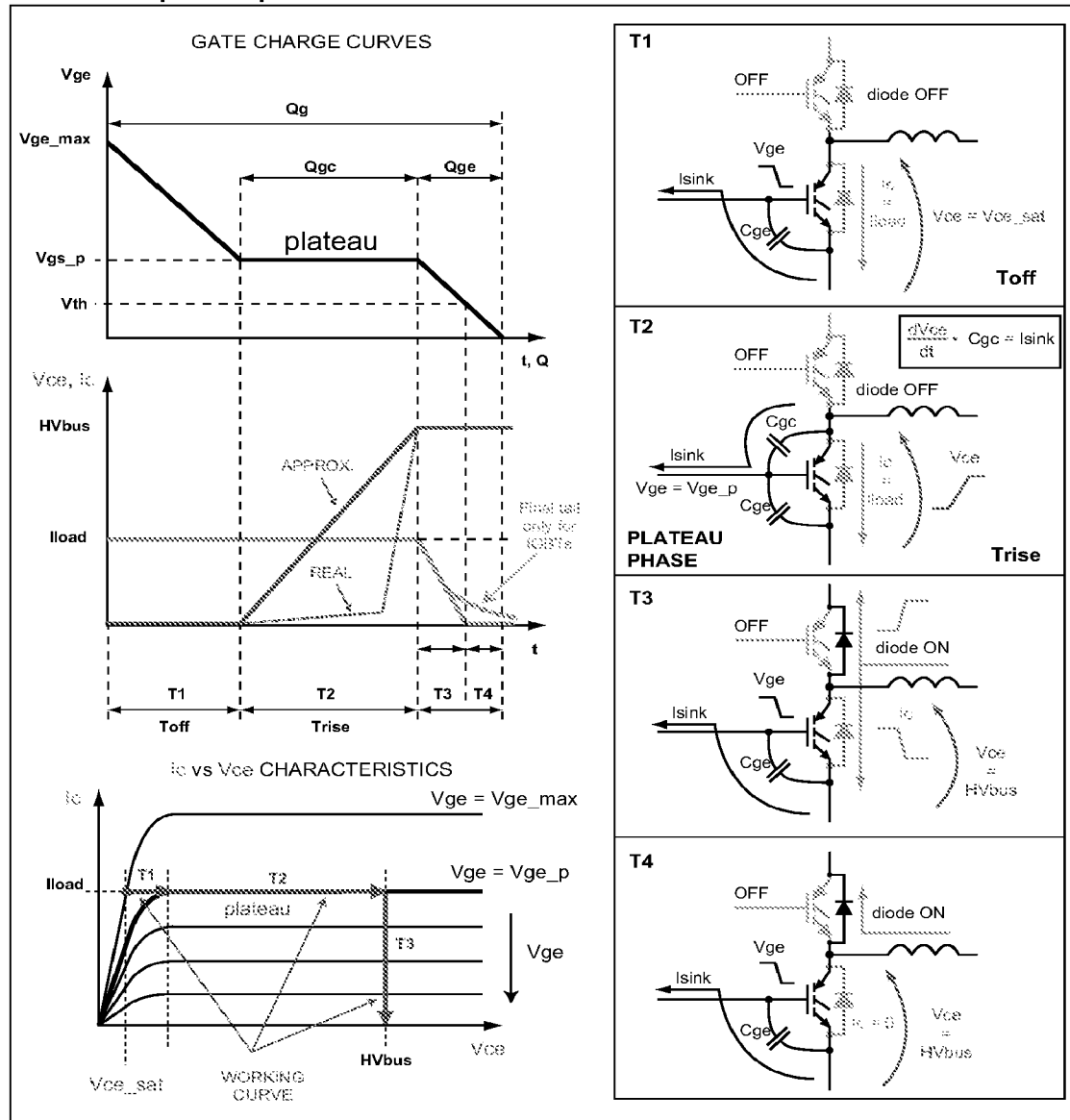
Equation 8

$$T_{FALL} = \frac{Q_{GC}}{I_{SOURCE}} = Q_{GC} \cdot \frac{(R_{DS(on)} + R_{GATE})}{VCC - V_{GEP}}$$

C_{ISSmin} is used because when the V_{CE} (or V_{DS}) of the power IGBT (or MOSFET) is maximum (equal to HV_{bus}), C_{ISS} (which depends on the V_{CE}) shows its minimum value.

During the turn-off of the power IGBT (or MOSFET), the behavior is the same as the turn-on, but in reversed time order. In Figure 28, the gate charge characteristics and V_{CE} vs. I_C curves are provided. The time required to sink the complete plateau charge is the T_{rise} time, while for T_{off} time is intended the time delay between the beginning of the gate charge and the increase of V_{ce} (or V_{ds}) voltage.

Figure 28. Turn-off hard switching details with induction load: gate charge and plateau phase



The rising dV_{OUT}/dt and T_{rise} can be calculated as follows:

Equation 9

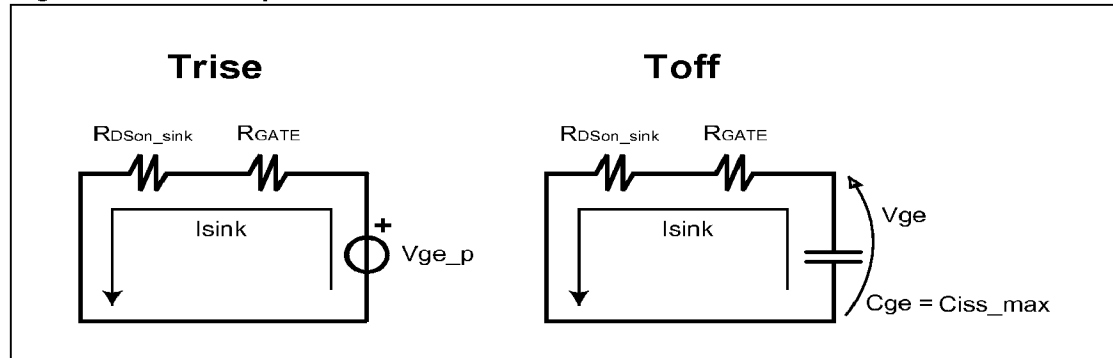
$$\left(\frac{dV_{OUT}}{dt}\right)_{RISE} = \frac{I_{SINK}}{C_{GC}}$$

Equation 10

$$T_{RISE} = \frac{Q_{GC}}{I_{SINK}}$$

As discussed above, the equivalent gate driver output circuit can be represented as in *Figure 29*:

Figure 29. Total equivalent circuit for the turn-off



Given the equivalent circuits in *Figure 29* and the approximations of the power IGBT (MOSFET) switching behavior, the timing equations are as follows:

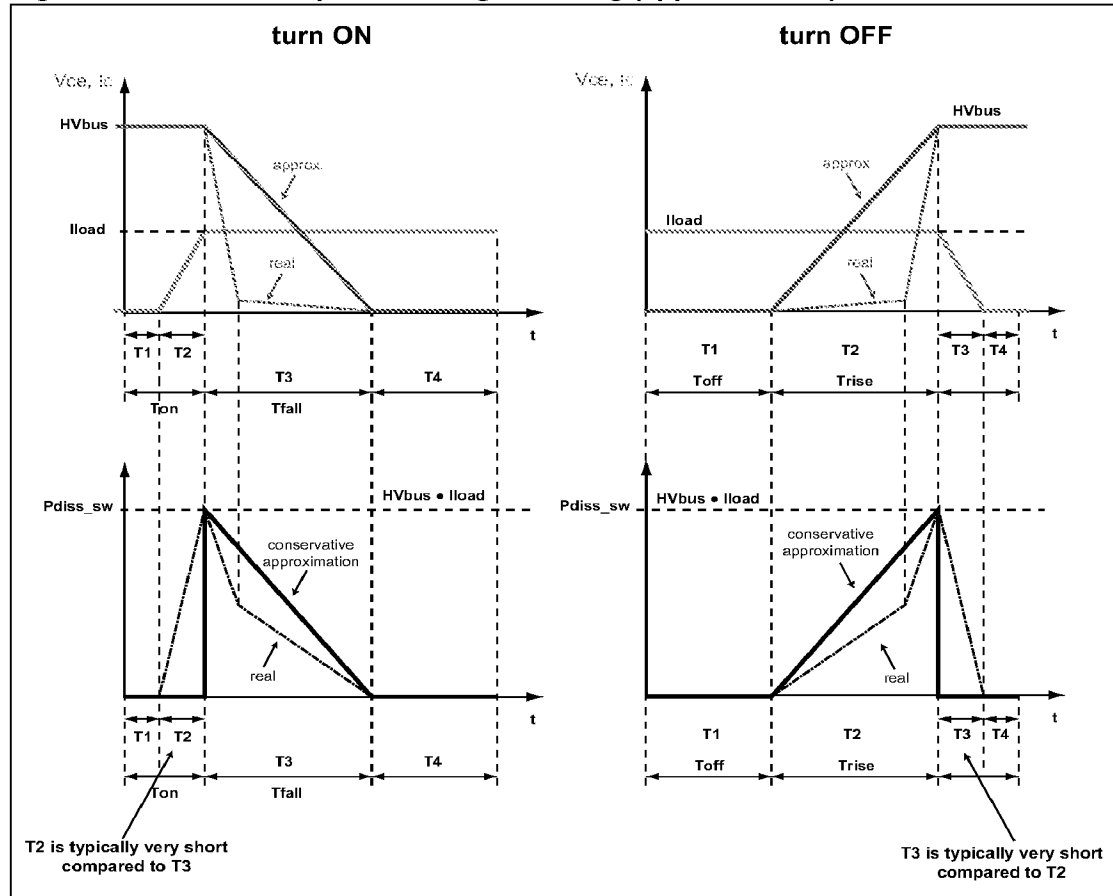
Equation 11

$$T_{OFF} = (R_{DS(on)} + R_{GATE}) \cdot C_{ISSmax} \cdot \ln \cdot \left(\frac{V_{GE_MAX}}{V_{GEP}} \right)$$

Equation 12

$$T_{RISE} = \frac{Q_{GC}}{I_{SINK}} = Q_{GC} \cdot \frac{(R_{DS(on)SINK} + R_{GATE})}{V_{GEP}}$$

As above, C_{ISS_max} is used because when the V_{CE} (or V_{DS}) of the power IGBT (or MOSFET) is minimum (equal to V_{CE_sat}) the C_{ISS} (which depends on the V_{CE}) shows its maximum value. Even if they are just approximations, the T_{fall} and T_{rise} are very important values to be estimated because they provide an indication of the power dissipation during the switching of the power IGBT (or MOSFET), which can be approximated as indicated in *Figure 30*.

Figure 30. Power dissipation during switching (approximation)

Since the T_2 time for the turn-on and T_3 time for the turn-OFF are much shorter compared to T_{fall} and T_{rise} , respectively, the approximated instantaneous amount of energy dissipated during the commutation can be calculated as the triangular area having as its base the T_{fall} , or T_{rise} time and as its height the product of the maximum V_{CE} voltage (equal to HV_{bus}) and the load current value:

Equation 13

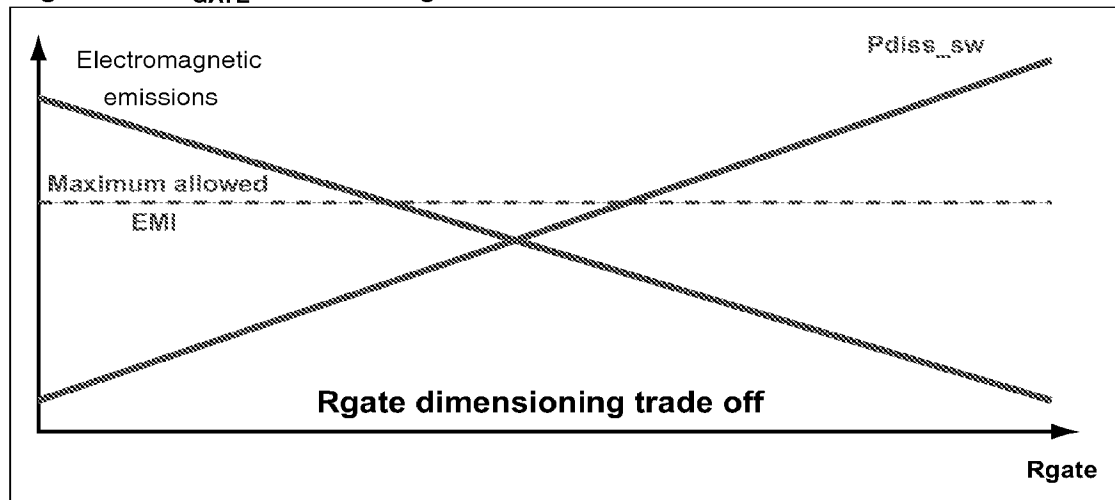
$$E_{DISS-SW} \cong \frac{(HV_{bus} \cdot I_{LOAD}) \cdot (T_{fall} + T_{rise})}{2}$$

Equation 14

$$P_{DISS-SW} \cong \frac{(HV_{bus} \cdot I_{LOAD}) \cdot (T_{fall} + T_{rise}) \cdot f_{SW}}{2}$$

The term f_{SW} represents the switching frequency of the power IGBT (or MOSFET).

Based the results obtained by these approximate calculations, it should be clear that the main consideration when dimensioning the gate resistor values is the trade-off between electromagnetic emission and the power dissipated during power IGBT (or MOSFET) switching, as shown in *Figure 31*.

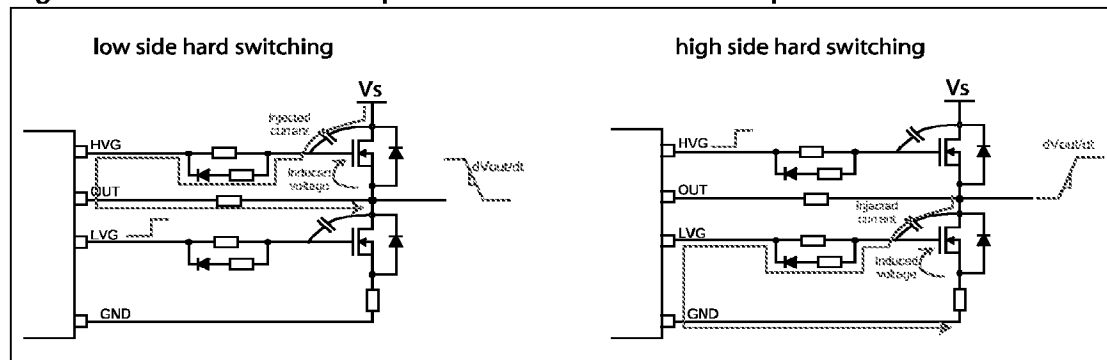
Figure 31. R_{GATE} dimensioning criteria

R_{GATE} influences the power dissipation on one side and the EMI effects on the other. Therefore, the best trade-off must be determined during the application design-in phase.

9 Induced turn-on phenomenon

One possible phenomenon to be analyzed is the induced turn-on that could occur on a turned-off power IGBT (or MOSFET) when its companion on the same half-bridge is switching on. This phenomenon is due to the current injected on the gate by the Miller parasitic capacitance (C_{GC}) which causes an undesired voltage increase on the gate of the power IGBT (or MOSFET) which should be kept well turned off. Induced gate voltage depends on the absolute value of the parasitic capacitance C_{GC} , its relative ratio with C_{GE} , the value of the dV_{OUT}/dt of the half-bridge and the value of the equivalent (turn-off) resistance between the emitter (or the source) and the gate. *Figure 32* illustrates this phenomenon.

Figure 32. Induced turn-on phenomenon - circuitual description



Note: Typically the induced turn-on phenomenon does not cause a complete turn-on of the power MOSFET, so it is rare to have destructive cross-conduction on the half-bridge during commutations. Nevertheless, weak conduction of the opened power switch could increase the power dissipation of the power stage, increasing the overall temperature of the power MOSFETs and reducing the efficiency. This is why this phenomenon deserves particular attention also in terms of thermal performance of the power application.

Some strategies to reduce this phenomenon are:

- a) **Reducing resistance path between gate and emitter (source).** Reducing as much as possible the gate resistance in which the injected current flows, the voltage drop on the gate becomes lower. The drawback is a possible increase of the dV_{OUT}/dt during turn-off of the power IGBT (or MOSFET) which commutates in hard switching, and then an increase of electrical noise and EMI issues (as explained in the previous paragraph, the power IGBT or MOSFET in hard-switching is always the one which has the channel in the same direction as the current). On the other hand, usually the dV_{OUT}/dt during turn-off is lower than during turn-on, because the plateau voltage is closer to the source voltage than the supply voltage of the driver. This results in a lower gate current for the discharge of the gate charge.

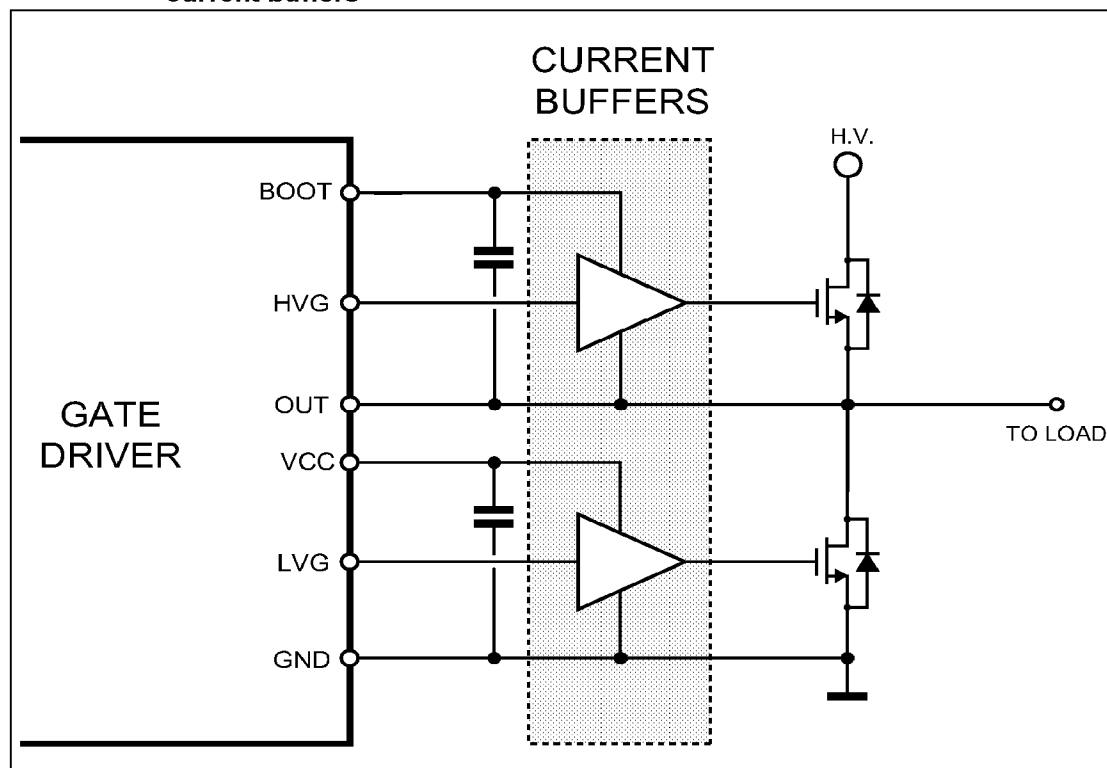
- b) **Reducing the maximum dV_{OUT}/dt .** This reduction can be achieved by increasing the gate resistor value which limits the gate current for the turn-on of the power MOSFET. The drawback is a consequent increase of the switching time which means dissipating more power during commutations.
- c) **Using a MOSFET with lower C_{GD}/C_{GS} (or C_{GC}/C_{GE} for IGBTs) ratio.** This solution is not always the simplest, but it could be the best strategy where the induced turn-on phenomenon is dominant.

10 How to increase the gate driver output current capability

In some cases, certain applications may require more gate driver output current capability. This requirement could be found in systems having a power rating higher than about 1 kW. In those applications, in fact, typically the power MOSFET/IGBTs have a large gate charge which contributes to a slowing down of the power switch transition (the dV_{OUT}/dt), increasing the power dissipation during each commutation. Note that also in applications with power rating higher than 1 kW, the output current capability of the L6390 could be enough if the power dissipation for commutation is acceptable (the power dissipation due to $R_{DS(on)}$ or V_{CESAT} is not dependent on the gate current). But if the limitation of this power contribution is a constraint, the dV_{OUT}/dt of each transition must be increased by enhancing the current capability of the gate driver.

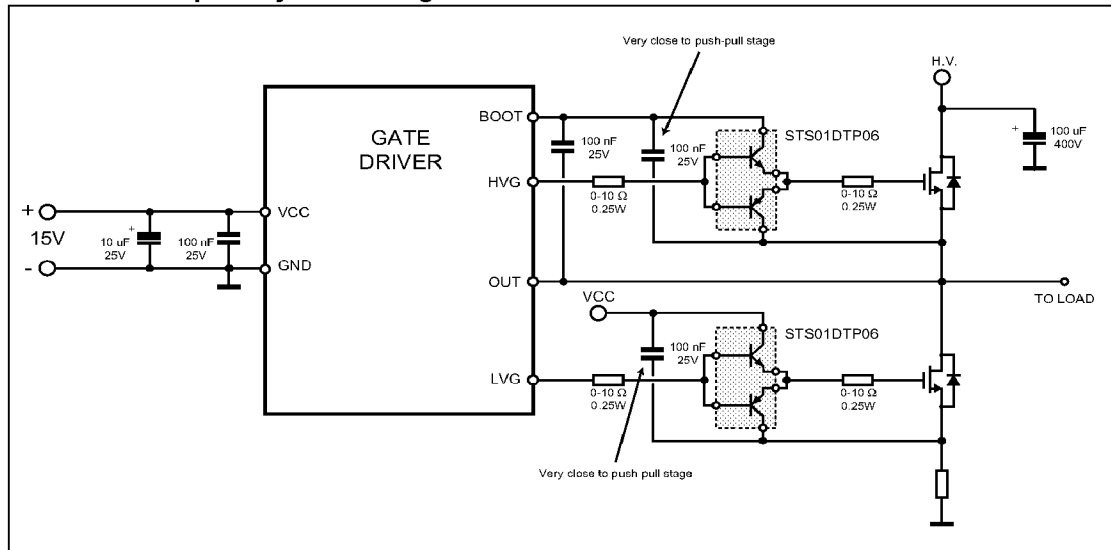
A simple way to increase the current capability of the gate driver outputs is to insert, in series with the two gate lines of one half-bridge gate driver IC, two external current buffers (Figure 33).

Figure 33. Block diagram of output current capability enhancement using external current buffers



Typically the current buffers are implemented using bipolar NPN-PNP push-pull non-inverting (emitter follower configuration) structures. In Figure 34, a typical circuit using a gate driver with bipolar push-pull current buffers is shown.

Figure 34. Example of a gate driving circuit with current buffers for current capability increasing



In the example above, the STMicroelectronics devices STS01DTP06 are dual NPN-PNP complementary bipolar transistors rated for 1 A of current and available in the small SO-8 package. The two push-pull structures are placed on each gate driving path and must be provided with a decoupling capacitor very close to device pins (see *Figure 34*).

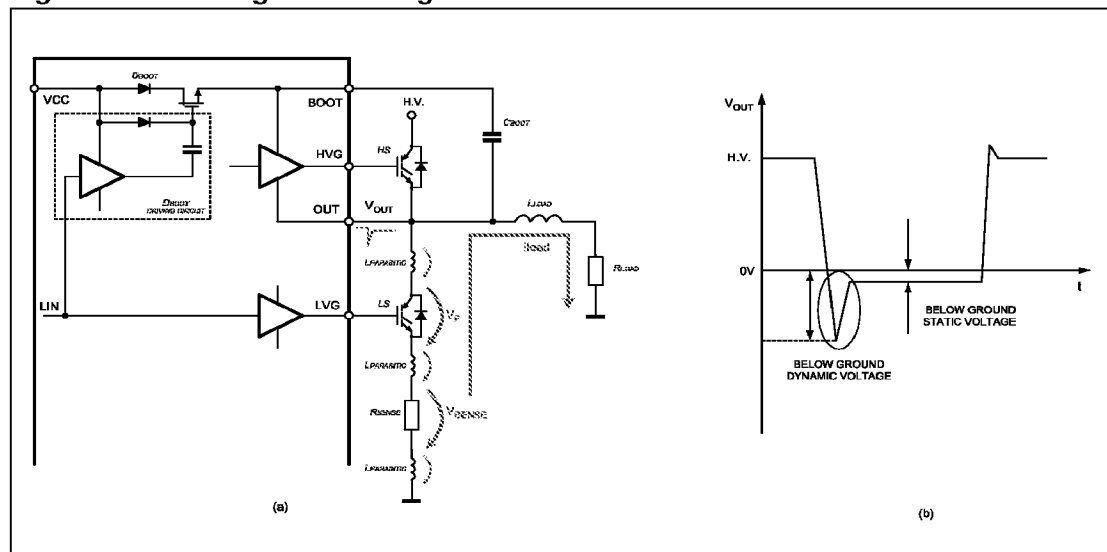
11 The below-ground voltage on the OUT pin

A typical phenomenon found in a great number of power applications is the below-ground voltage experienced by the OUT pin, which sometimes can be very high. Contrary to common opinion, the real problem of the below-ground voltage is not in the maximum absolute value of the OUT pin, but rather the voltage on BOOT pin and the over-charging of the bootstrap capacitance. In the paragraph that follows the root causes of the below-ground voltages will be explained in detail, and the description of the real issues which could result from the phenomenon will be provided.

11.1 The below-ground voltage phenomenon

In power applications using half-bridge topologies and typically driving loads with a significant inductive component, the output of the power half-bridge experiences, systematically, a below-ground voltage transition, which can be distinct in a dynamic contribution consisting of a greater undershoot spike and in a static contribution, which is a below-ground static voltage with lower absolute value (*Figure 35b*). This phenomenon occurs when the bridge carries out a so-called hard switching transition towards a low voltage level and the load current is outgoing (from the bridge to the load). When the high side switch turns off, the output current tends to remain quite constant due to the inductive component of the load, and then has to flow through the low side freewheeling diode, which turns on going from a high voltage reverse condition to a forward condition. It is evident that until the output bridge voltage has reached the "zero" value, the diode is turned off, so the output transition is dominated by the high side turn-off commutation. After the output voltage reaches the zero-voltage level, the diode can turn on and it begins to bring the entire load current in a very brief time, so the high di/dt causes the well-known forward peak voltage, which is the main contribution to the undershoot spikes. Other contributions to dynamic below-ground voltage are the spikes due to the high di/dt experienced by the parasitic inductances in series with the free-wheeling diode located along the turn-off current path of the half-bridge (*Figure 35*).

Figure 35. below-ground voltages in L6390



The overall below-ground voltage on the OUT pin can be calculated as follows:

Equation 15

$$V_{OUTmin_static} = -(R_{SENSE} \cdot I_{LOAD} + V_F)$$

Equation 16

$$V_{BGV_spike} = \underbrace{V_{FPK} + L_{PARASITIC} \cdot \frac{dI_F}{dt}}_{\text{dynamic contribution}} + \underbrace{V_{OUTmin_static}}_{\text{static contribution}}$$

where:

- V_{FPK} is the free-wheeling diode transient peak forward voltage, which depends mainly on the device technology and on the dI_F/dt of the current in the diode. Typical values could be from a few volts to more than 10 V. In *Figure 36*, V_{FP} vs. dI_F/dt of the STTH1L06 diode is shown.
- the dI_F/dt is the current slope in the low side IGBT/MOSFET and could have a value from a few tens to several hundreds of A/ μ s. The value depends mainly on the power switch characteristics and in part on the driving current.
- $L_{PARASITIC}$ represents the sum of all parasitic inductances on the current path and mainly depends on the PCB layout. Generally, during the design of the power application it is important to pay attention to the layout of the power bridges in order to limit this parameter. Typical values of a good layout are in the order of some tens of nH. Note that it is useful also to use R_{SENSE} resistors with low parasitic inductances for the same reason.
- $R_{SENSE} \cdot I_{LOAD}$ product is the value of the V_{SENSE} voltage and typically is less than 1 V, also for thermal dissipation issues on the same resistor.
- V_F is the forward voltage of the free-wheeling diode and it is usually less than 2 V.

11.2 How to reduce the below ground spike voltage

In order to reduce the below ground spike, the following action should be taken:

- Reduce the parasitic inductances (see *Figure 35*).
- Reduce the dI_F/dt by slowing down the turn-off of the high side IGBT/MOSFET.

In most of application the two previous strategies result to be enough to reduce properly the below ground spike voltage, increasing the robustness of power system and then the margin for safe operation of the application. On the other hand in some cases, where the below ground spike voltage is significantly higher, the above suggestions may be not sufficient to limit that value and then it could be useful to add some external components to improve further the noise robustness of the power stage section:

- Add a small resistor (2÷10 Ω typically) in series to the OUT line (see *Figure 37*). The series resistor has the positive effect of limiting the spike voltage on the BOOT pin, thanks to the filtering effect of such resistor coupled with the bootstrap capacitor. Note that, in order to obtain an effective filtering effect, the OUT resistor must be placed between the minus terminal of bootstrap capacitor and the output of the power stage, as indicated in *Figure 37*.

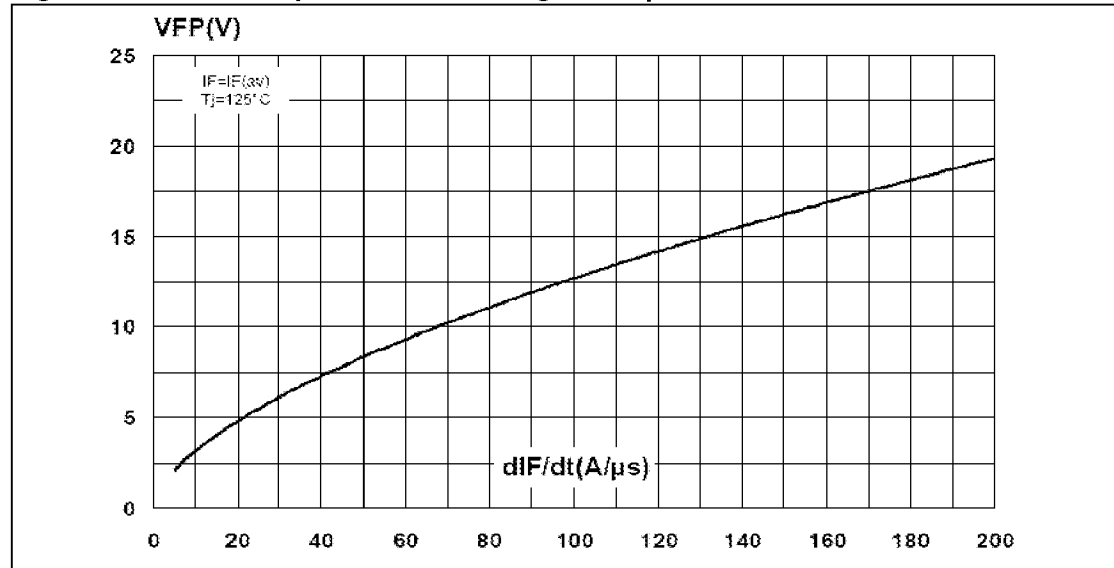
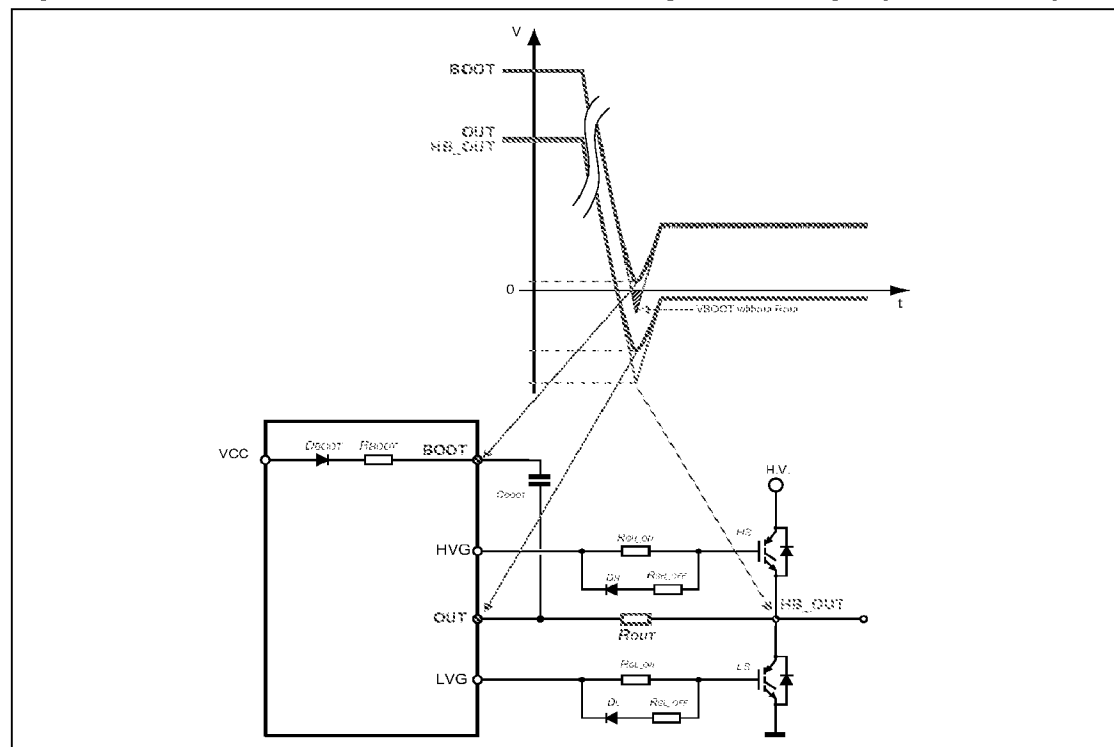
Figure 36. Transient peak forward voltage vs. di_F/dt of STTH1L06 diode

Figure 37. Use of OUT resistor to limit the below ground voltage spike on OUT pin

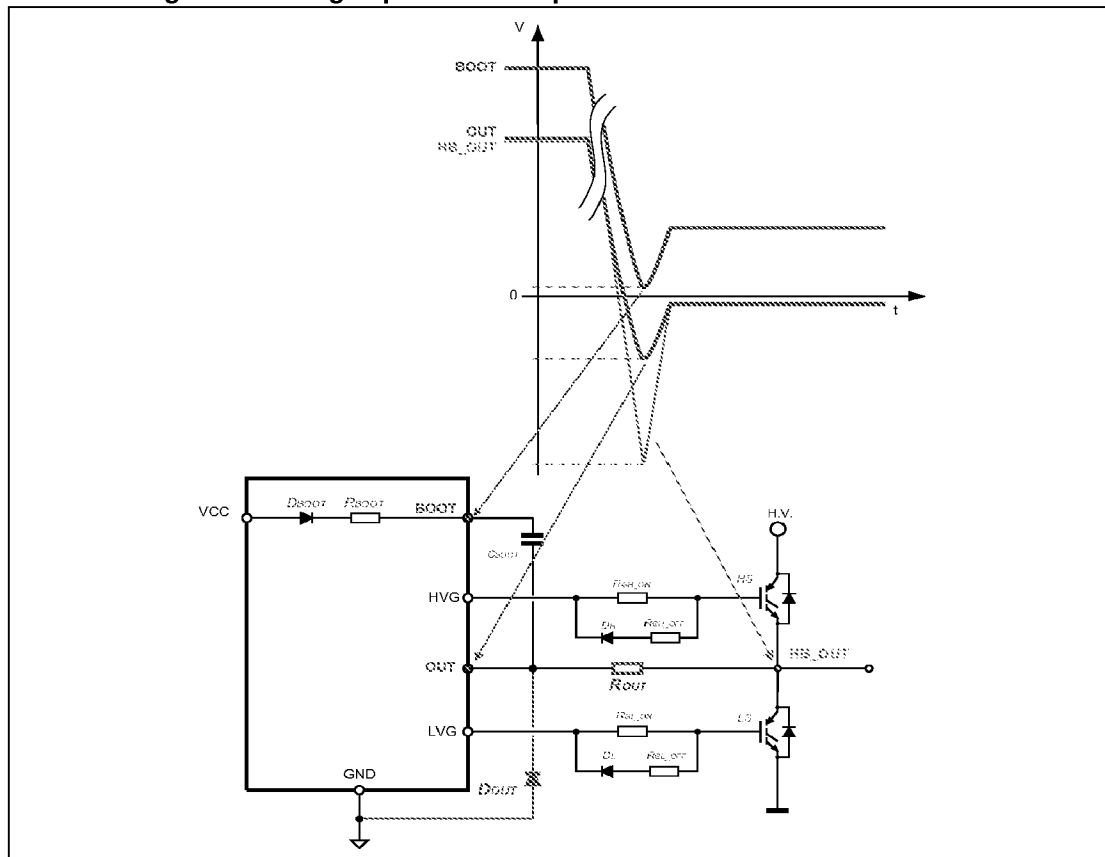


Note that this resistor is in series with both the turn ON and the turn OFF path, so it must be considered in the sizing of overall turn-ON and turn-OFF resistance.

In case neither the OUT resistor would not be enough to limit the below ground voltage, the following final resolving action could be taken:

- Add a high voltage fast diode (e.g. STTH1L06) between the GND pin and the OUT pin (very close to the device pins) in order to clamp directly on the gate driver OUT pin the below ground voltage spike. Note that, in any case, this diode must be used together with the OUT resistance suggested in the previous tip, because it must be avoided that the diode brings the very large load current during low side recirculation, in order to increase the clamping action of the diode itself (see following *Figure 38*).

Figure 38. Use of combination of OUT resistor and OUT diode to limit the below ground voltage spike on OUT pin



Note that R_{OUT} resistor is also in series with the charging path of the bootstrap capacitor. Its effects may be more evident during the first charge of the bootstrap capacitor, when it is completely discharged and a significant charging current may flow into the R_{OUT} resistor, producing an additional voltage drop between ground and OUT pin. Because during the bootstrap charging the HVG pin is set to low level, the OUT pin and the HVG pin are shorted together and have the same voltage, so the voltage drop due to R_{OUT} results directly transferred to the V_{GE} (or V_{GS}) of the high side IGBT (or MOSFET). Then the risk is that a weak turn ON of the high side power switch, when the low side power switch is already ON, could cause a cross-conduction in the power half bridge. Actually in most of cases this doesn't represent an issue for the proper working of the application, because the typical intrinsic resistance R_{BOOT} ($\sim 120 \Omega$) in series to the internal bootstrap diode is much higher than the R_{OUT} commonly used and the voltage drop on R_{OUT} is negligible.

11.3 Issues related to the below-ground voltage phenomenon

Contrary to common belief, the issues resulting from significant below-ground voltages on the OUT pin are not related to the maximum absolute voltage values that the OUT pin is able to withstand, but are instead related mainly to the voltage value of BOOT pin, which is indirectly bound to the OUT pin voltage. There are two main issues to be taken into consideration, all related to the absolute maximum ratings of the IC:

- VBOOT absolute minimum voltage
- VBOOT-VOUT absolute maximum voltage

11.3.1 V_{BOOT} voltage safe operating condition

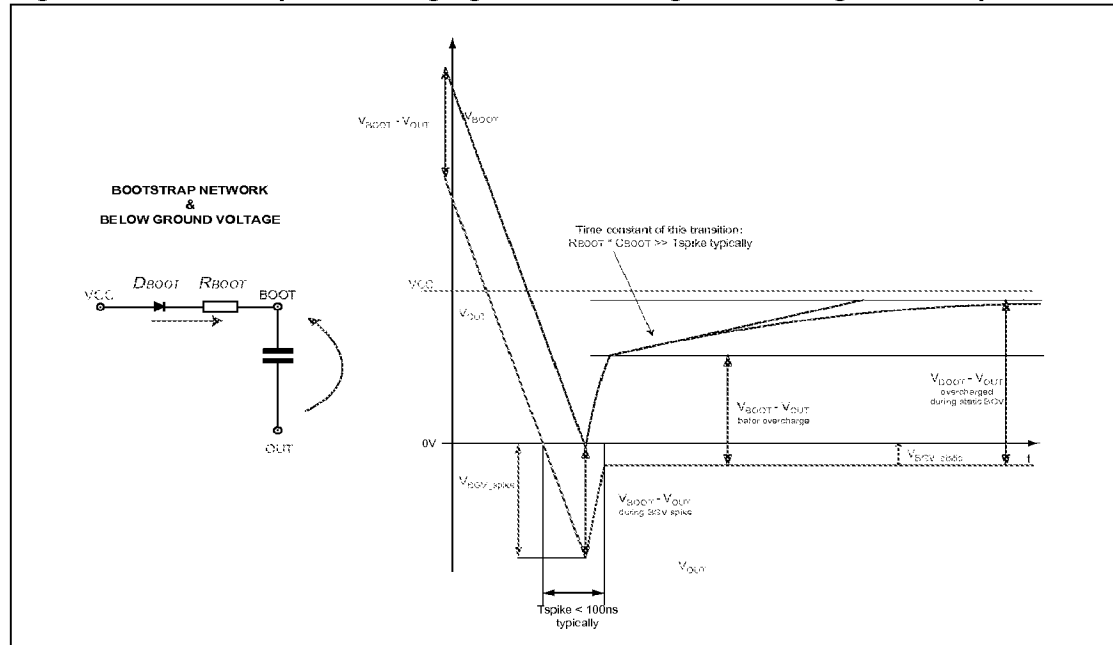
Electrically, the OUT pin could safely tolerate even many volts below ground without problems, but the BOOT pin cannot. The absolute value of the VBOOT voltage must not go steadily below -0.3 V, in order to avoid the turn-on of the built-in junction between BOOT pin and the IC gate driver substrate (connected to GND), which is normally in reverse condition. The turn-on of this junction could, in fact, cause a current large enough to damage the device.

11.3.2 Bootstrap capacitor over-charging

Another very important consideration is bootstrap capacitor over-charging. Actually, even before the OUT pin approaches the zero voltage value, the bootstrap diode (internal or external) tends to tie V_{BOOT} close to the VCC supply voltage. Since the OUT pin is below ground, the bootstrap capacitor is overcharged through the current coming from the bootstrap diode, and the V_{BOOT} - V_{OUT} voltage increases. It is very important that the bootstrap over-charge does not exceed the recommended maximum value for the V_{BOOT} - V_{OUT} voltage (see the relevant datasheet), as the high side floating section of the gate driver could be damaged.

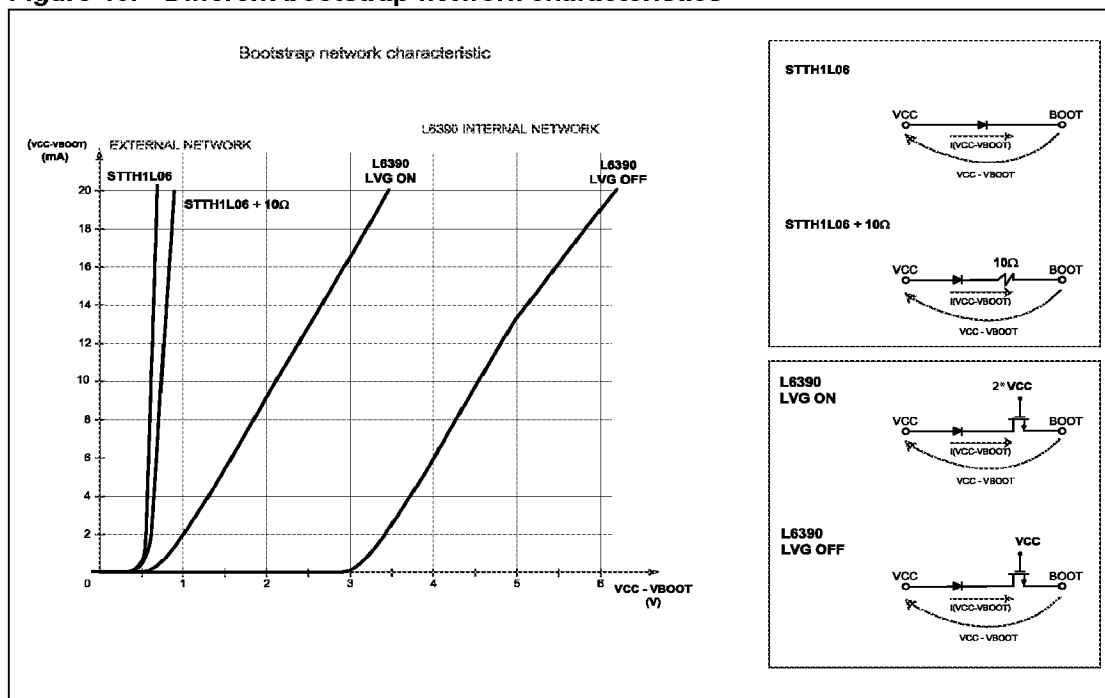
Note that a significant over-charging of the bootstrap capacitor is only possible through the static contribution of the below-ground voltage, because the dynamic below-ground voltage is very brief and the over-charging transition is limited by the time constant RC of the bootstrap capacitor and by the overall resistance in series with the bootstrap diode.

In *Figure 39*, the over-charging phenomenon is illustrated in detail.

Figure 39. Bootstrap over-charging due to below-ground voltage on OUT pin

Note that in the L6390 IC gate driver the internal DMOS in series with the integrated bootstrap diode is fully turned on only when LVG is ON. In fact, when the LVG is OFF, the gate of the bootstrap DMOS is biased at V_{CC} voltage. This means that if the V_{BOOT} is pulled down by the bootstrap capacitor (due to below-ground voltage on OUT pin) at about 3 V (typ) below V_{CC} voltage, the DMOS turns on again. Figure 40 shows some characteristics of different (internal and external) bootstrap networks.

Figure 40. Different bootstrap network characteristics

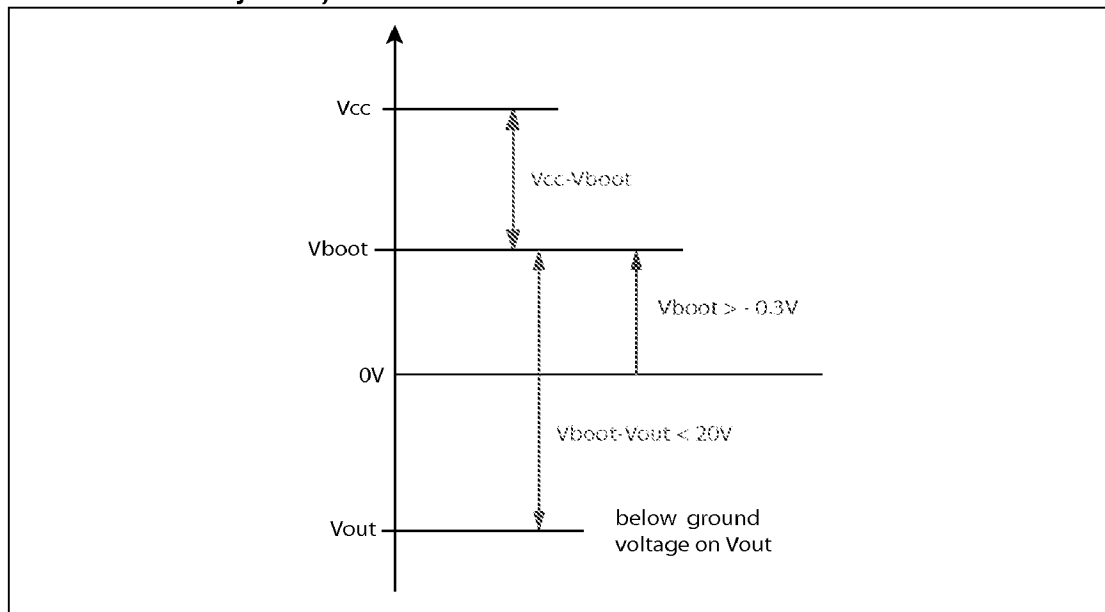


Regarding the below-ground voltage spike, typically this undershoot voltage has a very brief duration (less than 100 ns) and is not enough to further over-charge the bootstrap capacitor, because the charging time constant of the bootstrap is equal to the product of the C_{BOOT} and the R_{BOOT} resistance in series with the diode. Moreover, the higher this resistance, the lower the risk of bootstrap over-charging during BVG spikes.

For example, with a R_{BOOT} of 120 Ω and a C_{BOOT} of 100 nF, the associated time constant would be about 12 μ s, much higher than typical below-ground voltage spike duration. Using an internal bootstrap diode it is very difficult to over-charge the floating section up to dangerous voltage levels with the very short duration of the undershoot spike. More attention should be given to the below-ground voltage of the V_{BOOT} during this spike because, as explained previously, the internal junction V_{BOOT} -to-substrate could turn on; typically for very short time (a few tenths of nanoseconds), this is not a problem.

In Figure 41, the main conditions to avoid issues related to the below-ground voltage phenomenon in steady-state are resumed. As explained in paragraph 11.1, typically the static below-ground voltage is hardly higher than 2 V, so these constraints are usually not a limitation for most applications.

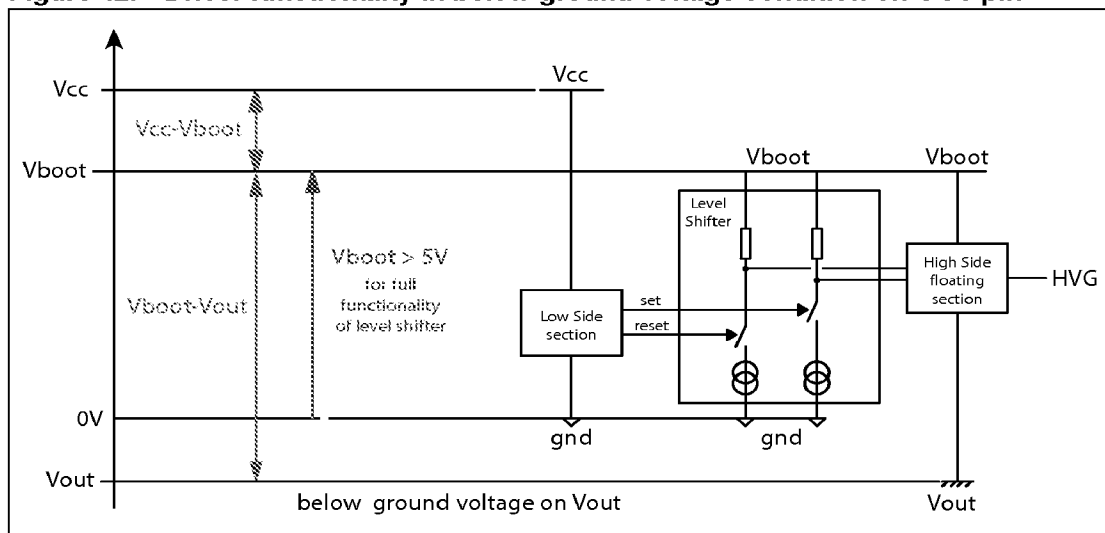
Figure 41. L6390 safe operating range when the OUT pin is below ground voltage (in steady state)



11.4 Functionality of L6390 outputs in below-ground condition

The L6390 IC gate driver makes use of a level shifter to send the set-reset information to the high side floating section. The level shifter, shown in *Figure 42*, is mainly composed of two high voltage switches, for set/reset signals, driven by the low voltage section and linked to two pull-up resistors connected to the BOOT pin.

Figure 42. Driver functionality in below-ground voltage condition on OUT pin



The two level-shifted signals are then fed into a logic latch in the high side floating section, providing the logic state for the high side gate driver (HVG output). The L6390 IC provides

full functionality of the switching operation of the high side section until the level shifter is operating. Because the two pull-up resistors are connected to the floating supply V_{BOOT} of the high side floating section, the driver still functions properly if the OUT voltage begins to go below ground. This means that the operating limit for the level shifter structure must be referred to the absolute value of the BOOT voltage relative to ground. In fact, the BOOT voltage can be considered the supply of the level shifter block. The minimum operating value of the level shifter supply is 5 V, so the V_{BOOT} value must be at least 5 V. The $V_{BOOT} - V_{OUT}$ voltage can be any value in the 12.4 V to 20 V range. If the BOOT voltage is between 0 V and 5 V, the functionality of the level shifter is not guaranteed, but internal structures should not be damaged. Some examples are given in the following section.

11.4.1 Steady state (DC) conditions

When the LVG is off, the bootstrap diode turns on only when the BOOT voltage is about 2 V below the VCC (worst case: minimum voltage drop value), so it could over-charge the bootstrap capacitor up to a maximum voltage of about $VCC - 2 V - V_{OUT}$. To avoid exceeding the value of 20 V on $V_{BOOT} - V_{OUT}$ voltage, the OUT pin could be forced permanently to the value of $VCC - 2 V - 20 V$ with the L6390 IC still operating safely, making the HVG output switching as the HIN logic input (see the table below).

Table 3. Minimum V_{OUT} in DC condition providing safe and full operation of the high side section

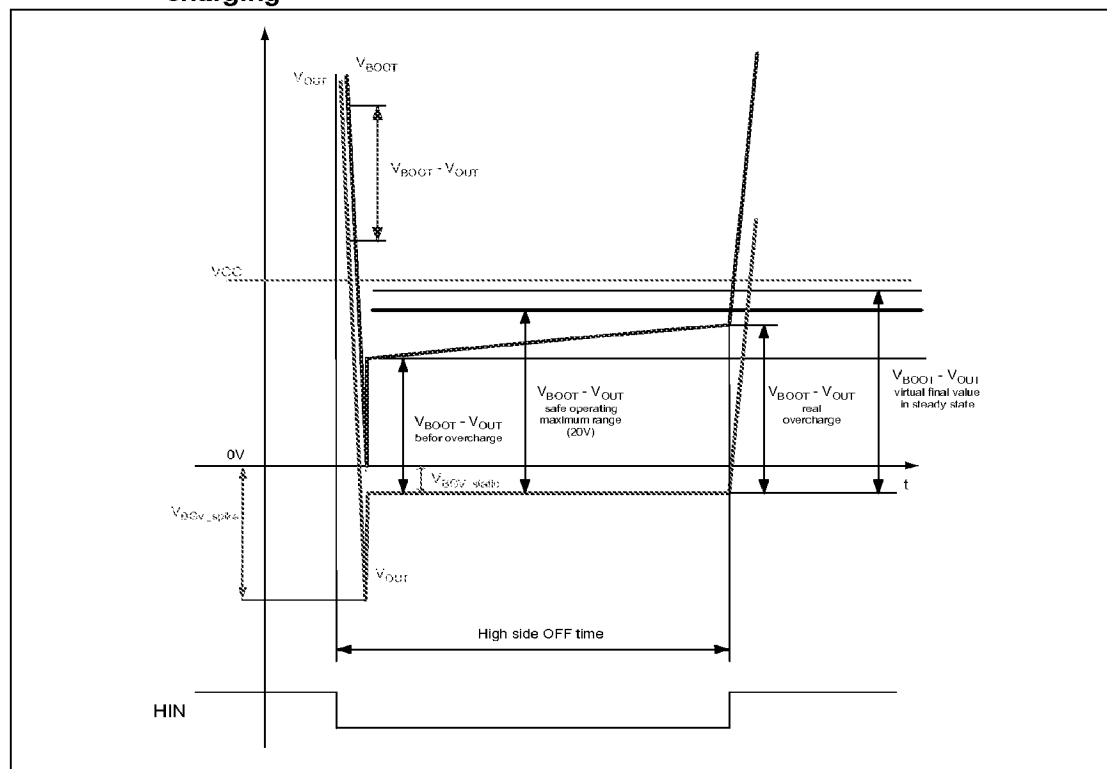
	Example 1	Example 2	Example 3
VCC	12.5	15	17
V_{BOOT}	10.5	13	15
$V_{OUT} \text{ (min)}$	-9.5	-7	-5
$V_{BOOT} - V_{OUT} \text{ (max)}$	20	20	20

It must be emphasized, as explained in the previous paragraph, how in most applications the static below-ground voltage of the OUT pin is very rarely lower than about - 2 V.

11.4.2 Transient conditions

If the time that the OUT pin goes below ground is limited, the bootstrap over-charge probably will not exceed the safe operating range (20 V) of $V_{BOOT} - V_{OUT}$, even if the V_{OUT} voltage is lower than the limit of $VCC - 2 V - 20 V$. In this case the gate driver and the high side section are fully operational if the V_{BOOT} voltage remains above 5 V, as explained previously, for proper functioning of the high side level shifter. The logic state is anyway held if the V_{BOOT} remains above ground.

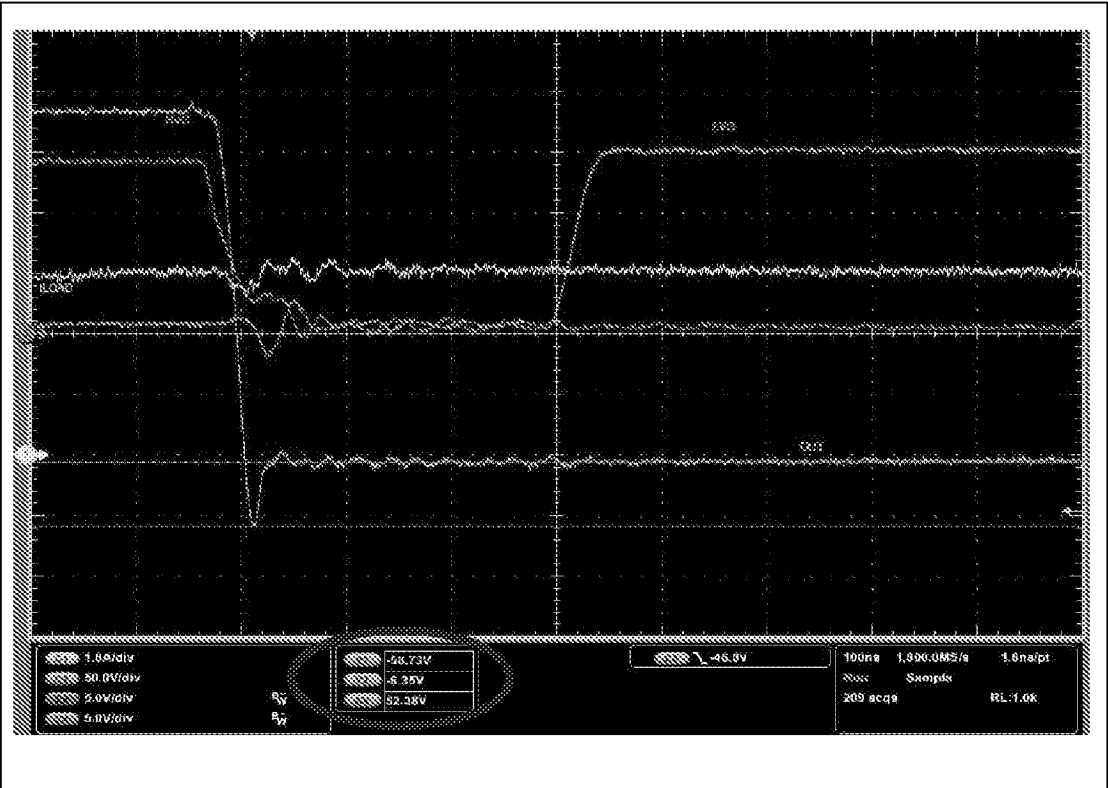
Figure 43. OUT below-ground voltage in transient conditions: limited boot over-charging



11.4.3 Below-ground voltage spikes

As previously stated, the duration of the dynamic contribution of the below-ground voltage on the OUT pin is not enough to over-charge the bootstrap capacitor. This fact usually removes the risk of exceeding the recommended maximum value of 20 V for $V_{BOOT} - V_{OUT}$ voltage, but it increases the danger of pulling down the BOOT pin below ground, thus violating its absolute minimum rating. As mentioned above, this phenomenon should be avoided for safe IC gate driver operation. However, note that applicative bench tests have shown the L6390 proper operation even with below-ground spikes on the OUT pin well above - 50 V (see *Figure 44*).

Figure 44. Example of below-ground voltage spike



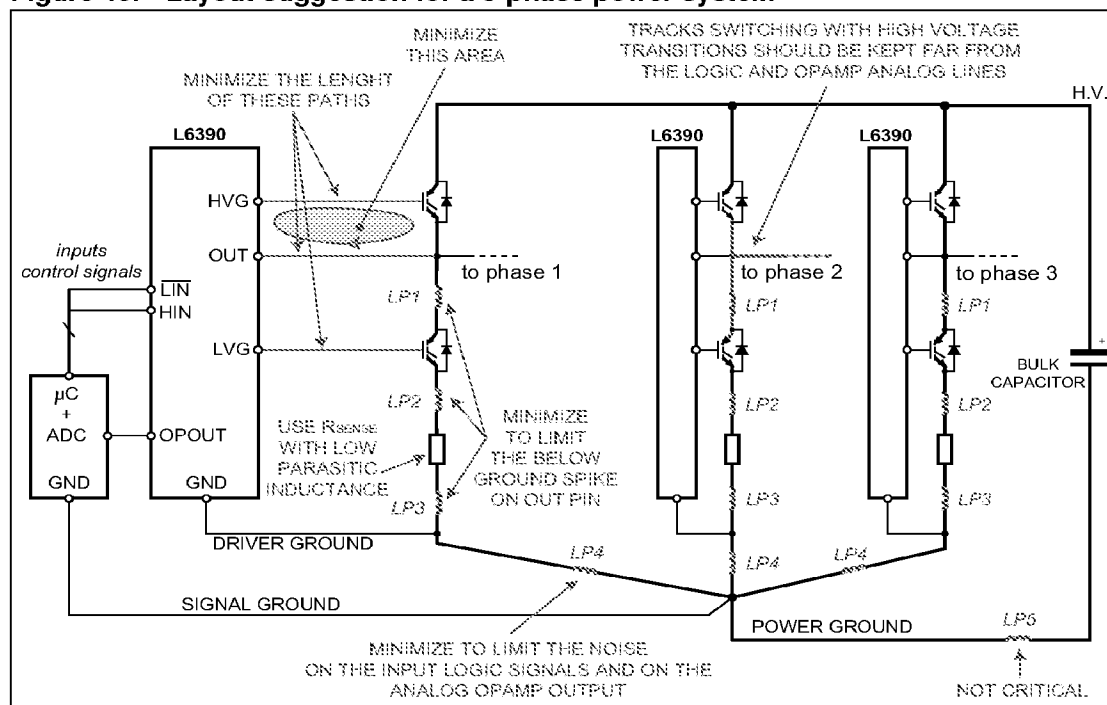
12 Layout suggestions

Typically, for power applications using high voltages and large load currents, the board layout of all circuits related to the power stage is important. Board layout includes several aspects, such as track dimensions (length and width), circuit areas, but also the proper routing of the traces and the optimized reciprocal arrangement of the various system elements and power sources in the PCB area.

Reasons to give particular attention to the PCB layout include EMI issues (both induced and perceived by the application) and over-voltage spikes due to parasitic inductances along the PCB traces, the proper connection of the sense blocks, the logic inputs and the analog outputs of the L6390 device. In fact, the L6390 IC not only has the function of driving the power stage, but also embeds analog sensing blocks such as comparator and op-amps. For example, especially regarding multi-phase power stages, it is important to keep the current reading, performed through the integrated op-amp, from ground noise.

In Figure 45 some layout guidelines and suggestions for a 3-phase application are provided.

Figure 45. Layout suggestion for a 3-phase power system



As explained in Section 8.9, the gate driving PCB traces should be designed to be as short as possible and the area of the circuits should be minimized to avoid the sensitivity of such structures to the surrounding noise. Typically, a good power system layout keeps the power IGBTs (or MOSFETs) of each half-bridge as close as possible to the related gate driver.

In Figure 45 a set of parasitic inductances related to the different circuit tracks is shown. The various groups of inductances may have undesired effects which should be limited as much as possible. Moreover, note that Figure 45 emphasizes parasitic inductances located on the lines usually managing high voltages and fast current transitions, which are very noisy.

The group of LP1, LP2 and LP3 parasitic inductances is located along the low side path of each half-bridge, between the OUT pin and the ground of the related driver and provides an undesired contribution to the issue of below-ground voltage spike on each OUT pin of the L6390 device (as described in *Section 11.1*). In fact, at the beginning of the current recirculation on the low side switches, the current may experience a high di/dt which may produce, on those parasitic inductances, significant voltage spikes. These spikes are summed with the voltage drop of the low side diode and with the R_{SENSE} and have a negative sign, so the overall voltage drop between OUT and GND may be significant. The recommendation is to limit as much as possible each contribution to this phenomenon by limiting the length of tracks LP1, LP2 and LP3 and using an R_{SENSE} resistor with low intrinsic inductance. More specifically, LP1 may be reduced by connecting the OUT line directly to the collector (or drain) of the low side IGBT (or MOSFET). LP2 may be reduced by placing the R_{SENSE} resistor as close as possible to the emitter (or source) of the low side IGBT (or MOSFET). The LP3 may be minimized by connecting the ground line (also called driver ground) of the related gate driver directly on the R_{SENSE} resistor.

LP4 represents the parasitic inductance located between the ground connections of each gate driver (driver ground) and the ground connection of the application controller (also called signal ground). Due to its location, this parasitic inductance introduces noise which is experienced by the input logic signals and the op-amp output analog signals. In fact, each phase of the bridge causes high currents (with high di/dt) to flow on these paths, resulting in voltage noise which drops between the gate driver ground and the controller ground. This noise between the two grounds is directly added to all logic and analog voltage signals between the gate driver and the microcontroller included the input logic signals and the analog output of the related gate driver op-amp. It is recommended to minimize this noise by reducing the distance between the signal ground and the driver ground (for each gate driver in the system) as much as possible. Generally, it is recommended to connect the signal ground to the three driver grounds through a star connection, in order to improve the balancing and symmetry of the 3-phase driving topology.

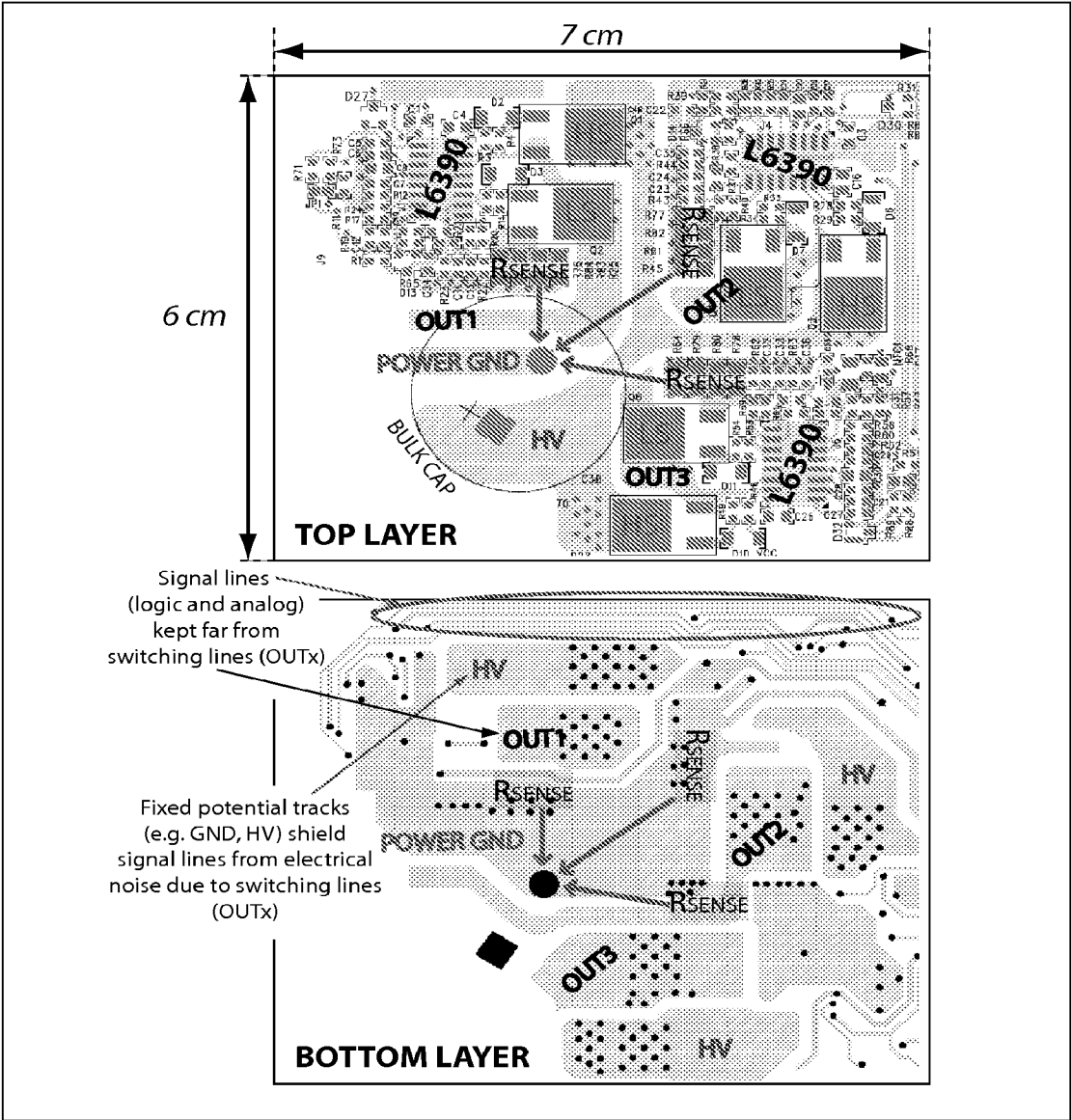
Note: It is very important to avoid any ground loop; only a single path must connect two different ground nodes.

The LP5 parasitic inductance is usually not critical, because it stands between the negative terminal of the bulk capacitor and the signal/power ground. The spikes on this parasitic element minimally influence other nodes of the system.

Another useful suggestion is to ensure some distance between the lines switching with high voltage transitions, and the signal lines sensitive to electrical noise. Specifically, the tracks of each OUT phase bringing significant currents and high voltages should be separated from the logic lines and analog sensing circuits of op-amps and comparators.

In *Figure 46* a practical example of the layout is provided. The example represents the 3-phase power stage section of the STEVAL-IHM021V1 demonstration board. It includes the L6390 gate drivers, the six IGBTs (or MOSFETs) in the DPAK package and the bulk capacitor. The layout suggestions described in this paragraph are implemented in the layout example in *Figure 46*. As described in *Figure 46*, the fixed voltage tracks, such as GND or HV lines, can be used to shield the logic and analog lines from the electrical noise produced by the switching lines (e.g. OUT1, OUT2 and OUT3). Each half-bridge ground is connected in a star configuration and the three R_{SENSE} resistors are very close to each other and to the power ground. Note also that the suggested 3-phase configuration occupies a modest amount of surface area (7 cm x 6 cm), and thanks to the SMD packages and board vias, no heat sinks are required.

Figure 46. Layout example from the STEVAL-IHM021V1 3-phase board



13 Revision history

Table 4. Document revision history

Date	Revision	Changes
02-Oct-2008	1	Initial release
03-Aug-2009	2	Removed chapter 11 The use of the resistor on the OUT line and added <i>Section 11.2: How to reduce the below ground spike voltage on page 40.</i>

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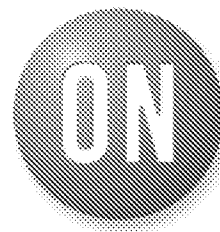
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Understanding the LLC Structure in Resonant Applications

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The resonant LLC topology, member of the Series Resonant Converters (SRC) begins to be widely used in consumer applications such as LCD TVs or plasma display panels. In these applications, a high level of safety and reliability is required to avoid catastrophic failures once products are shipped and operated in the consumer field. To face these new challenges, ON Semiconductor has recently released to new controllers, the NCP1395 (low-voltage) and the NCP1396 (high-voltage) dedicated to driving resonant power supplies, usually of LLC type. However, before rushing to design a converter of this type, it is important to

understand the resonant structure alone, object of the present application note.

The LLC converter

The LLC converter implies the series association of two inductors (LL) and one capacitor (C). Figure 1 shows a simplified representation of the resonant circuit where:

L_s is the series inductor

L_m is the magnetizing inductor

C_s represents the series capacitor

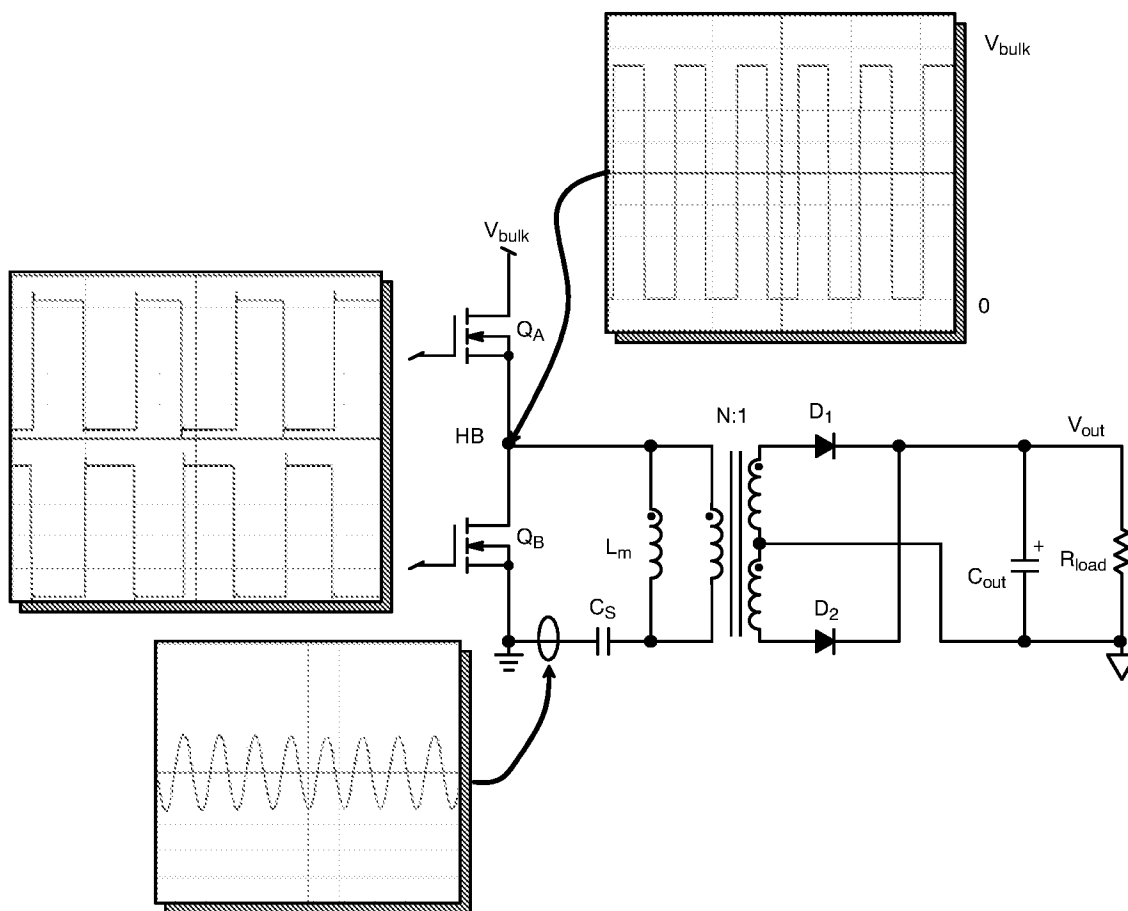


Figure 1. The LLC Topology Uses a Half-Bridge Configuration to Drive the Resonant Circuit

The operating principle is rather simple: a constant 50% duty-cycle switching pattern drives $Q_A - Q_B$ gates and a high-voltage square wave appears on node HB. By adjusting the switching frequency, the controller can control the power flow depending on the output demand. As a transformer is needed for isolation purposes, its magnetizing inductance plays the role of the second inductor L_m . The series inductor, L_s , can either be a separated element or physically lump into the transformer. In this case, a voluntary degradation of both primary and secondary coupling naturally increases the leakage inductance which can act as the series element. There are pros and cons to include the leakage element in the transformer. The cost and the absence of saturation play in favor of the integration but the difficulty to keep a precise value from lots to lots

associated with leaky transformers (radiated noise) has to be kept in mind when selecting the final configuration.

When studying the resonant converter, it is convenient to reduce the architecture to a passive element arrangement such as presented on Figure 2. The high-voltage square signal is replaced by its fundamental content thanks to the first harmonic approximation (the so-called FHA in the literature): because we operate a tuned LC filter, all harmonics can be considered as rejected and only the fundamental passes through. Of course, this statement holds as long the controller drives the resonating work in the vicinity of its resonant frequency. Figure 2 offers such a simplified representation of the resonant cell, actually pointing out a series impedance (L_s and C_s) with a parallel impedance (L_m and the reflected load).

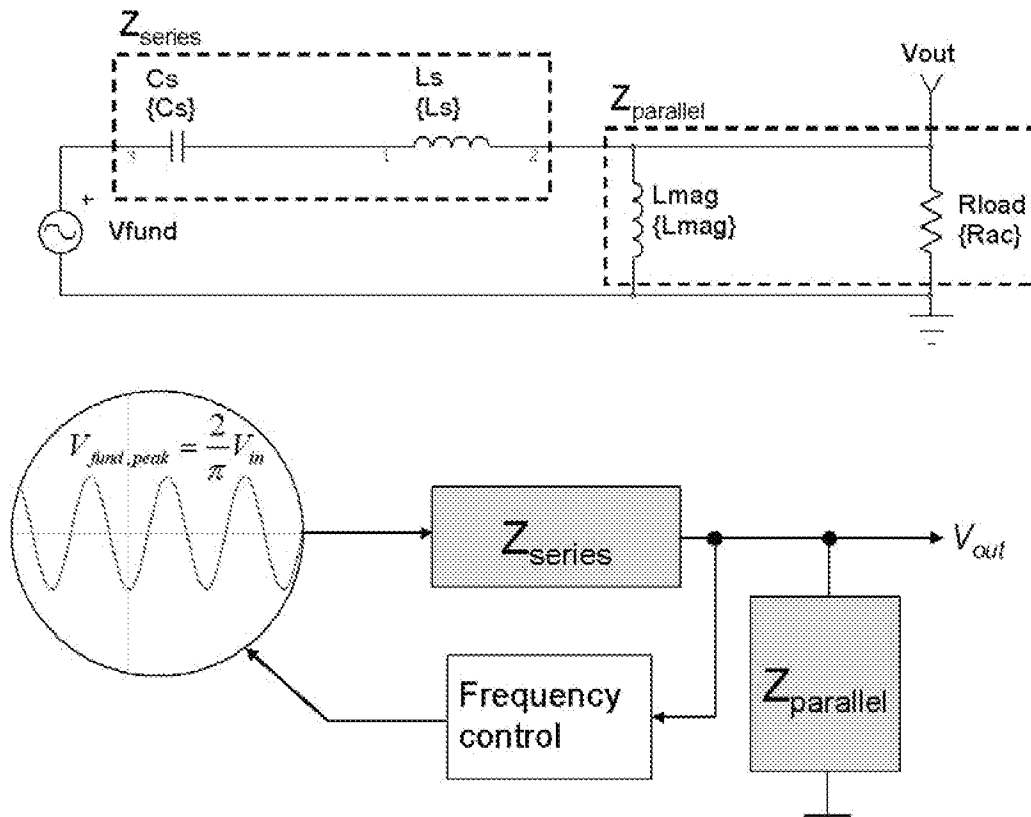


Figure 2. The Impedance Representation Makes the LCC Operation Easier to Understand

Depending on the loading, the network resonant frequency varies between two different values:

- $R_L = 0$, short-circuit, L_m disappears and Z_{series} becomes a short. The series resonant point for Z_{series} is thus

$$F_{max} = F_S = \frac{1}{2\pi\sqrt{L_S C_S}} \quad (\text{eq. 1})$$

At $F_{sw} = F_S$, Z_{series} becomes a short and the ac transfer function drops to 1 or 0 dB.

- $R_L = \infty$, light or no load condition, L_m appears in series with L_s and the whole network resonates to

$$F_{min} = \frac{1}{2\pi\sqrt{(L_S + L_m)C_S}} \quad (\text{eq. 2})$$

- $0 < R_L < \infty$, the resonance which combines L_m and L_s , shifts depending on the total quality coefficient.

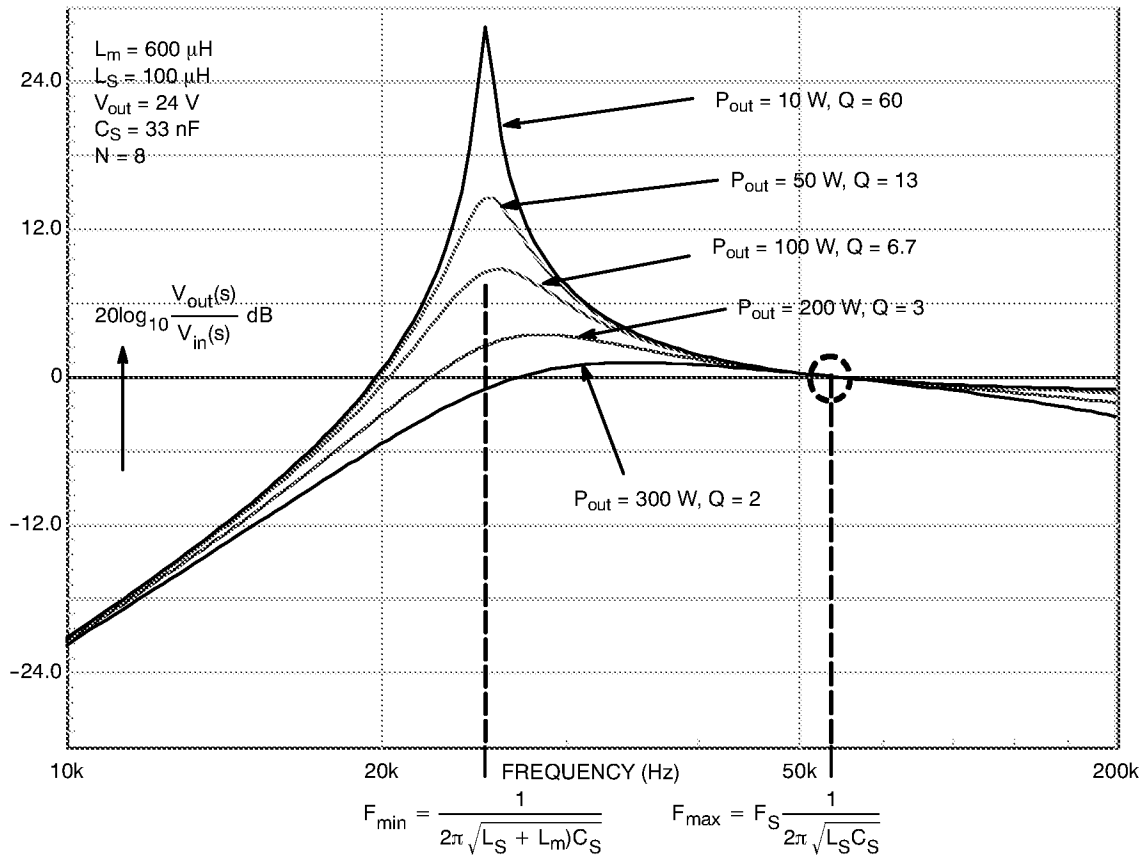


Figure 3. The AC Response of Figure 2 Circuit with Various Load Conditions

This is actually what Figure 3 plots suggest by showing the ac transfer function of Figure 2 as the load changes.

If we now study the impedance seen from the half-bridge node, we have an expression showing a series association of inductors and a capacitor. Sticking to Figure 2 sketch and writing the impedance seen between ground and Node 3, we have:

$$Z_{in} = Z_{L_s} + Z_{C_s} + Z_{L_m} \parallel R_{ac} \quad (\text{eq. 3})$$

$$Z_{in} = \left[\frac{(\omega L_m)^4 R_{ac}^2}{(R_{ac}^2 + \omega^2 L_m^2)^2} + \left(\omega L_s - \frac{1}{\omega C_s} + \omega L_m \frac{R_{ac}^2}{R_{ac}^2 + \omega^2 L_m^2} \right)^2 \right]^2 \quad (\text{eq. 4})$$

In the low frequency portion, the terms associated with inductors are of less importance and C_s dominates. The impedance is thus capacitive. As the frequency increases, the inductive portion starts to kick-in and the impedance goes up. This is what Figure 4 describes. As one can see, all the curves go through point A whose value is independent from the resistive loading. For the sake of a friendly exercise, we can solve Equation 4 with two different R_{ac} values and find the frequency at which input impedances equal. We obtain:

$$\omega_A = \sqrt{\frac{2}{L_m C_s + 2 L_s C_s}} \quad (\text{eq. 5})$$

If we substitute this value into Equation 4, the impedance at point A is:

$$Z_A = \frac{\frac{L_m}{2 L_s} L_s \omega_s}{\sqrt{\frac{L_m}{2 L_s} + 1}} \quad (\text{eq. 6})$$

If we define the ratio R by L_m/L_s , we can re-arrange equation 6:

$$Z_A = \frac{R}{\sqrt{2(R+2)}} \sqrt{\frac{L_s}{C_s}} = \frac{R}{\sqrt{2(R+2)}} Z_O \quad (\text{eq. 7})$$

Where Z_O represents the characteristic impedance of the series resonant network. Using the numerical values noted in the graphs, we obtain a frequency of 43.8 kHz and an impedance of 38.3 dBΩ (82.6 Ω).

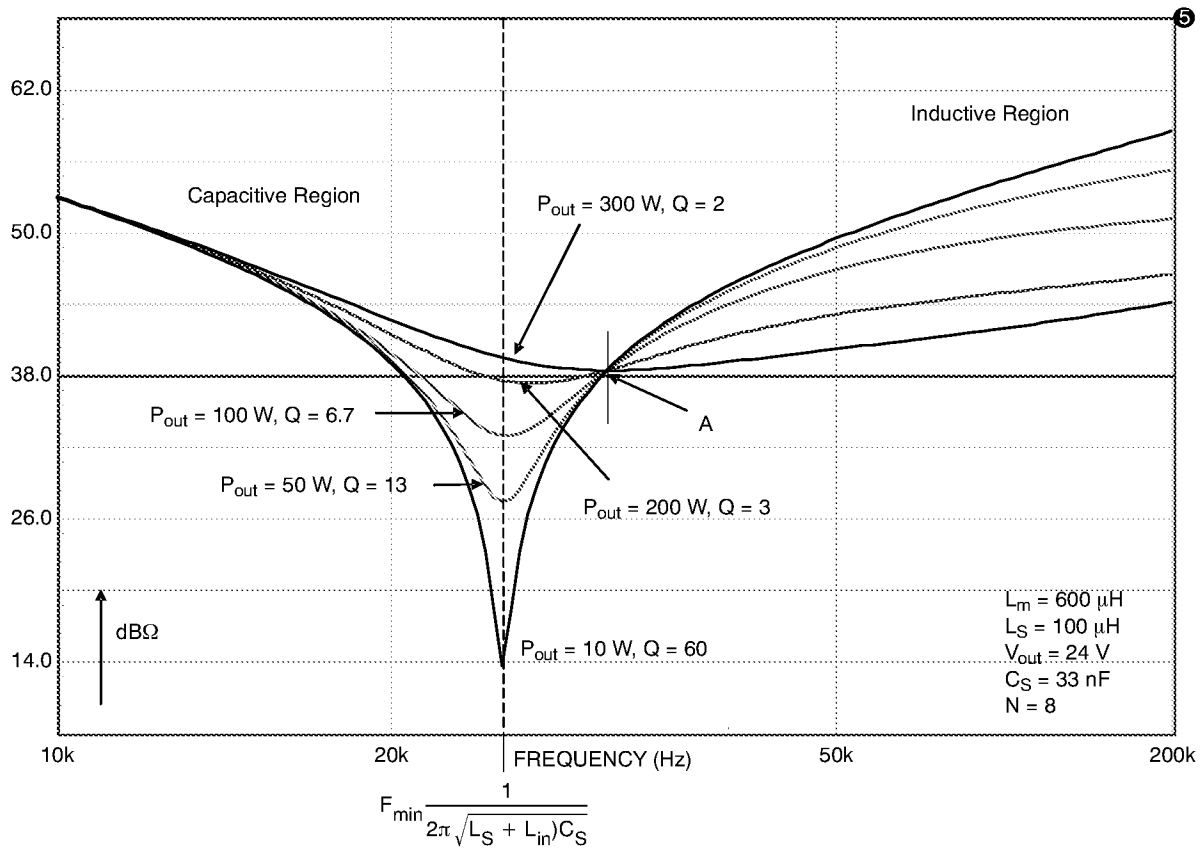


Figure 4. Impedance Plots at Various Power Levels

If we now observe the resonant current waveforms in a LLC converter working below or above the series resonance F_s , we have different types of operation:

- **Capacitive mode:** in this mode, where the current leads the voltage, the bridge MOSFETs operate in zero current switching (ZCS). ZCS means that power MOSFETs are turned-off at zero current. Back to figure 3, we can see that the output level goes up as the frequency increases.
- **Inductive mode:** in this mode, the current lags the voltage and the power switches are turned-on at zero volt (ZVS), virtually eliminating all capacitive losses. This operating way implies that a certain delay exists before operating the concerned MOSFET so that its

body diode turns on first. Observing figure 3, the output level goes down as the frequency increases.

Most of the LLC converters operate in the inductive region for the second bullet reason. Also, given the feedback polarity, if by mistake the closed-loop LLC enters the left side of the resonance, the control law reverses and a power runaway obviously occurs. It is thus extremely important to clamp down the lower frequency excursion in fault condition or during the startup sequence to avoid falling on the other slope of the characteristics.

The inductive region can be split into two other regions, depending where you operate compared to the resonant series frequency F_s , as defined by Equation 1. Figure 5 represents the classical set of curves often found in the dedicated literature:

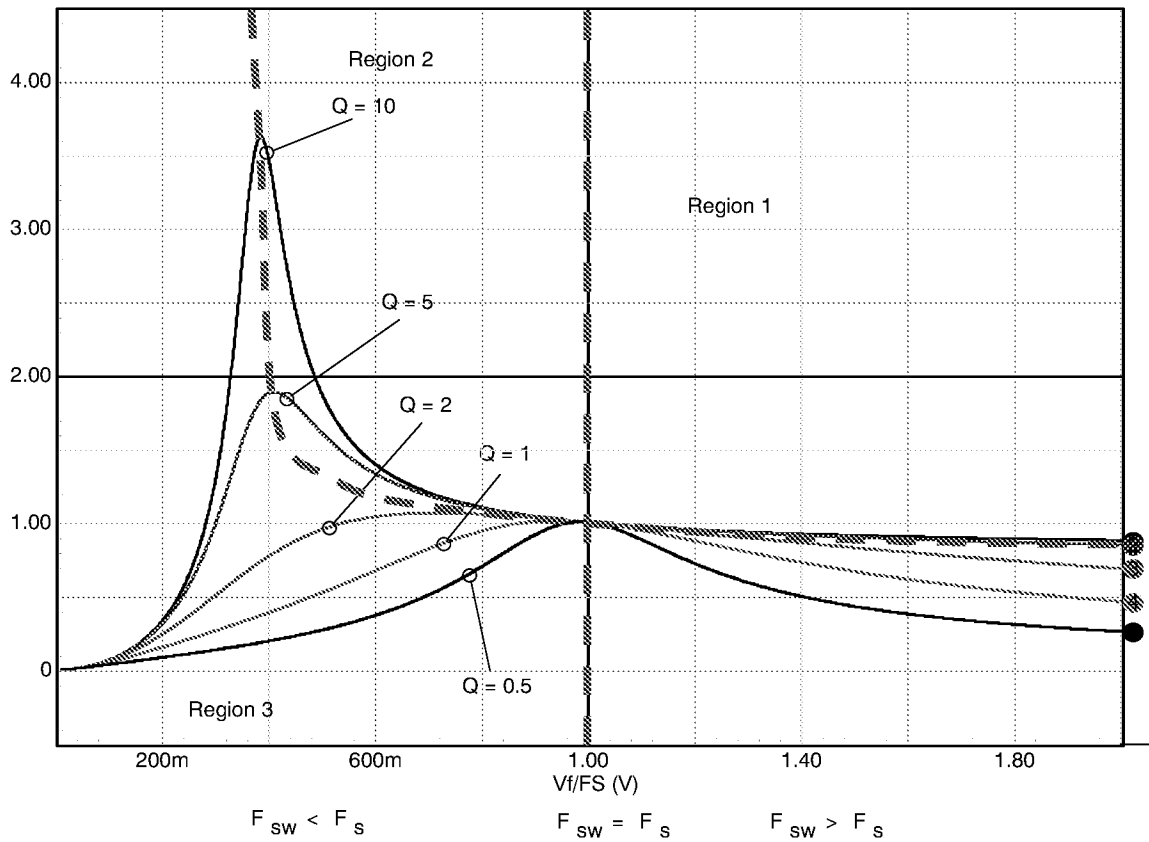


Figure 5. Typical Transmittance Curves with Various Loading Conditions, Highlighting Three Distinct Regions

Region 3 is the capacitive mode where you do not want to operate since ZVS is a wanted feature for the power switches. In regions 1 and 2, you still have ZVS on the power MOSFET's and the output diodes are operated in Zero Current Switching (ZCS), cancelling all associated losses at turn-off. Before discussing the benefits of a particular solution, let us have a look at the various operating phases the LLC converter is made of.

Operating Waveforms Below the Series Resonance, $F_{sw} < F_s$

For this example, we have selected a set of elements which operate the converter below the series resonance defined by Equation 1. The following value have been used:

$L_m = 700 \mu\text{H}$
 $L_s = 116 \mu\text{H}$
 $C_s = 28 \text{ nF}$
 $N = 8$

$$F_{\max} = F_s = \frac{1}{2\pi\sqrt{L_s C_s}} = \frac{1}{6.28 \times \sqrt{116\mu \times 28\text{n}}} = 88.3 \text{ kHz}$$

$$F_{\min} = \frac{1}{2\pi\sqrt{(L_s + L_m)C_s}} = \frac{1}{6.28 \times \sqrt{(116\mu + 700\mu) \times 28\text{n}}} = 33 \text{ kHz}$$

$F_{sw} = 70 \text{ kHz}$ at full load and nominal input voltage.

The converter delivers 24 V@10 A from a 380 Vdc input source and a simulation has been performed using the above values. Figure 6 shows the main waveforms obtained from the simulator. Let us study the switching events step by step to learn about the LLC behavior in this region.

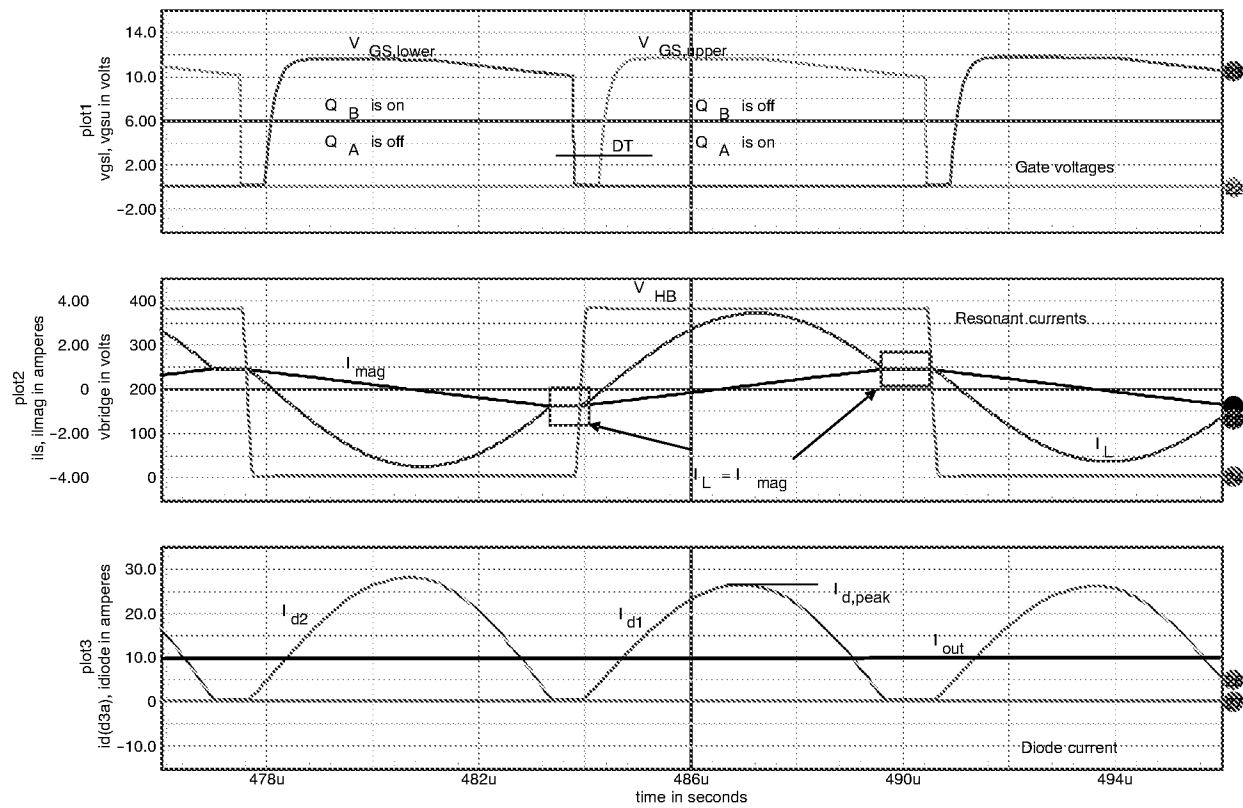


Figure 6. Waveforms Obtained for a Converter Operated Below the Series Resonant Frequency

Q_A is off, Q_B is on, D_2 is conducting :

The low-side MOSFET Q_B imposes a 0 V potential on the half-bridge node and the current circulates from its drain to source (first quadrant). The upper parasitic capacitor C_{ossA} is fully charged to the input voltage V_{bulk} since the HB node is grounded by Q_B . The secondary diode D_2 is conducting and imposes a voltage reflection $-NV_{out}$ over the magnetizing inductor L_m . Its current linearly decreases with a slope of $-NV_{out}/L_{in}$. As this inductor is dynamically shorted by the voltage reflection, it does not participate to the on-going resonance between L_s and C_s which deliver the output energy (the input source is out of the picture). The current flowing into the transformer primary side (given its theoretical representation, L_m associated to a perfect transformer) is the main current I_L minus the magnetizing current I_{mag} . D_1 is blocked and undergoes twice the output voltage given the transformer coupling. The circuit

resonates to F_s as L_m is shorted. Figure 7 depicts the situation during this period of time.

Q_A is off, Q_B is on, D_2 turns off:

As the network current I_L resonates in a sinusoidal manner, its amplitude peaks and then starts to dip towards 0. When it reaches a level equal to that of the magnetizing current, no current circulates in the transformer anymore: D_2 blocks and the voltage reflection over L_m disappears. The magnetizing inductor now comes back in series with L_s and C_s and changes the resonant frequency from F_s to F_{min} : the LLC converter is really a multi-resonant structure and the plateau - actually a small arch of a lower sinewave oscillation - in the current as it appears on figure 6 testifies for it. Both diodes are now blocked and this moment lasts until Q_B opens. Figure 8 represents the circuit during this time. As one can see, the output capacitor alone supplies the energy to the load.

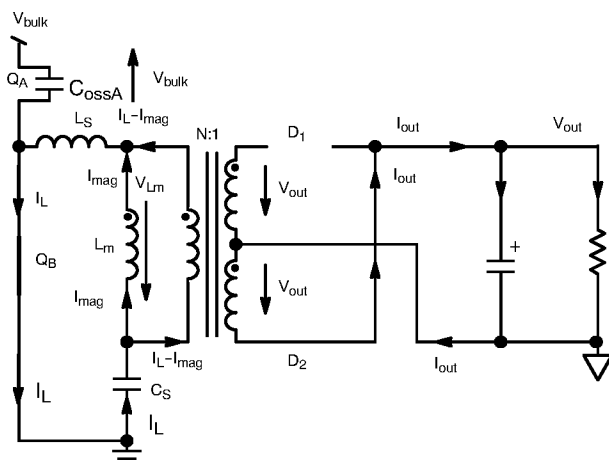


Figure 7. Q_A is Off, Q_B is On and Diode D_2 Conducts Current. L_m is Off the Picture as it is Dynamically Shorted by the Output Voltage Reflection.

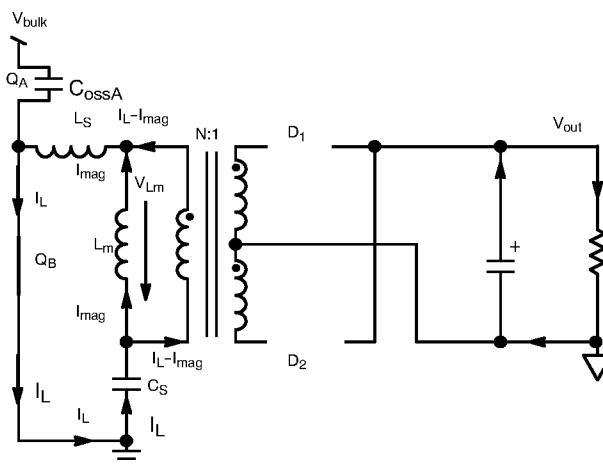


Figure 8. Q_A is Off, Q_B is On and Diode D_2 Blocked. L_m Comes Back Again in the Resonating Network and Changes the Resonant Frequency to F_{min} .

Q_A is off, Q_B is Off, Both Secondary Diodes are Blocked

Both transistors are now open, this is the dead-time period (DT on Figure 6). The dead-time is placed here to avoid cross-conduction between both MOSFETs but also to favor Zero Voltage Switching as we will see in a moment. Because the current was circulating from drain to source in Q_B , the circuit no longer sees an ohmic path when this transistor opens. The current strives to find a way through the parasitic drain-source capacitors C_{oss} of both Q_A and Q_B : C_{ossB} starts to charge (it was previously discharged by Q_B being on) and given the rise of V_{HB} towards the high voltage rail, C_{ossA} sees its terminals voltage going down to zero and then

reversing (Figure 9). At this moment, when the HB node reaches $V_{bulk} + V_f$, the body-diode of Q_A conducts and ensures energy re-cycling through the input source (Figure 10). You understand that this dead-time period must last a time long enough to allow for the complete discharge of C_{ossA} before re-activating Q_A so that its body-diode turns on first. If not, hard switching occurs and efficiency suffers.

As currents are oscillating, a time is reached where I_L and I_{mag} are no longer equal (end of the plateau) and a current circulates again in the primary side. D_1 starts to conduct and NV_{out} appears across L_m : the resonant frequency goes back from F_S to F_{min} . Figure 10 describes this moment.

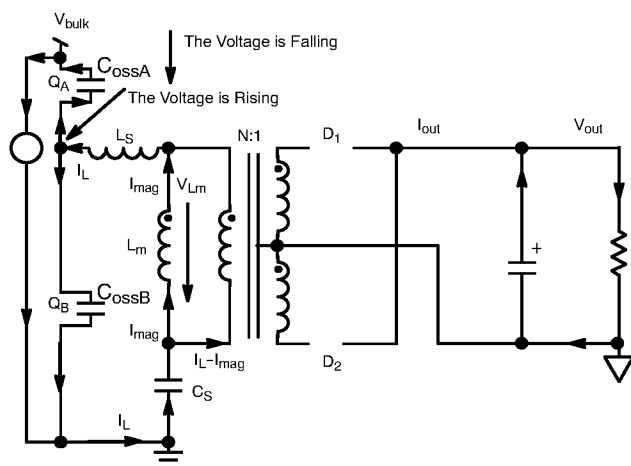


Figure 9. Q_A is Off, Q_B is Off. The Current Finds a Circulating Path Through Both Transistors C_{oss} , Both Secondary-Side Diodes are Off.

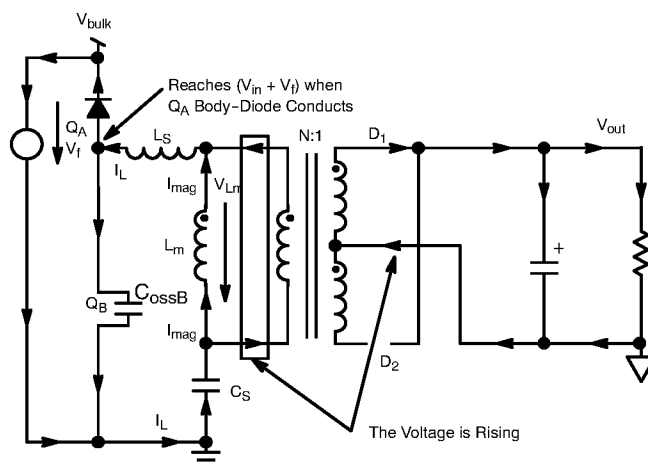


Figure 10. Q_A and Q_B are Still Off. The Current Finds a Circulating Path through the Upper-side Body Diode. D_1 Starts Conducting at the End of the Plateau when $I_L \neq I_{mag}$.

Q_A is on, Q_B is off, D_1 is on

Now that Q_A body-diode is conducting, we have a negligible voltage across its drain and source terminals: we

can therefore safely turn it on and benefit from Zero Voltage Conditions. As we have a sinusoidal waveform in the network, the resonating current reaches zero and reverses.

L_m is still dynamically shorted as D_1 is conducting. The energy is delivered by the source to the output load. This is illustrated by Figure 11.

Q_A is on, Q_B is off, D_1 turns off

The current I_L is moving down and reaches the magnetizing current level, we are the second plateau on

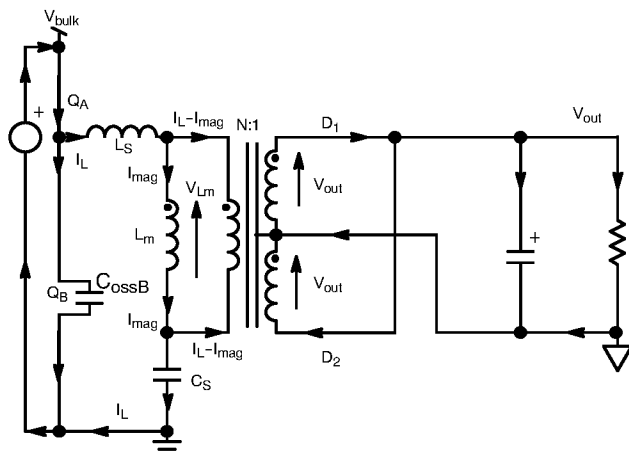


Figure 11. The Current is Now Flowing from the Source to the Output Via the Upper-Side Transistor Q_A .

Q_A is off, Q_B is off, both secondary diodes are blocked

At a certain time, both transistors block and only their drain-source capacitors remain in the circuit. The current keeps circulating in the same direction but C_{ossA} starts to charge: the voltage on the HB node drops and C_{ossB} depletes

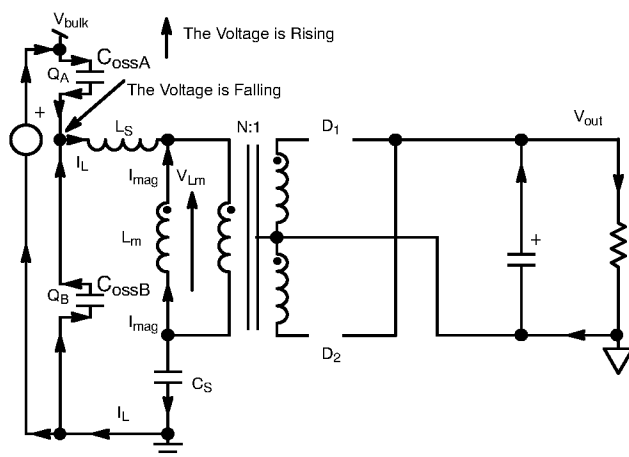


Figure 13. The Current is Still Flowing through the Source and Contributes to Discharge C_{ossB} .

The bridge voltage further dips and becomes negative until the body-diode of Q_B conducts. This is what Figure 14 suggests. At the end of the plateau, where $I_L = I_{mag}$, D_2 will start conducting, reflecting $-NV_{out}$ over the primary inductance. The energy comes from C_S and L_S , as the source

Figure 6. At this point, no current circulates in the transformer and D_1 naturally blocks. As explained before, the magnetizing inductor re-appears in the circuit since the output voltage reflection is gone. The resonant frequency changes from F_{min} to F_s and the energy to the load is delivered by the output capacitor alone. Figure 12 shows the circuit state during this event.

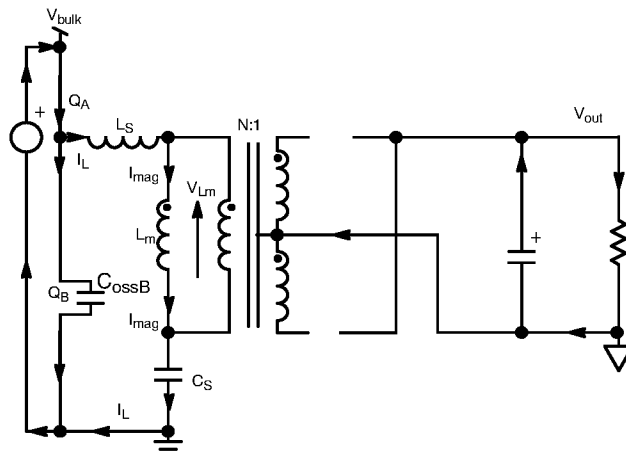


Figure 12. As Both Diodes are Off, the Network Includes the Magnetizing Inductance which Changes the Resonant Frequency.

towards ground. The drain falls down in a resonating manner, involving both C_{oss} in parallel and the equivalent inductor made of $L_S + L_m$. Figure 13 represents the circuit during this event.

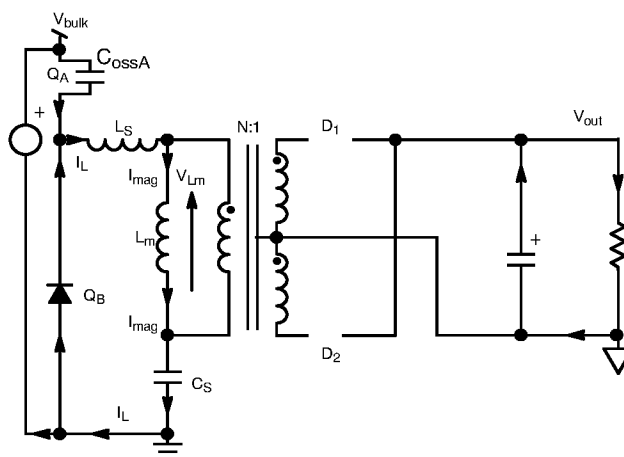


Figure 14. When the Voltage on the Node HB Swings Below Ground, Q_B Body-Diode Conducts.

is not playing any role here. The controller now activates Q_B in ZVS and the transistor conducts in its 3rd quadrant for a few moments, until the current reaches zero and swings negative: we are back at the beginning of the first phase.

Zero Voltage Switching

Figure 15 zooms on these ZVS events and show the various signals in play. The MOSFET current starts to be negative before the appearance of its gate-source bias: this is the body-diode conduction period. Then the MOSFET

turns-on at a V_F across its drain-source terminals but the current is still negative: we are in the 3rd quadrant conduction. Finally, the current becomes positive and flows from drain to source, back to the 1st quadrant.

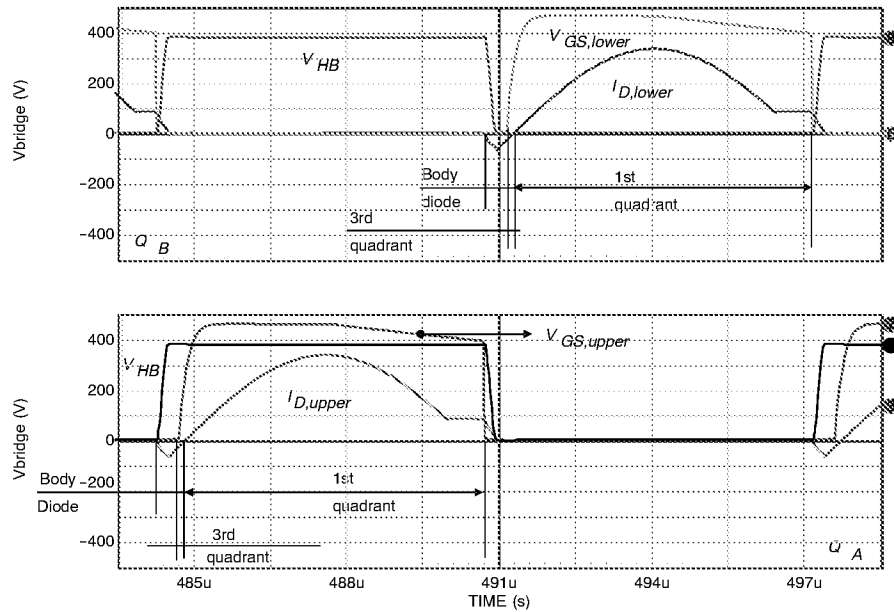


Figure 15. Simulation Results Zooming on the MOSFET Variables

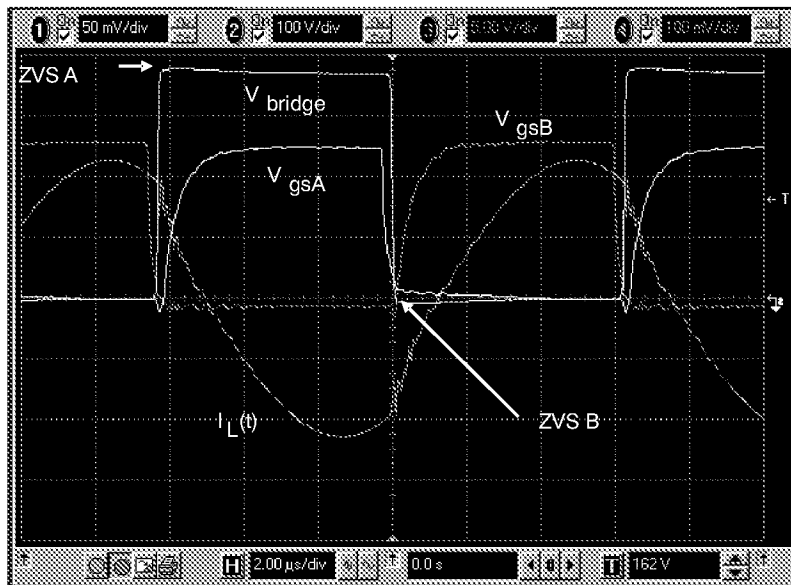


Figure 16. Measured Signals on a Demonstration Board Showing the ZVS Operation on QA.

The selection of a controller where the dead-time is adjustable therefore represents an important selection argument to fine tune the behavior and ensure a minimum conduction period of both body-diodes.

Zero Current Switching

By the term ZCS, we assume a natural blocking event when the current in the semiconductor is zero. When operating the LLC converter below F_s , as it is the case in this

example, both secondary-side diodes are operated in ZCS. The current in the concerned diode (D_1 or D_2) naturally reaches 0 when the magnetizing current I_{mag} equals the main

resonating current I_L . This is the plateau on figure 6. Observing the diode current in this particular mode gives smooth signals as shown on Figure 17.

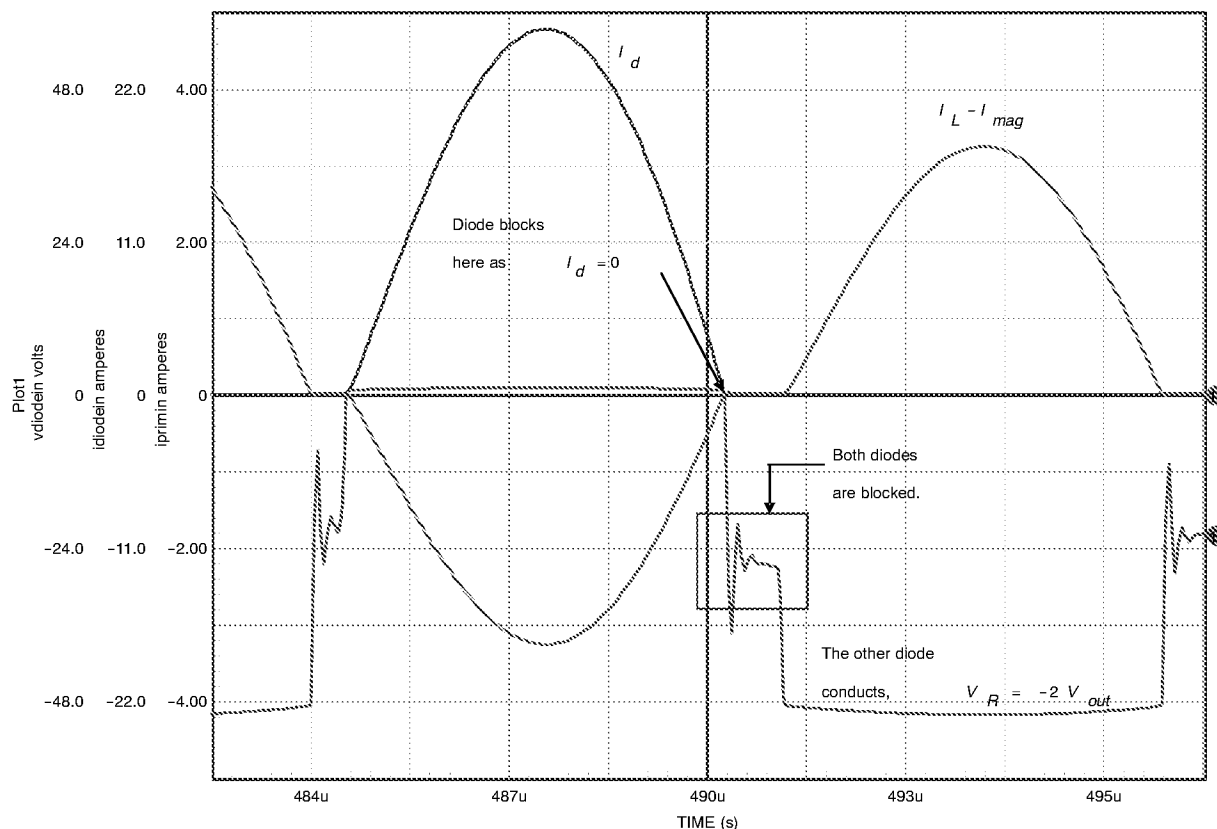


Figure 17. The Secondary-Side Diodes are Naturally Blocked When the Primary Current Vanishes to Zero

Startup sequence and short-circuit

During startup or short-circuit, the magnetizing inductor is shorted and the resonant frequency becomes F_s . Because we designed the LLC converter to operate at a frequency lower than F_s , the operating fault mode (lack of feedback) of the controller naturally lies below F_s . In other words, if the LLC converter quickly starts-up, without soft-start at all, the controller will quickly sweep from a high frequency value down to the minimum authorized in case of fault. The current in the network can therefore peak to a high value (at resonance, the LC impedance is only limited by ohmic losses) and destroy the power MOSFETs instantaneously. Figure 18a shows an oscilloscope shot captured on a LLC circuit started with a short soft-start period (≈ 20 ms): the current peaks to 6 A. Increasing the soft-start period to a few hundred of milliseconds clearly helps to smooth the peak and keep it below 4 A.

Short-circuit protection is more difficult to achieve given the resonating nature of the circuit. Some solutions exist like

differentiating the voltage across the capacitor C_s and routing the resulting voltage to a fast latch input. Figure 19 shows this solution where the component values must be adjusted to avoid false triggering in normal operating transients.

Reference [1] has experimented a solution where the resonating capacitor is split in two values - $C_s/2$ - and two high voltage diodes clamp the voltage excursion between ground and the bulk rail. As the voltage across the capacitor is limited, the resonant current is also clamped. The solution appears in Figure 20. There are several drawbacks associated to the usage of this diode arrangement such as a variable clamping level in relationship to the high-voltage rail. However, experience shows that this simple circuit brings an efficient protection to the converter experiencing a short-circuit. The diodes must be of fast types, MUR260 can be selected for this purpose.

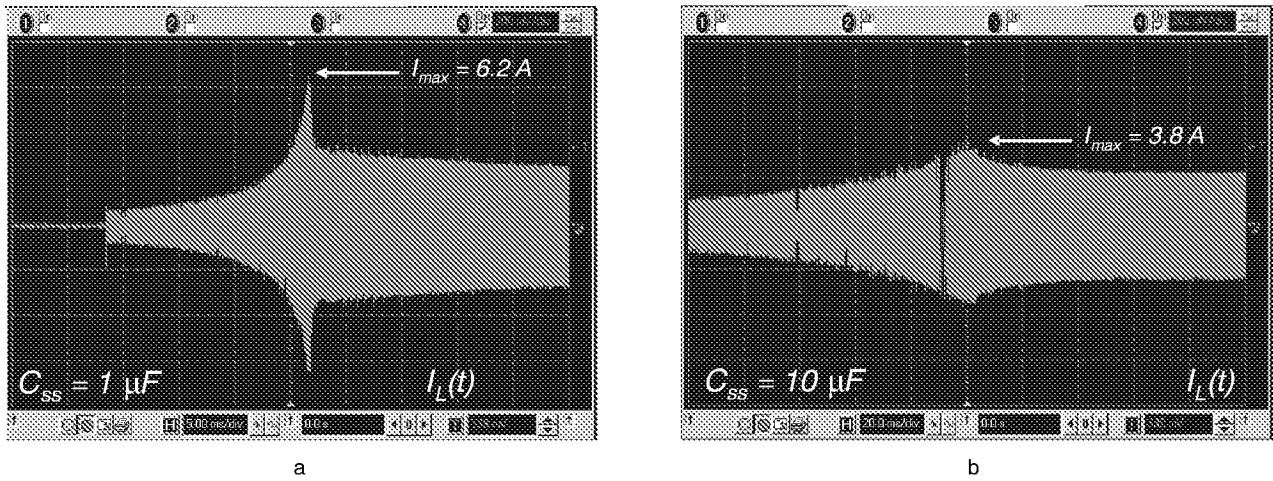


Figure 18. The LLC converter peaks to a high current if started too quickly. Increasing the soft-start sequence naturally calms down the current excursion.

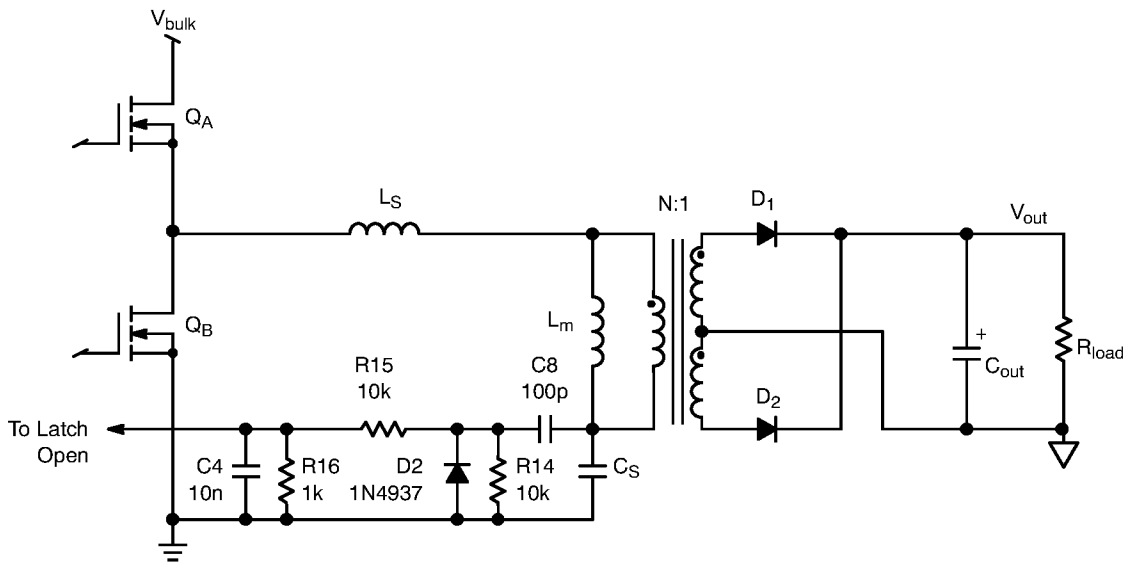


Figure 19. Differentiating the Voltage Across the Resonant Capacitor Gives an Indication of the Current Flowing Through it

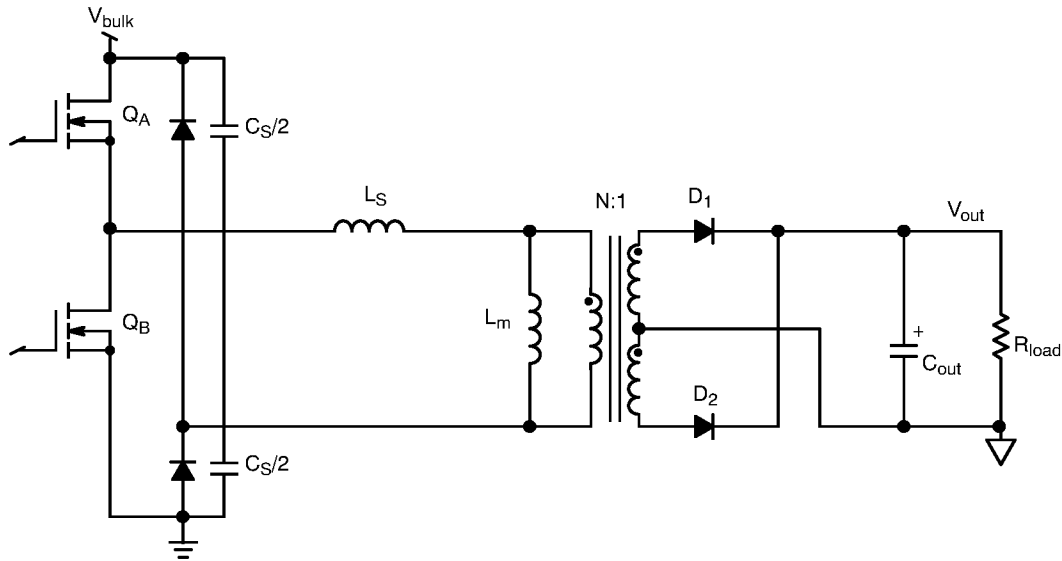


Figure 20. To Keep the Voltage Excursion on the Resonant Capacitor within Safe Limits, a Diode Network Forbids any Lethal Runaways

**Operating Waveforms Above the Series Resonance,
 $F_{sw} > F_s$**

For this example, we have selected a set of elements which operate the converter above the series resonance defined by equation 1. The following values have been used:

$L_m = 1.2 \text{ mH}$
 $L_s = 200 \text{ } \mu\text{H}$
 $C_s = 44 \text{ nF}$
 $N = 6$

$$F_{max} = F_s = \frac{1}{2\pi\sqrt{L_s C_s}} = \frac{1}{6.28 \times \sqrt{200\mu \times 44n}}$$

$$= 53.7 \text{ kHz}$$

$$F_{min} = \frac{1}{2\pi\sqrt{(L_s + L_m) C_s}}$$

$$= \frac{1}{6.28 \times \sqrt{(200\mu + 1.2m) \times 44n}} = 20 \text{ kHz}$$

$F_{sw} = 70 \text{ kHz}$ at full load and nominal input voltage.

The converter still delivers 24 V@10 A from a 380 Vdc input source and a simulation has been conducted using the above values. Figure 21 shows the main waveforms obtained from the simulator. There are several differences between this operating mode and the previous one:

1. In the previous mode, the magnetizing inductance was released at a point where both secondary-side diodes were blocked ($I_L = I_{mag}$). The resonant frequency was therefore moved from F_s to F_{min} during a certain time (the plateau on Figure 6). When operated above the series frequency F_s , the magnetizing inductance is always shorted by the reflected voltage NV_{out} or $-NV_{out}$ as one of the

secondary diode is always conducting. In other words, a single resonance occurs in this mode at full power, implying L_s and C_s only. L_m is out of the picture as long as the converter operates in continuous conduction mode (full load operation).

2. Observing Figure 21, we can see that the main resonant current I_L changes from a sinusoidal waveshape to a straight line, implying a change in the operating mode. This change occurs when a voltage discontinuity appears across L_s terminals. This discontinuity comes from the delay between the bridge signal V_{HB} and the reflected voltage polarity across the magnetizing inductor L_m . Figure 22 zooms on this particular moment where we can see that the bridge voltage goes down to zero via the body-diode activation of Q_B , but because there is still current flowing in the transformer primary side (I_L is different than I_{mag}), one of the secondary diode is still conducting, imposing a constant reflected output voltage across L_m . The voltage across L_s is up by one step which starts to reset it towards zero. This is the beginning of the linear segment, if we consider the voltage across L_s almost constant. When I_L reaches the magnetizing current I_{mag} , the conducting diode blocks and the primary current transitions to the second diode which now conducts. The voltage polarity across L_m reverses and the resonant current goes back to its sinusoidal shape. The next segment occurs when Q_B opens and the bridge voltage jumps to V_{in} via Q_A body-diode. This segment lasts until I_L reaches I_{mag} again.

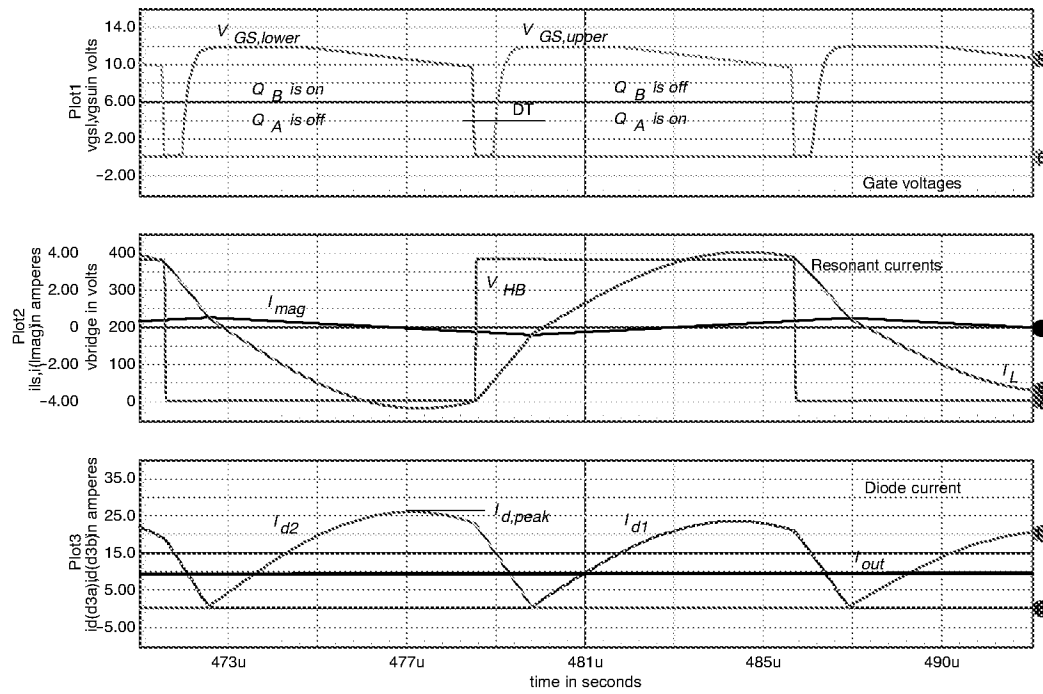


Figure 21. Figure 6 Waveforms Updated with a Converter now Operating Above the Series-Resonant Frequency F_s

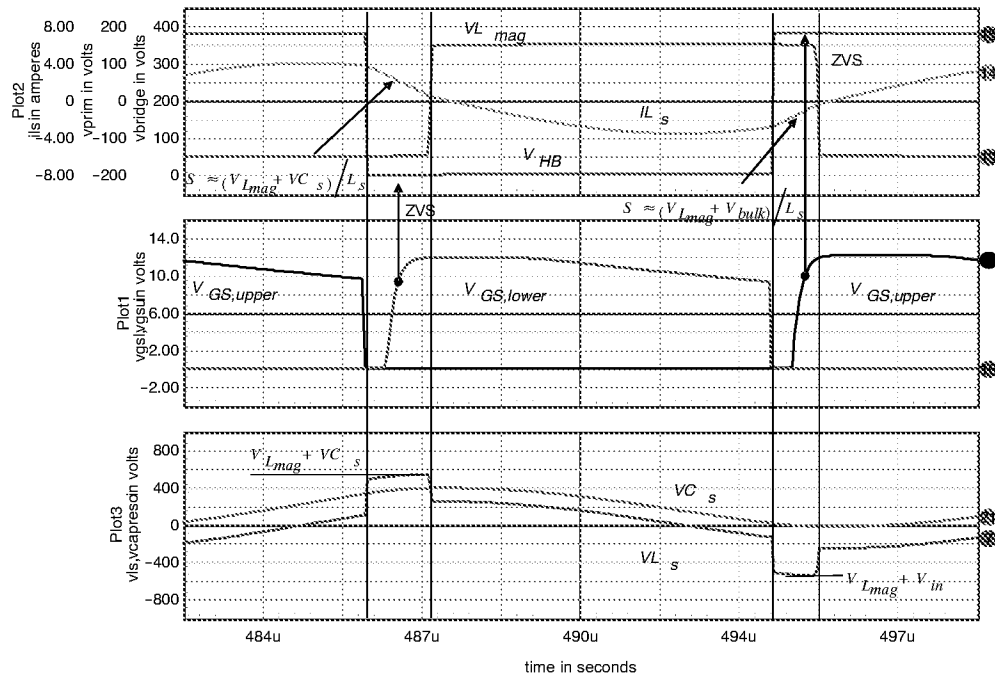


Figure 22. The Voltage Discontinuity Across L_s Induces a Linear Segment in the Resonant Waveform

1. The diode are still operated in ZCS despite a switching frequency above F_s . This is thanks to the linear reset taking place on the resonant current

(the segment on $I_L(t)$) which smoothly leads the concerned diode to a blocking state. Figure 23 illustrates this fact.

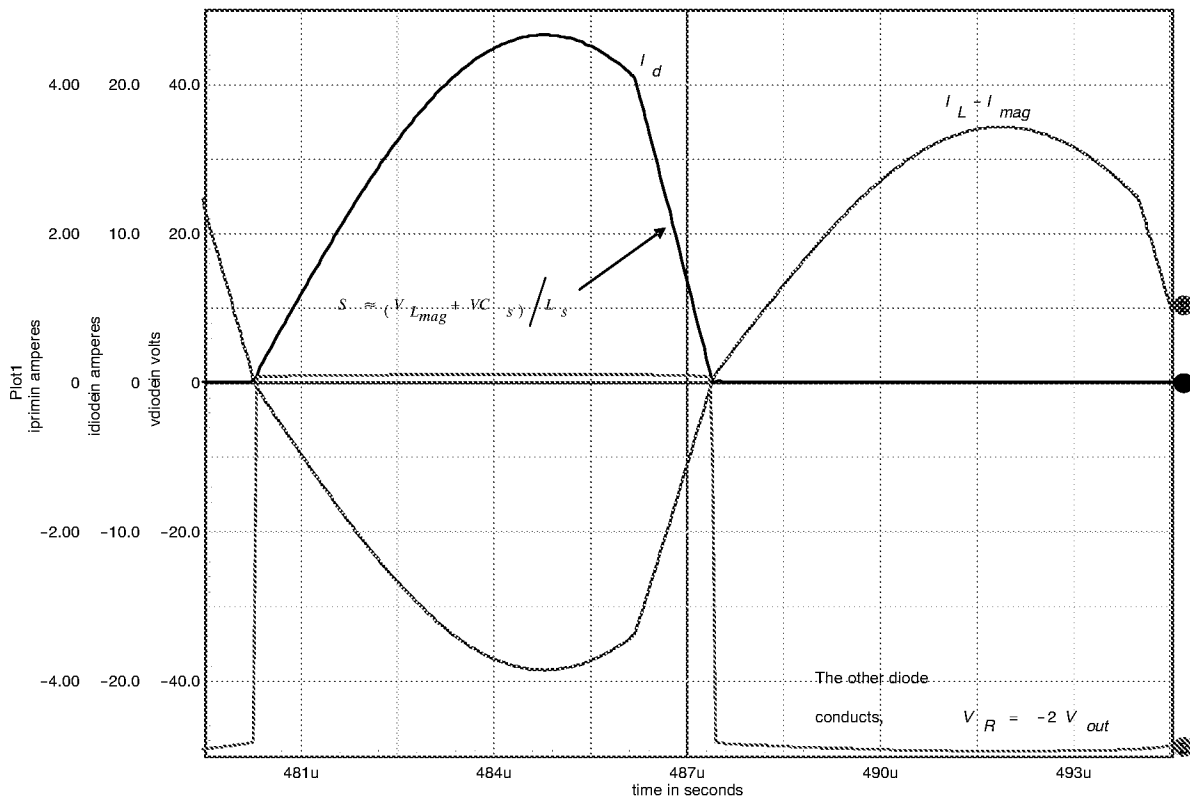


Figure 23. A Zoom on the Switching Diodes Reveal a ZCS Operation for F_{sw} greater than F_s

Operating Waveforms at the Series Resonance, $F_{sw} = F_s$

For this final example, we have selected a set of elements which operate the converter at the series resonance defined by Equation 1. The following values have been used:

$$L_m = 1.6 \text{ mH}$$

$$L_s = 277 \text{ } \mu\text{H}$$

$$C_s = 17 \text{ nF}$$

$$N = 8$$

$$F_{max} = F_s = \frac{1}{2\pi\sqrt{L_s C_s}} = \frac{1}{6.28 \times \sqrt{277\mu \times 17n}} = 73.4 \text{ kHz}$$

$$F_{min} = \frac{1}{2\pi\sqrt{(L_s + L_m)C_s}} = \frac{1}{6.28 \times \sqrt{(277\mu + 1.6m) \times 44n}} = 28.2 \text{ kHz}$$

$F_{sw} = 73 \text{ kHz}$ at full load and nominal input voltage.

When operated at the tank resonant frequency, the main current $I_L(t)$ is sinusoidal as confirmed by Figure 24.

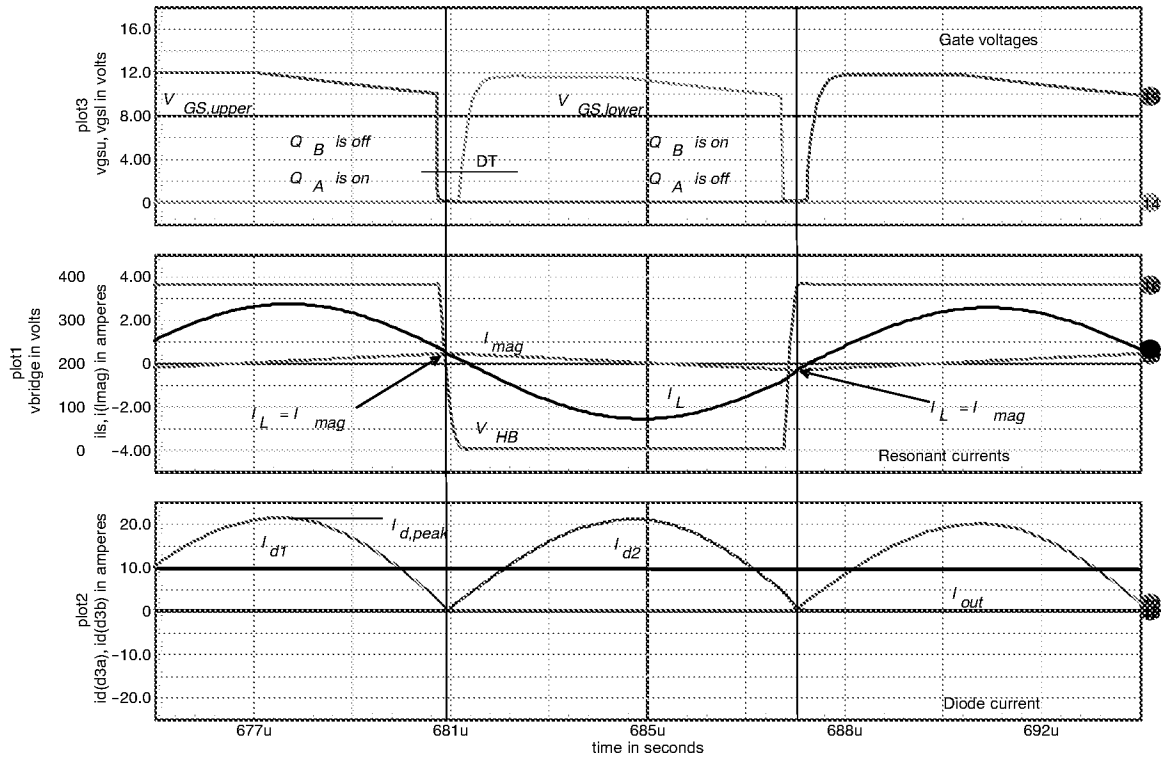


Figure 24. At the Resonant Frequency, the Main Current is Sinusoidal. Also, There is no Deadtime Between the Secondary-Side Diode Conduction Periods.

In this mode, the EMI signature is excellent as the distortion is least compared to the other modes. The secondary-side currents are at the boundary between the segment-like shape and the dead-time period, as respectively observed for $F_{sw} > F_s$ and $F_{sw} < F_s$. As we observed before, the diode block when the resonant current equal the magnetizing current I_{mag} .

Operating the LLC at the series resonant frequency offers another advantage. Back to Figure 3, we can see a point where all curves cross. This point, for which $V_{out}/V_{in} = 1$, is reached at the series resonance. When operated at this particular position, the LLC resonant network transfer function becomes insensitive to load variations. This characteristics is sometimes exploited when a LLC converter is designed to operate at a fixed switching frequency locked to F_s and the feedback loop drives the output voltage of a pre-converter.

The RMS current in the output capacitor is also at its lowest value as no discontinuity, or deadtime, exists as shown on figure 6 for $F_{sw} < F_s$. We can quickly derive its value in presence of sinusoidal signals:

$$I_{d,RMS} = \frac{I_{d,peak}}{\sqrt{2}} \quad (\text{eq. 8})$$

The total dc current contributed by both diodes in the output current delivered by the converter. This dc current can be linked to the equivalent full-wave rectification and equals:

$$I_{dc} = \frac{2I_{d,peak}}{\pi} \quad (\text{eq. 9})$$

We can then evaluate the ac current flowing into the output capacitor applying the following equation:

$$I_{Cou,RMS} = \sqrt{I_{RMS}^2 - I_{dc}^2} = \sqrt{\frac{I_{peak}^2}{2} - \frac{4I_{d,peak}^2}{\pi^2}} \approx 0.3 I_{d,peak} \quad (\text{eq. 10})$$

In the simulated example, if we have a diode peak current of 15.7 A at steady-state, the RMS current flowing in the capacitor is therefore 4.7 A. If we simulate a similar converter in a 100 kHz 2-switch forward configuration, the RMS current in the output capacitor reduces down to 0.5 Arms. That is one of the major disadvantage of the resonant operation. The switching losses are almost removed, allowing high-frequency operation, but conduction losses increase significantly.

Conclusion

This quick study of the LLC converter explores the various operating modes of the power supply, mainly dictated by the switching frequency value in relationship to the series resonant frequency F_s . Most of LLC designs are operated at the series resonance in full load and nominal input voltage conditions. The controller allows operation below F_s during an input voltage drop and lets the frequency exceed F_s in light load conditions. Despite sinusoidal

currents, care must be taken in the selection of the output capacitor given the high ac ripple. Compared to buck-derived applications, this is the penalty to pay with LLC converters, however, largely compensated by the reduction in switching losses on both the primary transistors (ZVS) and the secondary-side diodes (ZCS).

2. Bo Yang, "Topology Investigation for Front-End dc-dc Power Conversion for Distributed Power System", Virginia Tech Dissertation, 2003"
<http://scholar.lib.vt.edu/theses/available/etd-09152003-180228/unrestricted/>

References:

1. Bo Yang, Fred C. Lee, Matthew Concannon, Over Current Protection Methods for LLC Resonant Converter, IEEE Conference 2003

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POINT OF LOAD CONVERTERS - The Topologies, Converters, and Switching Devices Required for Efficient Conversion

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Abstract

Point of load (POL) or point of use (POU) converters are emerging as the popular solution for applications in which circuits require low voltages of 3.3 V and below. The demand for these types of voltage levels stems from the requirement for lower core voltages, and it is obvious that the current capability for these converters will increase even if the power capability stays the same. This has several implications for the power-supply circuitry, including the need to route low-voltage high current around a printed circuit board. This leads to relatively large voltage drops, higher power consumption, and large PCB tracks, and could result in poor output regulation. Furthermore, converting the voltage from 48 V to 0.9 V requires a very low duty cycle, which in turn results in low efficiency. Typically, there will be several voltage levels required across a PCB card, and these could range from 5 V down to 0.9 V, resulting in the need for multi-output converters or several POL converters.

Introduction

POL has many different topologies or configurations, and as of yet there appears to be no fixed conversion strategy for stepping down from 48 V to values that could be as low as 0.9 V. Figure 1 shows a typical architecture for supplying all the required voltage levels from a single "front-end" converter. This has a typical input voltage of 48 V, which is then stepped down, with transformer isolation to several outputs, to various voltage levels. These voltages then must be distributed to the part of the circuit that requires the power, which could be hundreds of millimeters distant. Figure 2 shows the schematic of the architecture for a distributed bus with a two-step conversion process. The second-step, or POL, converter takes the distributed bus voltage and converts it to the required voltage level at the point at which the circuit consumes the highest power. With this two-step approach, the front-end converter usually steps down to a fixed output voltage of either 8 V or 12 V. However, a new trend has been to convert the distributed bus down to a

value of 3.3 V. This voltage is then distributed around the PCB, or card, to a point of load converter, which is then used to supply the final voltage. Another option is to use a fixed duty cycle, typically resulting in a fixed ratio converter of 6:1, which is reportedly optimized to give a higher efficiency (for specified conditions) at a fixed duty cycle. However, that topology is beyond the scope of this paper.

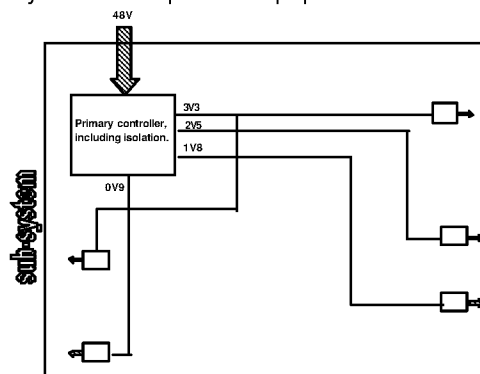


Figure 1. Block diagram showing typical architecture of a single front-end converter supplying all the voltage levels.

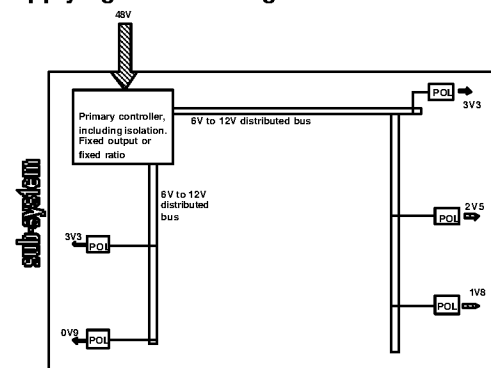


Figure 2. Block diagram showing typical architecture of a point of load (POL) distributed bus power system.

This paper explores front-end converter topologies, along with point of load converters, synchronous rectification, and the appropriate controllers and devices necessary to implement these converters. Using various semiconductor switches specifically designed for these

topologies, efficiency measurements are made and an objective assessment for these topologies in several applications and power ranges is given.

Background

Recent years have seen a move away from centralized ac-to-dc power supplies to decentralized (distributed) power systems that use dc-to-dc converters near their point of use. The major driving force has been the decrease in core voltages, and it can be seen from Table 1¹ that this trend is set to continue over the next decade. These low voltages at high currents are forcing the power conversion industry to re-evaluate conventional circuit topologies, component selection, and packaging concepts. The point of load requirement has arisen from computer hardware developments, but it is being seen throughout the semiconductor industry wherever DSPs are implemented, and it is present especially in telecom base stations and network infrastructure.

	Year							
	'01	'02	'03	'04	'05	'08	'11	'14
Process technology [μm]	0.15	0.15	0.15	0.12	0.12	0.09	0.06	0.06
Clock frequency [GHz]	1.8	2.1	2.5	2.9	3.5	7	11	15
Power consumption [W]	130	140	150	160	170	171	177	186
Static current requirement [A]	87	93	100	133	142	190	235	310
Min. voltage [V]	1.5-1.2	1.5-1.2	1.5-1.2	1.2-0.9	1.2-0.9	0.9-0.6	0.6-0.5	0.6-0.3

Table 1. Improvement in process technology and power requirements.

Before the point of load converter may be considered, it is necessary to investigate the converter that provides the distributed bus. Historically, the front-end converter has always stepped down to a fixed output to supply the majority of voltages needed by the sub-system. However, with a distributed architecture, the idea is to step down to a voltage between 6 V and 12 V. This ensures that the current is large enough to necessitate the need for large busbars around the system. Now a new trend is being seen in which the distributed bus voltage is to be stepped down to 3.3 V, thus providing the 3.3-V voltage requirement directly from the front-end converter and allowing the low voltages of 2.5 V and below to be provided by a POL.

The front-end controller performs several functions, including establishing a large step-down ratio and galvanic isolation. The entire power throughput will have to pass through the

front-end converter and conversely through the isolating transformer. In some circumstances this may be an off-the-shelf voltage regulator module (VRM), a custom module, or a discrete version designed in-house. Regardless of the route the manufacturer takes, the topology is usually determined by the power requirement of the system. The front-end converter must be highly efficient because all the power for the sub-system passes through it, and any inefficiency will be transferred through to the end output. For example, with a POL converter of 95% and a front-end converter of 75%, the resulting efficiency will be 71%. Therefore, great care must be taken to improve the front-end converter efficiency as much as possible.

There are several available topologies that could be considered for implementation, and Table 2 outlines several of them. Taking into consideration the power-requirement levels from Table 1, it is apparent that the forward converter, ignoring resonant topologies, is at present the most suitable topology. However, future power requirements may dictate the use of the half-bridge converter, which becomes more viable at higher power levels. This paper describes briefly both the half-bridge and forward converters, available from Vishay Siliconix, and presents efficiency results for the forward converter.

Topology	Common Characteristics				
	Power Level [W]	Voltage Stress [V]	Output Ripple Freq [Hz]	Max. Duty Cycle	Relative Cost
Flyback	5-150	$V_{in} + (N_p/N_s)V_{out}$	f_s	<0.5	Low
Forward	10-250	$2 \times V_{in}$	f_s	<0.5	Low
Push-Pull	15-150	$2 \times V_{in}$	$2 \times f_s$	<1.0	Med
Half Bridge	50-400	V_{in}	$2 \times f_s$	<1.0	Med
Full Bridge	200-2k	V_{in}	$2 \times f_s$	<1.0	High

Table 2. Characteristics of common converters.¹

The Half-Bridge Converter²

The half-bridge dc-to-dc converter configuration consists of two large, equal capacitors connected in series across the dc input, providing a constant potential of $\frac{1}{2} V_{in}$ at their junction, as shown in Figure 3. The MOSFET switches SW1 and SW2 are turned on alternatively and are subjected to a voltage stress equal to that of the input voltage. Because the capacitors provide a mid-voltage point, the transformer sees a positive and negative voltage during switching. This results in twice the desired peak flux value of the core because the transformer core is operated in the first and third quadrants off the B-H loop, and it experiences twice the flux excursion of a similar forward-converter core.

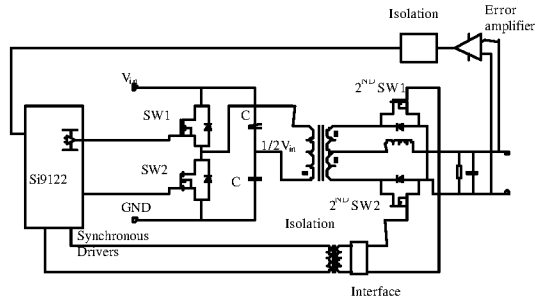


Figure 3. Schematic of the half-bridge dc-to-dc converter.²

The Forward Converter

The forward converter is very similar to the step-down dc-to-dc converter, with the transformer providing galvanic isolation and not being used to store energy. For the topology investigated in this paper, a simple reset winding is included to reset the magnetizing current in the transformer and prevent core saturation. The circuit used is a self-resonant reset circuit, which resets the magnetizing current and also recovers this magnetizing energy by charging it back to the input. This topology also allows for large ratio of input-to-output voltages.

The controller used is a Vishay Siliconix Si9118DY, and for simplicity the readily available 25-W demo board⁴ is used to evaluate a range of devices with varying parameters. Although the target requirement would be 100 W (from Table 1), the circuit investigated gives a valid comparison of switching devices. This demo board is used to determine the performance of two distributed buses, one with an output voltage of 12 V and another with an output voltage of 3.3 V. To obtain the 12-V output, the demo board^{3&4} was altered slightly. This included altering the number of turns on the transformer to 10 from 21, changing the feedback resistors, and using higher-voltage-output capacitors. The schematic of the circuit is shown in Figure 4. For the 3.3-V output, the only change to the demo board was to the feedback resistors.

Results: 12-V Output

Initially, two devices were chosen for the primary switch comparison, these being the Si9422DY and the Si4490DY. These gave a good trade-off between $r_{DS(on)}$ and gate charge (Q_g), with the Si9422DY having higher on-resistance and the Si4490DY having slower switching times. Table 3 shows the measured efficiencies of the converter using these two devices. It should be noted that the transformer has not been designed for the 12-V output – as this is beyond

the scope of this paper – and as such may contribute to higher losses than normally would be expected.

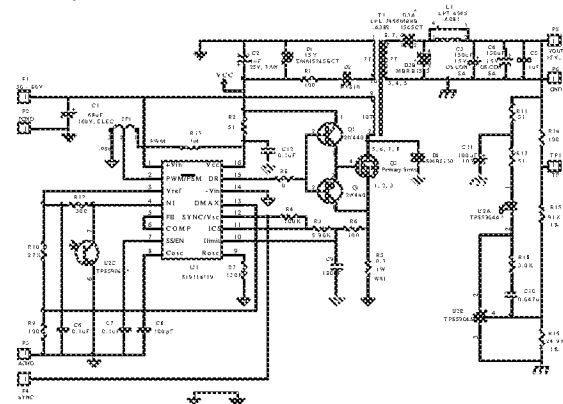


Figure 4. Schematic showing the circuit used to provide a 12-V, 2.5-A output.^{3&4}

The results in Table 3 show that increasing the switching frequency from 500 kHz to 1 MHz increases the efficiency in some conditions by 6%. However, increasing the switching frequency of the converter increases the switching losses of the primary MOSFET. Therefore, the losses in the transformer, in this instance, must contribute more to the total losses of the converter than the MOSFET, as increasing the switching frequency reduces the energy requirement or size of the core.

Table 3 also shows that the converter operated with a lower input voltage of 36 V is more efficient than when it is operated with a higher input voltage of 48 V or 60 V. This demonstrates that the switching losses are the dominant factor in the primary device. Using the equations in Table 5 and the idealized waveforms in Figure 6, it can be shown that by decreasing the input voltage from 60 V to 36 V, the switching losses will increase by approximately 40%, whereas the increase in conduction losses (due to the I^2 relationship) is approximately 70%. Therefore, if the losses were equally distributed, then changing the input voltage would have more of an impact on the conduction losses. Because this is not the case, the switching losses must be considerably more than the conduction losses.

The efficiency results also show the trade-off between conduction losses and switching losses for the comparison of the two MOSFETs. At the higher current levels, the lower $r_{DS(on)}$ device becomes more efficient, especially at the lower input voltage where the current will also be larger (36 V_{in}, 2.5A, 1 MHz). However, at the slightly lower current levels (60 V_{in}, 1.5 A, 1MHz), the

higher $r_{DS(on)}$ device is more efficient. This is due to the fact that reducing the $r_{DS(on)}$ at higher load currents counteracts the penalty of increased switching losses associated with higher Q_g .

DC input [V]	Switching Frequency								
	500kHz			750kHz			1MHz		
	Load current [A]			Load current [A]			Load current [A]		
9	36	75	77	78	80	82	83	81	83
4	48	70	75	76	75	78	81	76	79
2	60	67	71	75	71	75	77	72	76
2									
4	36	77	78	80	80	83	84	81	83
4	48	71	75	77	75	79	81	76	80
9	60	67	72	75	70	74	77	71	76
0									

Table 3. Efficiency of the Si9118DY 12-V converter using two different MOSFETs as the primary switch.

Three more devices were added to the comparison, and the efficiency results are shown in Table 4. The best efficiencies are achieved with the 1-MHz switching frequencies and with the Si4848DY. Again, this shows that the major losses are associated with the transformer. To obtain lower losses, the switching frequency must be increased. However, it is the combination of the fast switching times and low $r_{DS(on)}$ of the Si4848DY that provides the most efficient converter for a 48-V input and 1-MHz switching frequency. However, at the lower switching frequencies, the part with even-lower $r_{DS(on)}$, the Si4488DY, is more efficient.

Device	V_{ds} [V]	$i_{ds(on)}$ max [A]	Q_g typ [nC]	Switching frequency		
				500kHz	750kHz	1MHz
Si9420DY	200V	1	8.6	72% @ 2A	76% @ 2A	77% @ 2A
Si9422DY	200V	0.42	13	76%	81%	81%
Si4490DY	200V	0.08	34	77%	81%	80%
Si4488DY	150V	0.05	30	78%	81%	80%
Si4848DY	150V	0.085	17	77%	81%	82%

Table 4. Efficiencies at 48-V input, 12-V output, and load current of 2.5 A.

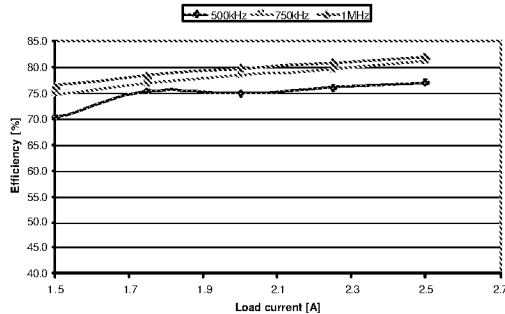


Figure 5. Efficiency for the 12-V converter, with 48-V input and with the Si4848DY MOSFET as the primary switch

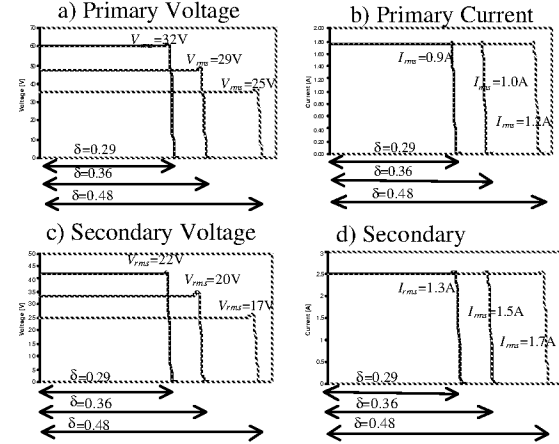


Figure 6. Idealized current and voltage waveforms for 12-V and 2.5-A output.

Key $\delta=0.29$ Vin 60V; $\delta=0.36$ Vin 48V; $\delta=0.48$ Vin 32V

Loss equations	
Primary Switch	$P_{con} = R_{ds(on)} I^2 \delta$ $P_{sw} = \frac{1}{2} V I (t_f + t_r) f_s$ $P_{gate} = Q_g V_g f_{sw}$
D1	$P_{con} = V_{sat} I(\delta)$
D2	$P_{con} = V_{sat} I(1-\delta)$
D1 + D2	$P_{rr} = V_{rr} \left(\frac{I_{rr}}{2} + I \right) f_s$

Table 5. Generic loss equations for the active devices in a forward converter.

Results: 3.3-V Output

As with the example of the 12-V converter, two devices initially were chosen for the primary switch comparison. Table 6 shows the measured efficiencies of the converter using the Si9422DY and the Si4490DY.

The highest efficiency occurs at a load current of between 2 A and 3 A, a range that ties in with the efficiencies of the 12-V output converter and the demo board⁴, which also had their highest efficiencies at 2.5 A. The efficiency appears to be dependent on current rather than power throughput, which may be due to the losses in the secondary-side Schottky diodes.

With the 3.3-V converter, the efficiency does not always improve with a lower input voltage. This means that the trade-off between switching losses and conduction losses is not as definitive as it was in the 12-V output case. Figure 8 shows the current and voltage waveforms for the 3.3-V converter, and it is seen that even though the rms values of the voltage will be lower, the peak values of the voltage will be the same for the 3.3-V output converter as for the 12-V converter. The primary current peak will be higher for the 3.3-V

converter with the same power throughput, resulting in higher switching losses for the same power. The conduction losses will also be higher because the rms current is higher for the same output power. Therefore, the losses for the 3.3-V converter will be greater than those for the 12-V converter for a given power output. This is not necessarily the case in this paper, due to the dominance of the modified transformer in the 12-V converter.

Device [Freq]	Vin [V]	Efficiency [%] @ Load current [A]							
		1	2	3	4	5	6	7	8
9422DY [500kHz]	36	78	80	81	80	79	78	77	75
	48	77	82	81	80	80	79	78	76
	60	75	81	82	80	79	79	77	76
9422DY [750kHz]	36	76	81	79	79	78	77	76	74
	48	75	80	82	79	79	78	77	75
	60	73	80	81	81	78	78	77	75
9422DY [1MHz]	36	72	79	81	78	77	76	75	74
	48	71	78	80	80	77	77	76	74
	60	69	77	79	80	77	76	75	74
4490DY [500kHz]	36	75	78	80	80	80	79	78	77
	48	71	79	79	79	79	79	78	77
	60	69	78	80	78	79	78	78	76
4490DY [750kHz]	36	76	79	80	80	79	78	77	76
	48	68	76	79	80	78	77	77	76
	60	65	74	77	78	77	77	76	75
4490DY [1MHz]	36	-	-	-	-	-	-	-	-
	48	75	74	76	76	76	76	75	73
	60	61	72	75	77	77	75	75	74

Table 6. Efficiency of the Si9118DY 3.3-V converter using two different MOSFETs as the primary switch.

Again, as with the 12-V converter test, three more devices were added for comparison (Table 7). In these tests the most efficient circuit has a 750-kHz switching frequency, which means that the transformer losses are not dominant, as they were with the 12-V output converter. In the 3.3-V converter, it is obvious that the switching losses of the primary device dominate the losses because the Si9422DY, with the lowest Q_g , is the most efficient device. However, at the higher loads (8 A, 750 kHz), it is the combination of the low switching and $r_{DS(on)}$ of the Si4848DY that provides the most efficient converter with an efficiency measurement of 79%.

Device	V _{ds} [V]	r _{ds(on)} max [Ω]	Q _g typ [nC]	Load [A]	Switching frequency [kHz]		
					500	750	1000
Si9420 DY	200V	1	8.6	3	-	80	78
				8	-	73 (@7)	67 (@7)
Si9422 DY	200V	0.42	13	3	81	82	80
				8	76	75	74
Si4490 DY	200V	0.08	34	3	79	79	76
				8	77	76	73
Si4488 DY	150V	0.05	30	3	79	77	76
				8	77	76	74
Si4848 DY	150V	0.085	17	3	80	79	77
				8	73	77	75

Table 7. Efficiencies using five different primary switches with a 48-V input and 3.3-V output.

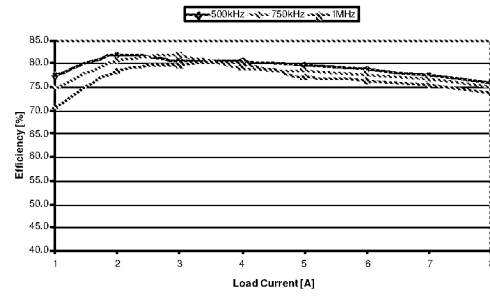


Figure 7. Efficiency for the 3.3-V converter

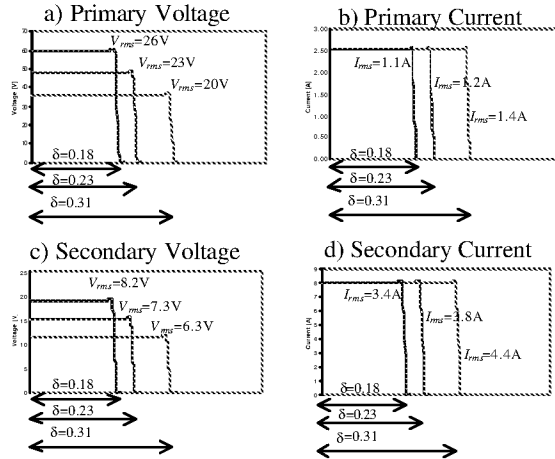


Figure 8. Idealized current and voltage waveforms for 3.3-V and 8-A output.

Key $\delta=0.18$ Vin 60V; $\delta=0.23$ Vin 48V; $\delta=0.31$ Vin 32V

Secondary Synchronous Rectification

One of the factors that affects the efficiency of the 3.3-V converter at the higher load currents necessary to obtain the same power output is the IV losses in the rectifier diodes (D3A and D3B in Figure 4). Therefore, these diodes can be replaced with MOSFETs to reduce conduction losses.

Self-driven secondary synchronous rectification is relatively simple to achieve with forward converters and is shown in Figure 9. However, care must be taken with the voltage levels that appear on the secondary-side transformer. For example, in considering the ideal voltage across the secondary transformer (Figure 8) for the 12-V converter, the minimum voltage is 25 V. However, with voltage overshoots this voltage often exceeds 30 V. Therefore, synchronous rectification was not considered for the 12-V converter due to the maximum gate-source voltages, which are typically 20 V. Furthermore, because the current levels in the 12-V converter are considerably lower than in the 3.3-V

converter, the benefits of introducing synchronous rectification are not as great. With an input voltage of 36 V, the 3.3-V converter secondary-side voltage was below 20 V (the maximum gate source voltage allowed for these devices). Therefore, the Si4888DY, which offers a good combination of Q_g and $r_{DS(on)}$, was used as the MOSFET for synchronous rectification. For an input of 48-V, the secondary-side voltage reached values higher than 20 V, so a capacitor was used in series with the gate of the MOSFET. This reduces the gate source voltage appearing on the MOSFET, but it has the disadvantage of slowing the device and increasing the gate losses (Figure 10).

Another option is to use a Schottky diode in parallel with the synchronous MOSFET to improve the reverse recovery characteristics. An advantage of this method is that the Schottky can have a greatly reduced current capability, as it will only be conducting for a small amount of the time.

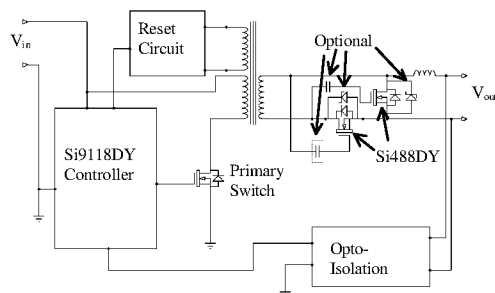


Figure 9. Schematic of the self-driven synchronous rectification.

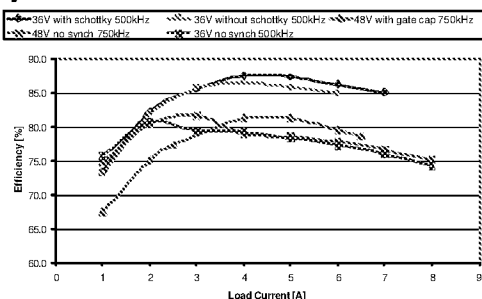


Figure 10. Synchronous rectification with 3.3-V output.

The Point of Load Converter

Although not considered in detail in this paper, the actual point of load converter has several important characteristics. First, it does not need to be isolated because the front-end converter has achieved this. Second, depending on the required voltage levels, the POL converters can

be synchronous buck or synchronous boost to provide the greatest efficiency. Finally, the controllers can also be simple to allow for the smallest size.

For the MOSFET devices used for these synchronous converters, there are also desirable characteristics, such as shoot-through protection to prevent spurious turn-on of the low-side switch. This has come about due to the need for increased switching times and hence smaller dead-times. Also, for the synchronous MOSFET (lower-side switch in buck), the $r_{DS(on)}$ characteristics are more important, and with higher power levels, the packages are becoming more critical in increasing the power density.

Conclusion

These results have shown that it is very difficult to predict the performance of a MOSFET in a switching converter. In some circumstances the major factor contributing to losses may be the $r_{DS(on)}$, and for another load point, the switching transients will be more important. Therefore, in some circumstances it may be more desirable to fix the parameters of the converter by fixing the output current rather than the output voltage. With the need to bus an intermediate voltage around the board, the voltage regulation can be accounted for with the point of load converter. In this case, the distributed intermediate voltage does not necessarily need to be a fixed voltage. Another issue is the requirement for the distributed bus voltage level. Lower voltage outputs mean higher load currents and more losses for the same power levels, but a 3.3-V distributed bus does have the advantage of having the 3.3-V output available when required. The efficiency can be improved dramatically with synchronous rectification, which may not be the case with the higher-output-voltage converter. Therefore, care must be taken in choosing the correct topology and MOSFET for a given application, and in some circumstances the devices and circuits should be chosen via practical testing.

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FIVR – Fully Integrated Voltage Regulators on 4th Generation Intel® Core™ SoCs

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Abstract—Intel’s® 4th generation Core™ microprocessors are powered by Fully Integrated Voltage Regulators (FIVR). These 140 MHz multi-phase buck regulators are integrated into the 22nm processor die, and feature up to 80 MHz unity gain bandwidth, non-magnetic package trace inductors and on-die MIM capacitors. FIVRs are highly configurable, allowing them to power a wide range of products from 3W fanless tablets to 300W servers. FIVR helps enable 50% or more battery life improvements for mobile products and more than doubles the peak power available for burst workloads.

I. INTRODUCTION

Intel’s® 4th generation Core™ microprocessors (code name Haswell) are powered by Fully Integrated Voltage Regulators (FIVR), the industry’s first large scale deployment of high current switching regulators integrated into a VLSI die and package. An overview of the scheme is given in Fig. 1(a). A first stage VR, which is on the motherboard, converts from the PSU or battery voltage (12-20V) to approximately 1.8V, which is distributed across the microprocessor die. The second conversion stage is comprised of between 8 and 31 (depending on the product) FIVRs, which are 140MHz synchronous multiphase buck converters with up to 16 phases. A simplified schematic for a two phase FIVR domain is shown in Fig. 1(b). The power FETs, control circuitry and high frequency decoupling are on the die, while the inductors and mid-frequency input decoupling capacitors are placed on the package. Each FIVR is independently programmable to achieve optimal operation given the requirements of the domain it is powering. The settings are optimized by the Power Control Unit (PCU), which specifies the input voltage, output voltage, number of operating phases, and a variety of other settings to minimize the total power consumption of the die.

FIVR is the enabling technology behind key improvements for Intel’s® 4th generation Core™ microprocessors including a 50% or more increase in battery life for mobile products, and a 2-3x increase in peak available power (which converts into burst performance). The motherboard voltage regulators eliminated by FIVR free up space that can be used to add platform features or reduce platform dimensions. Details are discussed in Section V.

A. Background

Intel’s® 2008 microprocessor microarchitecture introduced the Power Control Unit (PCU) [1], a microcontroller that monitored conditions across the die in real time, and dynamically adjusted a variety of settings to optimally manage power consumption and performance. One of the most important features controlled by the PCU were newly added high current power gates, which provided a significant improvement in CPU energy efficiency by eliminating large leakage losses on idle compute domains. The power gates on high current domains were introduced by adding gate transistors into thin ‘cracks’ between major functional blocks and represented a very small percentage of the total die area. The die bumps required to support the high current domains required a much larger area than the gate transistors themselves. The large “bump area” posed a hard barrier to productization until a scheme was devised to “borrow” bumps from surrounding circuitry using a thick, low loss routing layer. An extension of this scheme makes FIVR affordable.

A limitation of power gate is that all active domains still operate at the highest voltage required by any individual domain. To create separate voltage domains an entirely new regulator must be added to the motherboard, which adds cost, increases area, and requires extra package pins. An improvement suggested by recent research is the integration of high frequency buck regulators directly on the microprocessor package, or in the die itself [2] [3] [4] [5]. This allows a much larger number of independent power domains, each managed dynamically to match the local computational demand. For example, this would allow one CPU core to run at an elevated voltage and frequency to satisfy a heavy computational load, while other cores execute lower priority code at a much lower voltage and frequency to save power.

Each of the previous works cited had at least one issue making it poorly suited for broad, high volume deployment. The multi-chip approach taken by [2] resulted in an effective current density of 1.3A/mm² that would make it expensive to implement because of the silicon area required to support a full power microprocessor product. In [3] another

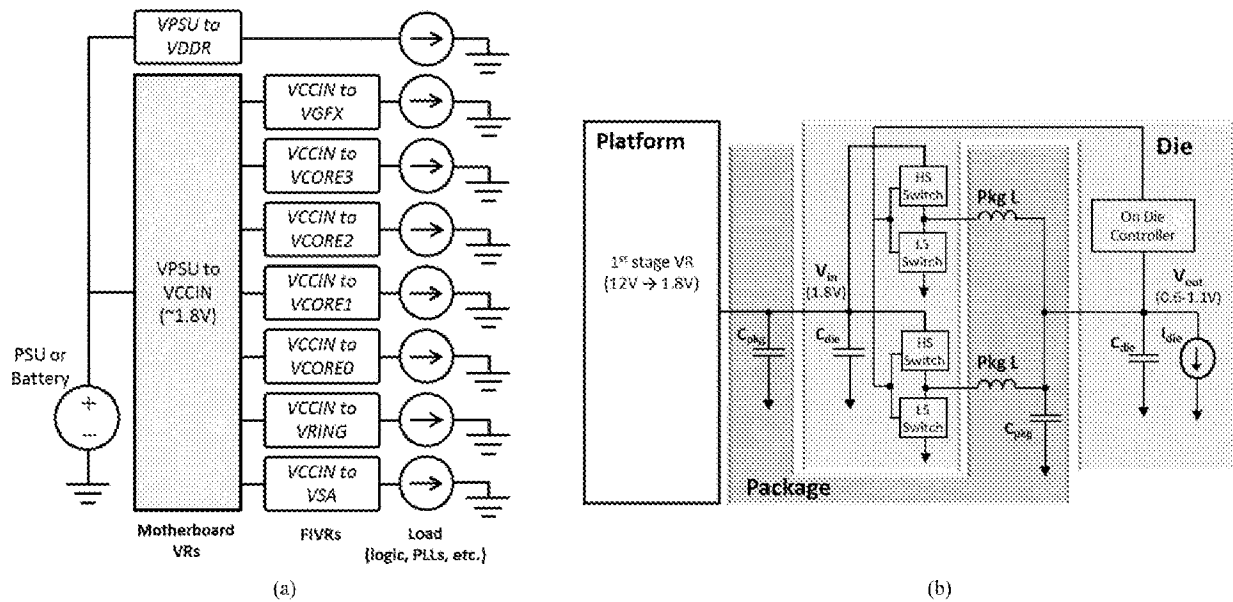


Figure 1. (a) Representative partitioning of the separate high current power domains on a 4th generation Core™ Microprocessor. (b) Simplified schematic of a single FIVR domain, showing the partitioning of the components between the die and the package.

example of the multi-chip approach is demonstrated. This work, which used a 90nm process and inductors integrated onto the die, achieved a higher current density (8A/mm²) but reported a relatively low efficiency of 76% (compared to 85% for 3.3V to 1.0V conversion in [2]). Instead of the multi-chip approach, the authors of [4] integrated the regulator directly into the die on a 45nm process, but still suffered from relatively low current density (1.7A/mm²). The authors also report an efficiency of 83% for 1.5V to 1.0V conversion due in part to the quality of the discrete inductors that were used.

FIVR builds on the VR designs presented in [2], while the implementation strategy that makes FIVR affordable, is an extension of the bump “borrowing” scheme developed for the high current power gates [1].

B. Motivation

This paper will show that FIVR addresses the issues in prior work that prevented broad deployment of integrated switching regulators in high volume products. Extending the earlier bump borrowing scheme yields the same current density increase, and corresponding cost decrease, that first made power gates affordable. Improvements in the inductors and transistors yield efficiency in the 90% range for typical high power workloads. The high unity gain frequency (up to 80MHz) allows FIVR to work with just on die MIM for output capacitance.

While these advancements are necessary, they’re insufficient to make a reasonable business case for FIVR. To pay the costs of developing and fabricating FIVR, it’s important to quantify the actual customer-visible benefits provided in a real implementation. At the start of the design FIVR’s expected benefits fell into half a dozen categories. FIVR delivered material benefits in every category, and some benefits were far larger than expected. The benefit categories were: battery life increase, increased available power (for increased burst performance), decreased power required for a given level of

performance (or almost equivalently, increased performance for a given power consumed), decreased platform cost and size, improved product flexibility and scalability. See section V for the detailed FIVR impact.

II. IMPLEMENTATION, DESIGN, AND SIMULATION

A. Circuitry

A block diagram representing the circuitry for a single FIVR domain is shown in Fig. 2. The buck regulator bridges are formed by replacing the power gates in previous products with NMOS and PMOS cascode power switches. The cascode configuration allows the power switches to be implemented with standard 22nm logic devices while still handling an input voltage of up to 1.8VDC [2]. This avoids the cost of extra processing steps for high voltage devices, while achieving excellent switching characteristics. The bridge drivers are controlled thru high-voltage level-shifters and support ZVS (zero-voltage switching) and ZCS (zero-current-switching) soft-switching operation. The gates of the cascode devices are connected to the “half-rail”, V_{ccdrv} , regulated to $V_{in}/2$. This is also the negative supply of the PMOS bridge driver as well as the positive supply of the NMOS bridge driver.

The area occupied by the power switches and drivers is small, so they are distributed across the die, immediately above the connection to their associated package inductor which minimizes routing losses. This is illustrated in Fig. 3(a), which shows the location of the package inductors under the die for a four core LGA part. The driver circuitry is interleaved with the power switches in an array which minimizes parasitics to allow for very high switching frequencies. This also allows the size of the bridge to be easily scaled based on the current requirement and optimization points for each power domain.

Each FIVR domain is controlled by a FIVR Control Module (FCM). The FCM contains the circuitry for generating the PWM signals using double-edge modulation, as indicated in

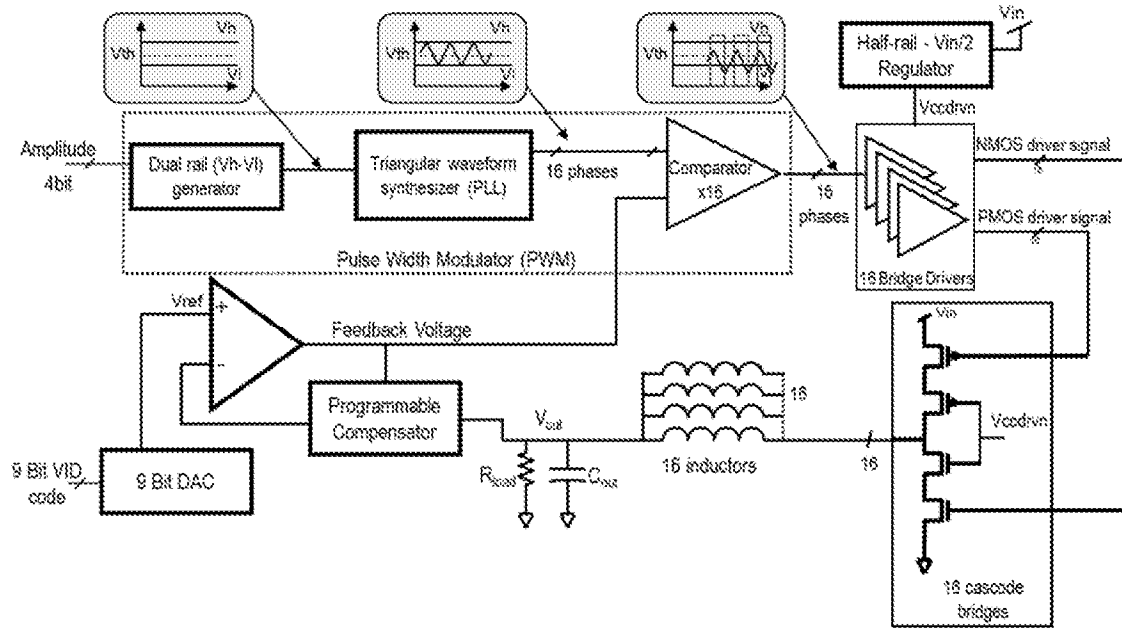


Figure 2. Simplified block diagram of the circuitry for a single representative FIVR domain

Fig. 2 by the dashed box. Separate circuitry not shown in Fig. 2 manages phase current balancing, and the resulting digital PWM signals are distributed from the FCM to individual bridges. The PWM frequency, PWM gain, phase activation, and the angle of each phase are all programmable in fine increments to enable optimal efficiency and minimum voltage ripple across a span of different operating points. Spread-spectrum is used for EMI and RFI (Radio Frequency Interference) control.

The FCM module also contains the feedback control circuitry (compensator). A high-precision 9-bit DAC generates a reference voltage for a programmable, high bandwidth analog fully differential type-3 compensator. Sense lines feed the output voltage back to the compensator. The endpoint of these sense lines is strategically placed to achieve minimum DC error and optimal transient response at an important circuit location in the domain. The compensator is programmed individually for each voltage domain based on its output filter, and can be reprogrammed while the domain is active to maintain optimal transient response as phase shedding occurs.

The key to making FIVR affordable was integrating the power devices directly into the microprocessor die. As with the power gate circuitry discussed in the introduction, the power switching circuitry for FIVR can be placed in small areas between major circuit blocks. The lower current handling of the die bumps makes the die bump area requirements for FIVR much larger than the actual die area required. Since FIVR is integrated into the microprocessor die, routing on the thick metal die layer allows extra bumps to be ‘borrowed’ from areas over other circuits, which avoids wasting any excess die area due to bump current limits. This makes the effective current density of FIVR 31A/mm², a 24x increase over the bump-limited 1.3A/mm² reported in [2].

B. Passives

In order to keep the buck output filter small enough to fit on the die and package it is necessary for FIVR to switch at a high frequency – 140 MHz in most cases. This allows the buck output filter inductors to be implemented using only the bottom metal layers of a standard flip-chip package. Power routing is constrained to the top layers of the package as a result, but the proximity of the inductors to the load ensures that minimal power is dissipated on these layers. The inductors are non-magnetic, i.e. Air Core Inductors (ACI). A representative ACI from an 8-phase domain of a product with an LGA package is shown in Fig. 3(b), including the connection points to the power switches, the DC current path through the inductor, and the connection of the inductor to the output plane. Package design rules allow the ACIs to be placed in to close proximity with one another. On a representative LGA package with four CPU cores, this allowed 59 inductors on 10 different voltage rails to be implemented in a 20mm x 8mm area. The package implementation also allows inductor designs to be customized on a per rail basis to meet efficiency, ripple, and transient response requirements.

Decoupling for the input rail is provided by a combination of ceramic package capacitors and on-die MIM capacitors [6]. The on package ceramic capacitors keep the output impedance of the input rail low from approximately 1 MHz to their self-resonance around 20 MHz. The MIM capacitors are on the die along with the power circuitry and provide high frequency decoupling, including at the switching frequency and its harmonics. Decoupling for the output rails is provided primarily by the MIM capacitors, which are sufficient to provide good transient response if wide bandwidth feedback control is used (see the results section). In some cases the MIM capacitors are supplemented with extra package ceramic capacitors. The comparatively low self-resonant frequency of the ceramic capacitors complicates the control loop design and

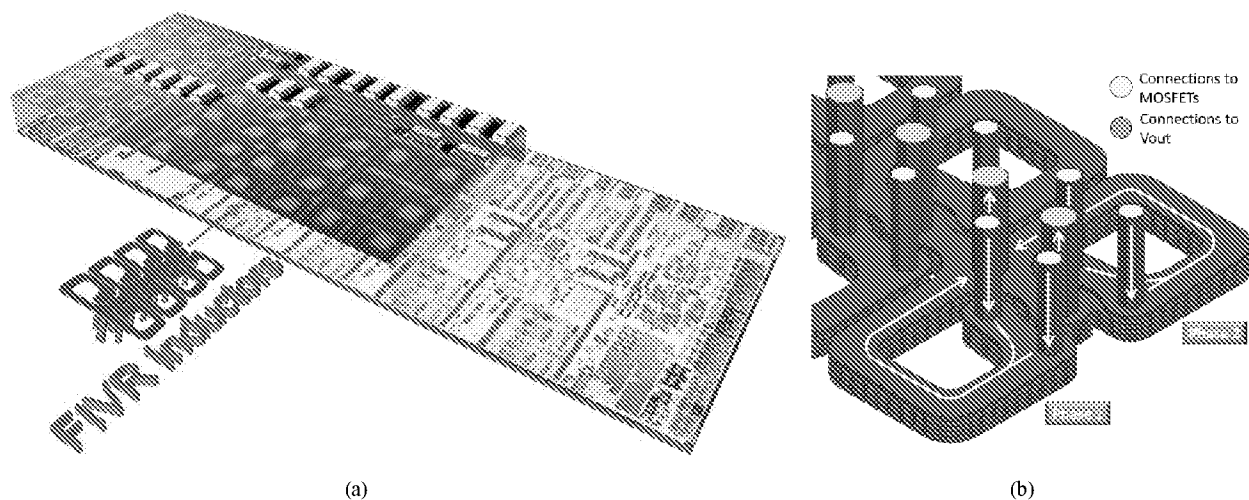


Figure 3. (a) The bottom of an Intel® 4th generation Core™ microprocessor LGA package is shown along with a picture of the corresponding die. A group of eight FIVR inductors is pulled off to the side. (b) An enlarged 3D view of two FIVR inductors is shown with current flow arrows.

does little to attenuate voltage ripple, but the ceramic capacitors can provide a net transient response benefit if they are robustly connected to the output power plane.

C. System Control

In order to minimize losses from FIVR, a modified version of the PCU [1] dynamically configures each FCM based on the current activity level of the domain. The PCU turns each rail on or off based on activity, and specifies an output voltage target to support the desired frequency. It also optimizes the settings discussed in section II.A for the anticipated operating conditions. These settings include the number of active phases (i.e. phase shedding to improve light load performance), the compensator settings (to maintain optimal transient response as the number of phases changes), and the timing of switch drivers (to ensure zero voltage switching at light loads). This allows each FIVR domain to operate at near peak efficiency across a wide range of load conditions from retention to Turbo. An example of the benefit this provides is shown in Section IV.A.

III. CHARACTERIZATION AND PERFORMANCE TESTING

Validating and optimizing a voltage regulator requires the measurement of key parameters such as voltage ripple, efficiency, power supply rejection ratio, transient response, and control loop stability margins. Due to FIVR's high level of integration and fast switching frequency many of these standard measurements are difficult or impossible to perform using off the shelf test equipment. For example, an IA Core voltage domain powered by a FIVR capable of supplying over 30A occupies less than 15mm² on the package, which is completely covered by the microprocessor die. This renders the attachment of an external load for a full current efficiency measurement impossible. This section describes some key Design For Test (DFT) features that are included in FIVR to allow accurate characterization.

A. Control Loop Transfer Function

To enable characterization of the control, a high frequency programmable signal generator is placed in the feedback network on every FIVR domain. The signal generator is activated in a test mode to inject a known, synchronized signal

into the on-die feedback loop. By controlling the test feature and monitoring the output voltage on the package with an oscilloscope, the response of the control loop is directly measured. Repeated measurements are used to tune the compensator to achieve fast response and good stability margins.

B. Load Transient Response and Rejection Ratio

Microprocessors require a nearly constant DC voltage in the presence of large load transients. For characterization purposes, however, it is difficult to create a well behaved step load using only the execution of code in normal operation. The authors in [7] instead use a scheme called Integrated Frequency Domain Impedance Meter (IFDIM), which gates the clock network on and off at a fixed frequency creating a large load transient. The frequency is programmable, the magnitude of the transient can be precisely calibrated using DC measurements, and the load step is known to occur within one clock cycle, so the microprocessor itself is effectively turned into an alternating current load. This feature is included on every FIVR domain, allowing the transient response to a known load to be measured. FIVR domains are characterized across a wide frequency range at multiple operating points for both output impedance and output load coupling across domains.

C. Efficiency

As was previously mentioned, the level of integration makes it impossible to connect a high current load directly to the output of a FIVR rail. An additional complication is that the circuitry on the die cannot be disconnected from the FIVR output, so whenever FIVR is powered some extra current draw due to leakage results. This required the development of a new technique for accurately measuring efficiency. A brief summary of the method is given here. First, a procedure using an external low current load and FIVR operating in a test mode is used to calibrate the leakage. The completely ungated clock tree is then operated at varying frequencies to create a large effective adjustable DC output current. An iterative series of measurements is then used to precisely calibrate the total current drawn by the clocks and the leakage, which allows the efficiency to be measured, when combined with conventional

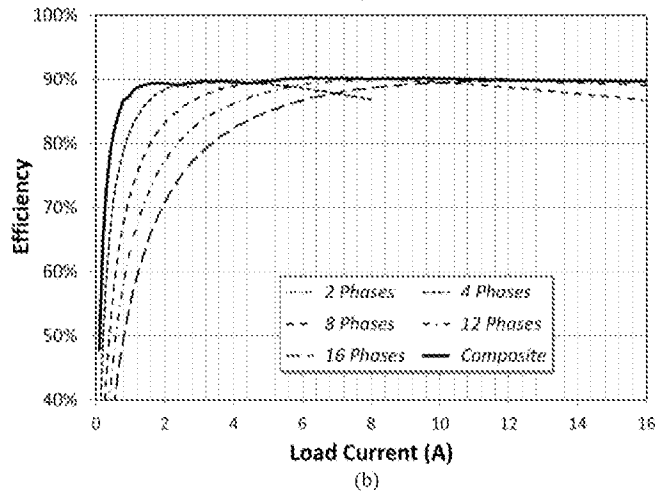
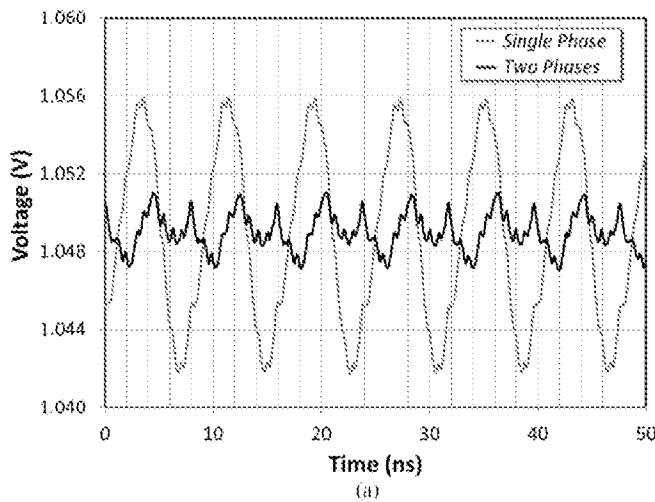


Figure 4. (a) Measured voltage ripple for a low noise domain for single phase and two phase operation (128 averages) (b) Measured efficiency for a voltage domain as a function of the number of active phases

measurements of the voltage and the current at the output of the first stage regulator.

IV. RESULTS

Due to the high switching frequency used, the performance of FIVR is sensitive to the layout of the die and the package which includes the inductors. Each combination of die and package is individually optimized and validated. The following section contains some key validation results from an Intel® 4th generation Core™ microprocessor with four microprocessor cores on an LGA package.

A. Measurements

Fig. 4(a) shows the voltage ripple for a low noise domain measured under the die near the connection of the ACI to the power plane. The measurement was averaged 128 times against the PWM clock (with spread spectrum clocking turned off). To achieve an accurate measurement, a controlled impedance differential sense line was routed on the package from the measurement location to a probe connection point with a matched termination. An active differential oscilloscope probe was then connected to the probe point. This ensures an accurate, wide bandwidth measurement is achieved, as opposed to

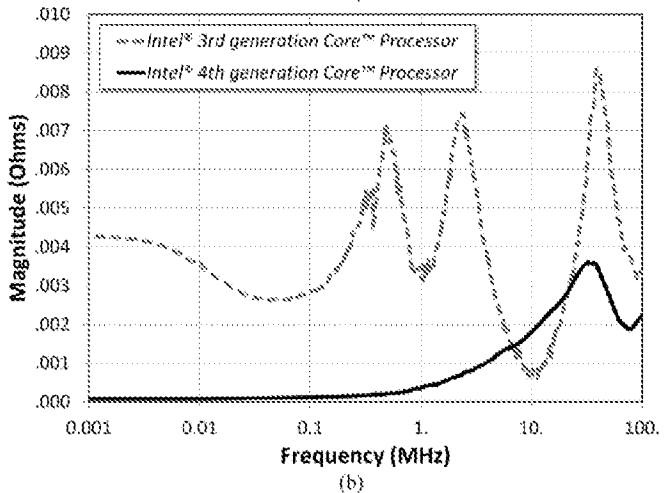
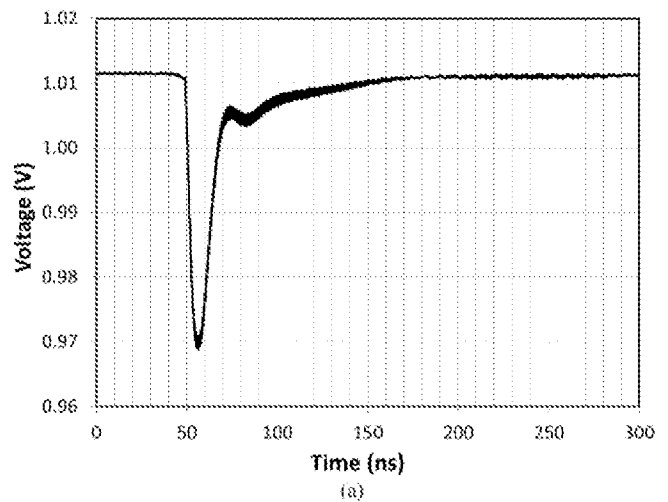


Figure 5. (a) Measured voltage droop on a graphics domain in response to an 8.5A step load (b) Comparison of the effective impedance profile for the graphics voltage domain on a 3rd generation Core™ microprocessor versus a 4th generation Core™ microprocessor

probing the package power planes directly or using a single ended sense line, which can substantially attenuate the measurement over a distance as short as a few millimeters. In two phase operation less than 4mV (less than 1% of the voltage set point) of ripple is achieved with a rail driven by air core inductors well under 2mm² in area.

Fig. 4(b) shows the efficiency as measured using the procedure in section III.C for 1.70V to 1.05V conversion with the bridges configured for hard switching. The efficiency measurement has been repeated for varying numbers of phases, in each case showing a peak efficiency of approximately 90% at 0.75A/phase. By employing a phase shedding scheme it is possible to keep the efficiency of the domain within a few percent of the peak efficiency of the domain from 1A to 15A. This is managed by the PCU which can phase shed when the efficiency can be improved, but also has the intelligence to avoid phase shedding when it could be problematic, for example when a large load transient is possible.

The measured output voltage (averaged 128 times) during an 8.5A load step generated by the IFDIM feature on the graphics voltage rail is shown in Fig 5(a). The measurement

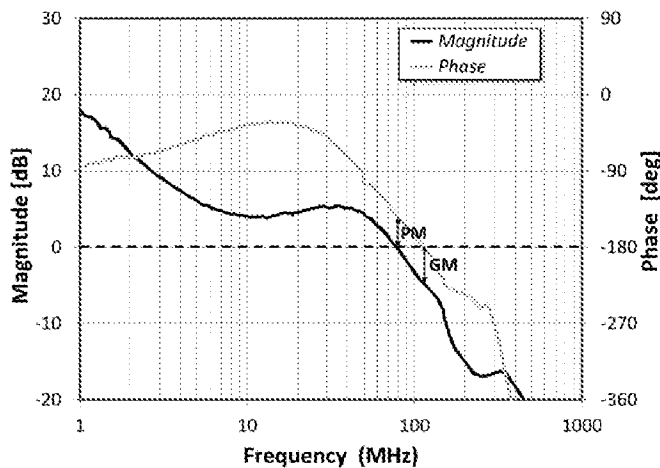


Figure 6. The measured open loop gain and phase of a FIVR showing 78MHz bandwidth with 40° phase margin

was performed with a similar probing configuration to that used for the voltage ripple measurement. The combination of a high bandwidth feedback loop and on die decoupling keep the voltage droop under 50mV despite a rise time for current step of under 1ns (orders of magnitude faster than normal graphics circuit behavior). The main droop event lasts under 30ns, and the DC voltage is restored within 100ns. A nonlinear control feature saturates the duty cycle when a large transient is detected (not active in Fig. 5(a)). The feature was found to provide up to a 25% reduction in voltage droop for step loads, but the benefit is significantly reduced for certain aperiodic load patterns. The effective output impedance profile for the same rail is shown in Fig. 5(b). The peak impedance demonstrates the fast bandwidth of the compensator. Because the inductors are located immediately below the actual area of the die that consumes current, the DC and low frequency load line is virtually zero. The figure also shows the impedance profile for an Intel® 3rd generation Core™ microprocessor graphics rail, which is powered by a platform VR. For this rail, a DC load line is required due to the distance between the VR and the die. Several resonant peaks occur from the various stages of decoupling capacitors on the motherboard and package, and the parasitic inductance between them and the actual point of current consumption on the die.

Fig. 6 shows the open loop transfer function for a FIVR domain, measured using the signal generator DFT. This rail demonstrates a unity gain bandwidth of 78 MHz while still maintaining 40° phase margin. Robust compensator circuit design and very small propagation delays were necessary to achieve this bandwidth, which, in turn, was required to maintain good transient response on rails with limited output capacitance. The high bandwidth also enables fast voltage transitions. A FIVR rail turning on and turning off are shown in Fig. 7. Both transitions are programmed to about half a microsecond for a full range transition – two orders of magnitude faster than a typical platform-based solution. The fast ramp rate translates into power savings for the system, as the voltage rails can be turned on, used, and turned off again almost instantly.

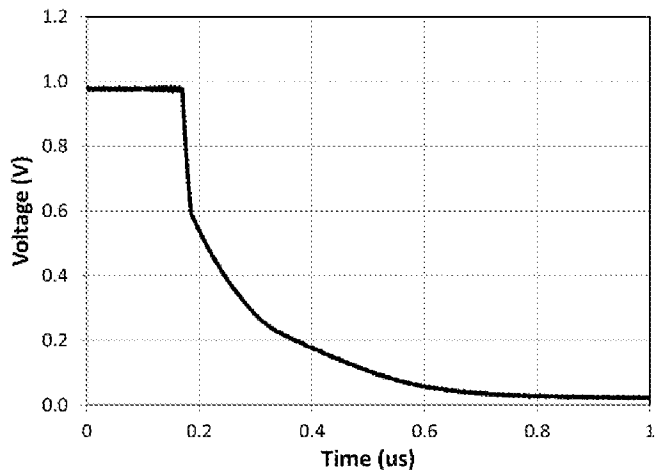
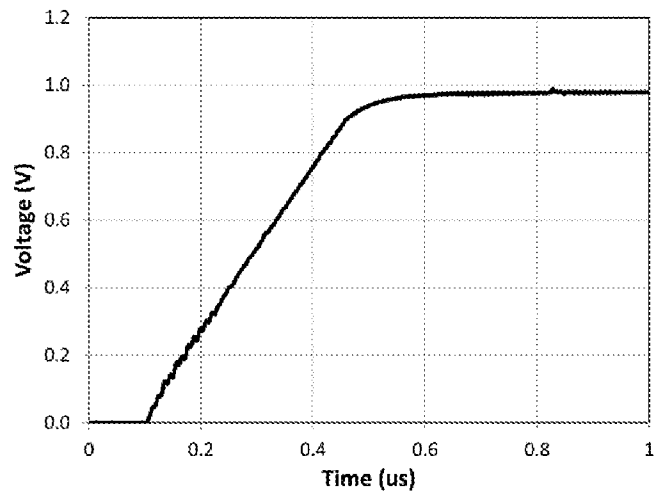


Figure 7. A FIVR rail ramping to its voltage set point from fully off, and then turning off again. Voltage transition times are programmable, but typically set for half a microsecond for a 1V transition.

A large number of additional measurements are taken for validation purposes that are not shown here due to space constraints. These include the output impedance of the V_m rail, audio susceptibility measurements, and the coupling noise due to load transients from one rail to another (particularly from very high current domains such as core and graphics to low current system agent domains). EMI/RFI characterization is also performed.

B. Comparison to Previous Work

Table I contains a comparison to previous works discussed in the introduction. FIVR operates at a higher switching frequency than previous works, which is enabled in part by very good gate charge characteristics for the MOSFETs. This allows up to 90% efficiency at a common conversion ratio.

V. FIVR IMPACT TO PRODUCTS

Battery life improvement: Sufficient battery life for a complete work day has long been desired from mobile products. FIVR, combined with power management architecture improvements, has enabled this for Intel® 4th generation Core™ products. Increases of well over 50% have been widely reported (for example, [8] and [9]). FIVR's battery life benefit comes by several means:

TABLE I. COMPARISON OF FIVR TO PREVIOUSLY REPORTED INTEGRATED VOLTAGE REGULATORS

Parameter	G. Schrom et al., 2010 [2]	T. DiBene et al., 2010 [3]	N. Sturcken et al., 2012 [4]	This Work
Process node	130 nm	90 nm	45 nm	22 nm
Switching Frequency	60 MHz	100 MHz	80 MHz	140 MHz
Unity Gain Freq	5 MHz	Not Published	Not Published	80MHz
Efficiency	85-88%, 3.3V:1.0V	76%	83%, 1.5V:1.0V	90%, 1.7V:1.05V
Total Output I _{max} capability	50 A	Limited by first stage and thermals (Up to 400 A)	1.2 A	Limited by first stage and thermals (Up to 700 A)
I _{max} /VR die area	1.3 A/mm ²	8 A/mm ²	1.7 A/mm ²	31 A/mm ²
Voltage rail count	4	20	1	8 to 31
Phase count	16	320	4	49 to 360
Integration level	MCM ^a	MCM ^a	Integrated into network die	Integrated into CPU die
Inductor technology	Package trace, & magnetic discrete	Magnetic thin-film on VR die	Discrete wire-wound air core	2D array of package trace
Capacitor type	Ceramic package caps	Ceramic package caps	Die Cap	Die Cap - MIM
Cout per Max Amp	2000 nF/A	not published	15 nF/A	7 nF/A

^a MCM – Multi Chip Module – the active circuitry is on a separate die assembled on the same package

- Standby current historically consumes a large fraction of the battery's stored energy. FIVR's fast bandwidth allows low frequency supply noise to be rejected, resulting in up to a 90% reduction in decoupling requirements. This allows both the first and second stages of regulation to be power cycled much faster than on previous products, enabling new deep sleep states with up to 20x lower standby power. With the lowered capacitance, power expended, and time wasted entering and exiting the states is similarly reduced. Reduced sleep-state entry/exit time also saves power by increasing sleep-state usage.
- FIVR's fast control loop and integration into the package result in one tenth the peak impedance of prior solutions (see Fig. 5(b)) in the sub-MHz stimulus range most relevant to the graphics architecture. The resulting low frequency supply noise reduction improves power at a given performance by up to 30%.
- FIVR increases the number of voltage rails, allowing each domain to be set at the minimum possible voltage that supports error-free operation, reducing both leakage and dynamic power.
- Replacing multiple high current voltage regulators on the motherboard with a single first stage regulator reduces the PCB footprint of the power delivery solution. This extra space can be used for a larger battery, with some examples demonstrating up to 10% growth.
- Trimming FIVR together with the microprocessor removes manufacturing guard-bands normally required to ensure that every VR will work with every CPU.

Increased available peak power: An illustrative example shows how FIVR can increase the peak power available to the microprocessor. A typical mobile processor platform using the prior generation power delivery scheme had two 30A, 1.1V VRs providing 33W for cores and 33W for graphics. Using the same power FETs and inductors for the FIVR's 1.8V input VR, the part has 108W power rail (30A/phase * 2 phases * 1.8V),

which can be dynamically allocated to a combination of FIVRs by the PCU. For core-only workloads, nearly the entire 108W can be allocated for the cores, increasing the available power ceiling by 3x. For graphics workloads, 36W can be partitioned to the cores with the remaining 72W going to graphics – more than double the power available from the 33W platform VR. Because power consumption scales as CV^2F and frequency scales with voltage, the increase in available power could be used to operate the graphics at up to 26% higher frequency than possible with the platform VR. A similar calculation yields a 44% higher core frequency in the core-only scenario. The duration of these scenarios is limited by the thermal capabilities of the platform, but translates into improved speed in many real scenarios.

Decreased power at a given performance: Intel's® Iris™ Pro graphics uses FIVR's higher available power to deliver high end graphics. FIVR's high unity gain bandwidth presents less than a tenth the peak output impedance provided by the prior generation's platform VR in the sub-MHz range important to the graphics load (see fig. 5b). Because FIVR has doubled the graphics power ceiling, few (if any) of our shipping parts would fit within the bounds of the older generation's platform. The higher currents typically imply hundreds of millivolt droops on the older platforms. The combination of high currents with high impedance peaks yields a hypothetical power tax in the 20-30% range (assuming one could, and actually would, ship these high current levels into the old platforms). FIVR avoids that tax.

Improved product flexibility and scalability: FIVR's ability to add voltage rails onto a common shared input rail without package growth or even platform changes brings significant flexibility and modularity into the design space that was not available before. New voltage rails can be added as needed, without any platform change. This ability allowed us to introduce the Iris™ Pro graphics into standard platforms even though new rails were needed for the EDRAM and its high speed OPIO link.

Platform size and cost reduction: Since four platform VR controllers are eliminated, along with the associated decoupling caps, power FETs and inductors, there's a clear platform size and cost advantage. FIVR's total platform BOM cost reduction is expected to be several billion dollars over the product lifetime. The power inductors feeding power to the CPU often show up in the critical thickness cross-section of small form-factor laptops and tablets, and trading FIVR's total component count reduction for a thickness reduction is straightforward. A platform phase count increase results in lower current per phase, and the lower phase current can be satisfied with a lower profile set of inductors.

In the prior generation platforms, some dual-sided PCBs have tall components like ICs and inductors on the primary side and lower profile discrete components on the secondary side. Often the secondary-side components are mostly high frequency decoupling, located immediately underneath front-side ICs, with sparsely populated areas between. In such cases, FIVR eliminates most of the secondary-side components, and frees up space on the primary side to accommodate the rest. The resulting populated PCB thickness is reduced by the height of the tallest removed backside components.

In small systems, the platform size tends to limit the feature set, leading to fewer connectivity options, smaller storage space, etc. FIVR's platform size reductions can provide more space to implement these features.

VI. CONCLUSION

Consumers expect every generation of mobile computer products to have more compute power, thinner and lighter form factors, and longer battery life than the last. The 4th generation Intel® Core™ power architecture using FIVR provides improvements in all three of these areas. To the author's knowledge, this is the first consumer product to make use of integrated switching regulators on this scale. Furthermore, FIVR's performance is improved versus previously reported prototypes. The authors therefore feel that FIVR is an important advancement in the field of power electronics.

ACKNOWLEDGMENT

We would like to acknowledge the following FIVR team members who were not already listed as authors: the FIVR silicon design team including George Geannopoulos, Keith Hodgson, Narayanan Raghuraman, Alex Lyakhov, Michael W Rogers, Ravi Vunnam, Lan D Vu, Mark S Milshtein, Chiu Keung Tang, Hong Yun Tan, Seh Leon Goh, Samie Samaan, Narayanan Natarajan, Rajan Vijayaraghavan, Ashish Khanna and Munish Chauhan; top-level integration by Pankaj Aswal; modeling and implementation development by Doug Huard and Alex Waizman; layout studies, inductor development and package designs by John Smith, Brad Larson, and Huong Do; mask design by Neal Tanksley and Galyna Burenkova; test and manufacturing support from RJ Hayes and Arun Krishnamoorthy; modeling support by Alex Levin and Anne Augustine.

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A Family of Zero Current Switching Switched-Capacitor DC-DC Converters

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Abstract— This paper presents a new zero current switching (ZCS) technique for a family of switched-capacitor dc-dc converters. Compared to the traditional ZCS switched-capacitor dc-dc converters by inserting a magnetic core in the circuit, these new ZCS switched-capacitor dc-dc converters employ the stray inductance present in the circuit as the resonant inductor and provide soft switching for the devices. These ZCS switched-capacitor dc-dc converters do not utilize any additional components to minimize switching loss and reduce the current and voltage spike, thus leading to high efficiency and reliable benefit over traditional switched-capacitor dc-dc converters. Moreover, the bulky capacitor bank existing in traditional switched-capacitor circuits for high power high current application to achieve high efficiency was reduced significantly. Small size, low capacitance, low ESR, high current rating and high temperature rating ceramic capacitors can be employed. Therefore, by using proposed ZCS technique, small size, high power density, high efficiency, high temperature rating, and high current rating switched-capacitor dc-dc converter could be built. Simulation and experimental results are given to demonstrate the validity and features of the soft switching switched-capacitor dc-dc converters.

I. INTRODUCTION

With the technological development of silicon carbide (SiC) and ceramic materials, very high temperature (250 °C) SiC switching devices and ceramic capacitors will be available. But the magnetic cores become dysfunctional at very high temperature. Therefore, magnetic-less switched-capacitor dc-dc converters operating at very high temperatures become possible and attractive with the adoption of natural air cooling by eliminating bulky heat sinks and magnetic cores. There are three main basic structures of traditional switched-capacitor dc-dc converters that is most popular, Marx generator type switched-capacitor dc-dc voltage multiplier shown in Fig. 1 [1-3], charge pump type multi-level modular switched-capacitor dc-dc converter shown in Fig. 2 [4, 5], and generalized multi-level type switched-capacitor dc-dc converter shown in Fig. 3 [6-9]. The derivation circuits of generalized multi-level type switched-capacitor dc-dc converter are also popular in automotive applications [10-14].

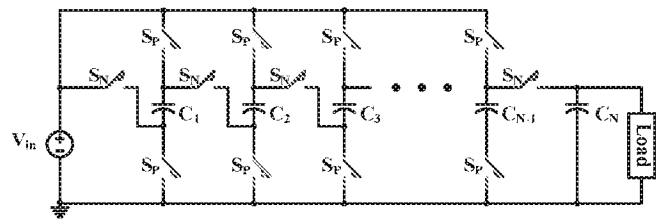


Fig. 1. Switched-capacitor dc-dc voltage multiplier.

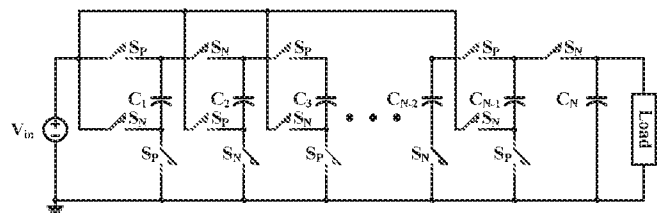


Fig. 2. Multilevel modular switched-capacitor dc-dc converter.

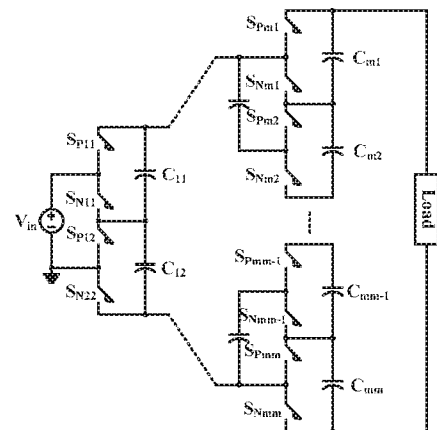


Fig. 3. Generalized multi-level switched-capacitor dc-dc converter.

However, there are many common drawbacks in traditional switched-capacitor dc-dc converters for high power and high current automotive application. A huge capacitor bank with high capacitance has to be utilized in order to

reduce the voltage ripple of the capacitors and achieve high efficiency, which will undoubtedly increase the size of the converter [7, 10, 13, 15]. Besides, with the increase of current rating, huge turn off current leads to unneglectable switching loss of the device. High voltage overshoot is also caused by large turn off current. Moreover, EMI problems caused by the di/dt and dv/dt generated during the switching transient are undesirable in automotive applications.

In order to reduce the switching loss, voltage spikes, and EMI, several ZCS methods have been proposed by inserting an inductor in series with the capacitor [16-21]. But these methods require a relatively big resonant inductor (larger than 1 μ H) which is not viable to be achieved by the circuit stray inductance as claimed. By inserting a magnetic core in the switched-capacitor circuit to achieve ZCS is a contradiction by itself. The switched-capacitor circuit with a magnetic core is not a switched-capacitor circuit anymore. Many good features of switched-capacitor will lose including good integration capability, small size and high temperature operation. Also, to utilize the parasitic inductance in the circuit, the switching frequency has to be pushed to tens of megahertz by using these methods, which is not technically feasible [22-24].

This paper presents a new ZCS strategy for a family of switched-capacitor dc-dc converters that is able to overcome the aforementioned drawbacks of traditional switched-capacitor dc-dc converters without inserting big magnetic cores or increasing switching frequency to megahertz level. This new strategy can achieve the ZCS for all the switches without losing any the good features of switched-capacitor circuit compared to the traditional ZCS strategy. This ZCS strategy employs the distributed parasitic inductance in the circuit resonating with the capacitors to provide zero current transition to the switching devices. So, the switching loss is minimized, voltage spike is reduced and EMI is limited. In case the stray inductance is not sufficiently large or equalized as expected, small air core inductors (less than 100n) or layout PCB wires can be used to meet the requirement of the stray inductance. By considering the influence of stray inductance, the power loss is only related to the conduction loss of capacitors and switching devices without considering gate drive and control loss, which is different from the traditional switched-capacitor circuits. Therefore, bulky capacitor banks with high capacitance to reduce voltage ripple and achieve high efficiency are no longer needed, while small size MLCC capacitors with low ESR and low capacitance can be adopted to increase power density and reduce the converter size. No extra components are added in the presented soft switching strategy, thus making it reliable and low in cost. A 160 W ZCS switched-capacitor dc-dc multiplier is analyzed and simulated to confirm the proposed ZCS strategy. A 12 V input 480 W ZCS voltage doubler prototype is also designed and built to confirm the operation, simulation and experiment results are provided.

II. PROPOSED ZCS SWITCHED-CAPACITOR DC-DC CONVERTER FAMILY

Fig. 4, Fig. 5, and Fig. 6 show the proposed ZCS switched-capacitor family utilizing the distributed parasitic inductance existing in the circuit. Fig. 4 shows the N-level ZCS switched-

capacitor dc-dc voltage multiplier. Fig. 5 shows the N-level ZCS multi-level modular switched-capacitor dc-dc converter (ZCS-MMSCC) which has already been discussed in detail in [25]. Fig. 6 shows the N-level ZCS generalized multi-level switched capacitor dc-dc converter. The parasitic inductance mainly includes the stray inductance due to the circuit layout, MOSFETs package parasitic inductance and the capacitor ESL. The stray inductance caused by circuit layout is usually the dominant part. By designing the circuit layout properly, the inductance needed for resonant can be achieved. In case the circuit layout is not well designed or equalized as expected, an air core inductor could be inserted to achieve the prerequisite distributed stray inductance. Because resonant inductors are distributed in the circuit, only small resonant inductance is needed for resonant. Therefore, ZCS of all the switching devices can be achieved by the resonant of the capacitor and the distributed stray inductance in the circuit.

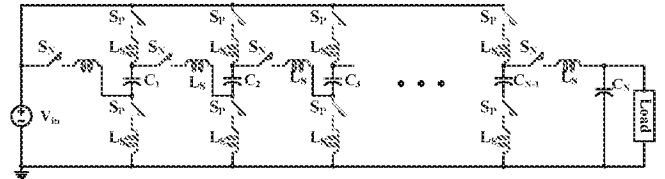


Fig. 4. ZCS switched-capacitor dc-dc voltage multiplier.

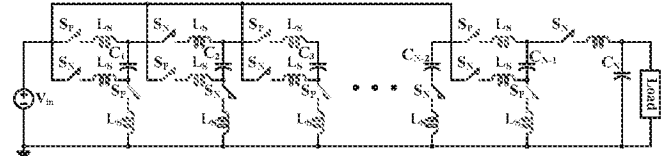


Fig. 5. ZCS multilevel modular switched-capacitor dc-dc converter.

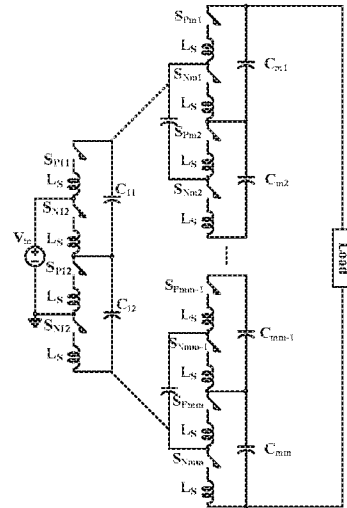


Fig. 6. ZCS generalized multi-level switched-capacitor dc-dc converter.

III. OPERATION PRINCIPLE

A. ZCS Switched-Capacitor DC-DC Voltage Multiplier

Fig. 7 shows the four-level ZCS switched-capacitor dc-dc voltage multiplier as an example. Similar to the four-level ZCS-MMSCC, this circuit also works as a four times step-up dc-dc converter. V_{in} represents the ideal input voltage source.

$L_{S1} - L_{S7}$ represents the equivalent stray inductance present in the circuit. L_{S2} , L_{S4} and L_{S6} are the equivalent stray inductance when the capacitor is charged in parallel. L_{S1} , L_{S3} , L_{S5} and L_{S7} are the equivalent stray inductance between each capacitor when the capacitor discharges in series. S_P and S_N are the same switching devices controlled complementary at 50% duty cycle. C_1 to C_3 are the capacitors with the average voltage of input voltage, while C_4 has the average voltage of four times of input voltage. L_S does not have to be in the position drawn in Fig. 7, it could be distributed anywhere in series with switches or capacitors in the circuit. Similar to the ZCS-MMSCC, the total equivalent stray inductance is the sum of the connection wire parasitic inductance, capacitor parasitic inductance and the MOSFETs package parasitic inductance. Usually the connection wire parasitic inductance is the major part of the stray inductance. For the analysis convenience, only one equivalent L_S is used to represent the total stray inductance present in each parallel resonant loop. And one equivalent L_S is used between two capacitors in the series resonant loop.

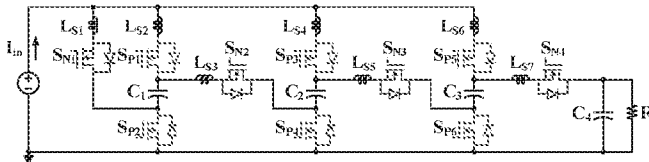


Fig. 7. Four-level ZCS switched-capacitor voltage multiplier main circuit.

Fig. 8 shows the idealized waveform of proposed ZCS switched-capacitor dc-dc voltage multiplier under steady-state conditions. V_{GS_SP} and V_{GS_SN} are the gate signal of switch S_P and S_N with 50% duty cycle. V_{DS_SP} and V_{DS_SN} are the drain source voltage of the switch S_P and S_N . Actually, the drain source voltage of different S_P and S_N are different in value but similar in shape due to the characteristics of voltage multiplier circuit. I_{SP} and I_{SN} are the drain source current of the switch S_P and S_N . All the switches of S_P or S_N have the same current waveforms. Assume input voltage is an ideal voltage source. By considering the stray inductance present in the circuit, when the switches are turned on, the current through the stray inductance, capacitors and the switch will begin to resonate from zero. By adjusting switching frequency to the resonant frequency, the current through switches S_P , S_N and stray inductance will decrease to zero when the switches are turned off. The half switching period is a half sinusoidal waveform. Therefore, the ZCS of all the switches is achieved in both turn on and turn off. The capacitor C_1 , C_2 and C_3 are charged in parallel in the half-period with the sinusoidal current waveform when S_P is on. And they are discharged in series with the input voltage source in another half-period also in the sinusoidal shape when S_N is on. $I_{C1,2,3}$ are the current through capacitor C_1 , C_2 , and C_3 , which is the sum of the current through the switch with the sinusoidal shape. When the capacitor is charged, the current is the positive part; when the capacitor discharges, the current is negative. The capacitor C_4 is charged when other capacitors are in series with input voltage when S_N is on. And it is discharged to the load when S_P is on. I_{C4} is the current through capacitor C_4 , which is a sinusoid waveform when S_N is on. When S_P is on, the current through C_4 is the negative dc load current with small ripple.

$V_{C1,2,3}$ is the voltage across the capacitor C_1 , C_2 and C_3 with the input voltage dc offset and a sinusoidal ripple. The voltage ripple is determined by the capacitor current and capacitance. V_{C4} is the voltage across the capacitor C_4 , with the dc offset four times input voltage and a voltage ripple also determined by the capacitor current and capacitance. The operation of this circuit can be described in two states shown in Fig. 9 and Fig. 10 with different switches turned on.

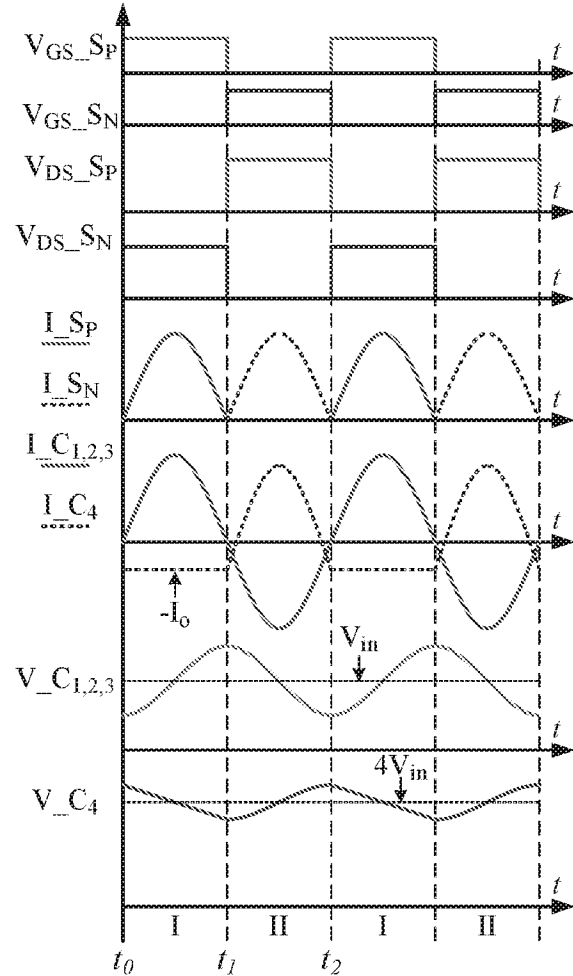


Fig. 8. Ideal waveforms of four-level ZCS voltage multiplier.

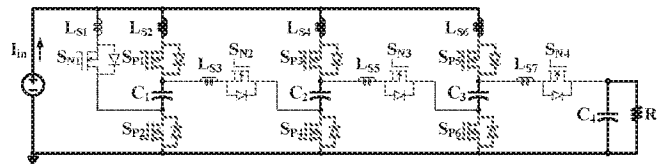


Fig. 9. Operation modes of state I when S_P is on.

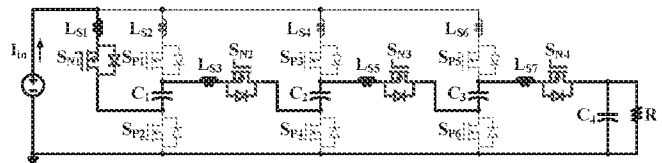


Fig. 10. Operation modes of state II when S_N is on.

1) State I [t_0, t_1]

Fig. 9 shows the state when S_P is turned on at $t = t_0$ while S_N is off. During this state C_1 , C_2 and C_3 are charged by V_{in} . Fig. 11 show the three simplified equivalent circuits of state I. Fig. 11(a) shows the situation when V_{in} , L_{S2} , S_{P1} , C_1 and S_{P2} form a resonant loop. Fig. 11(b) shows the situation when V_{in} , L_{S4} , S_{P3} , C_2 and S_{P4} form a resonant loop. Fig. 11(c) shows the situation when V_{in} , L_{S6} , S_{P5} , C_3 and S_{P6} form a resonant loop. Because of the presence of the stray inductance L_S , before the switch is turned on, the current through L_S already decreases to zero. The current through S_P will increase from zero when the switch is turned on, so S_P is turned on at zero current. For the case shown in Fig. 11(a), after L_{S2} and C_1 resonate for half cycle, the current through S_{P1} and S_{P2} falls to zero. Therefore S_{P1} and S_{P2} turn off at zero current. Similarly, for the case shown in Fig. 11(b) and Fig. 11(c) the current through S_{P3} , S_{P4} , S_{P5} and S_{P6} will realize zero current turn on and turn off too. So, ZCS is achieved on all the switches. The required stray inductance value is the same for the three circuit and is easy to be achieved by the circuit layout due to the symmetry of the circuit, assuming the capacitance are the same.

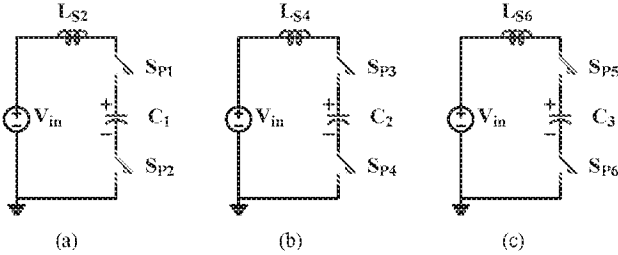


Fig. 11. Simplified equivalent circuits of state I when S_P is on.

Without loss of generality, the following assumptions have been made for the analysis: all the switches are ideal, i.e. no conduction resistance is considered; input voltage source is ideal, i.e. constant and no internal impedance; the capacitor ESR is zero. Assuming $C_1 = C_2 = C_3 = C_4$, so in order to have the same resonant frequency, $L_{S2} = L_{S4} = L_{S6}$. Assume $L_{S1} = L_{S3} = L_{S5} = L_{S7}$. The state equations of Fig. 11(a) are:

$$V_{in} = L_{S2} \frac{di_{S2}}{dt} + v_{C1} \quad (1)$$

$$i_{L_{S2}} = C_1 \frac{dv_{C1}}{dt} \quad (2)$$

The solutions are:

$$i_{S2}(t) = \frac{\pi P_o}{4V_{in}} \sin \omega_r t \quad (3)$$

$$v_{C1}(t) = V_{in} - \frac{\pi P_o}{4V_{in} C_1 \omega_r} \cos \omega_r t \quad (4)$$

Where V_{in} is the value input voltage, L_{S2} is the value of stray inductance, ω_r is the resonant frequency equals to $1/\sqrt{L_{S2} C_1}$, and P_o is the output power. After half cycle, the capacitor voltage is charged to:

$$v_{C1}(t_1) = V_{in} + \frac{\pi P_o}{4V_{in} C_1 \omega_r} \quad (5)$$

The capacitor voltage ripple is:

$$\Delta v_{C1} = \frac{\pi P_o}{4V_{in} C_1 \omega_r} \quad (6)$$

The state equations of Fig. 11(b) and Fig. 11(c) are similar to the state equations of Fig. 11(a). The voltage across C_2 and C_3 are the same as the voltage across C_1 . The voltage ripples of C_2 and C_3 are also the same as Δv_{C1} .

2) State II [t_1, t_2]

Fig. 10 shows the state when S_N is turned on at $t = t_1$ while S_P is off. During this state, C_4 is charged by the V_{in} , C_2 and C_3 in series. Assume the load is zero. Fig. 12 shows the simplified equivalent circuits of state II. It shows the situation when V_{in} , L_{S1} , S_{N1} , C_1 , L_{S3} , S_{N2} , C_2 , L_{S5} , S_{N3} , C_3 , L_{S7} , S_{N4} , and C_4 form a resonant loop. Similar as the last state, because of the presence of the L_S , the current through S_N will also increase its value from zero in a resonant manner. The zero current turn-on of S_N can be achieved. After stray inductance L_{S1} , L_{S3} , L_{S5} and L_{S7} in series resonate with capacitor C_1 , C_2 , C_3 and C_4 in series for half cycle at $t = t_2$, S_{N1} , S_{N2} , S_{N3} and S_{N4} will have a zero current turn-off when the current through them decreases to zero. Hence, zero current switching can be achieved on these switches.

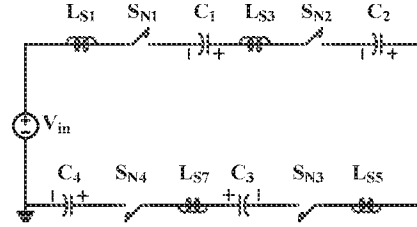


Fig. 12. Simplified equivalent circuits of state II S_N is on.

The state equations of Fig. 12 are:

$$V_{in} + v_{C1} + v_{C2} + v_{C3} = 4L_{S1} \frac{di_{L_{S1}}}{dt} + v_{C4} \quad (7)$$

$$i_{L_{S1}} = C_4 \frac{dv_{C4}}{dt} \quad (8)$$

The solutions are:

$$v_{C4}(t) = 4V_{in} - \frac{\pi P_o}{4V_{in} C_4 \omega_r} \cos \omega_r t \quad (9)$$

$$i_{L_{S1}}(t) = \frac{\pi P_o}{4V_{in}} \sin \omega_r t \quad (10)$$

After half cycle, the capacitor voltage is charged to

$$v_{C4}(t_2) = 4V_{in} + \frac{\pi P_o}{4V_{in} C_4 \omega_r} \quad (11)$$

The capacitor voltage ripple is:

$$\Delta v_{C4} = \frac{\pi P_o}{4V_{in} C_4 \omega_r} \quad (12)$$

If zero load is considered, the required stray inductance in the series resonant loop is the same as the parallel resonant loop shown in Fig. 11 which means $L_{S1} = L_{S3} = L_{S5} = L_{S7} = L_{S2} = L_{S4} = L_{S6}$. After considering the load effect with a negative dc offset in the capacitor C_4 current. The required stray inductance of series resonant loop will be a little bit smaller than the parallel resonant loop. Because the resonant frequency of a pure LC network is different with the LCR network.

B. ZCS Voltage Doubler

Fig. 13(a) shows the two-level ZCS generalized multi-level switched-capacitor dc-dc converter as an example. This circuit is called ZCS voltage doubler in short because the output voltage is twice as much as the input. V_{in} represents the ideal input voltage source. L_{S1} , L_{S2} , L_{S3} and L_{S4} can be considered as the distributed stray inductance in series with the switches in the circuit layout. S_P and S_N are the same switching devices controlled complementary at 50% duty cycle. C_1 and C_2 are the capacitors with the average voltage the same as the input voltage. Because of the symmetry of the circuit, the distributed stray inductance L_{S1} , L_{S2} , L_{S3} and L_{S4} can be replaced with an equivalent stray inductance L_S in the input side shown in Fig. 13(b) for the analysis convenience. Only one equivalent stray inductance L_S is needed to represent the total stray inductance in two resonant loop.

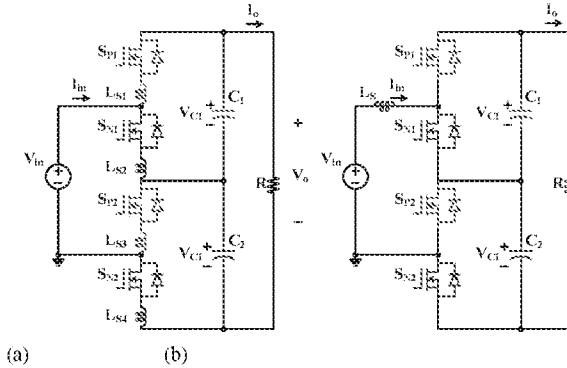


Fig. 13. ZCS voltage doubler and its topology simplification.

Fig. 14 shows the idealized waveform of proposed ZCS voltage doubler under steady-state conditions. V_{GS_SP} and V_{GS_SN} are the gate drive signal of switch S_P and S_N complementary with 50% duty cycle. V_{DS_SP1} to V_{DS_SN2} are the drain source voltage of switch S_P and S_N . I_{S_P} and I_{S_N} are the drain source current of the switch S_P and S_N . All the switches of S_P or S_N have the same current waveforms. Assume input voltage is an ideal voltage source. By considering the equivalent stray inductance in the input side, when the switches are turned on, the current through the stray inductance, capacitors and the switch will begin to resonate from zero. By adjusting switching frequency to the resonant frequency, the current through switches S_P , S_N and stray inductance will decrease to zero when the switches are turned off, which is the half period of the sinusoidal waveform. Therefore, the ZCS of all the switches is achieved in both turn

on and turn off. I_{C1} and I_{C2} are the current through capacitor C_1 and C_2 . The capacitor C_1 and C_2 are charged in each half-period with the sinusoidal current waveform when S_P or S_N is on. And they are always discharged in series with load current. So, the current through the capacitor C_1 , and C_2 , is the current through the switch subtracts the output current. V_{C1} and V_{C2} are the voltage across the capacitor C_1 and C_2 . V_o is the output voltage with the voltage ripple half of the voltage ripple of the capacitor. Because of the 180 degree phase shift of the capacitor voltage ripple, the capacitor voltage ripple will cancel out with each other. The operation of the circuit can be described in two states as shown in Fig. 15(a) and Fig. 15(b) with different switches turned on.

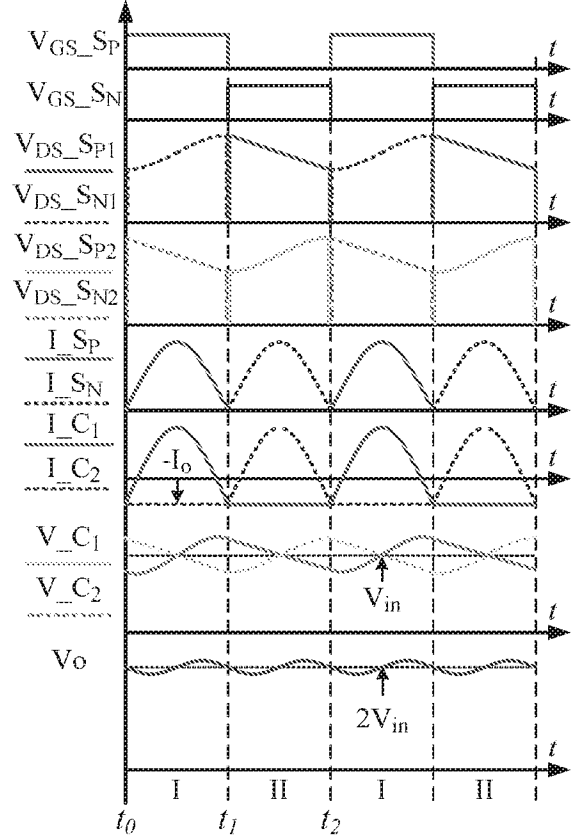


Fig. 14. Ideal waveforms of ZCS voltage doubler.

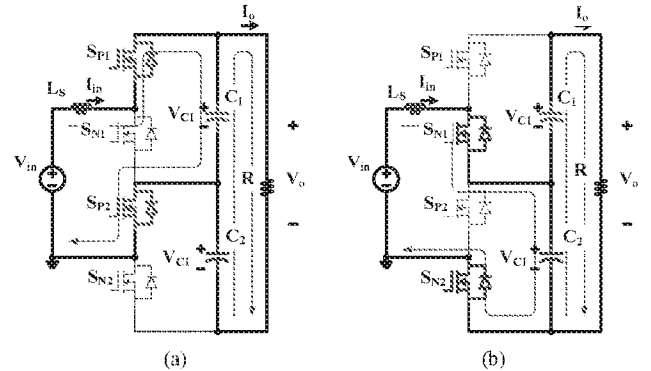


Fig. 15. (a) Operation modes of state I when S_P is on. (b) Operation modes of state II when S_N is on.

IV. DESIGN GUIDELINES

The 480 W ZCS voltage doubler design procedure is shown here as an example. The specification of the prototype converter is $V_{in}=12$ V, $V_o=24$ V, $P_o=480$ W, $f_s=44.25$ kHz.

A. Capacitance

Capacitance value should be chosen according to the voltage ripple of the capacitor using the following equation.

$$C = \frac{\pi P_o}{2V_{in}\Delta v_C \omega_r} \quad (13)$$

Because of the special structure of the circuit, the voltage ripple across all the capacitors are the same. And the value of the voltage ripple is determined by the output power, input voltage, capacitance and the resonant frequency. The capacitor voltage ripple should be chosen smaller than the input voltage to prevent the voltage across the capacitor from resonating to the negative region and lose the zero current switching. The output voltage ripple is about five times smaller than the capacitor voltage ripple because of the 180 degrees phase shift operation of the switches. And the capacitor voltage ripple will cancel out and the output voltage is two times bigger than one capacitor voltage. The output ripple is chosen around 8% of output voltage. So the capacitor voltage ripple is 40% of the capacitor voltage. According to the (13), the capacitance is chosen 47 μ F in the simulation.

B. Stray inductance

The circuit layout should be designed carefully, in order to make the stray inductance of each equivalent circuit equal. Assume all the stray inductance in the circuit is equalized. And the required stray inductance can be satisfied by using proper input inductance. Required stray inductance for ZCS can be determined by the resonant LCR network.

C. Switching frequency

Switching frequency should be set the same as resonant frequency in order to promise the zero current switching. Usually, the stray inductance of the circuit is already determined when the circuit layout is finished. The capacitance is also determined by the required voltage ripple. Assume the air core inductor is used for the input inductance, and the switching frequency can be determined by the input inductance value. If the input inductance is determined, the switching frequency can be twisted by monitoring the power device current when the switch is turned off. The switching frequency should be set exactly at the point when the switch current resonates to zero when the switch is turned off.

V. SIMULATION AND EXPERIMENT RESULTS

A. ZCS Voltage Multiplier

Fig. 16 shows saber® simulation waveforms of a 160 W four-level ZCS voltage multiplier, where V_p and V_n are the switch gate-source control voltage, V_{ds_Sp} and V_{ds_Sn} are the switch drain-source voltage, I_{Sp} and I_{Sn} are the switch drain-source current, $I_{C1,2,3}$ is the current through capacitor C_1 , C_2 and C_3 . I_{C4} is the current through capacitor C_4 . $V_{C1,2,3}$ is the voltage across the capacitor C_1 , C_2 and C_3 . V_{in} is the input voltage. V_o is the output voltage or the

voltage across the capacitor C_4 . The input voltage is 5 V. Switching frequency is about 70 kHz. Capacitance is 47 μ F and the value of the stray inductance $L_{S2} = L_{S4} = L_{S6} = 108$ nH, $L_{S1} = L_{S3} = L_{S5} = L_{S7} = 98$ nH. The series resonant stray inductance is smaller caused by the load effect. The simulation results is consistent with the theoretical analysis, which verifies the above analysis.

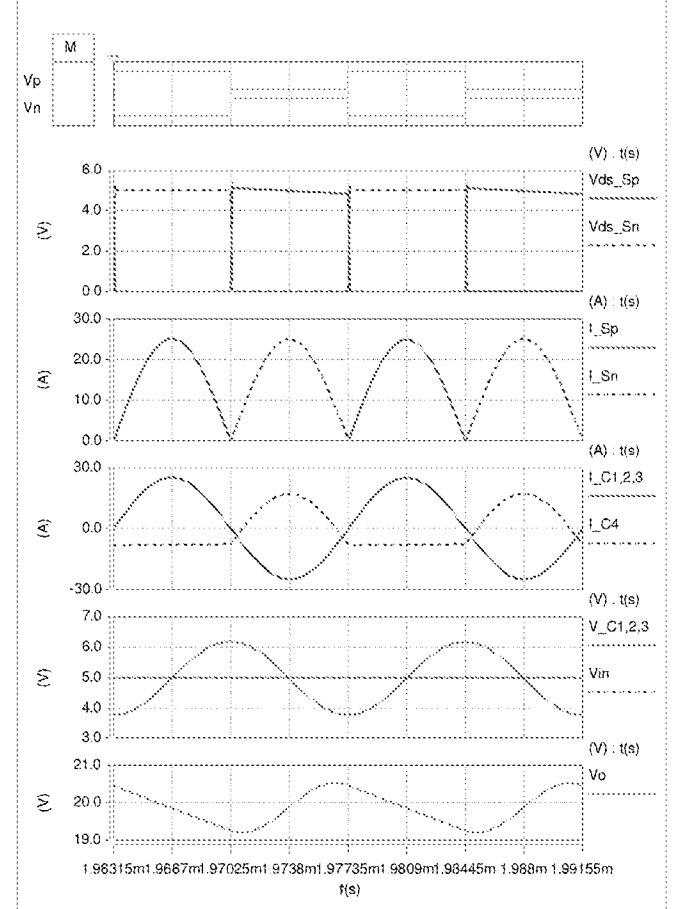


Fig. 16. Simulation results of 160 W ZCS voltage multiplier.

B. ZCS Voltage Doubler

Fig. 17 and Fig. 18 shows saber® simulation waveforms of a 480 W ZCS voltage doubler, where S_p and S_n are the switch gate-source control voltage, V_{ds_Sp1} , V_{ds_Sn1} , V_{ds_Sp2} , V_{ds_Sn2} are the switch drain-source voltage of switch S_{P1} , S_{N1} , S_{P2} and S_{N2} . I_{Sp1} and I_{Sn1} are the switch drain-source current of switches S_{P1} and S_{N1} . which is omitted here. The current through S_{P2} is the same as S_{P1} , the current through S_{N2} is the same as S_{N1} . I_{in} is the current through the input stray inductance L_S . I_{C1} and I_{C2} are the current through capacitor C_1 and C_2 . V_{C1} and V_{C2} are the voltage across the capacitor C_1 and C_2 . V_{in} is the input voltage. V_o is the output voltage. The input voltage is 12 V. Switching frequency is about 44 kHz. Capacitance is 47 μ F and stray inductance $L_S = 165$ nH. The simulation results is consistent with the theoretical analysis, which verifies the analysis.

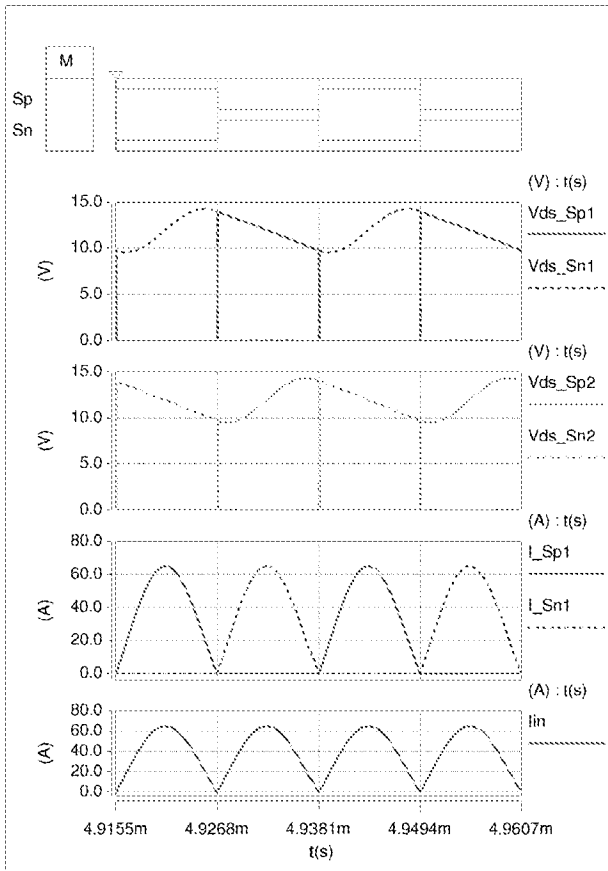


Fig. 17. Simulation results of 480 W ZCS voltage doubler.

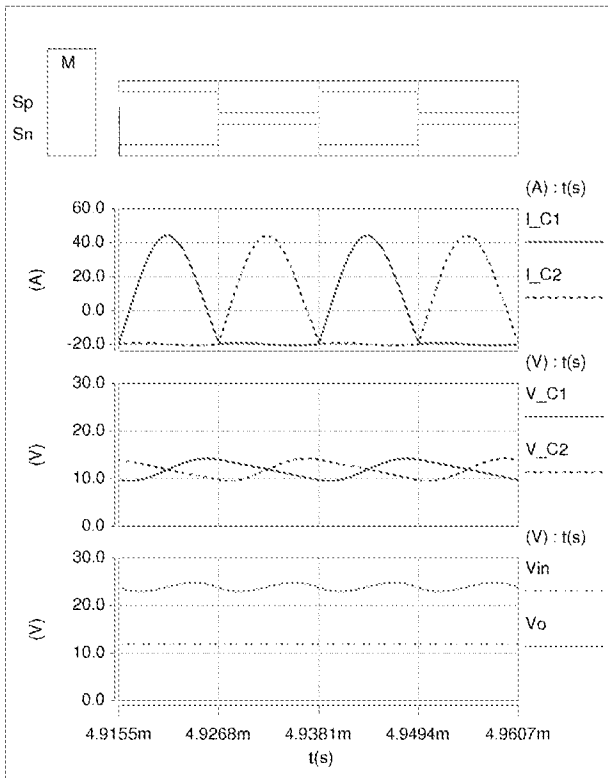


Fig. 18. Simulation results of 480 W ZCS voltage doubler.

In the 480 W ZCS voltage doubler prototype, the switching devices are two 30 V 180 A MOSFETs IPB009N03L from infineon connected in parallel. Resonant capacitor are ten 100 V 4.7 μ F MLCC capacitors C5750X7R2A475K from TDK connected in parallel. Input capacitor are twelve 16 V 470 μ F conductive polymer aluminum solid electrolytic capacitors PLG1C471MDO1 from nichicon connected in parallel. Gate drive voltage is 7 V. The switching frequency is about 40 kHz. The input stray inductance is around 200 nH.

Fig. 19 and Fig. 20 show the experiment waveforms of the 480 W ZCS voltage doubler prototype with the parts mentioned above. Fig. 19 shows the complementary gate drive signals of S_P and S_N with duty cycle about 49% due to some dead time. V_{GS_SP} and V_{GS_SN} are the complementary gate-source control voltage.

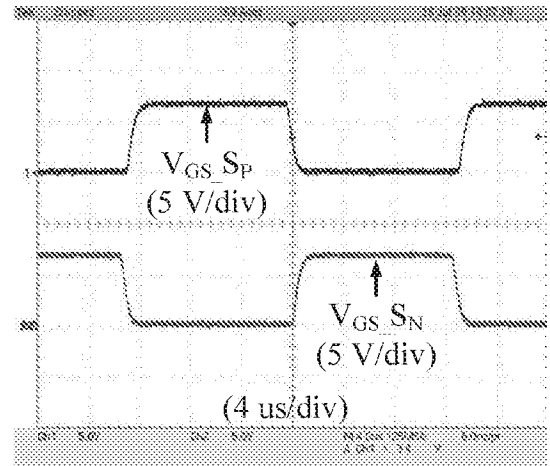


Fig. 19. Gate drive signal of ZCS voltage doubler.

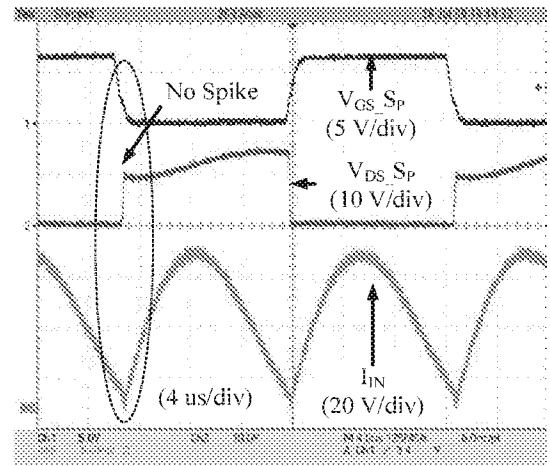


Fig. 20. Switch voltage and input current waveform of ZCS voltage doubler.

VI. CONCLUSION

In this paper, a family of zero current switching switched-capacitor dc-dc converters is proposed. By eliminating the bulky, lossy inductive component with a magnetic core, the

converter is able to operate at very high temperatures with high efficiency. By using the proposed soft-switching strategy, switching loss has been minimized and EMI has been restricted; the size of capacitors has been reduced, thus making the converter small and light. Owing to the utilization of distributed stray inductance or distributed air core inserting technique, the big inductor is avoided to achieve ZCS. Hence, the proposed ZCS switched-capacitor converter family shows great potential in high temperature, high power future automotive applications.

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Zero-Current-Switching Multilevel Modular Switched-Capacitor DC–DC Converter

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Abstract—This paper presents a quasi-resonant technique for multilevel modular switched-capacitor dc–dc converter (MMSCC) to achieve zero-current switching (ZCS) without increasing cost and sacrificing reliability. This ZCS-MMSCC employs stray inductance distributed in the circuit as the resonant inductor to resonate with the capacitor and provide low dv/dt and di/dt switching transition for the device. The ZCS-MMSCC does not utilize any additional components to achieve ZCS and meanwhile solves the current and voltage spike problem during the switching transition, thus leading to reliable and high-efficiency advantages over traditional MMSCC. Furthermore, the ZCS-MMSCC reduces the capacitance needed in the circuit to attain high efficiency. In this case, the bulky capacitor bank with high capacitance in a traditional MMSCC to reduce voltage difference and achieve high efficiency is no longer necessary. A 150-W four-level ZCS-MMSCC prototype has been built. Simulation and experimental results are given to demonstrate the validity and features of the proposed soft-switching switched-capacitor circuit.

Index Terms—Multilevel, switched capacitor, zero-current switching (ZCS).

I. INTRODUCTION

IN ORDER to obtain lightweight, small-sized, high-efficiency, high-power-density, and integrated dc–dc converters, switched-capacitor circuits have been investigated since 1970s [1], [2]. Conventional switched-capacitor dc–dc converters have some common drawbacks: 1) weak output regulation ability and structurally determinate voltage conversion ratio; 2) pulsating input current and high current spike; 3) high voltage spike across the switching device; 4) high electromagnetic interference (EMI); and 5) unidirectional power conversion ability. Many approaches have been proposed to achieve voltage regulation by using duty cycle control [2]–[4]. However, all these methods have to sacrifice efficiency to achieve the voltage regulation, which is only acceptable in low-power-conversion field. Several zero-current-switching (ZCS) techniques by inserting an inductor in the switched-capacitor circuit have been proposed to solve the voltage spike, current spike, and EMI problem [5]–[8]. However, adding a relatively big resonant

inductor into the switched-capacitor circuit to achieve ZCS is a contradiction by itself. A switched-capacitor circuit with a magnetic core is not a switched-capacitor circuit any more, and the good features such as high-temperature operation, good integration, and small size will be lost [9], [10].

Another type of switched-capacitor circuit, the flying capacitor multilevel dc–dc converter (FCMDC), has become more and more popular recently because of the potential automotive application [11]–[15]. However, there are several limitations of FCMDC, such as nonmodular structure, complex control strategy, and high-conversion-ratio unreachability. When the conversion ratio is larger than five or more than five levels have to be used, the control of FCMDC is becoming very complex. The high current stress of the switching device of FCMDC also prevents it from a high-current application.

In order to solve the problems of FCMDC in a high-voltage-conversion-ratio high-current automotive application, a multilevel modular switched-capacitor dc–dc converter (MMSCC) was proposed recently, as shown in Fig. 1 [16]. MMSCC has a modular structure, a complementary control scheme, capability of reaching high voltage gain, and reduced switch current stress. It shows good potential in automotive application [17]. Nevertheless, because MMSCC still uses hard switching, there still exist several problems limiting its power rating, such as high current spike, high voltage spike, high switching loss, and severe EMI noise. With the increasing of power rating or current rating of the MMSCC, the voltage spike and switching loss problems become more and more serious. In addition, in order to increase the total system efficiency by reducing the voltage difference between capacitors, higher capacitance electrolytic capacitors have to be used [13], [16]–[18]. The current rating of the electrolytic capacitor is small to some extent. In order to meet the high-current-ripple requirement, a huge electrolytic capacitor bank has to be used, which will increase the size of the converter significantly. Multilayer chip-type ceramic (MLCC) capacitor has much better performance than the electrolytic capacitor, such as equivalent series resistance (ESR), life time, and size in terms of current rating. However, the capacitance of MLCC capacitor is very small, so it is not feasible to use MLCC capacitor for the traditional MMSCC.

This paper presents a quasi-resonant technique for MMSCC that is able to achieve ZCS without increasing cost and sacrificing reliability. The proposed ZCS strategy employs distributed stray inductances present in each module of the circuit resonating with the capacitors to provide zero current turn-on and turn-off to the devices. Because MMSCC is particularly suitable for high-voltage-gain with high-input-current applications [16]–[21], the ZCS version also inherits all the benefits

Manuscript received November 13, 2009; accepted March 7, 2010. Date of publication September 7, 2010; date of current version November 19, 2010. Paper 2009-IPCC-331, presented at the 2009 IEEE Energy Conversion Congress and Exposition, San Jose, CA, September 20–24, and approved for publication in the IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS by the Industrial Power Converter Committee of the IEEE Industry Applications Society.

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Digital Object Identifier 10.1109/TIA.2010.2073432

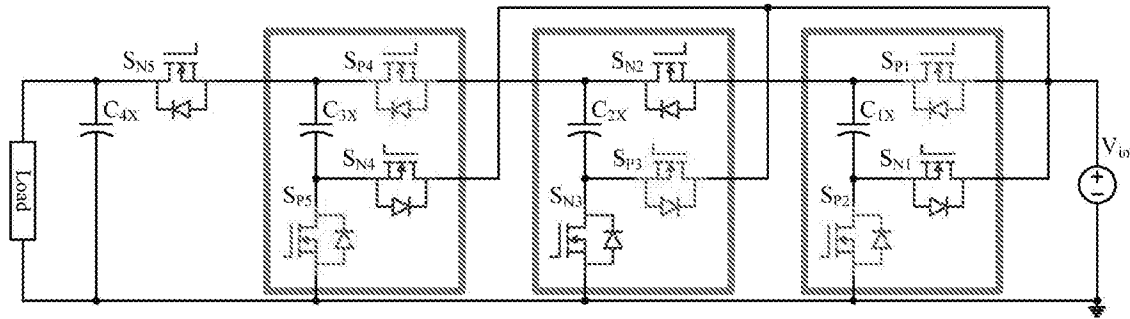


Fig. 1. Four-level MMSCC with three modular blocks.

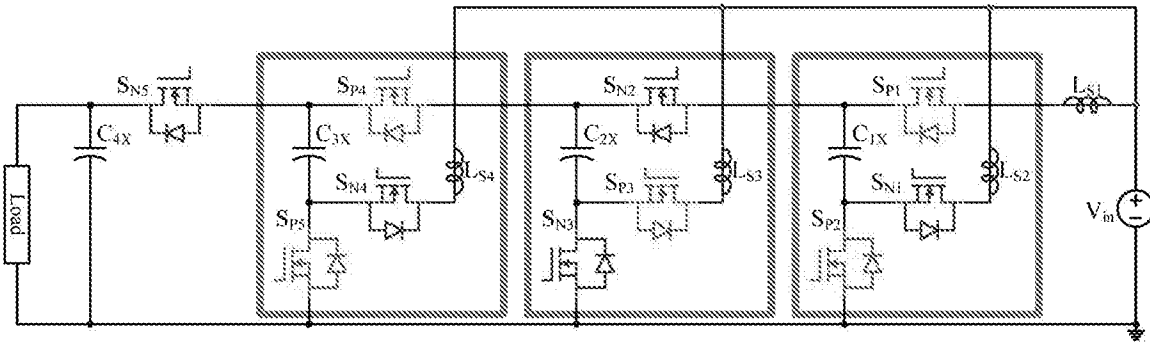


Fig. 2. Four-level ZCS-MMSCC with three modular blocks.

from it. By using the ZCS, the switching loss of the traditional MMSCC is minimized, and the current spike, voltage spike, and EMI are reduced. Aside from the benefits gained from ZCS and MMSCC, the proposed ZCS-MMSCC has other special features.

- 1) Different from other soft-switching techniques, this ZCS strategy does not use additional active devices or passive components, thus making it reliable and cost efficient.
- 2) Because of the module structure, the required resonant inductance is quite small for each module, thus making it feasible to be realized by the distributed stray inductance existing in the circuits.
- 3) The ZCS-MMSCC does not need high capacitance to achieve high efficiency like traditional MMSCC. In this case, a small-capacitance, high-current-rating, low-ESR, and small-size MLCC capacitor could be used. The total size of the converter is reduced significantly, and the power density and the converter current rating are increased correspondingly.

Hence, the proposed ZCS-MMSCC solves the aforementioned problems of traditional MMSCC and achieves ZCS without inserting big magnetic cores. For the same current rating and voltage gain, the ZCS version of MMSCC is smaller and has higher power density and higher efficiency than the traditional MMSCC. Therefore, the proposed ZCS-MMSCC is particularly suitable to the high-current high-voltage-gain automotive application. A 150-W four-level ZCS-MMSCC prototype was built to confirm the operation. Simulation and experimental results are given to demonstrate the validity and features of the proposed ZCS-MMSCC.

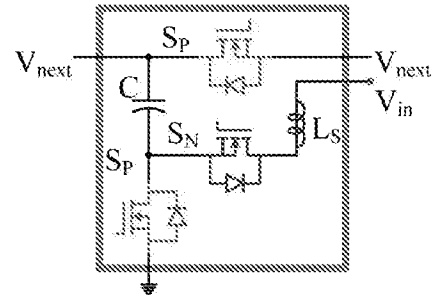


Fig. 3. Proposed ZCS-MMSCC modular block.

II. PROPOSED CIRCUIT TOPOLOGY

Fig. 2 shows the proposed four-level ZCS-MMSCC as an example. The circuit works as a four times step-up converter. Fig. 3 shows the proposed ZCS-MMSCC converter modular block. By using $n - 1$ modular blocks, a switch, and a capacitor, n -level ZCS-MMSCC with n times voltage gain can be generated. V_{in} represents the ideal input voltage source. L_S represents the equivalent stray inductance present in each module of the circuit. S_P and S_N are the same switching devices controlled complementary at 50% duty cycle. C_{1X} to C_{4X} are the capacitors with the average voltage from one time to four times of input voltage. L_S does not have to be in the position shown in Fig. 2; it could be distributed anywhere in series with switches or capacitors in the circuit shown in Fig. 4. L_S is the sum of the connection wire parasitic inductance L_{SW} , capacitor parasitic inductance L_{ESL} , and the MOSFETs package parasitic inductance L_{SP} . Usually, the connection wire parasitic inductance L_{SW} is the major part of the stray inductance, because all the stray inductances are in series when the current flows through. In each simplified equivalent circuit

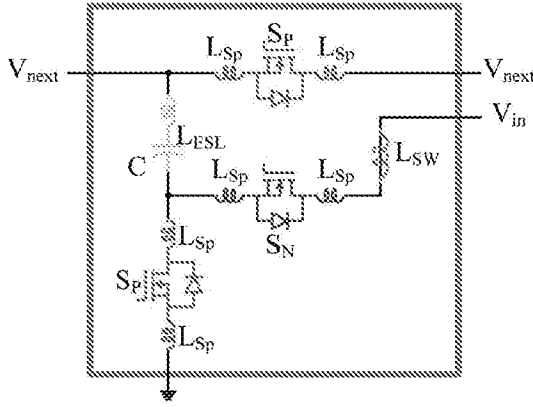


Fig. 4. ZCS-MMSCC modular block with all the stray inductances.

discussed in the next section, only one equivalent L_S is needed to represent the total stray inductance present in the circuit.

Due to the modular structure of the circuit, the stray inductance is distributed equally in each module. The required stray inductance for resonant is more than 100 times smaller (2–13.5 nH) than other ZCS strategies using the same switching frequency [6]–[8], [22], [23]. In case the stray inductance is not sufficiently large or equally distributed in each module, a small air core could be utilized to promise that each module will have the same stray inductance for resonant.

III. OPERATION PRINCIPLE

Fig. 5 shows the idealized waveform of the proposed ZCS-MMSCC under steady-state conditions. The gate signal of switches S_P and S_N is complementary, and duty cycle is 50%. Assume that the input voltage is an ideal voltage source. By considering the stray inductance present in the circuit, when the switch is turned on, the current through the stray inductance, capacitors, and the switch will begin to resonate from zero. By adjusting the switching frequency to resonant frequency, the current through switches S_P and S_N and the stray inductance will decrease to zero when the switch is turned off, which is the half period of the sinusoidal waveform. Therefore, the ZCS of the switch is achieved at both turn-on and turn-off. The capacitor is charged in half period with the sinusoidal current waveform and discharged in another half period also in the sinusoidal shape. Therefore, the current through the capacitor is the sum of the current through the switch, which is a sinusoidal waveform. When the capacitor is charged, the current is positive; when the capacitor discharges, the current is negative. The voltage across the capacitor has a dc offset with a sinusoidal ripple. The dc voltage offset is determined by the capacitor position in the circuit from one time input voltage to n times input voltage. The voltage ripple is determined by the capacitor current and capacitance.

The operation of the circuit can be described in two states, as shown in Figs. 6 and 7, with different switch turned on.

A. State I [t_0, t_1]

Fig. 6 shows the state when S_P is turned on at $t = t_0$ while S_N is off. During this state, C_{1X} is charged by V_{in} . C_{3X} is

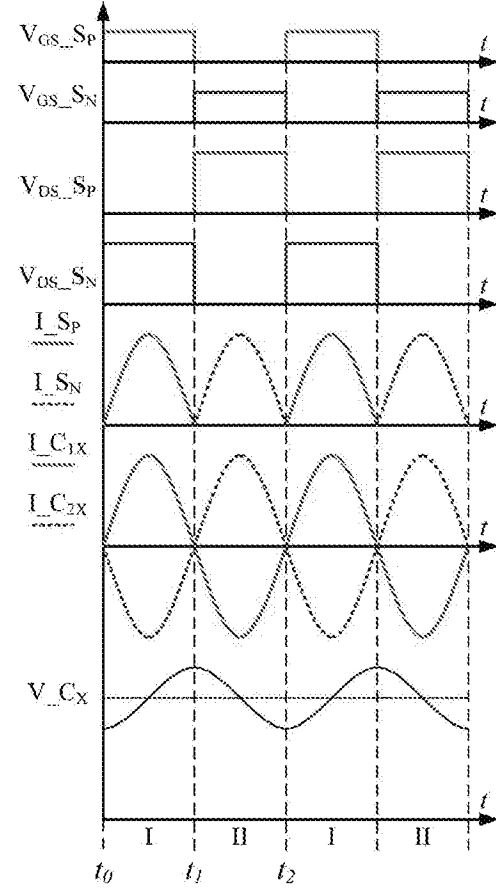


Fig. 5. Typical waveforms of the proposed ZCS-MMSCC.

charged by V_{in} and C_{2X} in series. Figs. 8 and 9 show the two simplified equivalent circuits of state I. Fig. 8 shows the situation when V_{in} , L_{S1} , S_{P1} , C_{1X} , and S_{P2} form a resonant loop. Fig. 9 shows the situation when V_{in} , L_{S3} , S_{P3} , C_{2X} , S_{P4} , C_{3X} , and S_{P5} form a resonant loop. Because of the presence of the equivalent stray inductance L_S , before the switch is turned on, the current through L_S already decreases to zero. The current through S_P will increase from zero when the switch is turned on, so S_P is turned on at zero current. For the case shown in Fig. 8, after L_{S1} and C_{1X} resonate for half cycle, the current through S_{P1} and S_{P2} falls to zero. Therefore, S_{P1} and S_{P2} should be turned off at this point so that the switches are turned off at zero current. For the case shown in Fig. 9, after L_{S3} resonates with C_{2X} and C_{3X} in series for half cycle, the current through S_{P3} , S_{P4} , and S_{P5} will decrease to zero, too. Therefore, zero-current turn-off is also realized on these switches. For the situation shown in Fig. 8, there is only one capacitor in the equivalent circuit other than two capacitors in series in other equivalent circuits. The required stray inductance value is two times smaller than that of the other circuits assuming the capacitances are the same. This will not be a problem in the real circuit design and layout; one only needs to pay attention to make sure that the stray inductance of all the other loops is twice as large as that of the first resonant loop.

Without loss of generality, the following assumptions have been made for the analysis: All the switches are ideal, i.e.,

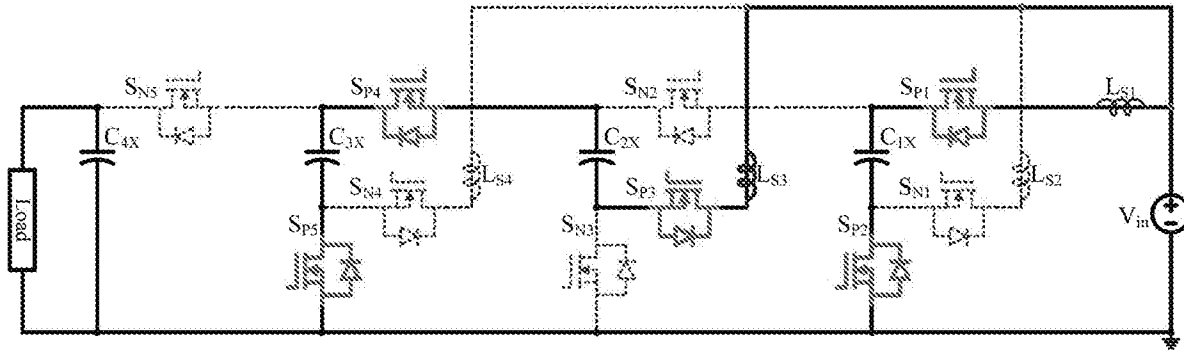
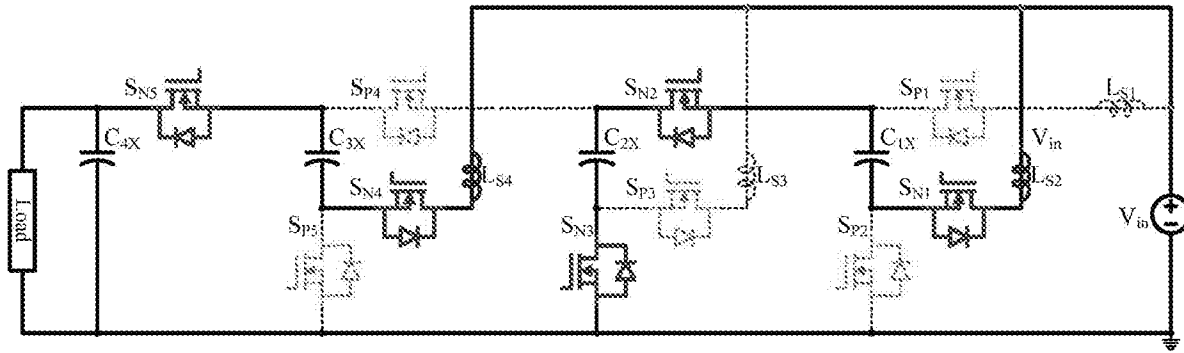
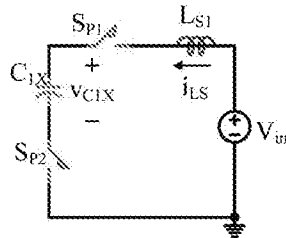
Fig. 6. Operation modes: state I- S_P is on.Fig. 7. Operation modes: state II- S_N is on.

Fig. 8. Simplified equivalent circuits of state I (a).

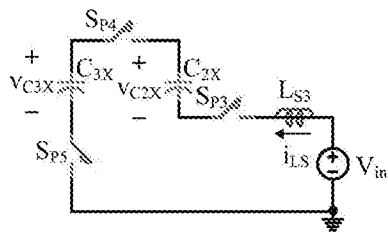


Fig. 9. Simplified equivalent circuits of state I (b).

no conduction resistance is considered; input voltage source is ideal, i.e., constant and no internal impedance; and the capacitor ESR is zero. Assume that $C_{1X} = C_{2X} = C_{3X} = C_{4X}$, so in order to have the same resonant frequency, $L_{S2} = L_{S3} = L_{S4} = 2L_{S1}$. The state equations of the simplified equivalent circuit shown in Fig. 8 are

$$V_{in} = L_{S1} \frac{di_{L_{S1}}}{dt} + v_{C_{1X}} \quad (1)$$

$$i_{L_{S1}} = C_{1X} \frac{dv_{C_{1X}}}{dt} \quad (2)$$

The solutions are

$$i_{L_{S1}}(t) = \frac{\pi P_o}{4V_{in}} \sin \omega_r t \quad (3)$$

$$v_{C_{1X}}(t) = V_{in} - \frac{\pi P_o}{4V_{in} C_{1X} \omega_r} \cos \omega_r t \quad (4)$$

where V_{in} is the value input voltage, L_{S1} is the value of stray inductance, ω_r is the resonant frequency equal to $1/\sqrt{L_{S1} C_{1X}}$, and P_o is the output power. After half cycle, the capacitor voltage is charged to

$$v_{C_{1X}}(t_1) = V_{in} + \frac{\pi P_o}{4V_{in} C_{1X} \omega_r} \quad (5)$$

The capacitor C_{1X} voltage ripple is

$$\Delta v_{C_{1X}} = \frac{\pi P_o}{4V_{in} C_{1X} \omega_r} \quad (6)$$

The state equations of the simplified equivalent circuit shown in Fig. 9 are

$$V_{in} = L_{S3} \frac{di_{L_{S3}}}{dt} + v_{C_{3X}} - v_{C_{2X}} \quad (7)$$

$$i_{L_{S3}} = C_{3X} \frac{dv_{C_{3X}}}{dt} \quad (8)$$

$$i_{L_{S3}} = -C_{2X} \frac{dv_{C_{2X}}}{dt} \quad (9)$$

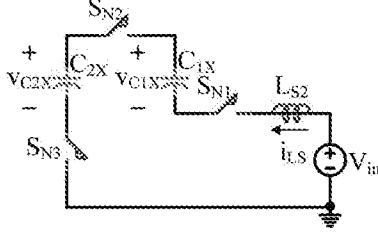


Fig. 10. Simplified equivalent circuits of state II (a).

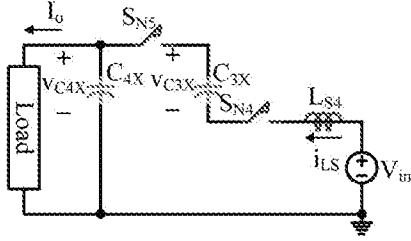


Fig. 11. Simplified equivalent circuits of state II (b).

Plug (17) into (7) and (9) to solve the following equations:

$$i_{LS3}(t) = \frac{\pi P_o}{4V_{in}} \sin \omega_r t \quad (10)$$

$$v_{C3X}(t) = 3V_{in} - \frac{\pi P_o}{4V_{in}C_{1X}\omega_r} \cos \omega_r t. \quad (11)$$

The capacitor C_{3X} voltage ripple is

$$\Delta v_{C3X} = \frac{\pi P_o}{4V_{in}C_{1X}\omega_r}. \quad (12)$$

B. State II [t_1, t_2]

Fig. 7 shows the state when S_N is turned on at $t = t_1$ while S_P is off. During this state, C_{2X} is charged by V_{in} and C_{1X} in series; C_{4X} is charged by V_{in} and C_{3X} in series. Figs. 10 and 11 show the simplified equivalent circuits of state II. Fig. 10 shows the situation when V_{in} , L_{S2} , S_{N1} , C_{1X} , S_{N2} , C_{2X} , and S_{N3} form a resonant loop. Fig. 11 shows the situation when V_{in} , L_{S4} , S_{N4} , C_{3X} , S_{N5} , and C_{4X} form a resonant loop. Because of the presence of L_S , the current through S_N will also increase its value from zero in a resonant manner. The zero-current turn-on of S_N is achieved. For the case shown in Fig. 10, after L_{S2} , C_{1X} , and C_{2X} resonate for half cycle at $t = t_2$, S_{N1} , S_{N2} , and S_{N3} will have a zero-current turn-off when the current through them decreases to zero. For the case shown in Fig. 11, after L_{S4} resonates with C_{3X} and C_{4X} in series for a half cycle, the current through S_{N3} , S_{N4} , and S_{N5} will also decrease to zero. Hence, zero-current turn-off is also realized on these switches.

The state equations of the simplified equivalent circuit shown in Fig. 10 are

$$V_{in} = L_{S2} \frac{di_{LS2}}{dt} + v_{C2X} - v_{C1X} \quad (13)$$

$$i_{LS2} = C_{2X} \frac{dv_{C2X}}{dt} \quad (14)$$

$$i_{LS2} = -C_{1X} \frac{dv_{C1X}}{dt}. \quad (15)$$

Plug (4) into (13) and (15) to solve the following equations:

$$i_{LS2}(t) = -\frac{\pi P_o}{4V_{in}} \sin \omega_r t \quad (16)$$

$$v_{C2X}(t) = 2V_{in} + \frac{\pi P_o}{4V_{in}C_{1X}\omega_r} \cos \omega_r t. \quad (17)$$

The capacitor C_{2X} voltage ripple is

$$\Delta v_{C2X} = \frac{\pi P_o}{4V_{in}C_{1X}\omega_r}. \quad (18)$$

The state equations of the simplified equivalent circuit shown in Fig. 11 are

$$V_{in} = L_{S4} \frac{di_{LS4}}{dt} + v_{C4X} - v_{C3X} \quad (19)$$

$$i_{LS4} = C_{4X} \frac{dv_{C4X}}{dt} + I_o \quad (20)$$

$$i_{LS4} = -C_{3X} \frac{dv_{C3X}}{dt} \quad (21)$$

where I_o is the output current. Since this converter is a four times boost circuit, $V_o = 4V_{in}$, so $I_{in} = 4I_o$. Plug (11) into (19) and (21) to solve the following equations:

$$i_{LS4}(t) = -\frac{\pi P_o}{4V_{in}} \sin \omega_r t \quad (22)$$

$$v_{C4X}(t) = 4V_{in} + \frac{\pi P_o}{4V_{in}C_{1X}\omega_r} \cos \omega_r t. \quad (23)$$

The capacitor C_{4X} voltage ripple is

$$\Delta v_{C4X} = \frac{\pi P_o}{4V_{in}C_{1X}\omega_r}. \quad (24)$$

The soft-switching condition in both states is $f_s = f_r$, where f_s is the switching frequency and f_r is the resonant frequency.

IV. DESIGN GUIDELINES

The 150-W four-level ZCS-MMSCC design procedure is shown here as an example. The specifications of the prototype converter are as follows: $V_{in} = 5$ V, $V_o = 20$ V, $P_o = 150$ W, and $f_s = 45$ kHz.

A. Capacitance

Capacitance value should be chosen according to the voltage ripple

$$C = \frac{\pi P_o}{4V_{in}\Delta v_C \omega_r}. \quad (25)$$

Because of the special structure of the circuit, the voltage ripple across all the capacitors is the same, and the value of the voltage ripple is determined by the input current, capacitance, and the resonant frequency. The capacitor voltage ripple should be chosen to be smaller than the input voltage to prevent the voltage across the first capacitor from resonating to the negative region and losing the ZCS. That means, the first capacitor should have a positive dc voltage offset. In fact, because of the special structure of the converter, the output voltage is n times

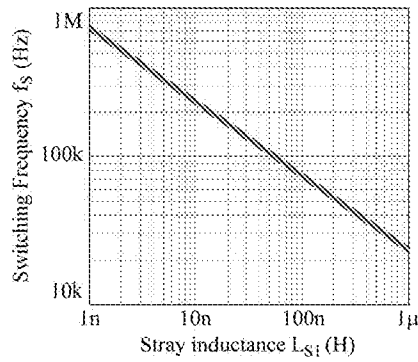


Fig. 12. ZCS stray inductance and switching frequency.

of the input voltage; even when the voltage ripple is as large as the input voltage, the voltage ripple on the last capacitor is still $1/n$. When the conversion ratio is large, the ripple will be small enough in the output side, and high capacitance is not needed to reduce the output voltage ripple.

The output ripple is chosen to be around 9% of the output voltage. According to (25), the capacitance is chosen to be $47 \mu\text{F}$ in the prototype. Another parameter to determine the capacitance value is the current rating of the capacitor; if the current rating of the capacitor is too small to meet the current requirements, then more capacitors in parallel have to be used. In this prototype, high-current-rating MLCC capacitors are used. Ten 100-V $4.7\text{-}\mu\text{F}$ C5750X7R2A475K capacitors from TDK are used in parallel. The current rating of each capacitor is around 3 A each.

B. Stray Inductance

The circuit layout should be designed carefully, in order to make the stray inductance of each equivalent circuit equal. In case the stray inductance is not equal or large enough, a small air core can be inserted to equalize the stray inductance of each module. The switching frequency should be adjusted according to the stray inductance present in the circuit if the circuit layout is already fixed. Required stray inductance for ZCS can be determined by the following equation, using the equivalent circuit shown in Fig. 8:

$$L_{S1} = \frac{1}{C_{1X} (2\pi f_s)^2}. \quad (26)$$

When the resonant capacitance is $47 \mu\text{F}$, Fig. 12 shows the curve of switching frequency versus stray inductance.

Table I shows the relevant required stray inductance according to the switching frequency, when the capacitance is $47 \mu\text{F}$. When the switching frequency is larger than 200 kHz, the required stray inductance is already smaller than the regular circuit layout stray inductance.

Obviously, for other equivalent circuits shown in Figs. 9–11, the stray inductance values of L_{S2} , L_{S3} , and L_{S4} for resonant need to be doubled. The capacitance value becomes half when two capacitors are connected in series in these equivalent circuits. The switching frequency can be easily determined by the actual stray inductance in the circuit. Around 100 and 200 kHz, the most often used switching frequencies in dc-dc converters,

TABLE I
REQUIRED STRAY INDUCTANCE ACCORDING TO SWITCHING FREQUENCY

Switching Frequency f_s	Stray inductance L_{S1}
500kHz	2.15 nH
400 kHz	3.37 nH
300 kHz	6 nH
200 kHz	13.5 nH
100 kHz	54 nH
50 kHz	215 nH

the needed stray inductance for resonating is less than 55 nH. This value is small enough to be realized by the stray inductance in the circuit layout or by a small air core. The inductance of a 10-cm-long wire or air core inductor is already about 50 nH. Only about 2-nH stray inductances are needed when the switching frequency is increased to 500 kHz. However, the stray inductance inside a normal MOSFET package such as TO-220 is already about 1–2 nH; when several MOSFETs are connected in series, this value could be higher considering the stray inductance of the MOSFET leg which means that the switching frequency does not need to be higher than 500 kHz to achieve ZCS.

C. Switching Frequency

Switching frequency should be set the same as the resonant frequency in order to promise the ZCS. Usually, the stray inductance of the circuit is already determined when the circuit layout is finished. The capacitance is also determined by the required voltage ripple. Assume that the stray inductance of each module is equalized. The switching frequency can be twisted by monitoring the power device current when the switch is turned off. The switching frequency should be set exactly at the point when the switch current resonates to zero when the switch is turned off.

V. POWER LOSS ANALYSIS

A. Power Device Loss

Conduction Loss: Assume that the input power is P_{in} , the power device turn-on resistance is R_{on} , the rms value of switch current is I_{S_rms} , the average value of input current is I_{in_ave} , the conversion ratio of the converter is N_T , and the conduction loss of power device is P_{S_con} . Thus, the power device conduction loss equation can be derived as follows:

$$I_{in_ave} = \frac{P_{in}}{V_{in}} \quad (27)$$

$$I_{S_rms} = \frac{I_{in_ave}}{N_T} \frac{\pi}{2} \quad (28)$$

$$P_{S_con} = I_{S_rms}^2 R_{on}. \quad (29)$$

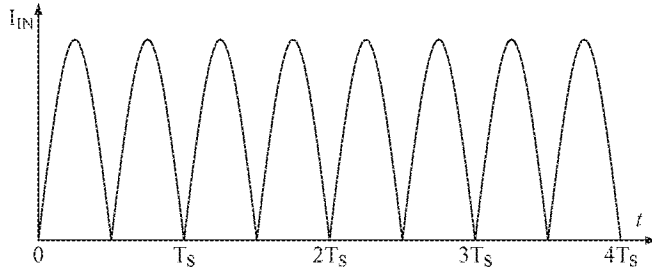


Fig. 13. Ideal input current waveform.

Plugging (27) and (28) into (29), the following conduction loss can be derived:

$$P_{S_con} = \frac{\pi^2 P_{in}^2}{4N_T^2 V_{in}^2} R_{on}. \quad (30)$$

Gate Drive Loss: Assume that the gate drive voltage is V_{GS} , the gate total charge is Q_g , and the gate drive loss of power device is P_{S_drv} . Thus, the power device gate drive loss can be derived using the following equation:

$$P_{S_drv} = V_{GS} Q_g f_s. \quad (31)$$

B. Capacitor Conduction Loss

Resonant Capacitor Conduction Loss: Assume that the capacitor rms current is I_{C_rms} , the capacitor ESR is R_{ESR} , and the capacitor conduction loss is P_{C_con} . Thus, the capacitor conduction loss can be derived using the following equation:

$$I_{C_rms} = \frac{\pi I_{in_ave}}{\sqrt{2} N_T} \quad (32)$$

$$P_{C_con} = I_{C_rms}^2 R_{ESR}. \quad (33)$$

Plugging (32) into (33), one can obtain

$$P_{C_con} = \frac{\pi^2 I_{in_ave}^2}{2N_T^2} R_{ESR}. \quad (34)$$

Plugging (27) into (34), one can obtain

$$P_{C_con} = \frac{\pi^2 P_{in}^2}{2N_T^2 V_{in}^2} R_{ESR}. \quad (35)$$

Input Capacitor Conduction Loss: Assume that the rms value of input current is I_{in_rms} , the capacitor ESR is R_{ESR_in} , and the conduction loss of input capacitor is $P_{C_in_con}$. The input current is shown in Fig. 13.

$$I_{in_rms} = \frac{\pi I_{in_ave}}{2\sqrt{2}} \quad (36)$$

$$P_{C_in_con} = I_{in_rms}^2 R_{ESR_in}. \quad (37)$$

Plugging (36) into (37), one can obtain

$$P_{C_in_con} = \frac{\pi^2 I_{in_ave}^2}{8} R_{ESR_in}. \quad (38)$$

Plugging (27) into (38), one can obtain

$$P_{C_in_con} = \frac{\pi^2 P_{in}^2}{8V_{in}^2} R_{ESR_in}. \quad (39)$$

C. Total Power Loss

There are $3N_T - 2$ switching devices and N_T capacitors in a N_T -level ZCS-MMSCC, so the total power loss P_{loss} is

$$P_{loss} = (3N_T - 2)(P_{S_con} + P_{S_drv}) + N_T P_{C_con} + P_{C_in_con}. \quad (40)$$

Plugging (30), (31), and (34) into (40), one can obtain

$$P_{loss} = (3N_T - 2) \left(\frac{\pi^2 P_{in}^2 R_{on}}{4N_T^2 V_{in}^2} + V_{GS} Q_g f_s \right) + \frac{\pi^2 P_{in}^2 R_{ESR}}{2N_T V_{in}^2} + \frac{\pi^2 P_{in}^2 R_{ESR_in}}{8V_{in}^2}. \quad (41)$$

Assume that η is the converter efficiency

$$\eta = \frac{P_{in} - P_{loss}}{P_{in}}. \quad (42)$$

Plugging (41) into (42), one can obtain (43), shown at the bottom of the page.

VI. SIMULATION AND EXPERIMENT RESULTS

A. 150-W Four-Level ZCS-MMSCC Prototype

Fig. 14 shows Saber simulation waveforms of a 150-W four-level ZCS-MMSCC, where V_{gs_Sp} and V_{gs_Sn} are the switch gate–source control voltages, V_{ds_Sp1} and V_{ds_Sn2} are the switch drain–source voltages, and I_{d_Sp1} and I_{d_Sn2} are the switch drain–source currents. The input voltage is 5 V. The switching frequency is about 45 kHz. Capacitance is 47 μ F, and stray inductance L_{S1} is about 250 and 500 nH for others. The simulation results are consistent with the theoretical analysis, which verifies the aforementioned analysis.

In the 150-W four-level ZCS-MMSCC prototype, the switching devices are two 30-V 180-A MOSFETs (IPB009N03L, from Infineon) connected in parallel. Resonant capacitors are ten 100-V 4.7- μ F MLCC capacitors (C5750X7R2A475K, from TDK) connected in parallel. Input capacitors are 14 6.3-V 470- μ F conductive polymer aluminum solid electrolytic capacitors (PLE0J471MDO1, from Nichicon) connected in parallel. Gate drive voltage is 15 V.

$$\eta = \frac{P_{in} - (3N_T - 2) \left(\frac{\pi^2 P_{in}^2 R_{on}}{4N_T^2 V_{in}^2} + V_{GS} Q_g f_s \right) - \frac{\pi^2 P_{in}^2 R_{ESR}}{2N_T V_{in}^2} - \frac{\pi^2 P_{in}^2 R_{ESR_in}}{8V_{in}^2}}{P_{in}} \quad (43)$$

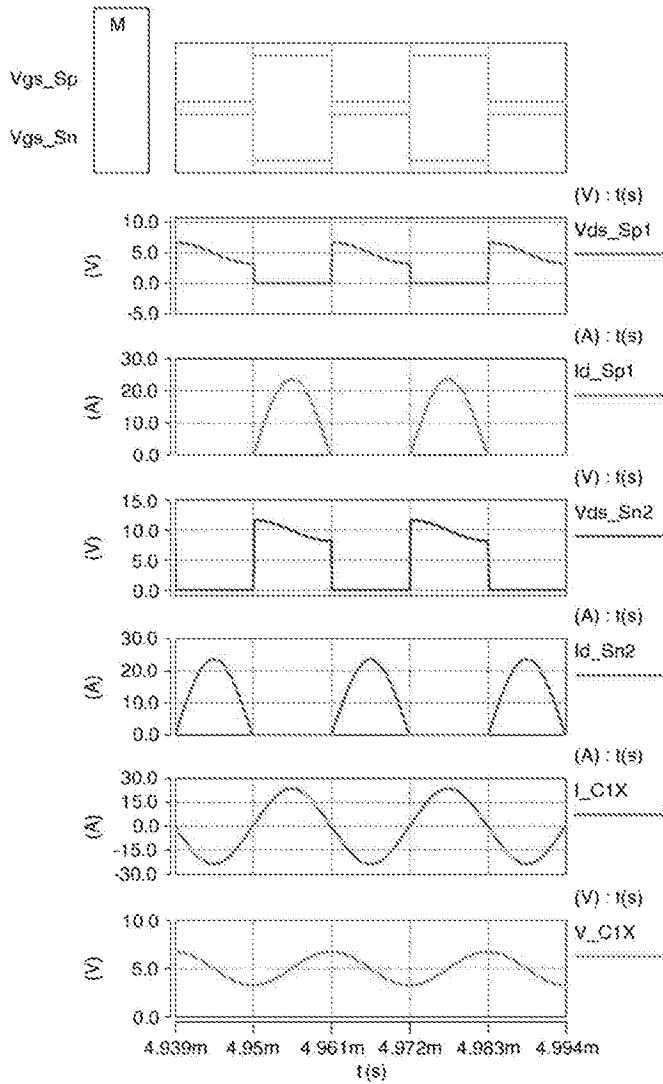
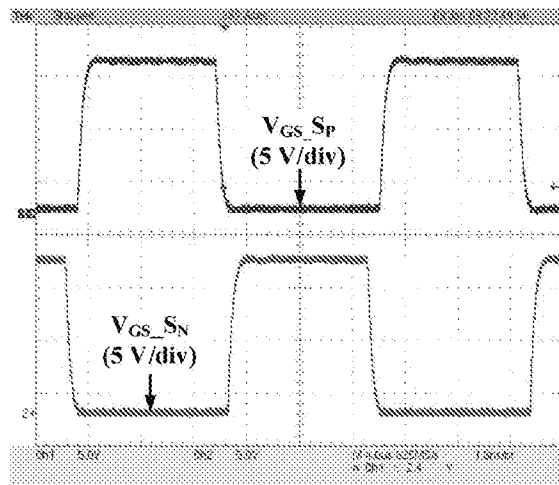


Fig. 14. Simulation waveforms of a 150-W four-level ZCS-MMSCC.

Fig. 15. Gate drive signals of S_P and S_N .

Figs. 15–18 show the experiment waveforms of the 150-W four-level ZCS-MMSCC prototype with the parts mentioned earlier. Fig. 15 shows the complementary gate drive signals of

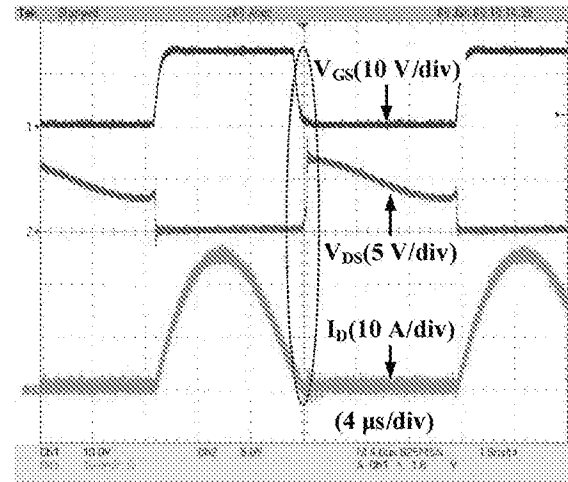
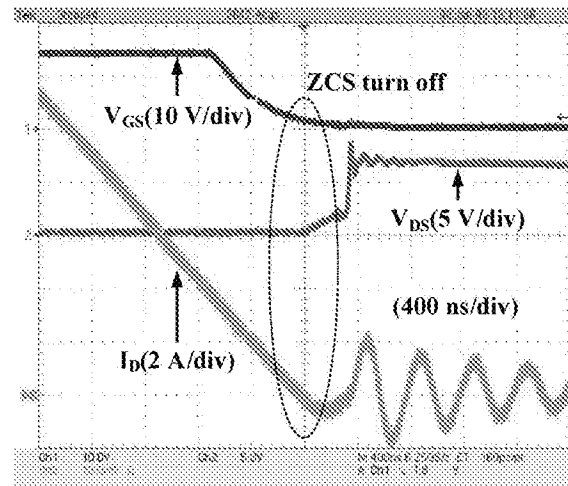
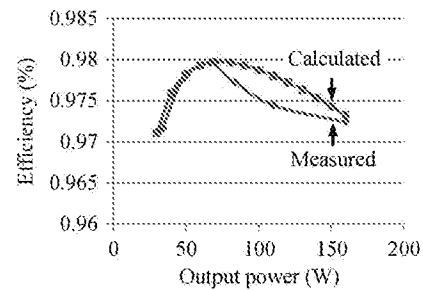
Fig. 16. Voltage and current waveforms of S_{P1} .Fig. 17. Zoomed-in voltage and current waveforms of S_{P1} .

Fig. 18. Measured efficiency and calculated efficiency.

S_P and S_N with duty cycle of about 48% due to some dead time. V_{GS_SP} and V_{GS_SN} are the complementary gate–source control voltages.

Fig. 16 shows the voltage and current waveforms of S_{P1} . V_{GS} is the gate–source voltage of S_{P1} , V_{DS} is the drain–source voltage of S_{P1} , and I_D is the drain–source current of S_{P1} . The resonant current waveform realizing ZCS is shown. Fig. 17 shows the zoomed-in waveforms of the switch S_{P1} turn-off transient, which indicates that the switch achieves ZCS during turn-off. From the experiment results, we can derive that the switch S_{P1} can achieve ZCS in both turn-on and turn-off. Other

switches have the voltage and current waveforms similar to that of S_{P1} which will not be shown here.

Fig. 18 shows the measured efficiency curve using Yokogawa WT1600 digital power meter of the proposed ZCS-MMSCC and the calculated efficiency using (43). The efficiency is above 97%, which is a big improvement than traditional MMSCC.

VII. CONCLUSION

The presented ZCS-MMSCC has all the advantages of ZCS and MMSCC, and it has some special features besides the features got from ZCS and MMSCC. All the features can be concluded as follows.

- 1) It reduces dv/dt , di/dt , and EMI because of ZCS.
- 2) It reduces the switching loss which improves efficiency.
- 3) It uses simple control strategy and possesses modular structure and bidirectional power conversion.
- 4) It increases the potential power rating.
- 5) It reduces the capacitance and capacitor size and lowers the cost.
- 6) It utilizes distributed stray inductance to achieve ZCS and improves reliability.

A new soft-switching operation strategy to the existing MMSCC has been proposed and validated by both simulation and experiment results. A 150-W 5-V input four-level converter prototype for automotive application has been built and tested based on the proposed circuit, and the efficiency of the prototype is higher than 97%. The ZCS-MMSCC overcomes the current and voltage spike problems of traditional MMSCC and provides a technically viable soft-switching strategy with circuit layout stray inductance. This technique could also be applied to the switched-capacitor circuits on a semiconductor chip with very small stray inductance in the circuit without pushing the switching frequency to megahertz. The proposed ZCS-MMSCC also shows great potential to high-voltage-gain and high-input-current applications.

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A 400 Amp fully Integrated Silicon Voltage Regulator with in-die magnetically coupled embedded inductors

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Thenus, Xiaobei Li Ph.D., Stephen W.
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- Intel Corporation

For APEC 2010 February 25th in Palm Springs CA



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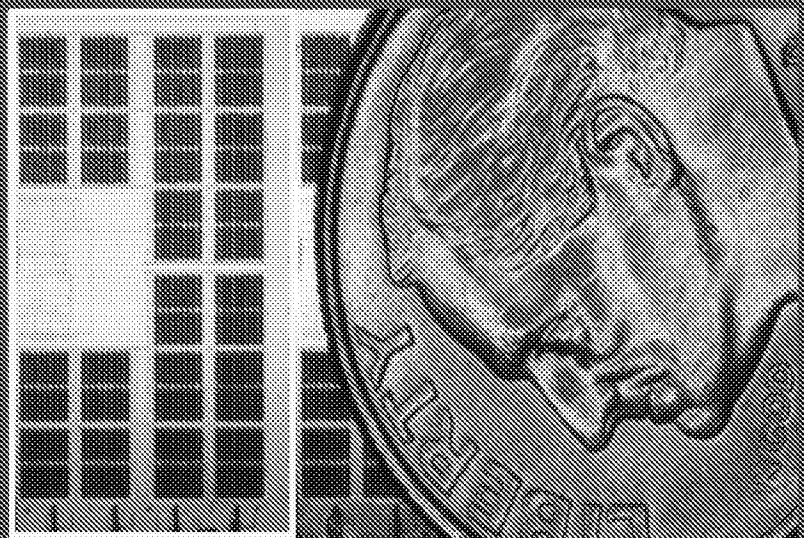
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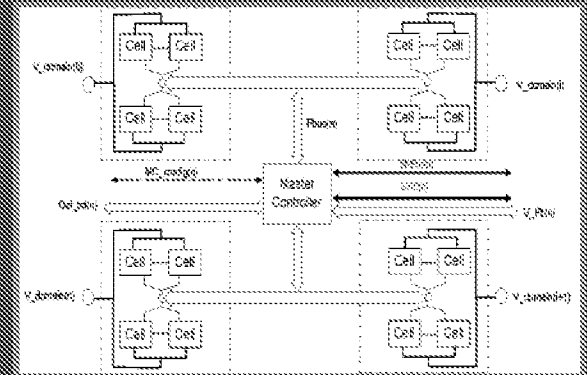
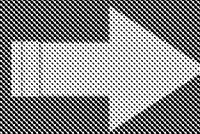
Agenda

- Architecture
- Circuits
- Results
- Conclusion

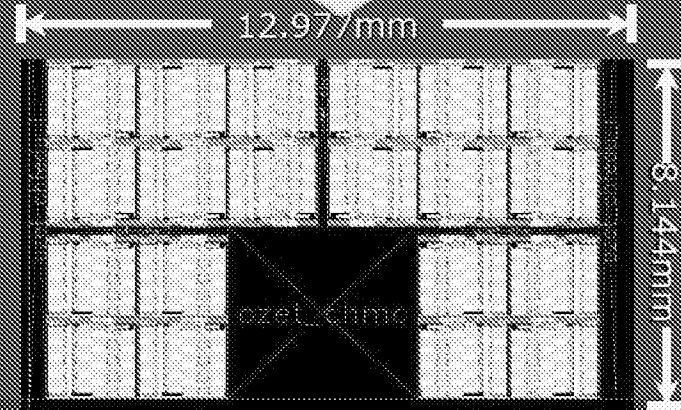


Integrated VR Technology

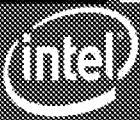
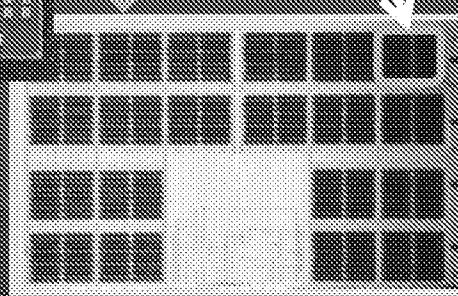
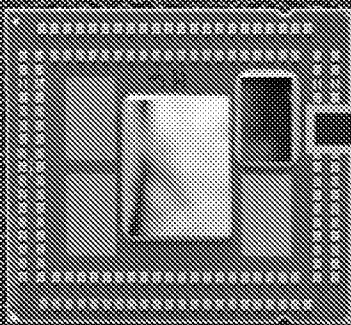
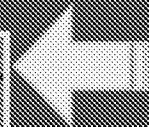
- 'Common Cell' Architecture – 20 cells
- Architecture supports flat efficiency curve
- Fine grain power management
 - Allows for multiple voltage rails
- Telemetry and Margining features
- Active Voltage Positioning for current sharing and balance
- Control features, including: JTAG, FPGA, Test/BIST



General Arch

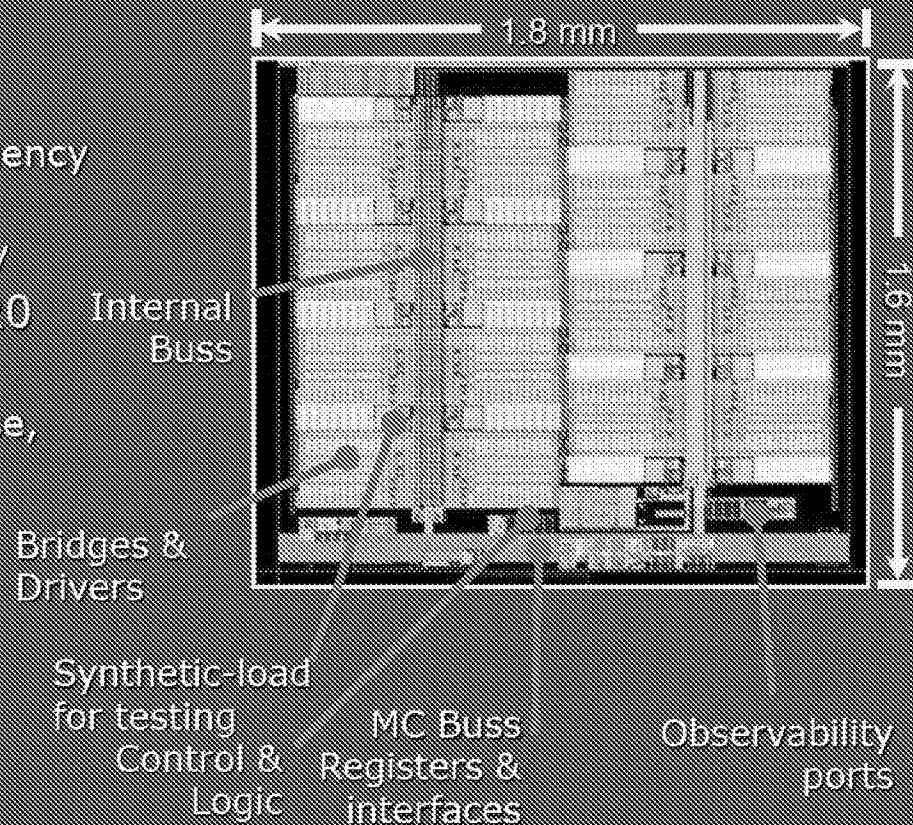


*Power cell
- 2.8 mm²*



Review: Power Cell Architecture

- Each Power cell = Mini VR
 - Up to 25A rating* - tested
 - Programmable switching frequency 30MHz to 140MHz
 - Ring coupled inductor topology
- 16 phases per power cell, 320 phases per chip
 - High phase count reduces noise, ripple
 - High granularity
 - Cell shedding
 - Bridge shedding
- BIST
 - Self-load and characterization system.

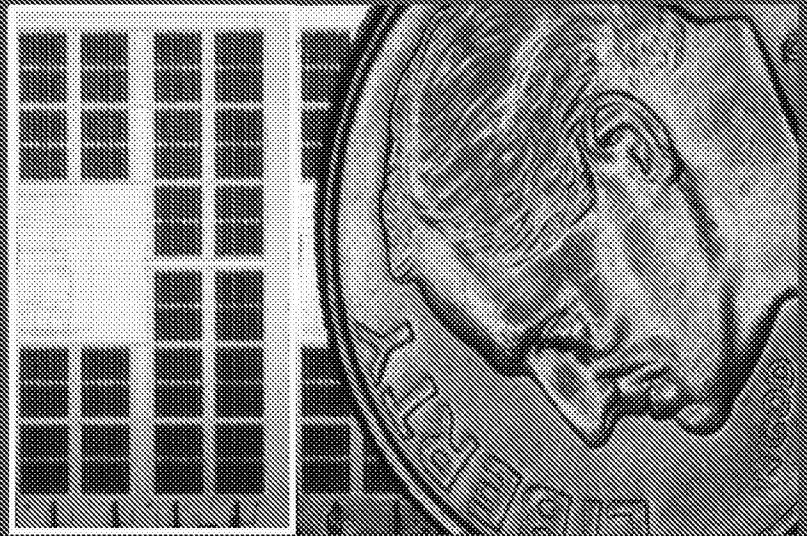


* Thermally constrained

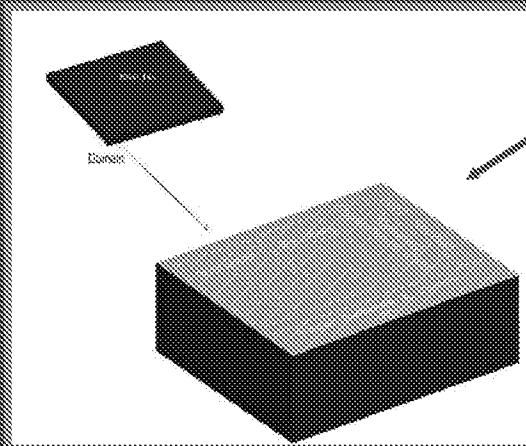


Agenda

- Architecture
- Circuits/Magnetics
- Results
- Conclusion



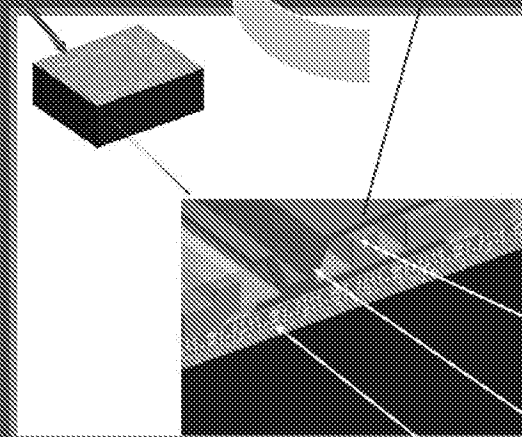
Thin-Film On-die Magnetics



Domain



Magnetic Cross section



L metal

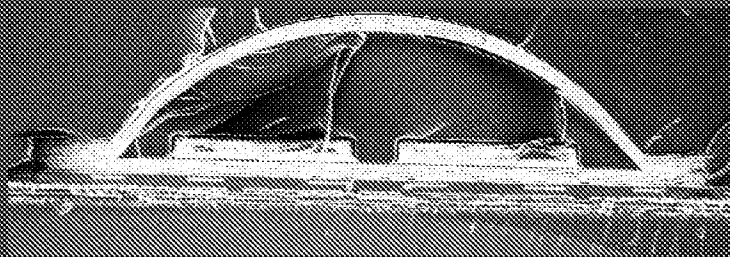
$\text{Ni}_{80}\text{Fe}_{20}$

m1-mX

- Technology targets & Stackup
- 90 nm technology for test devices
- 7-8 metal layers + thick metal(C4) + 2 Magnetic layers
- M1-m4/5 routing – bridge connections – decoupling capacitors
- M8 – inductor/transformer interconnect – $\sim 10 \mu\text{m}$ thick metal
- M7,M9 –magnetic material layers– $\sim 4 \mu\text{m}$ thick (laminated) $\text{Ni}_{80}\text{Fe}_{20}$



Gains from on-die magnetics



Magnetic SEM Cross section

- Energy density increased
- Volume shrinks
- Power Loss decreased



$$W_M = \frac{1}{2} \iiint_V \vec{B} \cdot \vec{H} dv \cong \frac{B^2 v}{2\mu_r \mu_0}$$

$$\frac{W_M}{W_A} \cong \mu_r$$

$$W_a \cong W_m \Rightarrow v_m \mu_r \cong v_a$$

Energy density in thin film magnetics volume compared with air core inductor is proportional to permeability μ_r which is typically > 1000

$$\frac{R_a}{R_m} = \frac{P_a}{P_m} \approx \frac{l_m \sqrt{\mu_r}}{l_a}$$

Thin-Film Magnetics in relation to VR Ckts

- $L_{\text{self}} \sim 17\text{nH}$ per phase. - $K \sim 93\%$
- 16 phase - 25A/cell I_{max}
- 2.8 mm^2 per cell

$$P_{L_c} \cong \Delta I^2 R_c = \frac{\Delta I^2 l_c}{\sigma w_c t_c}$$

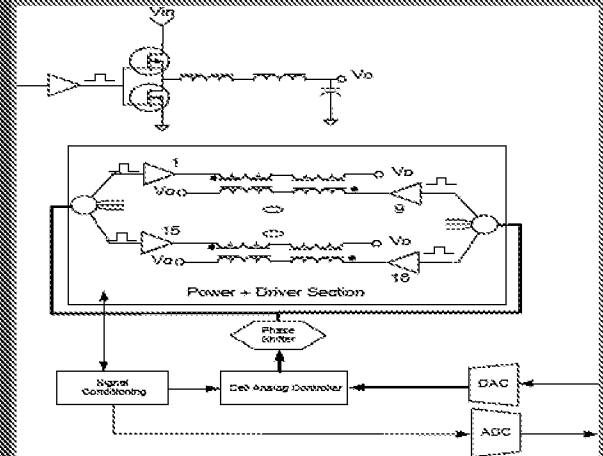
$$L \cong \frac{\Delta V \Delta T}{\Delta I} = \frac{\Delta V (1/F) D}{\Delta I}$$

$$* L_{\text{die}} \cong \frac{\Delta V (1/F) D}{\Delta I} \cong \frac{B_{\text{Sat}} w t_m}{\Delta I}$$

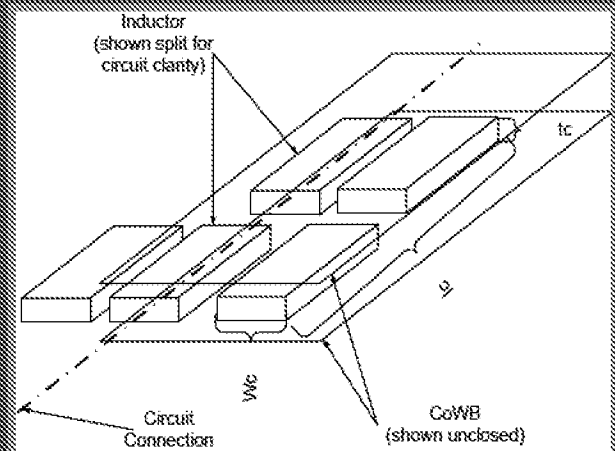
Single Magnetic Top View



Cell Circuit



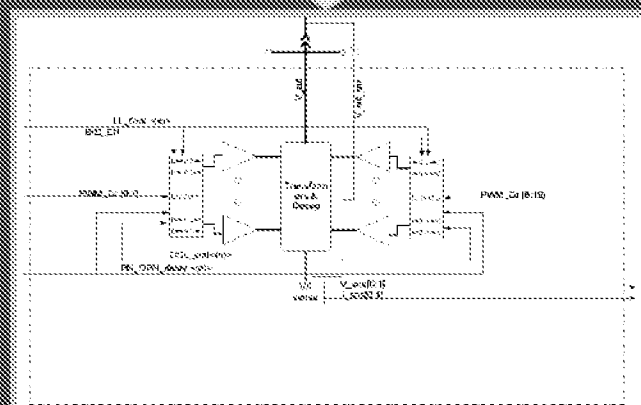
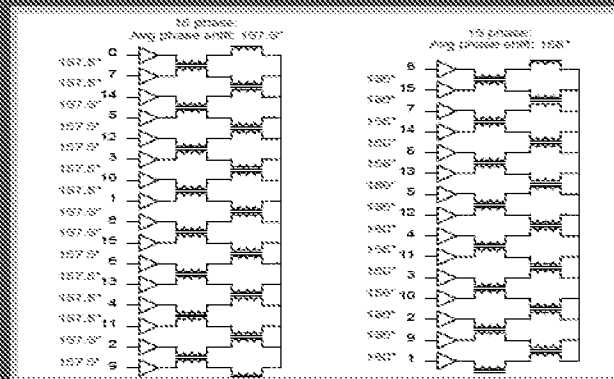
Inductor Physical



Power Train Architecture

- Cell level power train & Local Controller
- 16 phase 60-140 Mhz (per phase) coupled inductor
- Controller – type I analog
- Current Sense
- Flat efficiency with bridge shedding
- Loop programmable
- Register control between master controller and local cell controller
- Monitor and Observability thru pass-gate port design

16 phase VR



Power Cell Power Train Block

